Why global regularity for Navier-Stokes is hard

Terence Tao · 18 March 2007

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It is always dangerous to venture an opinion as to why a problem is hard (cf. Clarke's first law), but I'm going to stick my neck out on this one, because (a) it seems that there has been a lot of effort expended on this problem recently, sometimes perhaps without full awareness of the main difficulties, and (b) I would love to be proved wrong on this opinion :-) .

The global regularity problem for Navier-Stokes is of course a Clay Millennium Prize problem and it would be redundant to describe it again here. I will note, however, that it asks for existence of global smooth solutions to a Cauchy problem for a nonlinear PDE. There are countless other global regularity results of this type for many (but certainly not all) other nonlinear PDE; for instance, global regularity is known for Navier-Stokes in two spatial dimensions rather than three (this result essentially dates all the way back to Leray's thesis in 1933!). Why is the three-dimensional Navier-Stokes global regularity problem considered so hard, when global regularity for so many other equations is easy, or at least achievable?

(For this post, I am only considering the global regularity problem for Navier-Stokes, from a purely mathematical viewpoint, and in the precise formulation given by the Clay Institute; I will not discuss at all the question as to what implications a rigorous solution (either positive or negative) to this problem would have for physics, computational fluid dynamics, or other disciplines, as these are beyond my area of expertise. But if anyone qualified in these fields wants to make a comment along these lines, by all means do so.)

The standard response to this question is *turbulence* – the behaviour of three-dimensional Navier-Stokes equations at fine scales is much more nonlinear (and hence unstable) than at coarse scales. I would phrase the obstruction slightly differently, as *supercriticality*. Or more precisely, all of the globally controlled quantities for Navier-Stokes evolution which we are aware of (and we are not aware of very many) are either *supercritical* with respect to scaling, which means that they are much weaker at controlling fine-scale behaviour than controlling coarse-scale behaviour, or they are *non-coercive*, which means that they do not really control the solution at all, either at coarse scales or at fine. (I'll define these terms more precisely later.) At present, all known methods for obtaining global smooth solutions to a (deterministic) nonlinear PDE Cauchy problem require either

1. Exact and explicit solutions (or at least an exact, explicit transformation to a significantly simpler PDE or ODE);
2. Perturbative hypotheses (e.g. small data, data close to a special solution, or more generally a hypothesis which involves an ϵ somewhere); or
3. One or more globally controlled quantities (such as the total energy) which are both coercive and either critical or subcritical.

(Note that the presence of (1), (2), or (3) are currently *necessary* conditions for a global regularity result, but far from *sufficient*; otherwise, papers on the global regularity problem for various nonlinear PDEs would be substantially shorter :-) . In particular, there have been many good, deep, and highly non-trivial papers recently on global regularity for Navier-Stokes, but they all assume either (1), (2) or (3) via additional hypotheses on the data or solution. For instance, in recent years we have seen good results on global regularity assuming (2), as well as good results on global regularity assuming (3); a complete bibilography of recent results is unfortunately too lengthy to be given here.)

The Navier-Stokes global regularity problem for arbitrary large smooth data lacks all of these three ingredients. Reinstating (2) is impossible without changing the statement of the problem, or adding some additional hypotheses; also, in perturbative situations the Navier-Stokes equation evolves almost linearly, while in the non-perturbative setting it behaves very nonlinearly, so there is basically no chance of a reduction of the non-perturbative case to the perturbative one unless one comes up with a highly nonlinear transform to achieve this (e.g. a naive scaling argument cannot possibly work). Thus, one is left with only three possible strategies if one wants to solve the full problem:

1. Solve the Navier-Stokes equation exactly and explicitly (or at least transform this equation exactly and explicitly to a simpler equation);
2. Discover a new globally controlled quantity which is both coercive and either critical or subcritical; or
3. Discover a new method which yields global smooth solutions even in the absence of the ingredients (1), (2), and (3) above.

For the rest of this post I refer to these strategies as "Strategy 1", "Strategy 2", and "Strategy 3".

Much effort has been expended here, especially on Strategy 3, but the supercriticality of the equation presents a truly significant obstacle which already defeats all known methods. Strategy 1 is probably hopeless; the last century of experience has shown that (with the very notable exception of completely integrable systems, of which the Navier-Stokes equations is *not* an example) most nonlinear PDE, even those arising from physics, do not enjoy explicit formulae for solutions from *arbitrary* data (although it may well be the case that there are interesting exact solutions from special (e.g. symmetric) data). Strategy 2 may have a little more hope; after all, the Poincaré conjecture became solvable (though still very far from trivial) after Perelman introduced a new globally controlled quantity for Ricci flow (the *Perelman entropy*) which turned out to be both coercive and critical. (See also my exposition of this topic.) But we are still not very good at discovering new globally controlled quantities; to quote Klainerman, "the discovery of any new bound, stronger than that provided by the energy, for general solutions of *any* of our basic physical equations would have the significance of a major event" (emphasis mine).

I will return to Strategy 2 later, but let us now discuss Strategy 3. The first basic observation is that the Navier-Stokes equation, like many other of our basic model equations, obeys a *scale invariance*: specifically, given any scaling parameter $\lambda > 0$, and any smooth velocity field $u : [0,T) \times \mathbb{R}^3 \to \mathbb{R}^3$ solving the Navier-Stokes equations for some time T, one can form a new velocity field $u^{(\lambda)} : [0, \lambda^2 T) \times \mathbb{R}^3 \to \mathbb{R}^3$ to the Navier-Stokes equation up to time $\lambda^2 T$, by the formula

$$u^{(\lambda)}(t,x) := \frac{1}{\lambda} u(\frac{t}{\lambda^2}, \frac{x}{\lambda})$$

(Strictly speaking, this scaling invariance is only present as stated in the absence of an external force, and with the non-periodic domain $\mathbb{R}^3$ rather than the periodic domain $\mathbb{T}^3$. One can adapt the arguments here to these other settings with some minor effort, the key point being that an approximate scale invariance can play the role of a perfect scale invariance in the considerations below. The pressure field $p(t,x)$ gets rescaled too, to $p^{(\lambda)}(t,x) := \frac{1}{\lambda^2} p(\frac{t}{\lambda^2}, \frac{x}{\lambda})$, but we will not need to study the pressure here. The viscosity ν remains unchanged.)

We shall think of the rescaling parameter λ as being large (e.g. $\lambda > 1$). One should then think of the transformation from u to $u^{(\lambda)}$ as a kind of "magnifying glass", taking fine-scale behaviour of u and matching it with an identical (but rescaled, and slowed down) coarse-scale behaviour of $u^{(\lambda)}$. The point of this magnifying glass is that it allows us to treat both fine-scale and coarse-scale behaviour on an equal footing, by identifying both types of behaviour with something that goes on at a fixed scale (e.g. the unit scale). Observe that the scaling suggests that fine-scale behaviour should play out on much smaller time scales than coarse-scale behaviour (T versus $\lambda^2 T$). Thus, for instance, if a unit-scale solution does something funny at time 1, then the rescaled fine-scale solution will exhibit something similarly funny at spatial scales $1/\lambda$ and at time $1/\lambda^2$. Blowup can occur when the solution shifts its energy into increasingly finer and finer scales, thus evolving more and more rapidly and eventually reaching a singularity in which the scale in both space and time on which the bulk of the evolution is occuring has shrunk to zero. In order to prevent blowup, therefore, we must arrest this motion of energy from coarse scales (or low frequencies) to fine scales (or high frequencies). (There are many ways in which to make these statements rigorous, for instance using Littlewood-Paley theory, which we will not discuss here, preferring instead to leave terms such as "coarse-scale" and "fine-scale" undefined.)

Now, let us take an arbitrary large-data smooth solution to Navier-Stokes, and let it evolve over a very long period of time [0,T), assuming that it stays smooth except possibly at time T. At very late times of the evolution, such as those near to the final time T, there is no reason to expect the solution to resemble the initial data any more (except in perturbative regimes, but these are not available in the arbitrary large-data case). Indeed, the only control we are likely to have on the late-time stages of the solution are those provided by globally controlled quantities of the evolution. Barring a breakthrough in Strategy 2, we only have two really useful globally controlled (i.e. bounded even for very large T) quantities:

- The *maximum kinetic energy* $\sup_{0 \leq t < T} \frac{1}{2} \int_{\mathbb{R}^3} |u(t,x)|^2 \, dx$; and
- The *cumulative energy dissipation* $\frac{1}{2} \int_0^T \int_{\mathbb{R}^3} |\nabla u(t,x)|^2 \, dx dt$.

Indeed, the energy conservation law implies that these quantities are both bounded by the initial kinetic energy E, which could be large (we are assuming our data could be large) but is at least finite by hypothesis.

The above two quantities are *coercive*, in the sense that control of these quantities imply that the solution, even at very late times, stays in a bounded region of some function space. However, this is basically the only thing we know about the solution at late times (other than that it is smooth until time T, but this is a qualitative assumption and gives no bounds). So,

unless there is a breakthrough in Strategy 2, we cannot rule out the worst-case scenario that the solution near time T is essentially an *arbitrary* smooth divergence-free vector field which is bounded both in kinetic energy and in cumulative energy dissipation by E. In particular, near time T the solution could be concentrating the bulk of its energy into fine-scale behaviour, say at some spatial scale $1/\lambda$. (Of course, cumulative energy dissipation is not a function of a single time, but is an integral over all time; let me suppress this fact for the sake of the current discussion.)

Now, let us take our magnifying glass and blow up this fine-scale behaviour by λ to create a coarse-scale solution to Navier-Stokes. Given that the fine-scale solution could (in the worst-case scenario) be as bad as an arbitrary smooth vector field with kinetic energy and cumulative energy dissipation at most E, the rescaled unit-scale solution can be as bad as an arbitrary smooth vector field with kinetic energy and cumulative energy dissipation at most $E\lambda$, as a simple change-of-variables shows. Note that the control given by our two key quantities has worsened by a factor of λ; because of this worsening, we say that these quantities are *supercritical* – they become increasingly useless for controlling the solution as one moves to finer and finer scales. This should be contrasted with *critical* quantities (such as the energy for *two-dimensional* Navier-Stokes), which are invariant under scaling and thus control all scales equally well (or equally poorly), and *subcritical* quantities, control of which becomes increasingly powerful at fine scales (and increasingly useless at very coarse scales).

Now, suppose we know of examples of unit-scale solutions whose kinetic energy and cumulative energy dissipation are as large as $E\lambda$, but which can shift their energy to the next finer scale, e.g. a half-unit scale, in a bounded amount O(1) of time. Given the previous discussion, we cannot rule out the possibility that our rescaled solution behaves like this example. Undoing the scaling, this means that we cannot rule out the possibility that the original solution will shift its energy from spatial scale $1/\lambda$ to spatial scale $1/2\lambda$ in time $O(1/\lambda^2)$. If this bad scenario repeates over and over again, then convergence of geometric series shows that the solution may in fact blow up in finite time. Note that the bad scenarios do not have to happen immediately after each other (the *self-similar* blowup scenario); the solution could shift from scale $1/\lambda$ to $1/2\lambda$, wait for a little bit (in rescaled time) to "mix up" the system and return to an "arbitrary" (and thus potentially "worst-case") state, and then shift to $1/4\lambda$, and so forth. While the cumulative energy dissipation bound can provide a little bit of a bound on how long the system can "wait" in such a "holding pattern", it is far too weak to prevent blowup in finite time. To put it another way, we have no rigorous, deterministic way of preventing Maxwell's demon from plaguing the solution at increasingly frequent (in absolute time) intervals, invoking various rescalings of the above scenario to nudge the energy of the solution into increasingly finer scales, until blowup is attained.

Thus, in order for Strategy 3 to be successful, we basically need to rule out the scenario in which unit-scale solutions with *arbitrarily large* kinetic energy and cumulative energy dissipation shift their energy to the next highest scale. But every single analytic technique we are aware of (except for those involving *exact* solutions, i.e. Strategy 1) requires at least one bound on the size of solution in order to have any chance at all. Basically, one needs at least one bound in order to control all nonlinear errors – and any strategy we know of which does not proceed via exact solutions will have at least one nonlinear error that needs to be controlled. The only

thing we have here is a bound on the *scale* of the solution, which is not a bound in the sense that a norm of the solution is bounded; and so we are stuck.

To summarise, any argument which claims to yield global regularity for Navier-Stokes via Strategy 3 must inevitably (via the scale invariance) provide a radically new method for providing non-trivial control of nonlinear unit-scale solutions of arbitrary large size for unit time, which looks impossible without new breakthroughs on Strategy 1 or Strategy 2. (There are a couple of loopholes that one might try to exploit: one can instead try to refine the control on the "waiting time" or "amount of mixing" between each shift to the next finer scale, or try to exploit the fact that each such shift requires a certain amount of energy dissipation, but one can use similar scaling arguments to the preceding to show that these types of loopholes cannot be exploited without a new bound along the lines of Strategy 2, or some sort of argument which works for arbitrarily large data at unit scales.)

To rephrase in an even more jargon-heavy manner: the "energy surface" on which the dynamics is known to live in, can be quotiented by the scale invariance. After this quotienting, the solution can stray arbitrarily far from the origin even at unit scales, and so we lose all control of the solution unless we have exact control (Strategy 1) or can significantly shrink the energy surface (Strategy 2).

The above was a general critique of Strategy 3. Now I'll turn to some known specific attempts to implement Strategy 3, and discuss where the difficulty lies with these:

1. *Using weaker or approximate notions of solution* (e.g. viscosity solutions, penalised solutions, super- or sub- solutions, etc.). This type of approach dates all the way back to Leray. It has long been known that by weakening the nonlinear portion of Navier-Stokes (e.g. taming the nonlinearity), or strengthening the linear portion (e.g. introducing hyperdissipation), or by performing a discretisation or regularisation of spatial scales, or by relaxing the notion of a "solution", one can get global solutions to approximate Navier-Stokes equations. The hope is then to take limits and recover a smooth solution, as opposed to a mere global *weak* solution, which was already constructed by Leray for Navier-Stokes all the way back in 1933. But in order to ensure the limit is smooth, we need convergence in a strong topology. In fact, the same type of scaling arguments used before basically require that we obtain convergence in either a critical or subcritical topology. Absent a breakthrough in Strategy 2, the only type of convergences we have are in very rough – in particular, in supercritical – topologies. Attempting to upgrade such convergence to critical or subcritical topologies is the qualitative analogue of the quantitative problems discussed earlier, and ultimately faces the same problem (albeit in very different language) of trying to control unit-scale solutions of arbitrarily large size. Working in a purely qualitative setting (using limits, etc.) instead of a quantitative one (using estimates, etc.) can disguise these problems (and, unfortunately, can lead to errors if limits are manipulated carelessly), but the qualitative formalism does not magically make these problems disappear. Note that weak solutions are already known to be badly behaved for the closely related Euler equation. More generally, by recasting the problem in a sufficiently abstract formalism (e.g. formal limits of near-solutions), there are a number of ways to create an abstract object which could be considered as a kind of generalised solution, but the moment one tries to establish actual control on the regularity of this generalised solution one will encounter all the supercriticality difficulties mentioned earlier.

2. *Iterative methods* (e.g. contraction mapping principle, Nash-Moser iteration, power series, etc.) *in a function space*. These methods are perturbative, and require *something* to be small: either the data has to be small, the nonlinearity has to be small, or the time of existence desired has to be small. These methods are excellent for constructing *local* solutions for large data, or global solutions for *small* data, but cannot handle global solutions for large data (running into the same problems as any other Strategy 3 approach). These approaches are also typically rather insensitive to the specific structure of the equation, which is already a major warning sign since one can easily construct (rather artificial) systems similar to Navier-Stokes for which blowup is known to occur. The optimal perturbative result is probably very close to that established by Koch-Tataru, for reasons discussed in that paper.
3. *Exploiting blowup criteria*. Perturbative theory can yield some highly non-trivial blowup criteria – that certain norms of the solution must diverge if the solution is to blow up. For instance, a celebrated result of Beale-Kato-Majda shows that the maximal vorticity must have a divergent time integral at the blowup point. However, all such blowup criteria are subcritical or critical in nature, and thus, barring a breakthrough in Strategy 2, the known globally controlled quantities cannot be used to reach a contradiction. Scaling arguments similar to those given above show that perturbative methods cannot achieve a supercritical blowup criterion.
4. *Asymptotic analysis of the blowup point(s)*. Another proposal is to rescale the solution near a blowup point and take some sort of limit, and then continue the analysis until a contradiction ensues. This type of approach is useful in many other contexts (for instance, in understanding Ricci flow). However, in order to actually extract a useful limit (in particular, one which still solves Navier-Stokes in a strong sense, and does collapse to the trivial solution), one needs to uniformly control all rescalings of the solution – or in other words, one needs a breakthrough in Strategy 2. Another major difficulty with this approach is that blowup can occur not just at one point, but can conceivably blow up on a one-dimensional set; this is another manifestation of supercriticality.
5. *Analysis of a minimal blowup solution*. This is a strategy, initiated by Bourgain, which has recently been very successful in establishing large data global regularity for a variety of equations with a critical conserved quantity, namely to assume for contradiction that a blowup solution exists, and then extract a *minimal* blowup solution which minimises the conserved quantity. This strategy (which basically pushes the perturbative theory to its natural limit) seems set to become the standard method for dealing with large data critical equations. It has the appealing feature that there is enough compactness (or almost periodicity) in the minimal blowup solution (once one quotients out by the scaling symmetry) that one can begin to use subcritical and supercritical conservation laws and monotonicity formulae as well (see my survey on this topic). Unfortunately, as the strategy is currently understood, it does not seem to be directly applicable to a supercritical situation (unless one simply assumes that some critical norm is globally bounded) because it is impossible, in view of the scale invariance, to minimise a non-scale-invariant quantity.
6. *Abstract approaches* (avoiding the use of properties specific to the Navier-Stokes equation). At its best, abstraction can efficiently organise and capture the key difficulties of a problem, placing the problem in a framework which allows for a direct and natural resolution of

these difficulties without being distracted by irrelevant concrete details. (Kato's semigroup method is a good example of this in nonlinear PDE; regrettably for this discussion, it is limited to subcritical situations.) At its worst, abstraction conceals the difficulty within some subtle notation or concept (e.g. in various types of convergence to a limit), thus incurring the risk that the difficulty is "magically" avoided by an inconspicuous error in the abstract manipulations. An abstract approach which manages to breezily ignore the supercritical nature of the problem thus looks very suspicious. More substantively, there are many equations which enjoy a coercive conservation law yet still can exhibit finite time blowup (e.g. the mass-critical focusing NLS equation); an abstract approach thus would have to exploit some subtle feature of Navier-Stokes which is not present in all the examples in which blowup is known to be possible. Such a feature is unlikely to be discovered abstractly before it is first discovered concretely; the field of PDE has proven to be the type of mathematics where progress generally starts in the concrete and then flows to the abstract, rather than vice versa.

If we abandon Strategy 1 and Strategy 3, we are thus left with Strategy 2 – discovering new bounds, stronger than those provided by the (supercritical) energy. This is not *a priori* impossible, but there is a huge gap between simply wishing for a new bound and actually discovering and then rigorously establishing one. Simply sticking in the existing energy bounds into the Navier-Stokes equation and seeing what comes out will provide a few more bounds, but they will all be supercritical, as a scaling argument quickly reveals. The only other way we know of to create global non-perturbative deterministic bounds is to discover a new conserved or monotone quantity. In the past, when such quantities have been discovered, they have always been connected either to geometry (symplectic, Riemmanian, complex, etc.), to physics, or to some consistently favourable (defocusing) sign in the nonlinearity (or in various "curvatures" in the system). There appears to be very little usable geometry in the equation; on the one hand, the Euclidean structure enters the equation via the diffusive term Δ and by the divergence-free nature of the vector field, but the nonlinearity is instead describing transport by the velocity vector field, which is basically just an arbitrary volume-preserving infinitesimal diffeomorphism (and in particular does not respect the Euclidean structure at all). One can try to quotient out by this diffeomorphism (i.e. work in material coordinates) but there are very few geometric invariants left to play with when one does so. (In the case of the Euler equations, the vorticity vector field is preserved modulo this diffeomorphism, as observed for instance by Li, but this invariant is very far from coercive, being almost purely topological in nature.) The Navier-Stokes equation, being a system rather than a scalar equation, also appears to have almost no favourable sign properties, in particular ruling out the type of bounds which the maximum principle or similar comparison principles can give. This leaves physics, but apart from the energy, it is not clear if there are any physical quantities of fluids which are *deterministically* monotone. (Things look better on the stochastic level, in which the laws of thermodynamics might play a role, but the Navier-Stokes problem, as defined by the Clay institute, is deterministic, and so we have Maxwell's demon to contend with.) It would of course be fantastic to obtain a fourth source of non-perturbative controlled quantities, not arising from geometry, physics, or favourable signs, but this looks somewhat of a long shot at present. Indeed given the turbulent, unstable, and chaotic nature of Navier-Stokes, it is quite conceivable that in fact no reasonable globally controlled quantities exist beyond that which arise from the energy.

Of course, given how hard it is to show global regularity, one might try instead to establish finite time blowup instead (this also is acceptable for the Millennium prize). Unfortunately, even though the Navier-Stokes equation is known to be very unstable, it is not clear at all how to pass from this to a rigorous demonstration of a blowup solution. All the rigorous finite time blowup results (as opposed to mere instability results) that I am aware of rely on one or more of the following ingredients:

1. Exact blowup solutions (or at least an exact transformation to a significantly simpler PDE or ODE, for which blowup can be established);
2. An ansatz for a blowup solution (or approximate solution), combined with some nonlinear stability theory for that ansatz;
3. A comparison principle argument, dominating the solution by another object which blows up in finite time, taking the solution with it; or
4. An indirect argument, constructing a functional of the solution which must attain an impossible value in finite time (e.g. a quantity which is manifestly non-negative for smooth solutions, but must become negative in finite time).

It may well be that there is some exotic symmetry reduction which gives (1), but no-one has located any good exactly solvable special case of Navier-Stokes (in fact, those which have been found, are known to have global smooth solutions). (2) is problematic for two reasons: firstly, we do not have a good ansatz for a blowup solution, but perhaps more importantly it seems hopeless to establish a stability theory for any such ansatz thus created, as this problem is essentially a more difficult version of the global regularity problem, and in particular subject to the main difficulty, namely controlling the highly nonlinear behaviour at fine scales. (One of the ironies in pursuing method (2) is that in order to establish rigorous *blowup* in some sense, one must first establish rigorous *stability* in some other (renormalised) sense.) Method (3) would require a comparison principle, which as noted before appears to be absent for the non-scalar Navier-Stokes equations. Method (4) suffers from the same problem, ultimately coming back to the "Strategy 2" problem that we have virtually no globally monotone quantities in this system to play with (other than energy monotonicity, which clearly looks insufficient by itself). Obtaining a new type of mechanism to force blowup other than (1)-(4) above would be quite revolutionary, not just for Navier-Stokes; but I am unaware of even any proposals in these directions, though perhaps topological methods might have some effectiveness.

So, after all this negativity, do I have any positive suggestions for how to solve this problem? My opinion is that Strategy 1 is impossible, and Strategy 2 would require either some exceptionally good intuition from physics, or else an incredible stroke of luck. Which leaves Strategy 3 (and indeed, I think one of the main reasons why the Navier-Stokes problem is interesting is that it *forces* us to create a Strategy 3 technique). Given how difficult this strategy seems to be, as discussed above, I only have some extremely tentative and speculative thoughts in this direction, all of which I would classify as "blue-sky" long shots:

1. *Work with ensembles of data, rather than a single initial datum.* All of our current theory for deterministic evolution equations deals only with a single solution from a single initial datum. It may be more effective to work with parameterised familes of data and solutions, or perhaps probability measures (e.g. Gibbs measures or other invariant measures). One obvious partial result to shoot for is to try to establish global regularity for *generic* large data rather than *all* large data; in other words, acknowledge that Maxwell's demon might

exist, but show that the probability of it actually intervening is very small. The problem is that we have virtually no tools for dealing with generic (average-case) data other than by treating all (worst-case) data; the enemy is that the Navier-Stokes flow itself might have some perverse entropy-reducing property which somehow makes the average case drift towards (or at least recur near) the worst case over long periods of time. This is incredibly unlikely to be the truth, but we have no tools to prevent it from happening at present.

2. *Work with a much simpler (but still supercritical) toy model.* The Navier-Stokes model is parabolic, which is nice, but is complicated in many other ways, being relatively high-dimensional and also non-scalar in nature. It may make sense to work with other, simplified models which still contain the key difficulty that the only globally controlled quantities are supercritical. Examples include the Katz-Pavlovic dyadic model for the Euler equations (for which blowup can be demonstrated by a monotonicity argument; see this survey for more details), or the spherically symmetric defocusing supercritical nonlinear wave equation.
3. *Develop non-perturbative tools to control deterministic non-integrable dynamical systems.* Throughout this post we have been discussing PDEs, but actually there are similar issues arising in the nominally simpler context of finite-dimensional dynamical systems (ODEs). Except in perturbative contexts (such as the neighbourhood of a fixed point or invariant torus), the long-time evolution of a dynamical system for deterministic data is still largely only controllable by the classical tools of exact solutions, conservation laws and monotonicity formulae; a discovery of a new and effective tool for this purpose would be a major breakthrough. One natural place to start is to better understand the long-time, non-perturbative dynamics of the classical three-body problem, for which there are still fundamental unsolved questions.
4. *Establish really good bounds for critical or nearly-critical problems.* Recently, I showed that having a very good bound for a critical equation essentially implies that one also has a global regularity result for a slightly supercritical equation. The idea is to use a monotonicity formula which does weaken very slightly as one passes to finer and finer scales, but such that each such passage to a finer scale costs a significant amount of monotonicity; since there is only a bounded amount of monotonicity to go around, it turns out that the latter effect just barely manages to overcome the former in my equation to recover global regularity (though by doing so, the bounds worsen from polynomial in the critical case to double exponential in my logarithmically supercritical case). I severely doubt that my method can push to non-logarithmically supercritical equations, but it does illustrate that having very strong bounds at the critical level may lead to some modest progress on the problem.
5. *Try a topological method.* This is a special case of (1). It may well be that a primarily topological argument may be used either to construct solutions, or to establish blowup; there are some precedents for this type of construction in elliptic theory. Such methods are very global by nature, and thus not restricted to perturbative or nearly-linear regimes. However, there is no obvious topology here (except possibly for that generated by the vortex filaments) and as far as I know, there is not even a "proof-of-concept" version of this idea for any evolution equation. So this is really more of a wish than any sort of concrete strategy.
6. *Understand pseudorandomness.* This is an incredibly vague statement; but part of the difficulty with this problem, which also exists in one form or another in many other famous

problems (e.g. Riemann hypothesis, $P = BPP$, $P \neq NP$, twin prime and Goldbach conjectures, normality of digits of π, Collatz conjecture, etc.) is that we expect any sufficiently complex (but deterministic) dynamical system to behave "chaotically" or "pseudorandomly", but we still have very few tools for actually making this intuition precise, especially if one is considering deterministic initial data rather than generic data. Understanding pseudorandomness in other contexts, even dramatically different ones, may indirectly shed some insight on the turbulent behaviour of Navier-Stokes.

In conclusion, while it is good to occasionally have a crack at impossible problems, just to try one's luck, I would personally spend much more of my time on other, more tractable PDE problems than the Clay prize problem, though one should certainly keep that problem in mind if, in the course on working on other problems, one indeed does stumble upon something that smells like a breakthrough in Strategy 1, 2, or 3 above. (In particular, there are many other serious and interesting questions in fluid equations that are not anywhere near as difficult as global regularity for Navier-Stokes, but still highly worthwhile to resolve.)

Comments (826)



Dwight Walsh · 26 March 2022

Well, it has been a week now since I made my postings showing that your constructed q functions needed to be constrained by the Navier-Stokes equation in order to be a valid rebuttal to my proof. Now, I know you would like nothing better than to shoot down my arguments on this forum, but it seems you cannot do so.



Antoine Deleforge · 27 March 2022

No. I already explained to you in full details why our counter-example for q completely invalidates your proof in my last comment (13 February, 2022 at 9:57 am), regardless of whether it is constrained by NS, in a language that I believe should be understandable by a high-school pupil. I note you still haven't given any proper thoughts to this and keep displaying your misunderstandings over and over again. Again, if you are not able to follow this basic logic, I am afraid the situation is hopeless and there is no point in continuing writing here. Bye!



Dwight Walsh · 27 March 2022

There is something else you've overlooked in your constructed q function which I probably should have picked up weeks ago. The fact that this must be a smooth blowup means that there is a time t_b < T_b such that for all t in the semi-open interval t_b <= t T_b of |grad q(0,t)| = 0. For your constructed q, however, we have limit as t -> T_b of |grad q(0,t)| = \infty.

Now I realize this might not be the most pleasant, ego-building news, but I too have had my fair share of mistakes and disappointments. So let's be a sport about this.
— OK



Anonymous · 27 March 2022

Dwight, your proof does not even begin to address the difficulty of the problem. You're wasting everyone's time

Dwight Walsh · 27 March 2022

So fire me! (LOL)!

And don't tell me about wasting time until you have spent 10 years struggling unsuccessfully to rebuild a career after being laid-off for "lack of contract support" despite excellent performance ratings, spent the next 20 years unable to work due to heath problems resulting from the abuse of the first 10 years, and then dealing with a heart attack from it all!

But I did learn patience over these past 30 years. [Let's face it — my very survival depended on it!] Therefore I won't mind "wasting everyone's time" until I get some straight answers about where these errors are in my proof and why they are errors. At least they are getting paid. And so-called "global errors" which are red-flags at best and provide no such information are not sufficient. I already explained why false red-flags are likely with my particular proof.

Anonymous · 28 March 2022

If you so sure about the correctness of your proof, do you have a reasonable explanation how so many experts missed such a simple and elegant solution to the NS problem?

Dwight Walsh · 28 March 2022

I have an explanation though I doubt you would call it reasonable. It's the sort of thing that tends to happen when large amounts of credit and/or money is at stake. The current "experts" don't like competing with newcomers with new ideas that might actually work. I happen to be living proof of this, and it wouldn't surprise me in the slightest if others before me have had ideas similar to mine, but found themselves up against the same system that I am facing now. We simply don't hear about them.

Dwight Walsh · 28 March 2022

Antoine and anon,

Here's what you are missing in your postings of 10 February 2022 at 6:00 AM:

Since the blowup of q at |x| = 0 as t -> T_b must be a smooth blowup, there is a time t_b < T_b such that a global maximum in q occurs at this point
for all t in the semi-open interval t_b <= t T_b of |grad p| must equal 0. For every constructed q function shown on this page, however, this limit is infinite.

Now from equation (108) the function Q and therefore the scalar pressure p must also blow up at |x| = 0 as t -> T_b. But again, since this is a smooth blowup, we must have |grad p(0,t)| -> 0 as t -> T_b. This means that |grad p| is continuous on the entire CLOSED time interval [0,T_b], and therefore the time integral of |grad p| from 0 to T_b is finite, consistent with equation (120). At this point, equation (128) implies that the maximum of K at time T_b is finite, and therefore THERE IS NO BLOWUP AFTER ALL!

Oh well — Better luck next time!

Dwight Walsh · 29 March 2022

Antoine,

In your posting of 7 February 2022 at 2:14 AM, you stated:

“… the integral [in equation (118)] is unbounded at x=0 on the open-interval [0,T_b) (do the math, it’s pretty straightforward).”

In view of the facts presented in this current posting, do you still support your original statement that the integral in equation (118) is unbounded, or do you acknowledge that this statement is erroneous?



Antoine · 30 March 2022

Sorry but even after reading your comment 10 times I can’t make sense of it. There seems to be a display issue because the phrase “for all t in the semi-open interval t_b <= t T_b of |grad p|…" makes no sense.

It does not matter though. I can reiterate, with 100% certainty, that the counter example q that has been given to you by me and others constitutes a rigorous proof that the statement "(108) and (109) implies (118)" made in you paper is false in general, which invalidates your proof. The reason for this implication to be false has to do with an illegal implicit permutation of limit when you compute (116). But you do not seem to be able to understand this. That's why we prove it to you in the form of a counter example, which is (usually, for most people) easier to understand, while being a completely rigorous refutation.

It is amazing to me that you keep replying things like "but wait because your q has to verify this and this because of NS". It really shows that, after months, you still haven't understood our refutation on a fundamental, logical level. "Our q" does not have to verify anything that you see fit. As explained in details on my Feb. 1 comment, it is just a counter example to a specific, false logical implication made in your paper.

P.S.: Your recent comment where you write "The current “experts” don’t like competing with newcomers" show how sadly and completely misguided you are. I am not at all working on NS, I work in a completely different, remote field of applied math. You are not a "competitor" to me, or most likely to anyone writing in this comment section. In fact, most reasonable people don't waste their time trying to frontally tackle huge problems like NS. This is very well explained by Terence Tao in this blogpost: https://terrytao.wordpress.com/career-advice/dont-prematurely-obsess-on-a-single-big-problem-or-big-theory/.

I, for one, would be absolutely thrilled if a random, lone researcher like you would come up with the solution to a millennium problem. It just so happen that one basic mistake in your paper (pointed out by a previous anonymous commenter) was easy enough that I could understand it. I was then foolish enough to believe that I would be able to explain it to you clearly, hence saving your and everyone's time. Obviously I strongly overestimated you.



Anonymous · 30 March 2022

This article Ten Signs a Claimed Mathematical Breakthrough is Wrong is also relevant.



Dwight Walsh · 30 March 2022

In your posting on 5 February 2022 at 9:22 AM, you stated:

The counter example I gave for q *is* smooth. I never questioned or discussed smoothness in my argument, …

Perhaps we both should have thought a little more about what “smooth” means in the current context. What is happening as t ->T_b at |x|=0 is a SMOOTH BLOWUP in the q function. This means that q must have a smooth spatial profile at all points x in R^3 for t < T_b; and, as discussed in the paper, there must be a time t_b T_b is zero. Unfortunately, none of the constructed q function I have seen either here or in my email have this property, even though they may have a smooth temporal profile on the semi-open time interval [0,T_b) but clearly blow up as t->T_b.

Dwight Walsh · 30 March 2022

I see, after posting of course, that the same rendering problem is occurring now which happened in my last posting. If you see the combination t_b T_b in the posting, then probably the system dropped a less-than sign between the t_b and the T_b.

Dwight Walsh · 30 March 2022

Well this is incredible! I get thumb-downed for posting info on a rendering error. Some folks out there never cease to amaze me! (LOL)

Dwight Walsh · 30 March 2022

One important point I forgot to mention in this posting:

Even if you do find an acceptable q function, my posting of 28 March 2022 at 9:59 PM indicates that you are only a “hop and step” away from proving equation (120) which implies there is no blowup, thereby contradicting the conjectured q function.

Dwight Walsh · 30 March 2022

WHOOPS! I TAKE THAT BACK!

There is no contradiction! The fact that there is no blowup shows that by using an acceptable constructed q function, we come to the same conclusion that we did when using the actual q function. Therefore — no counter-example!

Case closed.

Anonymous · 31 March 2022

Jesus Dwight would you just stop already?! What is it you’re seeking? If you’ve truly discovered a complete solution, thus cracking NS, well good for you! If us mere mortals are too stupid to understand your solution why do you keep shoving it down our throats? Can’t you just revel in its beauty by your own? Isn’t that a reward in itself already?

Dwight Walsh · 31 March 2022

Come on, Anonymous — You know full well what I am seeking. You and your comrades are not claiming you don’t understand my proof. What you are claiming is that you have iron-clad proof that my proof is invalid. To make such a claim, however, you had better have a thorough understanding of what my arguments are and what invalidates them. But all you have been able to provide is a nebulous “global error” which is nothing more than a “red flag” at best. Then, on 31 March 2022, I show that even in playing by your rules, finding a so-called “counter-example” is not possible. None of your constructed q functions

meet the required constraints, and any others that do would lead to the same conclusion (ie. equation (120)) as the actual q function. So it appears that you don't even have those red flags anymore.

Therefore, I am seeking at least acknowledgement that you don't have conclusive proof that my proof of NS global regularity is invalid. Also, an apology for the insulting and offensive comments (such as your last one) would be most appropriate.



Anonymous · 31 March 2022

DW, you are fooling yourself by assuming the result to prove the result itself. Not to mention that you make numerous undergraduate level mistakes.

You do not understand the potential theory you quoted, and you do not at all understand the Poisson equation. This demonstrates that your claim of ever passing a course in PDE is false.

"Also, the proof we present requires only an undergraduate background in calculus, differential equations (ordinary and partial), potential theory, and vector analysis for a reader to follow it." As a matter of fact, the "paper" is so badly written that you fool yourself to think it is correct.

In the way you dispute Antoine's comment, you don't even have a sense of mathematical logic, which is an absolutely must for one's mathematical training. This is one of the reasons why you fool yourself.

If you love to fantasy your victory, feel free to do so. But PLEASE STOP POLLUTING THE COMMENT AREA.



Dwight Walsh · 31 March 2022

>DW, you are fooling yourself by assuming the result to prove the result itself.

Show me where I do this!

————————————————————

> Not to mention that you make numerous undergraduate level mistakes.

Give me just one example!

————————————————————

>You do not understand the potential theory you quoted, and you do not at all understand the Poisson equation. This demonstrates that your claim of ever passing a course in PDE is false.

This is immediately refuted with a quick look at my academic transcript. Not only that but I made extensive use of my PDE background in the electro-magnetics and quantum mechanics courses also on my transcript. Case closed!

————————————————————

>"Also, the proof we present requires only an undergraduate background in calculus, differential equations (ordinary and partial), potential theory, and vector analysis for a reader to follow it." As a matter of fact, the "paper" is so badly written that you fool yourself to think it is correct.

This doesn't jibe too well with the comment from one reviewer who stated "the proof is so easy that even an undergraduate reader could follow".
————————————————————

>In the way you dispute Antoine's comment, you don't even have a sense of mathematical logic, which is an absolutely must for one's mathematical training. This is one of the reasons why you fool yourself.

Yeah, somehow use of a false statement in a proof has always given me heartburn! Also, over the past few days I have shown that Antoine's constructed q functions (along with everyone else's) do not meet the required constraints. However, use of q functions that are consistent with the constraints leads to the same result (equation(120)) as the actual q function, thereby foiling any claims of a "global error" and preserving the credibility of the proof by your own rules!
————————————————————

>PLEASE STOP POLLUTING THE COMMENT AREA

Don't worry — I was hoping to receive a simple acknowledgement that your "counter-example" showing a "global error" in my proof is invalid. At this point, however, you've got your foot so far into your mouth I doubt that will be necessary.

Anonymous · 1 April 2022

Yes yes Dwight you got it. Congratulations and gtfo.

Dwight Walsh · 1 April 2022

Thank you for your acknowledgement that you do not have a valid counter-example for establishing a "global error" in my proof even if it was done begrudgingly and unprofessionally. But it seems that your comrades need a little more convincing.

Jas, the Physicist · 3 April 2022

Can you guys speak mathematics please this is really hard to read I don't know what you're saying.

Dwight Walsh · 1 April 2022

Antoine,

What about your version of q(x,t)? Do you now acknowledge that your constructed q(x,t) does not satisfy the smooth blowup constraint even though it may be smooth on the semi-open time interval [0,t)?

Anonymous · 2 April 2022

GTFO, DW. You proved the result you like and you should go publish it somewhere. Not here. You would probably also like to inform the medias as well regarding your breakthrough. Don't forget to prepare your undergraduate transcript since the journalists will certainly ask for it. Congratulations.

What others' objections do not matter. It is very much disgraceful, annoying and disgusting you bother others over and over. Stop. Go publish your result.

Dwight Walsh · 2 April 2022

I asked Antoine, not you!

>What others' objections do not matter. It is very much disgraceful, annoying and disgusting you bother others over and over.

That's because others stuck with the (false) "global error" claims over and over and refused to read the paper. I only defended the paper over and over. BTW — Since you are such an "expert" mathematician, how come you didn't find the actual (or "local") error in the proof, but instead jumped on the "global error" bandwagon? It seems to me that an expert would have found the error in less time than what it took you to post all of those insulting messages about me. Then, the matter would be settled, wouldn't it!?

Anonymous · 2 April 2022

DW, If you have a damn beautiful proof that you think correct, GO PUBLISH it. Or are you simply afraid of your "authorities" may easily point your mistakes if you submit your result anywhere? What the f* do you get by harassing readers in this blog?!

If your proof if ultimately correct, it does NOT make a damn difference if Antonie, or any other readers HERE objects you or not. You have no certification/credit to award even if every reader HERE says you are right. There are tremendously many online instructions for amateur mathematicians on writing and publishing papers.

If you don't care about publication, then post your preprints on arXiv like what a professional would do.

By all means, stop you $q(x,t)$ nonsense and live your life. You have done such thing for a half damn year.

Anonymous · 2 April 2022

"I only defended the paper over and over." This is not the right place to "defend" your paper, DW. Don't you understand?!

You could either get into a math PhD program in a graduate school so that you can write your masterpiece as a PhD dissertation and defend it in front of the thesis committee, or submit your "paper" to an academic journal.

Please, stop your smooth blowup drama here and leave.

Dwight Walsh · 3 April 2022

What makes you think I didn't already attempt everything you suggested? Those suggestions are actually two-party decisions and I am only one person. Therefore, they are not possible if I am simply ignored, which has been the case for 30 years now so I am ready for a change. Also, don't tell me about professionalism when my research career was done away with after just eight years!

So, while I got the most hostile reception ever on this forum, at least I wasn't ignored which is far better than what I got anyplace else.

 **Jas, the Physicist** · 3 April 2022

Which school did you want to go to but didn't get accepted in and why is your solution correct? I still don't even understand the problem do you want to see my solution?

 **Jas, the Physicist** · 3 April 2022

Go publish your result it has to be in a journal for two years before you can win the prize I think. You better do it or I'm gonna publish mine.

 **Anonymous** · 3 April 2022

Maybe they will let the two of you share the prize

 **Dwight Walsh** · 3 April 2022

This is not the right place to "defend" your paper, DW. Don't you understand?!

Don't you understand, Anonymous — This is not simply about defending a paper. At the time I announced the paper on this forum you claimed there was an error in equation (118) because I didn't consider q functions such as r^{-100} which blow up at r=0. You didn't seem to catch on to the fact that q was already defined in the proof, and therefore not adjustable at some future point in time. I clarified that this was not an error, but instead of acknowledging that I was only using q according to its definition, you simply went deeper into nonsensical claims. Eventually, it looked very much like you and your comrades were out to adjust the q function to your liking, thereby causing a fictitious blowup, and then claiming this blowup as a "counter-example" for proving a "global error". Can we say "strawman fallacy on steroids"?

At this point, I have a better (though not perfect) understanding of what you were trying to do. It is basically an error-in-proof detection technique based on the writings of Prof. Terence Tau. While I won't argue over its legitimacy, I will say that the user must thoroughly understand how to use it or the results can be very misleading, and I strongly believe this is EXACTLY what happened in the case of my proof. As I explained in previous postings, the constructed q functions used in these arguments were not consistent with the required constraints. I was able to show, however, that any correctly constructed q function would lead to the same long-term result as the actual q function, namely a finite time integral of |grad p| for all t > 0. This, of course, implies there are no "global errors". Nevertheless, you and your comrades tenaciously held onto the claim that my proof was erroneous despite the fact that you could not actually find any errors.

Now, as we all know, postings on an online forum can be read all over the world. This being the case, I do my utmost to keep my postings as clear and accurate as possible. If I do make a mistake, I do my best to correct it whether I discover it myself or if someone else informs me of it. Similarly, I expect

others to acknowledge their errors, and that includes erroneous claims made about my work or my ideas. Unfortunately, however, that part has been rather shortcoming lately, and I simply will not allow such claims to go unchallenged.

 **Anonymous** · 3 April 2022

Dwight.. if there's one thing you need to learn it's called the Dunning-Kruger Theorem. Go read it up – if you think deeply about it it just might help sort out your issues with NS.

But seriously, just look at what has transpired over all this time. You posted your purported proof, others tried to point out what's wrong with it and your subsequent posts basically amounted to shouting them down and accusing them of denying you the glory of solving this big-ass problem.

BUT YOU ARE WRONG! Jesus I have no background in hardcore PDEs and even I could understand what Antoine and others have been trying kindly to tell you, only to be met with frenzied denial. Really, stop thinking about PDEs for a while and try to understand the (rather basic, tbh) logic of Antoine's argument. Leave your ego at the door for just this once. Sit down and try to earnestly understand what all these reasonably intelligent people have got to say.

I already suspect you're fuming and foaming by now but here's the thing. You can proceed as I suggested – and I'm sincerely trying to help you – or you can continue shitposting here and wallowing in all that misery until TT blocks you. Be wise.

 **Dwight Walsh** · 6 April 2022

Well, except for the Dunning-Kruger Theorem, I've discussed all of these issues with you and your comrades a million and one times already, so there isn't much point to me repeating these arguments again. Evidently, you simply don't have the mathematical literacy to understand the concept of a smooth blowup which is central to the proof. Now, as I recall, you called yourself an "expert" in one of your posts, but I see no evidence of this whatsoever — not in mathematics anyway. Was this a self-appointed title?

BTW – It is ABSOLUTELY INCORRECT to place the lim as \epsilon -> 0 operation behind the integral sign in equation (117). In fact, you actually end up with an expression that doesn't even make sense since \epsilon appears only as the lower limit of integration. Therefore, in your version of the Green's function integral for solving the Poisson equation, you integrate over R^3 the lim as \epsilon -> 0 of a function that does not depend on \epsilon. Not only that, but now there is also a completely undefined \epsilon in the lower limit of the integral. Can we say COMPLETE UTTER NONSENSE! Now if I have completely misunderstood this "theorem" of yours, please send me or post a copy of it along with your proof complete with correctly rendered math symbols and equations. I won't divulge your email address to anyone without your permission.

Anyway, your comrade Antoine applied this "theorem", or at least attempted to do so, in his posting on 7 February 2022 at 2:14 AM to construct the version of

the q function used as a “counter-example” to my proof. Since the entire attempt was based on faulty claims from the start, however, I figured this was a moot point and didn’t raise the issue any longer.\

In regard to me “fuming and foaming”, all I ever did was react as a normal homo-sapient would when experiencing the type of frustrations I indicated in my last post on 3 April 2022 at 9:44 PM. While we are on the subject, however, I should add that you have used profanities over the last few posts whereas I have not.

Regarding Prof. Tau, he doesn’t seem to give a flip about what’s put up on this forum. He went on to bigger and better things over five years ago, probably after concluding that this problem truly was the “impossible problem” he described in this article. In fact, as I understand it, he is now looking for a counter-example to global regularity rather than proof of it. As I see it, however, he got stuck on the “infamous” nonlinear convective term and didn’t adequately consider other constraints on the solution such as those imposed by the smooth blowup conditions. I tried to inform him of this possibility on this forum a few times over the past 18 months, but got the same non-response that I have been accustom to for the past 30 years.

So, I have a suggestion for you. Since you seem to have lots of clout on this forum, perhaps you could contact TT, inform him of my ideas and my paper, and suggest that he get in touch with me. Although you are unable to comprehend the smooth blowup concept, perhaps he can. And, if I am correct, it could save him from decades of pursuing dead end leads. Now doesn’t that sound much more positive than claiming I will be wallowing in all that misery until TT blocks me?

 **Anonymous** · 6 April 2022

Dwight, you seem to have not looked at Tao’s progress toward disproving global regularity because he demonstrates the existence of what you call a “smooth blowup” for a slightly modified Navier-Stokes equation.

https://arxiv.org/abs/1402.0290

Your argument would just as well imply global regularity for this equation

 **Dwight Walsh** · 6 April 2022

>Your argument would just as well imply global regularity for this equation.

I’m afraid this statement needs proof. In addition to the smooth blowup constraints, my arguments use the relatively simple form of the exact NS equation to establish upper bounds on the fluid velocity and scalar pressure from the initial conditions. With Prof. Tao’s HORRENDOUSLY COMPLICATED “averaged” convective term, however, I’m not convinced this is even possible. So this time, the burden of proof is on you. And if you can’t follow even my relatively simple proof, you are in for one rough ride!

Dwight Walsh · 6 April 2022

>Your argument would just as well imply global regularity for this equation.

There is something I should have asked in my last post about this statement of yours. Since you don't understand my argument, how do you know it "would just as well imply global regularity" for Tao's "averaged" NS equation?

Anonymous · 10 April 2022

Dwight, something I think you don't appreciate is that there's nothing that special about the pressure. From an analytic standpoint it is almost the same thing is $u \otimes u$. If you take the averaged nonlinearity in Tao's paper and apply some order -1 operator to it, then you can call it p and apply the arguments from your paper (which I do understand. As you point out it's not difficult.)

Dwight Walsh · 10 April 2022

I'm not sure what you mean by "there's nothing that special about the pressure". This doesn't jibe very well with your statement on 22 September 2021 at 8:19 AM which reads as

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If you actually prove a bound on the time integral of grad p, concluding global regularity would be routine.

————————————————-

Also, if you follow equations (121)-(125) of the paper, you will see that grad p(x_b, t) is the only force (per unit volume) that can cause increase to the kinetic energy density K(x_b, t) at a potential blowup point x_b after this point becomes a global maximum of K. Therefore, it seems to me there is something quite special about the scalar pressure p. It's gradient is the only quantity that could possibly cause a blowup.

Finally, there is one important aspect to my paper that I don't believe you understand. In establishing upper bounds on the fluid velocity components u_i, the nonlinear (convective) term is used in its EXACT form instead of some sort of AVERAGE form. Now I don't follow at all how Dr. Tao comes up with his average nonlinear term, but in general, detailed information is lost in any averaging process as we choose a single value as "representative" of a bunch of values, and there is no single inverse operator that magically restores this lost information. I believe Dr. Tao calls this missing information the "finer structure" of the nonlinear term which he states is necessary for a positive solution of the Navier-Stokes global regularity problem. Anyway, it seems there is no suitable operator that can be applied to the averaged nonlinear term to restore the correct value of p. Therefore, the arguments from my paper cannot be applied to the average nonlinear term, and attempts to do so would produce meaningless results.

Anonymous · 6 April 2022

Look at what you're doing again Dwight. When people suggest something new, you dismiss it as 'horrendously complicated' and you're 'not convinced this is possible'. Be honest, have you really sat down and made attempts to understand what Prof Tao wrote? Or did you immediately dismiss it simply because it's completely beyond your reach?

Dwight Walsh · 6 April 2022

I've had a copy of this paper by Prof. Tao since before I even knew about this forum, so it is not exactly "something new" to me. However, since he and I are not working the same problem, it is irrelevant. So let's get back to the question as to what, if anything, is actually wrong with my proof of NS global regularity. And, since you can't understand smooth blowups or their implications, why don't you and maybe your comrades suggest to Prof. Tau that he get in touch with me?

Dwight Walsh · 9 April 2022

In the "Statement of main result" section of this paper, Prof. Tao states that

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"... any proposed positive solution to the regularity problem which does not use the finer structure of the nonlinearity cannot possibly be successful."

————————

In my paper, I use the nonlinear term in its EXACT form. How much more "finer structure" of this term is needed to resolve NS global regularity?

Jas, the Physicist · 3 April 2022

Okay, but what about that topology we are supposed to construct? Has anyone found the open sets yet? I remember thinking it was something very "not-smooth" like even fractal but can someone please correct me because I solved the problem in isolation and I don't think it's completely correct.

How would I turn a fractal into an open set anyway?

Jas, the Physicist · 3 April 2022

> Why is the three-dimensional Navier-Stokes global regularity problem considered so hard, when global regularity for so many other equations is easy, or at least achievable?

My first thought was: knots.

Knots are usually thought of as a one-dimensional object in three-dimensional space. Sometimes we refer to this as the knot having a codimension of 2 (3-1).

Codimension 2 turns out to make knots more interesting than they otherwise would be. In codimension 1, you don't have much freedom to deform your object–imagine a circle in a plane, you can certainly push and prod it into many shapes, but they're all the "same" in a sense, unless you cross the circle over itself, a transformation which isn't allowed. In codimension 3, where you have a one-dimensional object embedded in a four-dimensional space, you have too much freedom for anything interesting to happen–you can always unravel your knot.

As it turns out, however, we can talk about higher-dimensional knot theory, but we need to change what our idea of a knot is. That is, instead of imagining one-dimensional objects in higher-dimensional spaces, we want to consider codimension two objects like a two-dimensional surface embedded in four dimensions, and so on.

There is actually a freely available book appropriately called High-dimensional Knot Theory. It is likely unreadable to a layperson, but the entire book focuses on these codimension 2 embeddings.

Dwight Walsh · 23 April 2022

On 26 January 2022 at 10:08 AM, the following comment was posted by Anonymous:

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I gave you specific equation numbers and a counterexample for your inequalities for the pressure (equations (117)-(118)). Because you don't understand analysis well enough to pass an undergraduate course on the subject, you dismissed it immediately.

Everyone, please stop engaging with this guy. His paper is getting rejected from journals because it's wrong. It's clear as an expert reading his paper that there is no new idea, just an elementary analysis mistake.

————————————

Is that so!? Then perhaps you would like to explain how an article titled "A shorter solution to the Clay millennium problem about regularity of the Navier-Stokes equations" recently appeared in the Journal of Scientific Research and Studies [Vol. 9(1), pp. 8-16, February, 2022] in which the author, Konstantinos E. Kyritsis, proved Navier-Stokes global regularity using the EXACT same principles as I did, but with a somewhat different notation. Evidently, not everyone is buying your "counterexamples" or your "expertise", but I guess success is based on know-who and not know-how.

Anonymous · 24 April 2022

My comments apply just as well to that paper. You should contact them, Dwight, perhaps you can collaborate toward a proof of another millenium problem.

Dwight Walsh · 24 April 2022

Well then, I think this proves that just because a paper is "wrong" doesn't mean it will be rejected, right!? So let's not have any more nonsense about "his paper is getting rejected from journals because it's wrong". And, since you are such an "expert", why don't you fill me in a little on how an author can get a "wrong" paper accepted?

Dwight Walsh · 25 April 2022

In looking back over some of the recent comments, I noticed that you experts have been strangely silent about the question I posted on 09 April 2022 at 11:11 AM regarding how much "finer structure" of this notorious nonlinear term is needed to resolve NS global regularity. In my paper, I obtain EXACT upper bounds for the u_i based on the original form of the entire NS equation whereas everyone else (including Prof. Tao himself) uses approximations in the nonlinear term. Which would you say is more reliable? All right, experts — you can ignore this question now.

Anonymous · 19 November 2022

This just appeared in Quanta Magazine and is about a recent result that supposedly represents progress. It shows blowup in a restricted version of the Euler equation. I don't claim to understand it: https://www.quantamagazine.org/computer-helps-prove-long-sought-fluid-equation-singularity-20221116/

The preprint itself is at https://arxiv.org/abs/2210.07191 .

> **Anonymous** · 20 November 2022
>
> The importance of this proof is that it is the first (computer assisted) rigorous proof for a stable self-similar blowup (which gives some hope for its extension to the NS equations)

Anonymous · 30 November 2022

I just stumbled on this paper https://arxiv.org/abs/1308.2297 . On the Regularity for 3D Navier-Stokes Equation – Qun Lin. Did anyone of you ever have a look at it?

Anonymous · 8 December 2022

Hello Professor Tao and others. If one could prove that the Euler equations are regular would it then be easy to prove NS is regular? Are there any known results about this? Kind regards, Dr Hayes

> **Anonymous** · 9 December 2022
>
> Probably yes, but don't waste your time.
>
> **Terence Tao** · 11 December 2022
>
> If the Euler equations were shown to be regular in a sufficiently quantitative manner (so that one could control suitable spacetime norms of the solution in terms of the initial data) then I think this would rule out several possible types of Navier-Stokes blowup in which the viscosity term in the Navier-Stokes equation becomes small relative to the other components of the equation, at which point the dynamics become a perturbation of the Euler dynamics and perturbative techniques come into play. However, it is theoretically possible (though rather unlikely) that there could still be a blowup mechanism for Navier-Stokes in which the viscosity term is so large that it is no longer treatable as a perturbative correction. This is unlikely since if the viscosity term dominates then one would expect the solution to stay regular for all times; but there may be some strange intermediate (or "critical") regime in which neither the viscosity term nor the nonlinear terms dominate each other and so one cannot treat the dynamics as a perturbation of either the heat equation or the Euler equations. (We do have some results excluding several types of "critical" or "Type I" blowup from occurring, but they may not cover all possible intermediate scenarios.)

Robert Ferraro · 11 December 2022

In 2023 I will be submitting for publication a simple non-mathematical counterexample. It will be the "deep new ideas" the Clay Insitute is looking for. The problem as it stands is a red herring. There can be no solution in the current form of the equations except under a specific condition of fluid flow where smooth solutions do exist for all time. They are generally unsolvable because the fundamental physics underlying the equations are incorrect. Most of

the 17th thru 20th century physics are laced with these simple misunderstandings and time-wasting red herrings. Sorry to be the bearer of bad news.

Anonymous · 12 December 2022

Congratulations Robert. I am sure your claims are correct and will yield the Clay Prize for you.

Robert Ferraro · 12 December 2022

Lol. They are correct. 10 Thumbs down (so far). Wow, lots of skeptics. I assure you it is a red herring.

Jeff Dylan · 27 December 2022

You might want to check out this proof of Navier-Stokes global regularity before posting your counter-example. It is downloadable from:

http://www.navierstokessmoothsolutions.com

robertlferraro · 28 December 2022

If your proof is correct, why is it not accepted for publication? The problem, and this is why it is a red herring, is that the assumptions of the fundamental physics are incorrect. As I said above, there is one situation when the physics works. Incidentally, I will also be submitting a solution to the Upstream Contamination problem – https://en.m.wikipedia.org/wiki/Upstream_contamination – they are closely related. If you understand the fluid mechanics that causes this phenomenon you would understand why the fundamental assumptions of NS are wrong.

Jeff Dylan · 28 December 2022

Who says a paper has to be wrong to be rejected for publication? In fact, only two such rejections claimed mathematical errors on my part, and both of them were based on claims I did not make (ie. “straw-man arguments). Also, no one is claiming the physics to be right or wrong. We are only investigating the behavior of solutions to a mathematical problem — nothing more, nothing less. Whether it is physically meaningful is another matter.

Anonymous · 28 December 2022

Someone should tell Dwight Walsh that Jeff Dylan is taking credit for his extremely correct work!

Jeff Dylan · 30 December 2022

Yes, Anonymous. Why don’t you tell him? Of course, you will have to email him directly at

mathguy@navierstokessmoothsolutions.com

since he was blocked from further posting on this blog. If you do, however, you won’t be so anonymous to him anymore.

Anonymous · 30 December 2022

Cute idea, Dwight. Sorry you were banned.

Jeff Dylan · 1 January 2023

Dwight has another cute idea. For the past few months, he has worked on a small book that describes and documents the troubles he has had with journal editors and people on this blog. Just yesterday, he had the idea of posting an excerpt from this book, which is almost ready to go to print. In this excerpt, he used actual equations to show how the Poisson equation is solved using the 3D Green's function, and not just verbal arguments or claims of "cheating". It does get comical when he shows that the Anonymous-Deleforge-Viswanathan (ADV) theorem applied to electrostatics predicts that a point charge (eg. an electron) exerts a force on itself. Not only that, but it is an INFINITE force.

You can read and download this excerpt from

Click to access nsbook_excerpts.pdf

Happy New Year!

Anonymous · 22 September 2023

Greetings Navier-Stokes Folks,

I thought you might be interested in the fact that my book titled "The Navier-Stokes Millennium Problem: An Unwelcome Solution" in finally in print. It is available for purchase at Amazon.com. Sorry, no free downloads. I have heard it said that people generally better appreciate what they get when they pay a little something for it.

Anonymous · 13 October 2023

It looks like NS is still unsolved. I am still hopeful it can be solved in the next 10 years. We can't be cranky at whoever tries to solve it genuinely. Watch this space. Cheers, Dr Hayes

Anonymous · 13 October 2023

If indeed blowup is the case then we can probably show this using the string method (one of my vixra notes) and a big enough computer. Dr Hayes



porton · 13 October 2023

You can hire computing power for rent in Golem blockchain. As far as I know, it's much cheaper than to find a real super-computer.

Anonymous · 13 October 2023

Thanks man. But if this is some important secret that will reveal itself over time, might as well let the computer experts see what they can do with this. I hope they would not do it in some disguised way though. Cheers, Dr Hayes.

Anonymous · 14 October 2023

Dear Prof Tao. Just wanted to say if in the future you are wondering and interested someday in the string method and what people can do with it, then I would love to hear your thoughts on this in any way, you are welcome. That is all I want to say for now. Kindest regards,
Dr Hayes

Anonymous · 17 October 2023

Hello Professor Tao and others.
Let u be a weak solution of the 3D Navier-Stokes equations with $u_0 \in H$.
Is the function $\|\nabla u(t)\|$ absolutely continuous for all $t \in [0,T]$?



ppalme · 15 November 2023

Is the millenium problem statement the problem?

Looking at the problem statement of the Poincaré conjecture: “Every simply connected, closed 3-manifold is homeomorphic to the 3-sphere” which was solved by Perelman.

Comparing it to the input state statement of the Navier Stokes millenium problem of the Clay Institute: “In three space dimension and time given the initial velocity field…”.
I really wonder how concrete this problem statement with these initial conditions are in order to find a counter-example or a prove for beeing smooth and globally defined?
What happens if the space dimension is either 0 or indefinite or the initial velocity is 0 or indefinite. Or can their be an indefinite field defined without a boundary?

Anonymous · 15 November 2023

Maybe study some basic calculus and physics then look at the problem statement again.

Anonymous · 25 November 2023

Too many inconsiderate people in the world. Just look at how people drive. Don’t discuss any proposed proofs on this site!

Anonymous · 6 January 2024

An exact general solution to NS is worth way way more than the reward CMI offers. There might as well be no prize.
Cheers, Dr Hayes.

Anonymous · 7 January 2024

Dear Professor Tao,
Can you please remove the thumbs up, thumbs down thing from your blog. This is nonsense. Otherwise I will refuse to communicate on this site anymore.
Cheers, Dr Hayes

Anonymous · 7 January 2024

Before posting nonsense you need to take a basic class in PDE to understand that almost all of them do not have “general explicit solution”.

Anonymous · 18 February 2024

But if he does this, how are all of those people cheering him on supposed to express themselves? — Certain not from their mathematical knowledge or understanding of NS and PDEs in general

Anonymous · 7 January 2024

You need to be smashed over the head with the handbook of mathematical logic…

Anonymous · 9 January 2024

Dear Prof Tao,
Never forget you are regarded by many as one of the smartest persons on the planet. I am only smarter than the average bear. HNY24. Dr Hayes

Dwight Walsh · 17 February 2024

Greetings Navier-Stokes Folks,

Yes, it's been a while since I last commented here, and I thought you all might be interested in an update about a certain counter-example to my proof of Navier-Stokes global regularity that was posted by Antoine in early February 2022. In reviewing his comments concerning this counter-example over the past few weeks, I found a serious error which renders it false. Specifically, through his lack of understanding of how to evaluate the improper integral in the 3D Greens function solution for Poisson's equation, he introduced a non-existent singularity into the solution which allowed the time integral of |grad p| to be infinite when it could easily be shown to be zero. Also, his comments such as "it looks like your are implicitly permuting the limits in t and in \epsilon in your reasoning" simply don't make sense at all.

Below is a link to the document that explains and disproves this counter-example since the web forum seems unable to render equations correctly.

Click to access proposed_counter_example.pdf

Anonymous · 19 February 2024

I should add here that back in February 2022, I made some statements claiming that the sample function q needed to be consistent with the NS equation in order to be a valid counter-example to my proof of NS global regularity. At this point, however, I recognize that these statements were not strictly correct and offer my apologies to Antoine. One must be very careful, however, in altering function definitions in someone else's work. Sometimes, dependencies are obscured by the choice of notation, and errors will occur, but I don't believe that's what happened in this case.

In this particular case, we have easily shown, without "implicitly permuting limits", that \Lambda(0,t) = 0 for all t < t_b. Antoine, however, claimed that \Lambda(0,t) approaches infinity as t approaches t_b. So where is Antoine's error? Well, I think this time I will give him a chance to find it on his own. It might be a great learning experience and possibly help him reconsider comments such as "I am afraid the situation is hopeless and there is no point in continuing writing here". I will help only if he specifically requests it.

Therefore, Antoine's counter-example is still invalid, but not for the reasons I gave in Feb. 2022. Understand, however, that there were problems with the site not rendering equations correctly, and unfortunately, I did not review his work as carefully as I should have.

Well, maybe I'll give Antoine a little hint to help him find his error. I recognized what the trouble was when he claimed that equation (117) (ie. \Lambda(0,t)) blows up as 1/(t-t_b)^2.

Dwight Walsh
mathguy@navierstokessmoothsolutions.com

Dwight Walsh · 25 February 2024

Greetings Antoine,

It's been about a week since I posted my discovery of the error in your purported counter-example to my arguments showing NS global regularity, and was wondering if you had any luck in finding it yourself — or possibly with the help of others on this blog giving me down-votes. As I recall, you claimed that the integral in inequality (118), or equivalently the function \Lambda(x,t), blew up at time T_b in your counter-example at x=0 whereas I rigorously showed that it vanished! That's quite a gap and I would hope that you are at least mildly curious as to how it happened.

Also, it wasn't difficult for me to extend by arguments to any blowup that is radially symmetric about the blowup point. This alone means that all counter-examples proposed thus far in this forum fail. I am afraid that finding a valid counter-example to my proof is much more difficult than concocting some fancy r dependence for the Q function. It's not something that can be done "on the fly" to brush me off.

In fairness, however, it should be noted that Antoine's counter-example provides a valuable lesson in the mechanics of a possible NS blowup solution. It turned out that while the scalar pressure p grows arbitrarily large at the blowup point as t approaches T_b, the pressure GRADIENT (which is what actually drives the NS equation) starts at zero and remains at zero at this point. This could be an important stepping-stone to understanding my more general proof of existence and smoothness of NS solutions.

> **Anonymous** · 26 February 2024
>
> Would any of you nay-sayers care to make a constructive comment? Using what I would call intermediate calculus, I was able to rigorously prove that no radially symmetric blowups could occur. I didn't bother posting it, however, since I knew it would not be read. But if you are somewhat interested, I am including this proof in the upcoming Third Edition of my book titled "The Navier-Stokes Millennium Problem: An Unwelcome Solution". It will be available through Amazon.com and should be coming out in just a few weeks. Now it will cost you $25 USD + shipping to order it, but what the heck? Someone else could be collecting $1 million.
>
> > **Jas, the Physicist** · 27 February 2024
> >
> > I will read it that sounds interesting. Shouldn't you format your solution to post in a journal though? I thought that was one of the requirements to have it peer-reviewed. I recommend making it journal-ready and submitting it there too.

> **Jas, the Physicist** · 27 February 2024
>
> Thank you for taking the time to comment. Where do I find your solution to read? Have you published it in a journal or do you have it on your website?
>
> > **Anonymous** · 27 February 2024
> >
> > Thanks for your inquiry. My solution is posted on the website

http://www.navierstokessmoothsolutions.com

Also, I have submitted this paper to well over a dozen peer-reviewed journals, and they all rejected it, usually for reasons having nothing to do with mathematical correctness (eg. backlogging, appropriateness, or scope of the journal). In only two cases did the reviewers even claim I had an error in the proof, and those cases were based on "straw-man" claims that I didn't make.

Anyway with the problems I've had with the journal editors and the unprofessional comments toward me on this forum, I have more or less given up on the idea of getting my work accepted into a mathematics journal. Instead, I have decided to write a book about these troubles, and see if I can get the book to sell. It won't bring in $1 million, but maybe my efforts won't go completely to waste either.

So, you are welcome to download my "unwelcome solution" paper and email me if you have any questions.

Anonymous · 27 February 2024

I forgot to give you my email address in my last posting, although it may in other places on this blog. It is

mathguy@navierstokessmoothsolutions.com

Anonymous · 28 February 2024

Hello Dwight, the problem I had with your proposed proof in my opinion, when I looked at it, is that you assumed at the start what you set out to prove. Others pointed out that logic problem too I recall. Cheers, Dr Hayes

Anonymous · 28 February 2024

Dear Dwight, the reason people on this blog are unfriendly to you is because you accuse mathematicians of conspiring to suppress your paper. When people point out issues, instead of engaging with their criticisms in good faith, you assume it is part of a conspiracy to smear you and your work.

Anonymous · 28 February 2024

Greetings Anonymous,

Could you please explain to me what it is that I set out to prove which I assumed at the start. Also, what criticisms have I not engaged in good faith? Keep in mind, of course, that "good faith" does not require me to accept false theorems and false counter-examples in support of such criticism. Also, profanity is only offensive. It does not enhance their criticisms in any way.

Anonymous · 29 February 2024

The main thing you seem to look past (that I have read here) is that there can be a counterexample to the logic of one step in your proof in isolation, that is not a counterexample to the whole proof. In other words, your proof has a step of the form "there is an object A with properties X and X implies Y". Someone objects "here is a case

B with X but not Y" which calls into question the logic of the proof. It does not matter if B cannot come about by the same construction from which you got A.

Anonymous · 1 March 2024

I believe I addressed this logic issue in my post on 19 February 2024 at 1:47 AM, and fully concur with what you are saying. I also showed, however, that Antoine's specific counter-example is still invalid since I was able to easily prove that the time integral of |grad p| was zero whereas he claimed it to be infinite. That's quite a gap and you might note that he has not responded to this discovery I made about his "counter-example".

From there, it wasn't difficult to extend my arguments to all proposed radially symmetric counter-example blowups. My paper, of course, takes on the most general case of a proposed blowup, and proves by contradiction that it can't happen.

Dwight Walsh

Anonymous · 29 February 2024

Hello Dwight, that logic problem was probably explained best to you by a comment by rbcoulter a while back. Dr Hayes

Anonymous · 1 March 2024

Dear Anonymous,

I thought I should clarify the general strategy of my proof for you and at least one other who claims I have no strategy.

First, we note that the solution exists and is smooth at least initially. This much is given. This means that it must stay smooth for at least some finite time T_b prior to any blowup. Otherwise, it wouldn't make sense to even talk about derivatives at t=0.

Next, I prove that the scalar pressure p must exist and be smooth in the same sense as the fluid velocity u for any time t that u exists.

I then insert these results into the NS equation to show that the solution u,p must be smooth for as long as they exist. This means that u,p must be smooth at least on the semi-open interval [0,T_b). By "smooth" here, I mean spatially smooth in the same sense as the functions u,p were assumed to be initially, though perhaps with different coefficients.

Next, we show that the time integral of |grad p| must be defined at all points x in R^3 and all times. This, however, has been a point of controversy. Some claim that I must establish an upper bound for this time integral or my claim that it exists is invalid. This, however, is not true. For example, consider the function f(x)=1/x^2 if x not equal to 0, and 0 if x=0. This function is in fact defined and exists for all values of x, but is obviously not bounded near x=0.

At this point, we prove by contradiction that a blowup cannot occur at t=T_b. That is, we assume that a blowup ensues at some point x_b as t approaches T_b. For this to happen, of course, a global maximum in |u|^2 must occur at x_b immediately before the blowup. Otherwise, the blowup would occur somewhere else first, thereby contradicting our definition of x_b. Since this must be a "smooth" blowup, however, grad(|u|^2) must vanish at

x_b and the Laplacian of |u|^2 must be zero or negative at this point. Also, contributions from Prof. Tao's "infamous" nonlinear convective term vanishes at x_b. When all of this is inserted into the NS equation, we find that the grad p force acting on the fluid is the only term that could possibly cause a blowup at x_b. Therefore, a blowup can occur at x_b only if the time derivative of grad p also blows up at this point. But we already showed that this quantity is defined for all t>0, including t=T_b. Thus we see that the conjecture of a blowup solution u,p leads to a contradiction, and therefore no such blowup is possible.

Note that it is NOT circular logic to use the fact that the solution exists and remains smooth on the semi-open time interval [0,T_b) in order to predict the limiting value of the solution as t approaches T_b. All such computations assume values of t NEAR T_b and not t AT T_b. Therefore, we are allowed to assume a smooth solution u,p to predict progession toward a possible blowup and to evaluate the limit of that solution as t approaches T_b. Then, since we have already shown that the time integral of |grad p| (a function called \Lambda(x,t) in the paper) must be finite for all finite values of t, we may conclude that the u,p are also finite and smooth for all x as t approaches T_b, thereby contradicting the conjectured blowup.

Anonymous · 1 March 2024

Hello Dwight Walsh, with all this…., lets then see if Prof Tao and or CMI accepts your proposed proof. Dr Hayes

Anonymous · 1 March 2024

Wonderful! But can you get a response (positive or negative) from either Prof. Tao or the CMI? I CAN'T!

Dwight Walsh

Anonymous · 1 March 2024

I should have mentioned this above, but the CMI does not accept Millennium Problem solutions directly from the authors. In order to collect the $1 million prize, the author must first get the paper published in a peer-reviewed mathematics journal of international repute, and the solution must not be successfully challenged for a period of two years. Then and only then will the solution be put before some CMI sponsored committee that makes the final decision.

It would sure be a big "feather in the hat", if Prof. Tao even takes a look at the paper. It seems, however, that he has gone on to "bigger and better" things and doesn't respond either to my email or forum messages. Also, he has been totally passive about whatever mathematical nonsense gets posted. At one point in time (January 30, 2022 at 2:29 PM), in the peak of my frustrations with him and a few others, I posted a message that ended with WAKE UP AND SMELL THE COFFEE!! — Still no response.

Some believe he is now intent on finding a counter-example to rather than proof of NS global regularity. In view of my work, it would indeed be interesting if he claims success.

Dwight Walsh

Anonymous · 1 March 2024

Greetings Dr. Hayes,

You seem to be interested in my paper. I might suggest, however, that if you would like to correspond further about it, send me an email to the address below. That way, we won't unnecessarily "hog up" this forum and there would be more privacy to our communication. In fact, I am set up for the GnuPG if you would like to further secure our communications. If so, you can download my public key from the website at

https://www.navierstokessmoothsolutions.com

Please note that while I strongly believe I have a solution to the Navier-Stokes Millennium Problem, I am always open for suggestions as to how to make my arguments more presentable. Also, I am willing to share credit and authorship with those who are able to improvement to the point where it does get accepted for publication. Unfortunately, my research career met with a very early demise, and I have been "out of the loop" for nearly 30 years. Therefore, I have been unable to post my work on even a moderated archive. They all require references of some sort. This is why I opened my own "archive" for my solution to this problem.

Regards,

Dwight Walsh
mathguy@navierstokessmoothsolutions.com

Anonymous · 2 March 2024

Come on, you thumb-downers! Let's hear something constructive!

Anonymous · 8 March 2024

Dear Dwight,

If you would like an endorsement of your paper on arxiv, please provide the endorsement code.

Regards.

Anonymous · 9 March 2024

Thanks for your suggestion, but to whom do I send this code? I will need an email address which you could send to my mathguy account posted above.

Anonymous · 10 March 2024

Just put the code here, I can endorse papers in Analysis of PDEs.

Anonymous · 10 March 2024

OK — Here is the endorsement code, although I would feel a little more comfortable if I knew who I was sending it to.

QNRJTK

Anonymous · 16 March 2024

Dear Anonymous,

Thanks for your endorsement of my paper, but arXiv rejected it anyway. So like usual, its "back to the drawing board".

Anonymous · 8 March 2024

Hello Dr. Hayes,

It's been about a week since you contacted me about seeing if Prof. Tao and/or the CMI accepts my paper. I am still very interested in this prospect but have not heard back from you. Remember that I did say that neither one responds to my email, so any "seeing" will have to be done by you. Please get back with me and let me know what you want to do. You can use my email address mathguy@navierstokessmoothsolutions.com .

Dwight Walsh

Anonymous · 26 March 2024

Greetings Dr. Hayes,

I am still anxious to see what Prof. Tao and/or the CMI has to say about my proposed proof. As I have pointed out already, the proof is located on my website at

https://www.navierstokessmoothsolutions.com

and neither Prof. Tao nor the CMI have responded to any of my questions since my website opened back in 2020. Now in your message of 1 March 2024, you stated "with all this, lets then see if…", and since you used the word "lets" (contraction for "let us"), I take it that you plan this to be a joint effort between you and me. Well, I've done my part by making my solution available and sending them the website address. They, however, have chosen not to respond, ostensibly because of my lack of clout with them. That is where you need to come in. You need to convince them that my work is worthy of a fair chance in the mathematics community as opposed to being immediately dismissed simply because I am not a recognized "expert". Also, it deserves better than "straw-man" claims that I did not make or false counter-examples constructed on the fly for the sole purpose of giving me the brush-off.

I look forward to your reply and possibly hearing from Prof. Tao and the CMI.

Dwight Walsh, Ph.D. Physics

> **Anonymous** · 5 April 2024
>
> Dr. Hayes,
>
> Well, its been over a week since I made the above posting and I haven't received a word from either you, Tao, or the CMI. I take it then that you have not been successful in getting a response from these people. I didn't think you would be.
>
> Dwight Walsh, Ph.D.
>
> mathguy@navierstokessmoothsolutions.com

Anonymous · 9 April 2024

Greetings Navier-Stokes Folks,

I thought you might be interested in the fact that the Third Edition of my book titled "The Navier-Stokes Millennium Problem — An Unwelcome Solution" has been released and is available for order at Amazon.com. This new edition fixes some errors in my understanding

of Antoine's counter-example primarily due to mis-rendering of mathematical expressions on the webpage. With these issues cleared up, it was much easier to make "short work" of his arguments and show that the function Lambda(x_b,t) at the blowup point is a constant zero at all times instead of Lambda(x_b,t) blowing up at T_b as he claims.

Also, I inserted an additional chapter showing that it is NOT circular logic to use the GIVEN smoothness of the initial conditions to establish a limit on Lambda(x_b,t) as t approaches T_b. It must be remembered that we are given that a smooth solution exists at all points x in R^3 and all times t up to (but not necessarily including) time T_b. Therefore, we can assume smoothness as part of the given when evaluating the limit as t to T_b. We do NOT claim smoothness at T_b until we show (in equation (119)) that Lambda(x_b,t) must be finite for ALL values of t.

Dwight Walsh

mathguy@navierstokessmoothsolutions.com

Anonymous · 11 April 2024

So did you sumbit to any journals? How's peer review going?

Anonymous · 12 April 2024

Yes — Dozens of them! In most cases, the paper was either rejected without further comment or for non-mathematical reasons (eg. backlogging, scope of journal, appropriateness of topic). In two cases, the rejections were based on "straw-man" statements that I did not make.

In one case, the reviewer claimed I did have a valid proof of existence and smoothness, but the paper was still rejected over (fixable) typo errors and formatting issues. I re-submitted the paper about a year and a half later after acting on the reviewer's comments, but it was rejected again — this time for being "outside the scope" of the journal.

So, you asked "How's peer review going?" Answer: Quite well if the plan is to block any possible proof of Navier-Stokes global regularity.

Dwight Walsh

Anonymous · 13 April 2024

I should have added in my last comment that there is something about peer review that we must all understand. While peer review may be an important mechanism for maintaining credibility of publications, it can morph into ***competitor review*** if large amounts of recognition and/or money is at stake. In other words, since the reviewers must be knowledgeable in the same areas as the author, some may feel it would be their best interest that the problem remain "unsolved" for the time being so that they themselves have a chance of becoming the "first" to solve it later. ***Something to think about!***

Now, I am not accusing anyone, but the absurdities of the false counter-examples and false theorems presented in this forum to discredit my work is making me wonder. The best I can say for individuals using these arguments is that they are ***not*** the experts they claim to be.

For those who would like to understand more specifically what claims have been made against my work, and sometimes against me personally, you may wish to order my book titled The Navier-Stokes Millennium Problem — An Unwelcome Solution. Readers might be intrigued to learn that "theorem" used by Anonymous, Antoine Deleforge, and Gandhi Viswanathan (which I call the "ADV Theorem") to allow the possibility of a blowup at time T_b also predicts the infinite acceleration of a charged particle (eg. an electron) by its own self-field ***which is physically impossible***. This, I believe constitutes a *global error* in the ADV Theorem according Terence Tao's writings.

Dwight Walsh

mathguy@navierstokessmoothsolutions.org

Anonymous · 29 June 2024

Correction on email address. The correct address is:

mathguy@navierstokessmoothsolutions.com

Anonymous · 1 June 2024

humbly, in case you want to look https://math.stackexchange.com/questions/4925083/ask-questions-is-this-an-ok-answer-to-the-nse-smoothness-problem-i-have-a-m a fresh perspective containing your work.

Anonymous · 21 December 2024

Thanks for the awesome write up and framing!! I saw from the comments about knot theory. If there is any connection between knot theory and the creation of neutrinos, I would like any direction pointing from anyone here.

Since neutrinos are made in the radiative zone of the Sun, which is a high turbulent zone, the knots created could be from the mixing/stretching of knots of electrons during nuclear fission.

I am an enthusiast and would appreciate anything about knot theory and atomic structure.

Anonymous · 10 September 2026

So a dumb question in light of the recent AI progress (ignoring the controversy and assuming the AI counterexample is correct). So now there's a counterexample in R^3. Does the same counterexample necessarily work in the 3D torus? I'd guess yes since it's a finite time blowup with apparently some local stability, so you can suitably scale everything, but maybe I'm missing something. Is there something non-trivial that could leave the R3/Z3 setting still open? In an alternate world where someone proved no blowup in R3, could R3/Z3 then still be open or vice versa? Thanks.

Anonymous · 21 September 2026

So can I ask the current status of part a of the problem? Is it still open? It asks for a proof of no blowup in the absence of forcing, while part c (the part recently solved) was to show a blowup when forcing is allowed. The solution (and its antecedents by Córdoba and Martinez-

Zoroa) use the forcing term in an essential way. I'm guessing that type of forcing is what you meant by a "Maxwell demon" but maybe I'm wrong.

There's now known breakdown in unforced Euler. Could be that the proof doesn't work for NS because the slightest amount of viscosity is enough to damp out the breakdown?

 **R. Kh.** · 25 February 2019

As this is noticed by M Struve: O. A LADYZHENSKAYA would not have agreed with this formulation of the problem -corresponding regularity question – for the Sixht Problem of the Millenium – for 0. A – the MAIN PROBLEM IS EXISTENCE AND UNIQUENESS – OBVIOUSLY

R. KH; Z

 **victorivrii** · 25 February 2019

I am really puzzled by Mr Zeytounian comments (as long as I can decrypt his writings).

1) Indeed, Ch. Fefferman made an omission in the official description https://www.claymath.org/sites/default/files/navierstokes.pdf , not mentioning that in the periodic framework not only the gradient of the pressure but the pressure itself must be periodic.

However, this hardly warrants "very badli" statement, Furthermore, this omission was corrected on the 6th page of the official description [well, it was a patch; the original description originally contained 5 pages, with the 5th page having only few lines and then the 6th page was sloppily added 61/2 years later: it follows from it appearance that it was simply inserted into pdf file and looking at the shape of symbols one can see that it's author was a German mathematician)

2) This problem was not resolved as far as I know.

3) For Olga Ladyzhenskaya definitely the main question was the existence and uniqueness; however both of them are known, but in different classes. If the conjecture of the global regularity holds then the existence and uniqueness would be established in the class of smooth functions.

However, if the conjecture of the global regularity fails, the next problem indeed would be to find the class in which both existence and uniqueness hold.

>  **victorivrii** · 25 February 2019
>
> A bit lame analogy: for $u_t + (u^2/2)_x = 0$ uniqueness holds in the class of continuous functions, and the global existence holds in the class of bounded functions. However in the former class there is no global existence and in the latter there is no unicity.
>
> It was established 50+ y.a. (if I am not mistaken) that in the class of bounded functions there are both global existence and uniqueness, if to this equation dissipativity condition $(u^1/2)_t + (u^3/3)_x \le 0$ is added.

 **Bob McCann** · 8 May 2019

Physics perspective: Navier-Stokes is a model that applies to bulk fluids where the scale size is large enough that individual particle motion is not relevant. Nearing that scale it becomes necessary to switch models to something including distributions and collisions, e.g., the

Vlasov or Boltzmann equations. As an exercise in pure math, the Millennium problem is an excellent challenge for extending the reach of our analysis tool set. Here is an idea I have used in Plasma Physics. If we can bound the problem between two simpler problems that are better behaved, it may be possible to constrain the more difficult behavior. For instance, if we use a Fourier expansion with index "n" and the behavior of the first few terms is well behaved but high order terms are too difficult, then expanding in (1/n) and moving to the limit of n->infinity sometimes results in a well behaved equation that bounds the small scale behavior. Using analytic matching, analytic continuation, or the Wiener-Hopf technique (depending on the character of the small and large scale equations) can then provide tighter bounds on the more difficult intermediate scale behavior.

ferhatmohamed Ferhat · 16 May 2019

after 10 years of effort i think this puzzle requires to develope a new variational methods

Colin McLarty · 17 July 2020

Terrence: just a question about terminology. for use in a talk to historians and philosophers of math. I think that when you say "exact and explicit" you mean something like a closed form solution (where "closed form" is also not a precisely defined expression). Are you happy with using "exact solution" for a solution which is in no sense closed form?

> **Anonymous** · 17 July 2020
>
> I'm not Terence but I'd expect it to be possible to make a formal definition in terms of computable reals. E.g. an infinite series solution that converges to within epsilon after $f(\lfloor \frac{1}{\epsilon} \rfloor$ terms where f is computable using an explicitly stated algorithm, seems explicit to me.
>
> 
> **Terence Tao** · 17 July 2020
>
> Yes, in this text I mean solutions that are both exact (not approximate) and explicit (describable in terms of operations simpler than "solve the Navier-Stokes equations"), with the point being that it should be an easier task to determine the regularity or singularity of such explicitly described solutions than arbitrary (exact) solutions to the Navier-Stokes equations. On the other hand, explicit approximate solutions are typically not, by themselves, enough to settle these sorts of questions: just because an explicit approximate solution exhibits an explicit singularity at a point, it is often not feasible to obtain a sufficiently strong perturbative theory to establish that there is also an exact solution close to the approximate solution that also exhibits a singularity. (Perturbation theory tends to only work well when the functions involved are relatively small, but singular solutions by their nature tend to be quite large in various norms. One can sometimes get around this by working in multiple norms simultaneously, but it can be quite delicate.)

Christian · 20 August 2020

Dear Prof. Tao,

I wonder if it's possible to explain your words "the behaviour of three-dimensional Navier-Stokes equations at fine scales is much more nonlinear (and hence unstable) than at coarse scales" with estimates using $u^(\lambda)$ and $u.$

Many thanks for your effort.

Daniel · 4 December 2020

A proposed solution to the NS problem is at https://vixra.org/pdf/1911.0343v8.pdf

Anonymous · 27 December 2020

I recently came across this article, describing a fluid mechanics experiment (i.e. with actual water tanks and stuff) that seems to have uncovered some previously unknown conservation laws in near-turbulence that might extend into the turbulent regime. It probably doesn't reach the NS regularity question, but I wonder if it's been studied mathematically and is considered mathematically significant:

https://www.quantamagazine.org/an-unexpected-twist-lights-up-the-secrets-of-turbulence-20200903/

That's a popular-level article but it links to the research article that was in Science Magazine a while earlier.

Robert · 17 February 2021

If the Navier-Stokes problem cannot be solved by arguably the greatest mathematical mind on Earth, could that mean that the premise, i.e., the formulation of the Navier-Stokes equation is in itself incorrect?

Can anyone tell me whether proving that the premise is incorrect would be considered sufficient as a counter example?

In Professor Tao's article he speaks of:
"Discover a new globally controlled quantity which is both coercive and either critical or subcritical"

If this was discovered, then this would render the premise incorrect since this new globally controlled quantity is not in the equation. Am I right or wrong here? I am just a layman in the field of mathematics and physics but I like to solve problems.

> **Anonymous** · 17 February 2021
>
> 1. It is a valid interesting problem regardless of whether Terry (or any other expert) can solve it.
> 2. A hypothetical new globally controlled quantity would already be "in the equation," just not yet identified by mathematicians.
>
> > **Dwight Walsh** · 28 December 2021
> >
> > As "just a layman", you seem to have much insight into what's happening in regard to this millennium problem. To answer your question, I believe the problem is correctly formulated in the Official Problem Description which came out in the year 2000. Traditionally, however, actual attempts at proving existence and smoothness of solutions to the NS equation have generally started with the work of the French mathematician Jean Leray in 1934 in which he showed that a certain class of "weak" solutions to the NS equation always exist. Unfortunately, it seems that this tradition has carried over to the

point to where the mathematical establishment will not view any proposed solution to the NS millennium problem as credible if it is not based on Leray's work.

Although Prof. Tao may have a great mathematical mind and genius level IQ, I believe he has followed with this tradition and made some bad mistakes in the process. The problem with his arguments is that he fails to constrain div u = 0. In each of his online forums and presentations on the NS equation, he makes an assertion at some point that we can neglect the scalar pressure since it only keeps div u = 0. Unfortunately, however, "neglecting" p and grad p is the same as setting them equal to zero, and from that point on, he no longer mentions p, grad p, or div u. Instead, he only makes "re-scaling" arguments with the nonlinear transport term to claim that energy could be redistributed into successively smaller scale sizes, thereby resulting in a blowup in energy density. In general, his analysis is vague, intuitive, and doesn't prove anything.

By failing to constrain div u = 0, Dr. Tao's arguments support only an under-determined version of the problem where there are more unknowns than equations. This, of course, gives him (and the entire mathematics community) great freedom to find "counter-examples" to existence and smoothness that are not actually solutions to the NS equation. In this case, it would be ABSOLUTELY IMPOSSIBLE for the mathematics community to recognize a valid proof of existence and smoothness of solutions to the NS equation.

Therefore, while I believe the actual formulation of the NS millennium problem is correct, there have been serious errors in pursuing solutions. And, unless the mathematics community approaches this problem with open minds, it won't be solved for a long, long time.

 **Dwight Walsh** · 29 December 2021

I forgot to tell you in my above posting that I started my own chain of comments which are a little way down the page. In particular, I posted my own solution to the NS millennium problem to the website at

http://www.navierstokessmoothsolutions.com

As you may imagine, however, peer reviewed journals don't seem to want to touch it with ten foot pole, probably because my approach of leaving the NS equation intact breaks with tradition. Thus far, it has been rejected by about a dozen such journals. A few readers have suggested that I post the paper to a moderated archive, but to do that, I still need to be sponsored by a reputable mathematician, and after 30 years of no professional contacts, no such individual would even know me.

And so, the mathematics community moves forward with their attempted "solutions" to this millennium problem unaware that their most impressive discoveries (including the fact that they can "prove" existence and smoothness in two dimensions but not three) are merely artifacts of their failure to properly account for p and grad p.

 **Polihronov** · 25 April 2021

Dear Prof. Tao,

Thank you for a very informative blog, I have read it many times!
I wanted to ask a question about the viscosity and NSE scale-invariance. In the blog you

discuss one of the strategies and then mention that pressure gets rescaled too; then you have said that the viscosity ν remains unchanged (under rescaling).

Would you be able to share your thoughts on the meaning of this. In the Millennium problem, how is the viscosity ν set to behave under rescaling.

My attempt to understand it:
In the dimensionalized NSE, the viscosity ν is a fixed constant; however, since it has dimension/units; then, ν must be rescaled when the NSE is rescaled because the units are rescaled.

Coming to the nondimensionalized NSE: here the viscosity ν is replaced by the Reynolds number Re. Since in the Millennium problem, the viscosity is a fixed parameter, it would then follow that the Reynolds number is a fixed number as well.
About the scalability of Re:
It appears that one can look upon the nondimensionalized NSE in its own merit, and written in the nondimensionalized variables x, y, z, t, it can be thought of having scale-invariance in these variables. Leray's self-similar solutions seem to suggest this as well. They are written for the nondimensionalized NSE; their name, "self-similar" refers to rescaling. Their rescaling always yields the proper rescaling of the velocity u and pressure p. In other words, x, y, z, t must be scalable in the nondimensionalized NSE. Is then the Reynolds number scalable just like the viscosity is scalable?

Would you be able to comment on this. Would there be a mathemathical reason to set ν or Re as non-scalable. I'm wondering, since there is no unified viscosity theory, and the molecular origins of viscosity are unclear.



Terence Tao · 26 April 2021

Mathematically, there is no distinction between a "rescaleable" or "non-rescalable" quantity (or a "dimensional" or "dimensionless" quantity), as long as we are describing quantities such as position coordinates x_1, x_2, x_3, time t, and viscosity ν in terms of real numbers. (If one wants to describe such quantities using $\mathbf{R}$–torsors instead, then there are distinctions of this type, but this is not usually how the mathematical equations are formulated.)

In particular, one is free to apply any (smooth, invertible) change of variables one wants to the fields u, p to transform the Navier-Stokes equation to another equation. Sometimes when one does so, the transformed equation is the same as the original equation (with no change in the viscosity parameter); in other cases, it transforms to a Navier-Stokes equation with a different viscosity parameter; and in yet other cases it transforms to a completely different equation which is not of Navier-Stokes type. All three such types of transformations are mathematically useful, though in any specific context one such transformation may be more useful than another.

More specifically: for any $\lambda > 0$, the change of variables $(u, p) \mapsto (u^{(\lambda)}, p^{(\lambda)})$ defined by

$$u^{(\lambda)}(t,x) := \frac{1}{\lambda} u(\frac{t}{\lambda^2}, \frac{x}{\lambda}); \quad p^{(\lambda)}(t,x) := \frac{1}{\lambda^2} p(\frac{t}{\lambda^2}, \frac{x}{\lambda})$$

transforms solutions (u, p) of the Navier-Stokes equation at a given viscosity ν to solutions $(u^{(\lambda)}, p^{(\lambda)})$ of the Navier-Stokes equation with the same viscosity ν. This gives

a scale-invariance to the set of solutions to such an equation (with fixed viscosity ν) which is very useful for understanding the structure of the space of such solutions, as is discussed in this post.

More generally, though, one can consider dimensional scalings $(u,p) \mapsto (u^{(T,L)}, p^{(T,L)})$ for any $T, L > 0$ defined by

$$u^{(T,L)}(t,x) := \frac{L}{T} u(\frac{t}{T}, \frac{x}{L}); \quad p^{(T,L)}(t,x) := \frac{L^2}{T^2} p(\frac{t}{T}, \frac{x}{L})$$

which transform a solution (u,p) to Navier-Stokes at a given viscosity ν to a solution $(u^{(T,L)}, p^{(T,L)})$ to Navier-Stokes at viscosity $\frac{L^2}{T}\nu$. The previous scaling is the special case in which $(T,L) = (\lambda^2, \lambda)$. This more general scaling is useful when one wants to analyse a specific time scale T and a specific length scale L, or if one wishes to investigate limits when the viscosity tends to zero or to infinity. It is not used in this post, but this does not mean that it is not useful in other contexts.

As an example of the third type of transformation, the transformation $(u,p) \mapsto (u,\omega)$, where

$$\omega(t,x) = \nabla \times u(t,x)$$

is the vorticity field, transforms the usual formulation of Navier-Stokes into the vorticity-stream formulation (consisting of the vorticity equation and the Biot-Savart law), in which the pressure is absent. This is a useful formulation for many purposes (particularly in two dimensions, when vorticity becomes a scalar transported by the velocity field).



Polihronov · 27 April 2021

Dear Prof. Tao,

Thank you for your reply! Yes, then the solutions

$\latex u^{(\lambda)}(t,x):=\frac{1}{\lambda}u(\frac{t}{\lambda^2},\frac{x}{\lambda})$;
$\latex p^{(\lambda)}(t,x):=\frac{1}{\lambda^2}p(\frac{t}{\lambda^2},\frac{x}{\lambda})$

would fall into the scenario of Energy-supercritical NSE as you had said above.
These solutions are scalable; they are well-known as Leray's self-similar solutions. One arrives at them when the viscosity $\latex \nu$ is not changed by rescaling, as you mentioned. Their form is

$\latex u(t,x):=\frac{1}{\sqrt{t}}u(\frac{x}{\sqrt{t}})$;
$\latex p(t,x):=\frac{1}{t}p(\frac{x}{\sqrt{t}})$
and are known not to be strong solutions (Cannone and Planchon /1996). It would not be feasible then to use Beale-Kato-Majda to study them, since they are not strong.

Also, the Millennium problem requires smoothness of solutions at $\latex t=0$. Leray's self-similar solutions are not defined at $\latex t=0$ and are divergent at the origin (or, along the coordinate axes).

To sum up, if viscosity $\latex \nu$ is not changed by rescaling, we arrive at an energy-supercritical NSE with scalable solutions that are not strong solutions; and are divergent at the initial moment and along the coord. axes. Such approach to show NSE regularity would be bound not to succeed.

If the viscosity $\latex \nu$ is allowed to scale, all of the above issues are resolved. Namely, smoothness at $\latex t=0$, along the axes and criticality are no longer a problem. Beale-Kato-Majda show strong solutions at $\latex T=\infty$. Would this be an acceptable approach to the Millennium problem?

 **Polihronov** · 3 May 2021

Dear Prof. Tao,

If I could add this detail –
The solutions $u^{(\lambda)}(t,x) := \frac{1}{\lambda}u(\frac{t}{\lambda^2},\frac{x}{\lambda})$; and $p^{(\lambda)}(t,x) := \frac{1}{\lambda^2}p(\frac{t}{\lambda^2},\frac{x}{\lambda})$ form a group of scaling transformations. It is a Lie group, and the solutions $u^{(\lambda)}; p^{(\lambda)}$ have important invariant properties. From these properties, one can derive their functional form

$u^{(\lambda)}(t,x) = \frac{1}{\sqrt{t}}F(\frac{x}{\sqrt{t}})$

and

$p^{(\lambda)}(t,x) = \frac{1}{t}f(\frac{x}{\sqrt{t}})$

where F and f are arbitrary functions. Here $u^{(\lambda)}$ and $p^{(\lambda)}$ are derived under the assumption that the viscosity ν is unchanged under scaling. One can see that u, p are not defined at $t = 0$ and are not smooth functions.

It is true that supercriticality is an obstruction to proving NSE regularity; although supercriticality comes to view at a deeper level of analysis. It seems that one has more urgent issues coming from the NSE scaling transformation itself. What would then be the justification of analyzing NSE regularity through $u^{(\lambda)}$, $p^{(\lambda)}$? How could they be useful for understanding the structure of the space of NSE solutions.

>  **Terence Tao** · 4 May 2021
>
> The rescaled solutions $u^{(\lambda)}, p^{(\lambda)}$ are not required to equal the original solutions u, p; typically, they are a different solution to the Navier-Stokes equations, with a different set of initial data (a rescaling of the original set). So they do not need to take the scale-invariant form you describe.

 **Polihronov** · 5 May 2021

The Lie scaling transformation preserves the NSE; If ν is to remain unchanged, the transformation is

$u^{(\lambda)} = \frac{1}{\lambda}u$

$p^{(\lambda)} = \frac{1}{\lambda^2}p$

$(x_1,x_2,x_3)^{(\lambda)} = \lambda(x_1,x_2,x_3)$

$t^{(\lambda)} = \lambda^2 t$

where $a^{(\lambda)}$ denotes the transformed entity a.

Then

$\left(\frac{1}{\sqrt{t}}F(\frac{x}{\sqrt{t}})\right)^{(\lambda)} = \frac{1}{\sqrt{\lambda^2 t}}F(\frac{\lambda x}{\sqrt{\lambda^2 t}}) = \frac{1}{\lambda}\frac{1}{\sqrt{t}}F(\frac{x}{\sqrt{t}}).$

u and p transform as expected. $u^{(\lambda)}$, $p^{(\lambda)}$ are rescaled versions of the original u, p.

 **Polihronov** · 10 May 2021

I would add this as well –

The questions arise due to the complexity of Lie's invariant theory. When speaking of invariants, the assumption is

$u^{(\lambda)} = u,$

while it is not the case; the correct invariance is

$u^{(\lambda)} = \frac{1}{\lambda} u.$

Understanding differential invariants in detail is crucial in them being highly relevant to the NSE.
This is true specifically on the functional form of the various differential invariants.

Charles L. Bouton, a Harvard Math professor has written an excellent article on this subject:

Invariants of the General Linear Differential Equation and Their Relation to the Theory of Continuous Groups

American Journal of Mathematics, Vol. 21, No. 1 (Jan., 1899), pp. 25-84

http://www.jstor.org/stable/2369876

This article settles the matter; and of the importance of the viscosity ν in NSE regularity. Which importance goes even further in posing the question on how ν behaves at various scales in fluids.

 **J. Yong** · 8 July 2021

There is a paper by A. Tsionskiy and M. Tsionskiy with the title "Existence, uniquess and smoothness of a solution for 3D Navier-Stokes equations with any smooth initial velocity. A priori estimate of this solution" published in Adv. Theor. Math. Phys.19 (2015), 701-743. Is there any comment on that? Does it mean that the problem has been solved? Or the paper has some problems?

 **Anonymous** · 8 July 2021

It's clear at a glance that this paper has basically no chance of being correct. Surely no expert has wasted their time checking it.

 **Grigori Rozenblum** · 30 November 2021

This reference to Tsionskiy is fake. In the Journal, on these pages, different papers are published. https://www.intlpress.com/site/pub/pages/journals/items/atmp/content/vols/0019/0003/index.php

 **Dwight Walsh** · 6 September 2021

In January 2020, I came up with a proof of existence, smoothness, and uniqueness of solutions to the incompressible, zero driving-force Navier-Stokes equation, consistent with statement (A) of the Official Problem Description for the Navier-Stokes Millennium Problem. The interesting part is that I was able to show this result using only vector analysis, Poisson's integral, and some elementary and partial DEs. Thus far, however, my proof has been

rejected by eight journals for reasons not related to mathematical correctness. In fact, one reviewer stated “the proof is so easy even an undergraduate reader could follow”. Now, my Ph.D. is in physics and not mathematics, but I have worked some with the Navier-Stokes equation in one of my research project while I was still employed. Also, it sometimes takes a newcomer to shed new light on a problem that has had the experts stumped for a long time.

In my paper, I was able to show that any solution of the NS equation that is initially smooth must remain smooth for as long as it exists. Therefore, if a blowup occurs in finite time, it must be a “smooth blowup” where the solution remains spatially smooth at all times, even when approaching blowup. Also, this blowup point must be a global maximum, and I was able to show that the fluid velocity at such a point cannot reach infinite velocities even in infinite time. Thus, this smooth blowup cannot happen and the solution remains smooth for all time.

If you would like to review my paper, you can download it at

http://www.navierstokessmoothsolutions.com

since you may not see this paper in a journal for good long time.



Anonymous · 8 September 2021

There are some problems with the assumptions you make about t_b, x_b, and K_b; however the biggest flaw with this paper is that you assume integral |grad p| dt is finite up to the blowup time. This is only necessarily true for every t *less* than the blowup time as it is a supercritical quantity (see Tao’s article above).



Anonymous · 8 September 2021

*subcritical, sorry



Dwight Walsh · 11 September 2021

I do not assume that the time integral of |grad p| is bounded up to the blowup time. I PROVE IT. Please re-read the section titled “Existence of Pressure Gradient Integral over Time”, and let me know if you find any errors in the proof.



JacquesM · 7 February 2022

This message alone is enough for the people here to stop reading your paper. You asked people to review it, but anytime somebody points you a flaw or at the very least a problem, you insist that they should read more carefully. Send your paper to Annals of Mathematics or Acta Mathematica if you really think you have a proof of what you claim. Simple as that.



Dwight Walsh · 9 January 2022

I should have recognized this months ago, but your very use of the term “blowup time” implies that you are assuming there is a blowup in finite time. Isn’t it rather difficult to prove a smooth solution when a blowup is one of the “givens”?



Dwight Walsh · 11 September 2021

From the comments I have received from various sources, I sense there is one crucial point I need to emphasize. That is, this paper shows that the time integral of |grad p| is actually BOUNDED on the ENTIRE open interval (0,T). The first step in proving

this is to show that the spatial integral of the function q over R^3 (or any subset thereof) is bounded above. This is done in the discussion preceding inequality (108). In this discussion, the only assumption made is that the u_i remain smooth according to inequality (79). No mention is made about how large the u_i values become. Therefore, inequality (108) remains valid even as our smooth solution approaches blowup. Then, since all spatial integrals of q are bounded (regardless of the size of the u_i), it follows that the time integral of |grad p| is also bounded over all points in the open interval (0,T). This, as we show later, becomes the final argument proving that a "smooth blowup" is not possible.

 **Dwight Walsh** · 12 September 2021

Would any of you "thumbs down" folks care to leave a constructive comment on what you find erroneous about my paper, or do you not understand the problem that well yourselves?

 **victorivrii** · 12 September 2021

1. In your "proof" that |\nabla p| is bounded, after (117) you wrote "Since the function q(x, t) is bounded and continuous" . Why this function, defined by (112) is bounded and continuous?

2. There is a simple criteria to distinguish proofs which may be correct, from those which are wrong for sure: the latter "work" in any dimension. "Elementary methods" like yours, which do not use even imbedding theorems, definitely fail this criteria.

 **Dwight Walsh** · 13 September 2021

I did not claim that |grad p| is bounded. I stated and proved that the time integral of |grad p| is bounded over the entire open time interval (0,T_b) for which the fluid velocity remains smooth. This, of course, implies that a "smooth blowup" is not possible.

In regard to your second comment, I quite frankly don't know what you are talking about. The fact is that despite all of the journal rejections and less than positive feedback my paper has received on this site, at least one reviewer seemed to believe I had a valid proof. You, however, have not attacked my paper with a single factual statement. Also, I have seen lots of theorems proven but have never heard of "imbedding theorems" before.

 **Gandhi Viswanathan** · 13 September 2021

I am a theoretical physicist, not a professional mathematician. So my knowledge of real analysis is admittedly limited. But even I know the importance of the Sobolev embedding theorems. If you have not heard of Sobolev spaces, then it will be very difficult for you to be taken seriously by researchers in the field. Try to see first whether your "proof" works for simpler model systems, such as the 2D viscous quasi-geostrophic equations. Or even simpler models, such as the 3D or even 1D Burger's eq. with a fractional laplacian. And if others don't understand or agree with your point of view, try not to blame them. Instead, try to see their point of view. After all, professional mathematicians have devoted many years of their lives to learn their trade.

Daniel Hayes · 29 November 2021

Why would a proposed proof of global regularity for NS in any dimension be wrong for sure?

Daniel · 12 September 2021

Equation (6) is wrong

Dwight Walsh · 13 September 2021

What do you mean “equation (6) is wrong”. It’s part of the problem description. If you have issues with that, you will need to take them up with Charles Fefferman or the CMI.

Anonymous · 13 September 2021

Asking renowned mathematicians on their website to review a result is simply bad form – rather you should submit your result to a journal

victorivrii · 13 September 2021

Actually, no: the partial derivatives in (6) should be with respect to x,y,z respectively.

Dwight Walsh · 13 September 2021

In response to various naysayers about my work with the NS Millennium Problem, I have posted a new version of my paper to my website at

https://www.navierstokessmoothsolutions.com

In this new version, I went through the section titled “Existence of Pressure Gradient Integral over Time” with a fine-toothed comb and believe I have left little doubt that the time integral of |grad p| is not only finite, but also is bounded uniformly over the entire open time interval (0,T) by a single upper bound. Every statement I make in this section about the time dependence of q and |grad p| is backed by previously proven equations, and the result is that that the time integral of |grad p| is bounded regardless of how large the fluid velocity becomes. I simply don’t know how I can possibly make the proof any clearer.

Anyway, those who are genuinely interested but were unable to follow this section in my previous paper are welcome to download this new version and see if they get a better understanding.

Also, I apologize to victorivrii and Ghandi Viswanathan for coming up with a proof that does not use Sobolev embedding theorems and for which the result falls out nicely in three dimensions.

Finally, in regard to the comment from Anonymous, I must say that just because I am not renown does not mean I am not capable. My career, while it lasted, was primarily with the US defense industry, and therefore I had little opportunity to publish, especially in peer reviewed journals. One of my research project did involve the NS equation, but it only resulted in a few conference papers. When the “Cold War” ended in the early 1990s, so did my research career, and I have been professionally ignored ever since that time. I

submitted my paper to eight journals so far, and have received nothing but flat rejections or possibly rejections having nothing to do with mathematical correctness, which seems quite odd for a problem of such “importance”.

Now, as I see it, for someone with my degree and capabilities to be so ignored for the past thirty years is also “bad form”. So what do you suggest I do!?



victorivrii · 14 September 2021

In you new version the second equality in (117) is obviously wrong. Please find the error by yourself.



Dwight Walsh · 14 September 2021

Thanks so much!



Dwight Walsh · 19 September 2021

For those still interested in my work and possible contributions to solving the NS millennium problem, I just made a major breakthrough in regard to the time integral of |grad p|. As I stated in previous comments, much of the negative feedback I received concerning my proposed solution to the Navier-Stokes Millennium Problem has been claims that I have not adequately proven that this time integral is finite and in fact uniformly bounded over the entire open time interval immediately preceding the blowup. That is, there may be agreement that the time integral of |grad p| is finite at all times prior to the blowup, but may not be bounded as t approaches the blowup time. Well, in my break-through, I was able to show that the time integral of |grad p| is in fact bounded above by E_0 / (2\pi\nu) on the entire time interval 0 < t < T_b, where E_0 is the initial energy of fluid motion, \nu is the normalized viscosity coefficient of the fluid, and T_b is the blowup time. Now, I still believe my previous versions were generally correct, though perhaps not as well pulled together. In view of how nicely my new writing of the proof came together with arguments that are easy to follow and verify along with a clear, simple final result; I would dare claim that the "naysayers" would have one monster of a challenge in debunking it.

Again, for those interested in my work, and even those who would like to shoot it down (LOL), you can download my latest version of the paper (Version 26) from my website at

https://www.navierstokessmoothsolutions.com

Now, I acknowledge that Dr. Tao far surpasses me in mathematics education and intellect as do probably most others making comments in this blog. Nevertheless, a proof is a proof, regardless of whether elementary or highly advanced concepts are used, or what educational background the author has. Also, arguing that the time integral of |grad p| cannot be uniformly bounded (with a single upper bound) because it is a "supercritical" quantity is invalid since this term applies only to "weak" solutions of the "averaged" NS equation instead of actual solutions to the original one. In general, I believe that the whole idea of "weak solutions" has been far overused, and too many "important" results are actually artifacts of adjustments made to the equations in order to arrive at "solutions" of some sort. Discussion and sometimes an entire lingo then centers around these results rather than those of the original equations, and individuals trying to bring new insight to the problem are not well received by the established "experts". This, I

sense, is what I am up against and has very much curtailed any actual progress in solving the NS millennium problem.

> **Anonymous** · 20 September 2021
>
> In order to reduce the (understandable) doubts of Math journals editors toward an "outsider" mathematical credibility, you may try to register to a university as a mathematics student, and then (as a mathematics student) try to gradually publish your results as a series of well-written small papers (with "less ambitious" titles).

> **Anonymous** · 20 September 2021
>
> This inequality you claim is dimensionally inconsistent...normalizing nu=1, integral grad p dt has dimensions of length^-1, while the energy has units of length. This issue is the essence of Tao's above article about supercriticality.



Dwight Walsh · 29 September 2021

In my most recent effort to established an upper bound on the time integral of |grad p|, I came up with some interesting results. While I found it impossible to actually pin down a value for this upper bound (which I would expect depends on initial conditions), I believe I was able to prove its existence and that it must be finite for all t and as t approaches infinity. This would in turn prove existence and smoothness of solutions to the NS equation for all t > 0.

Anyway, those interested are welcome to download my most recent paper from the website at

http://www.navierstokessmoothsolutions.com

and check on the arguments leading to equation (113).



Dwight Walsh · 30 September 2021

In view of the fact that my postings seems to have drawn some highly unprofessional responses, I won't be posting anything further unless specifically requested by another participant of this thread. My website at

http://www.navierstokessmoothsolutions.com

is up and running again after being down for a few days, and those who are genuinely interested in my ideas about the NS Millennium Problem are welcome to visit the site, download my most recent version of the paper, and send me their comments with the posted email address.



Polihronov · 12 September 2021

Sorry to interject, I did not receive constructive feedback either...

Perhaps Prof. Tao could elaborate what, as an example, would be any other forms of $u(x,t)$.

For the folks that left a "thumbs up", I appreciate this; if you would comment on what part of the argument appealed to you as correct; it would be a great part of this discussion.

 **rbcoulter** · 20 September 2021

I can't follow the reasoning starting at (26). It seems as all the effort in the proof is in showing that there exist a system of curves (meeting the Clay criteria) that bound the solution (and its derivatives) of the Poisson equation. However, I can't see how this proves the time evolution of a flow scenario that is initially bounded by the Clay criteria, and stays bounded for all t. It is though some sort of circular logic is going on which involves assuming the derivatives in the NS equations exist for all time. Most of the effort in the paper seems dedicated to constructing the bounding curves. This is trivial task if one first assumes the derivatives don't blow up.

 **victorivrii** · 21 September 2021

Out of 7 Clay Millenium Problems one was solved and 6, including NSE, remain unsolved. There were several proofs published by well-established mathematicians and containing subtle (or not so subtle) crucial errors, several pranks (https://vixra.org/author/jorma_jormakka, for NSE the author exploited the lack of peridicity condition with respect to spatial variables for a pressure in the original Clay text) and many "solutions" and "proofs" written by cranks and freeks.

It is not dissimalar to Fermat's Last Theorem: even after Wiles' proof there is a flow of simple and elementary "proofs".

One did not need a crystall ball to predict this side effect of the Millenium Problems project.

😀

 **Dwight Walsh** · 21 September 2021

I acknowledge I made a goof in equation (115) in Version 26 of the paper, and it will probably be a few days before I have a corrected version posted. However, I still believe I had this part correct in previous versions, but with the repeated rejections, I keep trying to find ways to make my arguments shorter and simpler without compromising validity. But, as we can see, I make mistakes like everyone else. Anyway, I am still working on clarifying the argument that the time integral of |grad p| is uniformly bounded on the entire interval 0 <= t < T_b.

 **Dwight Walsh** · 22 September 2021

I would like to address the concerns that some individuals are having in regard to possible circular logic in which we assume existence and smoothness of solutions in order to prove existence and smoothness of solutions.

In this problem, we are given through the CMI criterion that a smooth solution exists initially, but we don't know how long it stays that way. So, in equations/inequalities (17)-(79), we show that the CMI criterion implies that a solution u must be smooth IF IT EXISTS (ie. remains finite). This would preclude functions u from evolving such as sin[1/(x − X)] which may be bounded but still show wild behavior at one or more points. Such functions would violate the ranges of the higher spatial derivatives imposed by the CMI smoothness criterion. A smooth solution, however, can still approach infinite values in finite time unless proven otherwise. Stated more precisely, we can get the fluid velocity to be arbitrarily high by choosing t sufficiently close to T_b.

Therefore, we are NOT assuming that the derivatives in the NS equation or anything else exists for all time. In equations/inequalities (80)-(125), the idea is to see whether a smooth solution could could evolve into a blowup at some point which we call x_b and (finite) blowup time which we call T_b. If such a blowup does occur, the fluid velocity function u must still stay smooth at all points and all times on the interval 0 <= t < T_b since u is still finite at those times and therefore smooth according to equations (75)-(79). This is what is often called a "smooth blowup" and it is the only kind of blowup we can have in this problem.

Anticipating the possibility of such a smooth blowup, the section titled "Existence and Smoothness of Solution over Time" explains that in a smooth blowup, a global maximum must first form at the blowup point (which we call x_b) at a time t_b where t_b < T_b. The equations and discussion then shows that for the blowup to occur, the time integral of |grad p| at this point must become infinite at T_b. Therefore, the "crux" of my proof is in showing that this time integral of |grad p| remains finite and in fact uniformly bounded on the entire time interval 0 <= t < T_b, and there is certainly some controversy at this point over whether I have adequately shown this.

Anyway, the important point to realize concerning "circular logic" in my arguments is that smoothness (over space) does not imply binding of the solution over time. They must be established separately.

> **Anonymous** · 22 September 2021
>
> If you actually prove a bound on the time integral of grad p, concluding global regularity would be routine. However it is extremely unlikely such a bound exists for the reasons listed above in Tao's article...
>
> > **Dwight Walsh** · 13 January 2022
> >
> > Why don't you just read my paper and see for yourself if I have shown a bounded time integral of grad p. If you have any questions, just send me an email.

Dwight Walsh · 23 September 2021

I just posted Version 27 of my paper to the website at

https://www.navierstokessmoothsolutions.com

which I believe fixes the obvious error in equation (115) of Version 26 and does some minor tweaking of arguments and explanation to hopefully make them more understandable. It is the section titled "Existence of Pressure Gradient Integral over Time" leading up to inequality (117) that contains my argument for claiming that this integral is finite for all times the fluid velocity remains smooth, regardless of how high the fluid velocity gets. You might want to go to this website and check it out.

Dwight Walsh · 25 September 2021

For those interested in my work in establishing a uniform bound on the integral of |grad p|, please check out Version 28 of my paper at

https://www.navierstokessmoothsolutions.com

In this version, I completely overhauled the section titled "Existence of Pressure Gradient Integral over Time", giving much stronger, clearer arguments showing uniform binding of

the time integral of |grad p|, which in turn is part of the proof that “smooth blowups” in this problem are not possible.



Dwight Walsh · 26 September 2021

To the one giving me the “thumbs down”:

Would you care to give a material response to my comment, stating what error(s) you have found in my new arguments for the time integral of |grad p| being bounded? As I see it, this is a blog where individuals interested in the NS problem can share ideas on this topic, and not a political rally.



Gandhi Viswanathan · 26 September 2021

The reason somebody (not me) probably gave you thumbs down is that you seem not to be willing to accept that your paper was rejected by the journals and by the expert mathematicians for a good reason. You are insisting on an argument that is unconvincing. Let me ask you: Assuming that your proof is correct, you should be able easily to tell us what is the controlling norm. What is it? Does this norm depend on time? If yes, how so? Is it a conserved quantity, or does it satisfy an exponential or double exponential bound etc.? Why doesn’t it blow up? Does it scale correctly under scale transformations? And what is the new idea that you had that allowed you (but nobody else before you) to find this controlling norm? Without convincing answers to such questions, many people will assume that your “proof” is not correct (and hence not worth going through). Extraordinary claims require extraordinary evidence, as they say.



Anonymous · 26 September 2021

This is well said, Gandhi



Dwight Walsh · 22 December 2021

Actually, most of your questions and claims are easily addressed. First, no “controlling norm” has even been defined in this proof. We simply don’t use that particular concept — whatever it is. Expecting me to come up with such a “norm” in this paper could be compared with expecting me to use calculus and differential equations to balance a checkbook. BTW, you are the only who has thrown the “controlling norm” term at me thus far.

Also, if you read the paper, you will see why the solution (or “controlling norm”) doesn’t blow up. Assuming I am communicating with professional mathematicians, they should be able to readily follow my more “elementary” arguments and point out actual flaws in the proof without posting anonymous “thumbs down”. These and personally insulting comments only convince me more than ever that my proof is correct.

In regard to “scale transformations”, this becomes an issue only if we neglect the scalar pressure (and its gradient) which Dr. Tao does in this this article. (Please see his quote “…, but we will not need to study the pressure here.”). Therefore, it is no mystery to me “Why global regularity for Navier-Stokes is hard”. In my paper, where I correctly take into account the pressure gradient as well as the nonlinear transport term, I show that a (smooth) blowup is not possible.

Finally, your statement that extraordinary claims require extraordinary evidence is applicable only if my evidence is read in an unbiased manner. Otherwise, my efforts constitute a waste of time.



Anonymous · 24 December 2021

It seems that the only way to convince the experts that your proof is correct, is by showing its correctness via an (objective!) automatic formal proof verification (using a standard software for such formal proof verification). Such formal verification is known to be much more reliable than the usual (human!) peer review verification.



roland5999 · 26 September 2021

Hello Dwight Walsh !!
i follow this thread for 10+ years and occasionally put some comments here as well (sometimes less and sometimes more politely; now washed away by the kind winds of time).
I learnt quite a couple things from this (and elsewhere) in the meantime.
It is worthwhile to consider the kind remarks of Gandhi Viswathan carefully. Basically, this thread here is the WRONG one for you (and myself).
You are (with an obvious passion) pointing to something relevant to the physics-realm.
Such arguments are not fit to be uttered in this forum,
which is dedicated to the Feynmannian “dry water” version of the NS formulation.
Pressure as such does not even exist there-in (i.e. can be eliminated by a mathematical trick).
i am willing to interact with your pointed website and talk elsewhere.

Dear other writers/readers and dear administrator,
please ignore these offtopical words.
with kind regards: nobody.



victorivrii · 30 September 2021

Do we need a flamewar? Yes, I already explained why there is no chance that the simple and elementary solution of NSE is correct. Decision to ban any user belongs to the master of the blog, not to us. And it would be fair to issue the warning first.

Unfortunately (but not unexpectedly) the master of the blog does not seem to care about this part of the blog anymore. I am not giving up following this topic because I hope that rather soon the real mathematician will publish solution (positive or negative) somewhere (probably in arXiv first) and post a link here.



Dwight Walsh · 12 December 2021

I realize I stated on Sept. 30 that I would not be posting anything further to this thread, but since no one else has posted anything since that date either, I thought that some readers might be interested in my latest developments — maybe even those who don’t particularly like me.

In my latest paper (posted on navierstokessmoothsolutions.com), I show from three different angles that the time integral of |grad p| must be finite at all points in R^3. For example, if

there is some point x_b where the scalar pressure p blows up at time T, then grad p(x_b,t) approaches zero as t approaches T. This is due to the fact that in a smooth blowup, the spatial max/min values of p can only occur at points where grad p=0, and since blowups can evolve only from global maximums, it follows that grad p smoothly approaches zero as the blowup happens. Therefore, |grad p| at x_b remains finite over the entire finite closed time interval [0,T].

Also, I should point out that in re-examining previous versions of the paper, I believe my arguments were correct in version 24 and even earlier, although there may have been some room for improvement on clarity. In fact, one reviewer of version 5 seemed to believe I did have a valid proof of existence and smoothness, although the paper was rejected over grammatical and formatting issues. It seems, ironically, that my difficulty lately has not been with the mathematics but with the "experts" who simply don't believe my work is worth their time. This is clearly demonstrated in their comments to me in which they have jumped to conclusions and made claims about my arguments that are way off base. Unfortunately, this has resulted in me making a dozen more revisions of the paper, hoping to address the concerns they wouldn't have if they had only read the paper in the first place. And since my arguments are "elementary", they should have no trouble following them.

Anonymous · 12 December 2021

Hello Dwight, The key error is in equation (118). This is finite for any epsilon individually but you should expect the integral to be unbounded as epsilon -> 0. You mention the Dirac delta as the worst-case behavior at 0, but really you should imagine something like r^{-100}, say.

Dwight Walsh · 12 December 2021

No, this is not an error. In this section we are still within the realm of a smooth, existing solution and do not introduce the concept of (smooth) blowups until the next section. Therefore, the function q must still be smooth, and r^{-100} is not smooth at r=0. In the arguments following equation (118), however, I show that even if this smoothness requirement for q is relaxed to allow delta function singularities (which are the strongest singularities allowed by inequality (109)), the time integral of |grad p| is still finite at the singularities.

I decided to include these arguments in the paper not because they are actually needed in the proof, but to ease the minds of several readers who seemed to believe I was using "circular logic" in assuming a smooth solution to prove a smooth solution.

Anonymous · 12 December 2021

I only meant to make the philosophical point about why the delta function is the wrong heuristic. If you want a literal example, suppose q is a smooth function which is =delta^(-3) when |x|2delta. (Note importantly that this is consistent with your (109).) When epsilon << delta, the integral in (117) is approximately delta^(-2).

The fact that we can satisfy (109) but find counterexamples to (117) and (118) is the exact point made by Tao in this article. (109) is a supercritical bound while (117) and (118) are subcritical.

Anonymous · 12 December 2021

Sorry I meant: q is a smooth function such that q=delta^(-3) when |x| 2delta.

Anonymous · 12 December 2021

It won't render it for some reason. Again: q is a smooth function such that $q = \delta^{-3}$ when $|x| \leq \delta$ and $q = 0$ when $|x| > 2\delta$.

Dwight Walsh · 12 December 2021

I'm afraid I don't quite understand what you are talking about, but equations (109), (117), and (118) all arise as necessary conditions for the fluid velocity and scalar pressure functions to satisfy the non-compressible Navier-Stokes equation. Therefore, counter-examples to (117) and (118) could not be actual solutions to the Navier-Stokes Millennium Problem, but only "weak" solutions.

Anonymous · 14 December 2021

The example q I give above is not a counterexample of the Navier-Stokes problem, it is a counterexample of the deduction of (118) from (117), with the added feature that it is still consistent with (109). Since the choice of δ is arbitrary, you can take it to be, say $\delta = 1/(T_* - t)$ where T_* is the hypothetical first blowup time.

Dwight Walsh · 15 December 2021

Wrong. In obtaining (118) from (117), all we did was take the lim as epsilon -> 0 indicated in (117), which was a rather trivial step since smoothness of u (and therefore q) prior to blowup was already proven in the discussion leading to inequality (79). Your "counterexample" however, assumes a (highly unlikely) time dependence of the solution that has nothing to do with the Navier-Stokes equation.

Anonymous · 10 January 2022

I'm not going to argue with you Dwight. You like to claim that no mathematician has looked at your work but it seems the real problem is you refuse to listen when they point out an error...

Dwight Walsh · 11 January 2022

> I'm not going to argue with you Dwight.

Good! You are making absolutely zero sense. And if you are so confident in your claims, why do you not put your name on them?

Dwight Walsh · 21 January 2022

>... but it seems the real problem is you refuse to listen when they point out an error ...

Wrong! You are the one not listening and making errors. I am doing my utmost to listen, but find it impossible to make sense out of any of your arguments. Where, for example, did you get the notion of a time dependence like 1/(t – T_b)

or a space dependence like 1/r^(100)? Did it, by any chance, slip your mind that we are working with solutions to the Navier-Stokes equation? I'm afraid that if you can't tie in your bizarre time and space dependencies with actual solutions to the Navier-Stokes equation, then your arguments are meaningless.

 **Anonymous** · 12 December 2021

If the proof is sufficiently simple, is it possible to formalize it and then try to give it a formal verification (as done e.g. for the extremely complicated proof of the Kepler conjecture)?

 **Dwight Walsh** · 14 December 2021

I'm not sure what you mean by "formalize" the proof and then try to give it a "formal verification". Are you referring to getting it posted in an archive of some sort where other mathematicians can review it for free? If so, then the answer is a plain simple "no". With my career fizzle 30 years ago, no mathematician nowadays would sponsor me. In fact, look at the insults and thumbs down I have had on this forum alone.

 **Anonymous** · 14 December 2021

I meant that if the proof is sufficiently simple, there are several software programs that can give it (after translating the proof into a symbolic input syntax which is needed by the automatic verification program) automated(!) rigorous(!) computerized verification.

Several complicated proofs (too complicated to manually(!) verify all their details) were already given such automated(!) computerized verification – see e.g. the Wikipedia article on the Kepler conjecture and the automatic symbolic verification of its extremely complicated proof.

 **Dwight Walsh** · 15 December 2021

I'm afraid you would know much more about computerized verification than I do. In fact, the only time I heard anything about computer-aided proofs was with the proof of the four-color map conjecture in 1976, and in that case there was some controversy over the validity of such a proof. Setting those issues aside, however, I don't sense that the NS Millennium Problem would lend itself as well to such an analysis since it involves only a few logical arguments and not imassive numbers of logical statements to be evaluated.

 **Dwight Walsh** · 18 December 2021

Dear Dr. Tao,

I read you comments concerning the maximum kinetic energy and cumulative energy dissipation, and I understand that these two quantities cannot by themselves establish existence and smoothness of solutions to the NS equation. From my study of this problem, however, I believe that the global maximum of the kinetic energy density K = u^2(x,t)/2 provides the "missing link" that allows proof of existence and smoothness. First note that a "smooth blowup" is occurs iff this value becomes infinite in finite time. But then, since we are working with a viscous fluid, it is possible to show that no maximum point can attain infinite velocity in finite time. If you are interested, please load my paper from the website at

http://www.navierstokessmoothsolutions.com

Anonymous · 19 December 2021

Is energy subcritical, supercritical or critical in 3D ? What about in 2D ? Professor Tao's post is partially about this issue. Once one understands the concept of how quantities scale, it becomes clear why kinetic energy cannot be a controlling norm.

Dwight Walsh · 19 December 2021

I realize that total kinetic energy cannot be a controlling norm by itself. What I am saying is that knowledge of the maximum kinetic energy density (ie. joules/meter^3), which is a global but time-dependent quantity, along with the two supercritical quantities indicated by Prof. Tao (total kinetic energy and cumulative dissipation energy) is sufficient to establish existence and smoothness of solutions to the incompressible NS equation.

I did not consider any 2-dimensional version of the problem since the NS equation is formulated in 3 dimensions, not 2, and the Green's function used in many of my arguments applies only to 3 dimensions.

Also, how quantities scale is completely independent of existence or smoothness of solutions to any equation. The rule is that one chooses a scale size (or equivalently, a system of units) that is most convenient for the problem at hand, and uses it consistently.

Dwight Walsh · 19 December 2021

Dear Prof. Tao

Could you please explain why "we will not need to study the pressure here"? The grad p term cannot be negligible compared to the nonlinear transport term since the sum of these two terms must have a zero divergence, and in the analysis that follows, you discuss the possibility of a blowup occurring at some time T as a result of energy being shifted to successively smaller space scales.

In the paper I posted at http://www.navierstokessmoothsolutions.com, however, I show that when grad p is correctly taken into account, then no such blowup is possible.

Antoine Deleforge · 29 December 2021

I am just a random researcher in acoustics & signal processing so most of the theory surrounding Navier-Stoke goes way above my head, but even I can tell that you are completely misguided when you say that Prof. Tao is "neglecting the grad p term". He literally wrote "we will not need to study the pressure here", how is that equivalent to "neglecting the grad p term" according to you? Are you a native English speaker (genuine question, I am not) ? The two expressions have basically nothing to do with each other. There is a huge difference between choosing *not to study* a particular quantity and *neglecting a term* in an equation. You see this, right?

Dwight Walsh · 29 December 2021

I'm afraid that in this problem, not studying the pressure is the same as setting the pressure to zero. Even if the scalar pressure p is totally unknown, the grad p term must still be carried in the NS equation simply for mathematical correctness. Dr. Tao,

however, never once mentions this term after making the assertion that “we will not need to study the pressure here”. Instead, he discusses the problem only in terms of “rescaling”, “supercritical and subcritical quantities”, and “coercive and noncoercive quantities”. The presence or absence of the grad p term does not affect his arguments one iota. Therefore, by not including this term in his analysis, Dr. Tao is by default setting it equal to zero. Hence, his methods do not in general apply to solutions of the incompressible NS equation

 **Dwight Walsh** · 29 December 2021

I should also point out that in the comment by Robert (17 Feb. 2021), he quotes Prof. Tao’s proposed strategy

“Discover a new globally controlled quantity which is both coercive and either critical or subcritical”

and correctly points out that discovery of such a quantity would render the formulation of the problem invalid since this new quantity was not part of the original Official Problem Description. Fortunately, Dr. Tao at least recognizes that his arguments are not complete, but the problem is not some missing coercive quantity. It is failure to maintain the incompressibility constraint (div u = 0).

In my paper at http://www.navierstokessmoothsolutions.com, however, I correctly included the grad p term in the NS equation, and there no issues related to an incomplete problem description.

 **Daniel** · 18 May 2022

Interesting that your symbol matches that of Anonymous.

 **victorivrii** · 30 December 2021

> I’m afraid that in this problem, not studying the pressure is the same as setting the pressure to zero.

Absolutely wrong. One of the completely valid strategies is to **exclude** pressure. How to do it: if the system has a form
$u_t - A[u] = \nabla p$, $\latex \nabla \cdot u =0$ where $\latex p$ belongs to a prescribed class (f.e. in the *periodic* settings it must be periodic with respect to spatial variables). Here A is the nonlinear operator.

Consider projector $\latex P$ in the space of $\latex L^2(\mathbb{R}^3;\mathbb{R}^3)$ on the subspace orthogonal to the space formed by $\latex \nabla p$ for all such $\latex p$. Then since $\latex \nabla \cdot u =0$ we see that $Pu=u$ and then the problem becomes
$u_t - PA[u] = 0$ with $\latex u:\ \nabla \cdot u =0$. That’s all. This problem is **equivalent** to the original problem.

On the other hand, simply putting $p = 0$ would force us to drop condition $\nabla u = 0$, so the problem would be $u_t - A[u] = 0$ which is not equivalent to the original problem. But nobody suggests to do this. **Dwight Walsh** stop fighting with the straw man.

Remark If we exclude non-linear part from $\latex A$, taking $\latex A=\nu \Delta u$ then $\latex PA[u] =A[Pu]$ and then indeed we would have equation $u_t - \nu\Delta u = 0$ with a compatible to it condition $\latex \nabla \cdot u =0$ but it is not the case.

> “Discover a new globally controlled quantity which is both coercive and either critical or subcritical” and correctly points out that discovery of such a quantity would render the formulation of the problem invalid since this new quantity was not part of the original Official Problem Description.

Again wrong. This quantitity would be a tool. F.e. considering heat equation $u_t = \Delta u$ one can observe that for solution $\latex \iiint u dx $ is constant and $\latex \iiint u^2 dx$ decays but those quantities are not the poart of the original problem description.

I hope this LaTeX works



Dwight Walsh · 30 December 2021

Very good! Now would you like to explain exactly how Prof. Tao implements all of this to maintain a zero divergence fluid velocity and actual solutions to the NS equation as opposed to “weak” solutions? Keep in mind, of course, that he does not use equations of inequalities in this article — just vague intuitive arguments that work for compressible as well as incompressible fluids.



Anonymous · 30 December 2021

Hi Dwight, if you want to learn about the basic theory of the Navier-Stokes equations, I would recommend you start with Tao’s lecture notes on the subject. Start here: https://terrytao.wordpress.com/2018/09/16/254a-notes-1-local-well-posedness-of-the-navier-stokes-equations/

In particular, sections 2 and 4 will dispel your misunderstanding about the pressure. Hopefully this text will be accessible to you.



Dwight Walsh · 30 December 2021

No, I believe I am well acquainted with the NS equation including the role of the scalar pressure and how it is obtained. Have you read my paper at http://www.navierstokessmoothsolutions.com where I prove existence, smoothness, and uniqueness of solutions to this equation? I may not use the same notation as Prof. Tao, but the concepts are still there.

My concern with this current article by Dr. Tao is what seems to be arguments based solely on “re-scaling” of the solution to claim the possibility of energy being transferred to successively smaller scale sizes, eventually resulting in a blowup. His arguments are vague and intuitive, but he does assert at one point that we do not need to study the pressure here. This leads me to believe that we will not be considering the scalar pressure in whatever it is we are doing, and in fact, Dr. Tao doesn’t mention pressure again in this article. In this case, however, how do we maintain incompressibility of the fluid? In general, WHAT IS HE DOING AND HOW DO WE KNOW IT IS APPLICABLE TO THE NS EQUATION?

Finally, I should note once again that in the paper I posted at

http://www.navierstokessmoothsolutions.com

grad p and the nonlinear transport term are correctly taken into account, and regularity is established without such issues arising.

 **victorivrii** · 31 December 2021

> Now would you like to explain exactly how Prof. Tao implements all of this to maintain a zero divergence fluid velocity and

This is a common knowledge; procedure was well understood even before T. Tao was born

> actual solutions to the NS equation as opposed to “weak” solution

It works for both smooth solutions and week solutions as well. Simply smooth/wee solutions of the reduced problem correspond to smooth/week solutions of the original problem. Please read materials suggested to you.

 **Dwight Walsh** · 31 December 2021

>This is a common knowledge

This is a common cop-out for people who don’t know themselves.

 **victorivrii** · 31 December 2021

Explaing:
Consider reduced equation $u_t - PA[u] = 0$. Applying $(I - P)$ (remember, P is an orthogonal projector) we get
$(I - P)u_t = 0$. Thus if $latex(I-P)u=0$ as $t = 0$, it will be so for any $t > 0$ as well. And $latex(I-P)u=0$ is equivalent to $\nabla \cdot u = 0$ (Calculus II).

Happy

 **Dwight Walsh** · 1 January 2022

Yes, and I believe this is equivalent to how I propagate the NS solution in my paper, although I do not use the Leray projector notation. Unfortunately, however, Prof. Tao’s vague writing style (often without equations) makes it extremely difficult for a newcomer to the NS equation to follow, and is wide open for misunderstanding. Anyway, after a few more reviews of his arguments about the fluid energy being transferred to successively smaller scale sizes, I believe I have a somewhat better understanding what he is saying, and will not press issues on his arguments not being consistent with div u = 0 or the NS equation in general.

I do, however, have a much more serious concern about his belief that a blowup solution is always theoretically possible unless a new quantity is discovered that is coercive and critical or subcritical. This, of course, would change the problem statement for the NS global regularity. In my paper, however, I show that as any “smooth” blowup ensues, the work done by the (retarding) viscous forces acting on a volume element at the blowup point exceeds the work done by grad p forces on this volume element. Hence, no such blowup would be possible. This, I believe, is an EXTREMELY important result that Dr. Tao misses in his equation-free arguments.

victorivrii · 31 December 2021

> No, I believe I am well acquainted with the NS equation including the role of the scalar pressure and how it is obtained

You are presumably well with the acquainted how NSE is derived but your knowledge of mathematical aspects is marginal at best. And this ignorance is aggravated by your arrogance. Sorry for blunt assessment but

When, O Catiline, do you mean to cease abusing our patience?

Dwight Walsh · 31 December 2021

I will continue to do whatever it takes to get the word out about my solution to the Navier-Stokes Millennium Problem unless I am shown to be wrong mathematically. And if you don't like it, that is your problem.

victorivrii · 31 December 2021

🤣🤣🤣🤣🤣🤣

Anonymous · 31 December 2021

"NS Don Quixote"

Dwight Walsh · 4 January 2022

I should indicate at this point that several statements I made over the past week or two concerning Dr. Tao's are probably not accurate due to my misunderstandings of what he is trying to show. This includes my claims of his possible inconsistency with the NS equation and incompressibility. Unfortunately, however, his mostly verbal and intuitive descriptions were extremely difficult for me to follow. Therefore, I no longer have the same issues about his arguments that I recently posted in my comments.

My current concern about his article in not so much in what he has but what he is missing. I agree that the nonlinear terms appearing in the transport and grad p terms could channel fluid energy into smaller scale sizes and could lead to a blowup if unchecked, but does this actually happen? Dr. Tao seems to recognize that the currently known coercive, critical, and super-critical quantities along with scale-size arguments is insufficient to establish global regularity. However, his suggestion of finding a new coercive/critical parameter would change the statement for the NS global regularity problem (ie.NS Millennium Problem), and claims it to be an "impossible problem" in his final paragraph.

What I believe Dr. Tao is missing in his arguments is the fact even a blowup solution must be smooth until the actual blowup time. Therefore, the blowup point must originate from a global maximum, and since the fluid kinetic energy density is still a smooth function, this maximum point must also be a level point. Additionally, the Laplacian of this function at the maximum point must be negative or zero. In my paper at

http://www.navierstokessmoothsolutions.com

these properties of a smooth blowup are used to show that no such blowup is possible for solutions of the NS equation.

Anonymus · 7 January 2022

Shush

Dwight Walsh · 14 January 2022

WOW! What gems of wisdom!

Anonymous · 10 January 2022

Mathematicians have known that solutions stay smooth until the blow up time since Leray's thesis in 1934.

Dwight Walsh · 11 January 2022

Yes, I realize this and in fact proved it my paper, assuming of course there is a blowup. Unfortunately, however, many reviewers seem to have DEFINED a blowup time T, and claimed that I have only proven a smooth solution on the semi-open interval [0,T) while there still could be a blowup at time T. My question to them is exactly what happens at this time T? How did they choose it? Could it be that they think of it as an actual time at which the fluid velocity becomes infinite at some point, as they seem to be calling it? If so, then it's no wonder the NS regularity problem has not been solved. Proving existence and smoothness when blowup is a given would be a monumental task!

Anonymous · 12 January 2022

The blowup time is defined as the supremum over all finite nonnegative times t for which there is a smooth solution on [0, t). you need to show rigorously(!) that this supremum is $+\infty$

Dwight Walsh · 13 January 2022

Good! Now can you look at equation (128) and find a particular value T of t such that K_b(t) -> infinity as t -> T? If not, then there is no blowup time and we have a smooth solution. Right? Remember, of course, that in the discussion leading to equations (119) and (120), we showed in two ways that the time integral of |grad(p)| has a finite upper bound that applies for all t > 0.

Dwight Walsh · 13 January 2022

Also, I should add that all of my arguments and equations/inequalities hold on the semi-open time interval [0,T) where the solution exists and is smooth, and make no claims about what happens at t = T. To prove existence and smoothness of a solution, I show that there is no value of T such that u(x_b,t) becomes arbitrarily large for t sufficiently close to T, but still less than T. Therefore, THERE IS NO CIRCULAR LOGIC WHATSOEVER in my arguments, and I strongly believe that one would be very hard pressed to come up with a presentation that is clearer and more rigorous.

Dwight Walsh · 21 January 2022

I guess that after a week, you are still not up to the challenge of finding a blowup time T such that K_b(t) -> infinity as t -> T. I didn't think you would be!

Anonymous · 22 January 2022

We've found the error in your paper, now please stop commenting here.

Dwight Walsh · 22 January 2022

Then where exactly is this error in my paper? And I want an answer in terms of page, paragraph, and equation numbers. Also, you failed to provide an answer to my question about blowup time. Finally, I will continue making comments and asking questions until I get some good answers.

anon · 23 January 2022

Isn't it quite clear already that you won't be finding the kind of discussion you are looking for here?

Dwight Walsh · 24 January 2022

>Isn't it quite clear already ...

Unfortunately, you are quite correct! Evidently, the "experts" at this Navier-Stokes global regularity problem are exempt from proof or being disproved.

Anonymous · 26 January 2022

I gave you specific equation numbers and a counterexample for your inequalities for the pressure (equations (117)-(118)). Because you don't understand analysis well enough to pass an undergraduate course on the subject, you dismissed it immediately.

Everyone, please stop engaging with this guy. His paper is getting rejected from journals because it's wrong. It's clear as an expert reading his paper that there is no new idea, just an elementary analysis mistake.

Dwight Walsh · 30 January 2022

Don't comment on my analysis ability when you can't tell the difference between a solution and non-solution to the Navier-Stokes equation!

Dwight Walsh · 23 January 2022

If you actually found an error in the paper, you would be able pinpoint it and make it clear to others why it is an error. But you can't. Now, I thought mathematics was supposed to be very precise and rigorous. WHAT HAPPENED!!??

Anonymous · 25 January 2022

Try to submit again your proof to a leading journal and see their response

Dwight Walsh · 29 January 2022

And where exactly is this "counterexample" of yours?

Polihronov · 10 January 2022

Hello Prof. Tao and Everyone,

Thank you for your post of May 4, 2021. I have spent time reading your replies as well as reading sections of the main blog post. I do appreciate this web site, the comments and ideas presented and the concept of criticality through which the NSE solutions are analyzed; and have followed the blog posts made through the years. The information I posted in 2021 above was given in parts; some references to articles were made; for this reason, if you would allow me, I would like to summarize it in a more contained form; as well as to request feedback.

In the article "T. Tao, Proceedings: Current Developments in Mathematics International Press, 255 (2006)" we find the definitions of X-critical, X-subcritical and X-supercritical PDEs. All of them require a scaling law for a solution u; and the criticality of the value X then depends on the scaling behavior of u.

As I understand it (please correct me if I'm wrong), NSE criticality is estimated through

$(x', y', z') = k(x, y, z)$

$t' = k^2 t$

$(u', v', w') = \frac{1}{k}(u, v, w)$

$p' = \frac{1}{k^2} p$

because it is part of its definition.

According to Lloyd, "S. P. Lloyd, Acta Mech. 38, 85 (1981)", this transformation is a Lie group of scaling transformations. Charles L. Bouton, an American mathematician and a Harvard mathematics professor was a student of Sophus Lie; Bouton wrote a number of articles on Lie transformations and their invariants; however Bouton's doctoral thesis is especially interesting. It is published in the literature as "C. L. Bouton, Am. J. Math. 21, 25 (1899)".

In his article, Bouton examines a very similar scaling transformation. He determines the functional form of the invariants of this transformation. Repeating Bouton's analysis for the case of the NSE shows that under the above Lie transformation, the NSE solutions must have the form

$u(x, y, z, t) = \frac{1}{\sqrt{t}} u(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}})$

$p(x, y, z, t) = \frac{1}{t} p(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}})$

This conclusion follows from Bouton's first theorem and is a result of his analysis. In his article, he lists a number of characteristics of the Lie invariants and they are well defined mathematically.

All this seemed interesting to me since it links together NSE criticality, the invariance of u and the possibility of scaling-induced blow-up.

I am very interested to hear other's opinions on the invariance of u.



victorivrii · 10 January 2022

There may be an invariant solution (which is non-smooth as $t \to +0$) but it does not help to prove ordisprove that any solution smooth as $t = 0$ is smooth for all $t > 0$.



Polihronov · 10 January 2022

Dear Prof. Ivrii,

Thank you for replying! (U of T is awesome :)
It would be very interesting to see if the Bouton-form of the solutions

$u(x,y,z,t) = \frac{1}{\sqrt{t}} u(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}})$

$p(x,y,z,t) = \frac{1}{t} p(\frac{x}{\sqrt{t}}, \frac{y}{\sqrt{t}}, \frac{z}{\sqrt{t}})$

is the only form $u(x,y,z,t)$ and $p(x,y,z,t)$ can have when we use the scaling transformation shown above. According to Bouton's first theorem, this must be the case since they should both transform identically to the jet-space variables u and p of the Lie scaling transformation. Once this is established, one can find solutions that would not be subject to blow-up; I have a link with more information on how to accomplish this; but would first need Prof. Tao's permission to post such a link here in his blog.

I was respectfully seeking feedback on the above expressions for $u(x,y,z,t)$ and $p(x,y,z,t)$; the fact that they must have this form would have strong implications on how one estimates NSE criticality.

Many thanks.



Dwight Walsh · 21 January 2022

On several occasions, I mentioned the idea that the global maximum of the kinetic energy density could be the missing globally controlled quantity that is needed to establish global regularity of solutions to the (incommpressible) Navier-Stokes equation. Thus far, however, I have only been booed and hissed for suggesting this possibility. Since no one is willing to give my paper a fair chance, I decided to try getting my point across one line at a time.

First, we note that if a (smooth) blowup occurs at some point x_b, it must start with a global maximum of the kinetic energy density (or equivalently, the fluid velocity |u|) forming at x_b. The reason for this is that if the blowup occurs without a global maximum ever forming at x_b, then the blowup would occur first at other points x such that K(x) > K(x_b), thereby contradicting the definition of a blowup point.

Now, does this feel alright to everyone? Remember that if you disagree, you are welcome say so, preferably indicating what you find wrong with this result. There is no need for thumb-downs. In fact, I tend to view them as indicating that you are afraid of where I may be going with these questions.

That's all for this posting.



Dwight Walsh · 30 January 2022

Dear Anonymous,

When are you going to learn that you can't concoct a counterexample by arbitrarily choosing space and time dependencies? You must show that your counterexample is in fact a solution to the Navier-Stokes equation. Otherwise, all you have is garbage! In my paper, I stick with the Navier-Stokes equation and don't go half-cocked into dependencies that have nothing to do with the NS equation. And it doesn't matter how many times you post your "counterexample", in the end it will still be garbage!

Dwight Walsh · 30 January 2022

Dear Prof. Tao,

You have been strangely silent about all of my postings, especially those having to do with making the Navier-Stokes regularity not so "hard". Instead, you simply allow the insults and bizarre counterexamples to my work (on the part of your comrades) to continue unchallenged. Is this the type of mathematics you want on your forum!? WAKE UP AND SMELL THE COFFEE!!

Dwight Walsh · 1 February 2022

If you all are willing to give me a few minutes of your attention without insults or a know-it-all attitude, I have some ideas that may help resolve our differences concerning the Navier-Stokes Millennium Problem

In my paper, I believe I successfully showed existence and smoothness of solutions to the (perfectly) incompressible Navier-Stokes equation as specified in the Official Problem Description (OPD) by Charles Pfefferman. Also, I should note that in most cases when my paper was rejected, the journal either did not give me any reason why or turned the paper down over issues having nothing to do with mathematical correctness (eg. backlogs). One reviewer did, however, claim that I had a valid proof and stated "the proof is so easy even an undergraduate reader could follow". So why did they reject the paper? — Ask them!

So, with the scrutiny I have given this paper, which includes debunking false claims about it, I don't believe there is much doubt that I proved existence and smoothness of solutions to the NS Millennium Problem as defined in the OPD. BUT IS IT THE PROBLEM WE REALLY WANT TO SOLVE? The OPD assumes a perfectly uniform fluid which is not true in 99%+ of practical cases. Also, viscosity and scalar pressure gradients would be the only forces acting between the fluid elements. Surface tension also plays a role in the case of water splashing and forming bubbles. These bubbles, of course are much less dense than the surrounding water and therefore splashing would not be possible under the OPD for the NS Millennium problem.

In conclusion, it seems that the lessons to be learned in solving the Navier-Stokes Millennium problem as currently stated are few and far between and not very interesting in regard to fluid motion. The fact that existence and smoothness of solutions to the NS Millennium Problem fall out as easily as I have shown implies that this model cannot predict turbulence.

> **Anonymous** · 2 February 2022
>
> Note that any relativistic version of NS has the fluid velocity bounded by the velocity of light.
>
> > **Dwight Walsh** · 2 February 2022
> >
> > Physically speaking, you are correct. For practical applications on earth, however, non-relativistic mechanics generally does the job well enough. For stars and black holes, however, it's a whole new story that I know little about. The equivalent of the NS equation would have to track energy and momentum instead of fluid velocity since these quantities can approach infinity even though any velocity would have a

strict upper bound of c. General relativistic effects (ie. curvature of space-time) also makes the task especially formidable.

> **Dwight Walsh** · 2 February 2022
>
> Well, well well ... I see that some mentally challenged individual simply couldn't pass up the opportunity for another thumbs-down despite the fact that I wasn't even talking about solving the NS equation. Very impressive!!

Dwight Walsh · 2 February 2022

I have a special message for those who still insist that I made a mistake in equations (117) and (118) by not choosing a strong enough spatial singularity. Actually, I have been over this time and time again, but it doesn't seem to stick! If you read the paragraphs following these two equations, you will see that they apply only before any blowup, and therefore the fluid velocity and the function q are still SMOOTH. [I presume, of course, that I am working with a literate society even if we can't do mathematics].

It is not until the next section that we consider the possibility of a (smooth) blowup. To see if this can happen, we see if there exists a time T such that the SMOOTH fluid velocity reaches an infinite value somewhere as t -> T. If no such T exists, then we have proven existence and smoothness.

Therefore, the solution is SMOOTH up to and including equation (120). COMPRENDO!?

> **Antoine Deleforge** · 4 February 2022
>
> Dear Dwight,
>
> I will try to re-explain to you the error in your paper because it seems that you have misunderstood it (to be fair it was not very well formulated by the previous anonymous commenter – who was not me by the way).
>
> Your paper contains a hole in the mathematical reasoning. Specifically, you claim that the integral in (117) is bounded before the blowup time, but you do not have a rigorous proof of it. The argument you give is essentially that the "strongest singularity" that q could have while respecting (109) would be to approach a Dirac delta function when approaching the blowup time T_b. You argue that if this happens only at a finite number of points and if q remains smooth before the blowup time, then everything is fine and (117) remains finite.
>
> However the term "strongest singularity" is not mathematically rigorous, and your claim is false in general. Indeed, one can build infinitely many counter examples where q verifies (109) *while* making (117) unbounded. For example, q(x,t) could be any smooth function equal to 1/|t-T_b|^3 for |x|2|T-T_b|. If you do the computation, you'll see that (109) remains bounded (in fact constant) as t approaches the blowup time while (117) is unbounded (it increases like 1/|t-T_b|^2).
>
> Now your reply to this is that these counter examples for q do not obey the Navier Stokes equations. Well, of course they don't! Otherwise we could submit these counter-examples to the Clay Institute and become millionaires :-).
>
> But it doesn't matter that they don't. They are here to show you that there is a missing proof in your paper, namely, you need to rigorously prove that *any* counter examples

one could find for q that verifies (109) while making (117) infinite does not obey NS. The burden of proof is on you, since you are the one making the extraordinary claim of having solved NS. If you manage to obtain such proof, assuming all the rest of your paper is correct, then congrats, you will have solved the NS millenium problem!

Unfortunately, according to your own paper, proving this is equivalent to solving NS, which is notoriously a very difficult problem [see this very blogpost by Terence Tao] that have baffled thousands of mathematicians over the past hundred years. So your chances of success are extremely thin and I would strongly recommend focusing your energy on something else.

Best regards,
Antoine

Antoine Deleforge · 4 February 2022

Sorry, there are some display issues: " $q(x,t)$ could be any smooth function equal to $1/|t-T_b|^3$ for $|x|2|t-T_b|$."

Antoine Deleforge · 4 February 2022

Again: "q(x,t) could be any smooth function equal to 1/|t-T_b|^3 for |x| lower than |t-T_b| and to 0 for x larger than 2|t-T_b|."

Dwight Walsh · 4 February 2022

>Specifically, you claim that the integral in (117) is bounded before the blowup time, but you do not have a rigorous proof of it.

WRONG! Equations (1) – (79) are dedicated to showing that any solution remains smooth for as long as it remains finite (ie. exists). Those coefficients in equation (79) are analogous to set used by Charles Pfefferman in specifying the initial conditions of the spatial derivatives of u, and are essentially the definition of "smooth". Therefore, smoothness is proven at time t if we show that these coefficients exists (which we do!). So smoothness is proven for as long as the solution remains finite. [Hint: Try reading the paper before trying to shoot it down!]

Dwight Walsh · 5 February 2022

Since we have proven smoothness of the fluid velocity, it then follows from equations (80)-(118) that the integral in (117) is finite. Again, READ THE PAPER!

Antoine Deleforge · 5 February 2022

I genuinely do not understand why you talk about smoothness here. The counter example I gave for q *is* smooth. I never questioned or discussed smoothness in my argument, I just pointed out that you do not have any proof that (117) remains bounded before the blowup time.

Dwight Walsh · 6 February 2022

>I just pointed out that you do not have any proof that (117) remains bounded before the blowup time.

What's your definition of blowup time? My understanding is that it is the time T (assuming it exists) when the fluid velocity and its spatial derivatives become

infinite at some point(s) in R^3. Before that time, u and its spatial derivatives are finite at all points in R^3, which implies that q must also be finite at all points in R^3. Now, examining the integral in equations (117) and (118), we see that we are integrating q (a bounded function) over a finite range of the primed spherical coordinates. HOW COULD THE VALUE OF THIS INTEGRAL POSSIBLY BE INFINITE?

 **Antoine Deleforge** · 7 February 2022

I think some confusion arises from conflating “finite/infinite” and “bounded/unbounded”. A function cannot, strictly speaking, takes “infinite values”, this is not mathematically rigorous. It is always “finite” where it is defined. It can however have singularities, i.e., points where it is not well defined.

On the other hand, a function f can be *unbounded* on a (possibly *open*) set S, which means that for any Ω gt 0 there exists $x \in S$ such that $f(x)$gt Ω, or in words, the function can take arbitrarily large values over this set (i.e., a “blowup”). In your paper, what you really want to prove is that $\Lambda(x,t)$ is bounded over the right-open interval $[0, T_b)$ for all $x \in R^3$ (Note: right-open means that T_b is *not* included here) in order to obtain eq. (120).

We gave you an example of a function $q(x,t)$ that is smooth for all $x \in R^3$ and t in $[0, T_b)$, that verifies eq. (109), but for which the limit when t goes to T_b of the limit when ϵ goes to 0 of (117) is $+\infty$, i.e., the integral is unbounded at $x = 0$ on the open-interval $[0, T_b)$ (do the math, it’s pretty straightforward).

As Gandhi Viswanathan explained to you, it looks like your are implicitly permuting the limits in t and in ϵ in your reasoning, which you cannot do in general. By the way you often drop the t dependency of q in your equations, which could explain why you missed this point.

Unfortunately, you no longer have a proof of (120) due to this issue.

 **Dwight Walsh** · 7 February 2022

[Please disregard my previous posting which did not render correctly]

>We gave you an example of a function q(x,t) that is smooth for all ...

Again, you are not following the fundamental logic of the proof. Although I have hammered on this multiple times, I will do it once more here. It was proven in the discussion leading to equation (79) that a smooth, bounded solution exists for at least a finite time, and that solution remains smooth for as long as it exists (ie. remains finite for all x in R^3). Therefore, if a blowup does occur at some finite time T_b, it must be a “smooth” blowup where the fluid velocity remains smooth for all t 0.

Since the value of T_b doesn’t even exist, it would seem that your arguments are “snafu”! OK, the ball is in your court now.

 **Dwight Walsh** · 7 February 2022

[I will try once more to get this posting to render correctly]

>We gave you an example of a function q(x,t) that is smooth for all ...

Again, you are not following the fundamental logic of the proof. Although I have hammered on this multiple times, I will do it once more here. It was proven in the discussion leading to equation (79) that a smooth, bounded solution exists for at least a finite time, and that solution remains smooth for as long as it exists (ie. remains finite for all x in R^3). Therefore, if a blowup does occur at some finite time T_b, it must be a "smooth" blowup where the fluid velocity remains smooth for all t < T_b. In order to determine whether such a blowup can occur, we ask the question: (cont'd)

 **Dwight Walsh** · 7 February 2022

[cont'd]

"Is there a finite time T_b such that the (smooth) fluid velocity (or equivalently the energy density K) becomes arbitrarily large for t sufficiently close to T_b?"

[cont'd]

 **Dwight Walsh** · 7 February 2022

[cont'd]

This is the question we consider in equations (121)-(128), and since u and K are smooth functions, all results from previous sections apply here. The final result, given in equation (128), shows that K has an absolute upper bound for all x in R^3 for each value of time t. This implies that the conjectured blowup time T_b does not actually exist and that we have proven existence and smoothness for all t > 0.

Since the value of T_b doesn't even exist, it would seem that your arguments are "snafu"! OK, the ball is in your court now.

 **Dwight Walsh** · 7 February 2022

[cont'd]

This is the question we consider in equations (121)-(128), and since u and K are smooth functions, all results from previous sections apply here. The final result, given in equation (128), shows that K has an absolute upper bound for all x in R^3 for each value of time t. This implies that the conjectured blowup time T_b does not actually exist and that we have proven existence and smoothness for all t > 0.

[cont'd]

 **Dwight Walsh** · 23 February 2022

>As Gandhi Viswanathan explained to you, it looks like your are implicitly permuting the limits in t and in \epsilon in your reasoning, which you cannot do in general.

It seems to me that both you and Gandhi could use a refresher lesson in how to solve the Poisson equation using the 3D Greens function. What you need to

do is put the ENTIRE non-homogeneous term (q in our case) under the integral sign, exclude from the region of integration a small sphere of radius \epsilon centered at the field point, do the integral, take the limit as \epsilon -> 0, and that's it. There are no "implicit limits" nor any other limits than the one just mentioned. Nor is there any issue concerning "commuting limits" since there is only one.

But I really don't want to discuss singularities any further. It was probably a mistake for me to mention them in the first place since my proof only involves smooth solutions up to the blowup time, assuming it exists. I was, however, trying to briefly explain what I thought was an interesting side point and also thought I had clearly explained this. But as things turned out, most readers focused on the singularities instead of my main arguments, thereby completely misunderstanding what I was claiming and how I was proving it.

 **Anonymous** · 23 February 2022

> *What you need to do is put the ENTIRE non-homogeneous term (q in our case) under the integral sign, exclude from the region of integration a small sphere of radius \epsilon centered at the field point, do the integral, take the limit as \epsilon -> 0, and that's it. There are no "implicit limits" nor any other limits than the one just mentioned. Nor is there any issue concerning "commuting limits" since there is only one.*

WRONG!

This is fundamentally WRONG. Don't try NSE before knowing the basic real analysis.

Qualifying exam not passed. Case close.

 **Dwight Walsh** · 7 February 2022

Again, you are not following the fundamental logic of the proof. Although I have hammered on this multiple times, I will do it once more here. It was proven in the discussion leading to equation (79) that a smooth, bounded solution exists for at least a finite time, and that solution remains smooth for as long as it exists (ie. remains finite for all x in R^3). Therefore, if a blowup does occur at some finite time T_b, it must be a "smooth" blowup where the fluid velocity remains smooth for all t 0.

Since the value of T_b doesn't even exist, it would seem that your arguments are "snafu"! OK, the ball is in your court now.

 **Dwight Walsh** · 4 February 2022

>The argument you give is essentially that the "strongest singularity" that q could have while respecting (109) would be to approach a Dirac delta function when approaching the blowup time T_b.

WRONG! I may have mentioned what certainly seems to be the strongest singularity, but it's NOT the basis of my argument. I'll talk some more on singularities later. My argument here is that for all equations leading to (120) we are considering only the solution prior to any blowup. And recall from equation (79), there is at least a finite time over which such a smooth exists. The possibility of a (smooth) blowup is not addressed until the next section.

 **Dwight Walsh** · 4 February 2022

Now let's consider the issue of singularities. In writing equation (118), it was easiest for me to picture a delta function as the singularity in q, and in fact I carried this notion in actually writing the equation. However, my arguments apply equally well to ANY singularity occurring at the field point, and I made a minor revision in my paper this evening to reflect this.

Now, in evaluating the improper integral arising from solving the Poisson equation with the 3D Green's function, we first exclude from the integration region a small spherical region (of radius epsilon) centered at the field point x. We then take the limit as epsilon -> 0 to get the solution of Poisson's equation. In taking this limit, however, we completely neglect the behavior of q at the field point x. THIS MEANS THAT THE VALUE OF THE INTEGRAL IN EQUATION (118) IS TOTALLY UNAFFECTED BY ANY SINGULARITY IN q.

So, play around to your heart's content with whatever wild singularity at the field point strikes your fancy. It won't affect anything.

 **Dwight Walsh** · 4 February 2022

> ... and I would strongly recommend focusing your energy on something else.

After putting about three nails in the coffin concerning your arguments, I can understand why you would make such a suggestion.

 **Gandhi Viswanathan** · 5 February 2022

Dear Dwight, if you are having trouble understanding what Antoine Deleforge has written, consider your own example of a Dirac delta function singularity. Recall that the Dirac delta is not a function (it is a tempered distribution), but it can be expressed as a limit of a sequence of functions, such as ever-narrower (and taller) Gaussians. In your paper, you take the epsilon-to-zero limit after eq 118. But you do this AFTER you have taken the implicit limit to the Dirac delta. This is wrong. The two limits do not commute. You must first take the limit of epsilon to zero after eq 118, and only then consider what happens if you have a Dirac delta. Otherwise, you are conveniently ignoring what happens just outside or around the point of the singularity. But this is "cheating", you are not allowed to ignore this region. If you do the limits in the right order, the singularity cannot be neglected. Your "proof" is indeed wrong.

 **Dwight Walsh** · 5 February 2022

>In your paper, you take the epsilon-to-zero limit after eq 118.

You bet I do! That is how the improper integral is defined for solving the Poisson equation with the 3D Green's function.

————————————————-

>But you do this AFTER you have taken the implicit limit to the Dirac delta.

WRONG! I don't do anything with the Dirac delta function nor any other type of singularity at the field point. Whether q has an actual value at the field point or if it forms a singularity at this point, it has no effect on the integral in equations (117) and (118) because it is excluded from the integration region.

Since you are a theoretical physicist, you should be able to follow the analogy I make with electrostatics. The mathematics is identical to what I am doing here. Does the self-field of a charge (even a point charge) cause its own acceleration? I don't think so! Similarly, a singularity in q at the field point does NOT affect the integral in equation (117). This then leads to the result in equation (120) that the time integral of grad p must also be finite.

 **Dwight Walsh** · 6 February 2022

Also, don't forget that by hypothesis, q is finite and smooth in all equations up to equation (120). Recall that we proved in equation (79) that a smooth solution exists for at least a finite time T. Therefore, we have the right to claim that equations (117) and (118) apply for t < T, in which case q has no singularities anywhere, including the field point. It is not until the following section that we consider whether such a smooth solution could evolve into a blowup. THIS REASON BY ITSELF IMPLIES THAT THE INTEGRALS IN (117) AND (118) ARE FINITE. This is actually my primary argument. My arguments involving the possibility of a singularity in q was included as "icing on the cake" and to (hopefully) satisfy those who tend to believe there is circular logic in my main arguments.

 **Dwight Walsh** · 6 February 2022

And while I am on the topic of circular logic, you and a bunch of others keep claiming that I do not assume a strong enough singularity in q, and that the integral in (117) must be based on such a singularity. But isn't that circular logic? You tell me — How am I supposed to prove existence and smoothness when in fact a singularity is one of the "givens"?

 **Dwight Walsh** · 17 February 2022

Correction to comment made on 6 February, 2022 at 10:44 AM.

In my most recent review of comments made on this forum, I discovered that I was somewhat confused as to exactly which points were covered in each section. It turns out that equations (115) and (118) both follow from inequality (109) and some elementary analysis involving integration in spherical coordinates.



Dwight Walsh · 17 February 2022

Also in this above comment, the all-caps sentence “THIS REASON BY ITSELF …” and everything following it can be deleted.



Dwight Walsh · 19 February 2022

… these counter examples for q do not obey the Navier Stokes equations. Well, of course they don’t! Otherwise we could submit these counter-examples to the Clay Institute and become millionaires :-).

This is exactly my point! In solving the Navier-Stokes equation (or any first-order in time PDE), the values of all dependent variables, including q, are only set initially (ie. at t=0). For all future times t>0, the values of the dependent variables are determined ONLY by the time evolution of the NS equation, and can no longer be altered at the discretion of the reader.

Therefore, to show a valid counter-example to my proof, you must do MUCH MORE than simply construct a q function that blows up but still satisfies inequality (109). You must prove that there exists a set of initial data for the NS equation such that your desired form of q eventually arises from the time evolution of the u_i according to the NS equation. And, as you said yourself, this would be nothing short of solving the Navier-Stokes Millennium Problem by counter-example, after which you can claim the glory and prize money,



Dwight Walsh · 22 February 2022

> … you need to rigorously prove that *any* counter examples one could find for q that verifies (109) while making (117) infinite does not obey NS.

Actually, you have proven this yourself! Your “counter-examples” reset q and therefore the fluid velocity components u_i, thereby rendering them as no longer solutions to the NS equation. Hence, the “does not obey NS” clause of your statement is proven. QED



Jeff Dylan · 15 May 2023

Your “counter-example” is totally WRONG! First, after wading through the equation mis-renderings in this blog, it seems that you messed up in determining the integral of q/r^2 as indicated in equation (117). In fact, you didn’t do an integral at all. You merely took the (constant) spatial integral of q and divided it by R^2, where R=|t-T_b| is the upper limit of the integration variable r.

Aside from that, however, your grad q at the “blowup point” is zero at all times. This implies that the time integral of grad p must also be zero, and therefore equation (117) still stands. Hence, you have a prime-example, not a counter-example of \Lambda being defined at all times.



Gandhi Viswanathan · 9 February 2022

Dear Dwight, try to slow down and think more carefully. You are too quick to dismiss others’ criticisms. Let me try to explain yet again the main point in a slightly different language. If you know how to extremize functionals subject to constraints, use your eq.

109 as a constraint to maximize the LHS of 117 or 118. You will see that the maximum can be made arbitrarily large. This means that the maximum velocity and also the maximum kinetic energy density cannot be bounded, so that your whole argument falls apart. This error has been explained by several others here already. Insisting that you are correct (even using capital letters) will not fix the error. I am also very sorry to tell you that there is no part of your paper that is salvageable, because what is correct is old and what is new is irremediably wrong. There is, moreover, no minimally plausible high-level strategy to your argument. I know that this is not what you want to hear and I am indeed sorry to give you this bad news. My advice to you: go back and relearn undergraduate and graduate-level courses with the open-minded attitude of the eager student. Read the seminal books, do the exercises and study the classic papers. It is never too late to learn. Focus on enjoying your studies. Then, as icing on the cake, you might even discover interesting new (and publishable) results.



Dwight Walsh · 9 February 2022

You didn't answer my question!

How am I supposed to prove existence and smoothness when in fact a singularity is one of the "givens"?



Dwight Walsh · 20 February 2022

>If you know how to extremize functionals subject to constraints, use your eq. 109 as a constraint to maximize the LHS of 117 or 118. You will see that the maximum can be made arbitrarily large. This means that the maximum velocity and also the maximum kinetic energy density cannot be bounded, so that your whole argument falls apart.

You seem to have forgotten another constrain on the left-hand side of equation (118), namely the Navier-Stokes equation. I'm afraid it takes MUCH MORE than pointing out that "the maximum can be made arbitrarily large" to show that my "whole argument falls apart". You must prove that there exists an actual initial data set for the fluid velocity such that your desired forms for this function eventually arise from solution of the NS equation.

In my proof of existence and smoothness, I maintained consistency with the Navier-Stokes equation each step of the way. Unfortunately, I don't believe the same claim can be made for your arguments.

Furthermore, the patronizing comments can be skipped in your future posts.



Gandhi Viswanathan · 22 February 2022

Dear Dwight,

I think you are getting ahead of yourself.

Remember that you cannot in general obtain for this problem a coercive (or controlling) critical or subcritical estimate from a supercritical estimate.

Professor Tao explains supercriticality and subcriticality etc. very clearly in his original post.

For instance, you claim that you are able to use supercritical quantities such as the total kinetic energy to bound the maximum kinetic energy density (which is subcritical). This is a red flag.

See Remark 52 in the following post of Professor Tao:

> 254A, Notes 1: Local well-posedness of the Navier-Stokes equations

 **Dwight Walsh** · 22 February 2022

>For instance, you claim that you are able to use supercritical quantities such as the total kinetic energy to bound the maximum kinetic energy density (which is subcritical).

WRONG! At no time do I use total kinetic energy (nor any other "supercritical" quantity) to argue that maximum kinetic energy density is bounded. That comes only from equation (85) (which follows directly from the Navier-Stokes equation in just a few steps), equation (119) (the hotpoint of contention about my proof), and an understanding of how a smooth blowup proceeds. (See equations (121)-(128))

 **Gandhi Viswanathan** · 23 February 2022

Dear Dwight, you are again mistaken. See, your estimates and bounds at the start of the paper are all supercritical. But by the end of your paper you have, miraculously, obtained a coercive subcritical estimate. But as I explained earlier, you cannot do this for the Navier-Stokes problem. If you still really do not understand why, consider this useful metaphor: turning water into wine. Transforming a supercritical estimate miraculously into a subcritical estimate (let alone a coercive subcritical estimate) for the 3D NS problem is analogous to turning water into wine. Yet such a miracle is exactly what you claim in your paper, because by the end of your paper you have a bound on the maximum kinetic energy density, which is subcritical. Do you see the problem here? You might as well also claim that you can turn water into wine!!!!!

 **Dwight Walsh** · 24 February 2022

What do you mean "miraculously"!? I've sweat over this problem for nearly 20 years. Could it be that maybe I discovered something that no one else had, including Prof. Tao? Maybe I came up with a much better way of accounting for nonlinear convective term that Prof. Tao still sweats over. This method would eliminate the need for complicated "scaling" arguments, averaged bi-linear operators, or any other approximate form of this nonlinear term. Or, maybe I discovered that a global maximum must first form at any potential blowup point, and in knowing this, it's easy to show that a blowup does NOT occur after all, thereby proving existence as well as smoothness. — But I guess we will never know for sure if no one gives the paper serious consideration because of their blind faith those "red flags".

 **Dwight Walsh** · 25 February 2022

>See, your estimates and bounds at the start of the paper are all supercritical.

WRONG! The initial conditions are as subcritical as you can get since they are defined at each and every point. Also, they are as coercive as you can get since they are EXACTLY SET as opposed to being estimated by time evolution.

 **Gandhi Viswanathan** · 25 February 2022

Dear Dwight, you are wrong on both counts. You write that "the initial conditions are as subcritical as you can get since they are defined at each and every point," which means that you do not understand at all what is meant by supercritical and subcritical. I suspected that this was the case, but now it is clear that you do not have a minimal understanding of such concepts. Subcriticality is not a property relevant to initial conditions!!! We only care about the subcriticality etc. of quantities that give control over blow-ups, e.g. perhaps certain global norms. Similarly, you write that "they are as coercive as you can get since they are EXACTLY SET as opposed to being estimated by time evolution." Again, you are mistaken and seem quite confused actually. If you had read and understood the original post of Professor Tao, you would know that the term "coercive" relates to a quantity that gives control over blow-ups or in some sense keeps the solution within some region of the function space. For example, in 2D, total energy is coercive and critical, which rules out finite-time blow-ups. In 3D, energy is supercritical.

You either did not carefully read the original post of Professor Tao or else you did not minimally understand it. Either way, I am giving up on further discussion with you. The "impedance mismatch" for communication between us is just too large.

 **Dwight Walsh** · 25 February 2022

All right, so maybe I don't understand Prof. Tao's post very well. But since "subcritical", "supercritical", "coercive", "noncoercive", "scalability", etc. seems to be the only lingo spoken here, I do best I can.

Now, the first sentence of Prof. Tao's Remark (52) reads as

"These heuristics can also be used to locate errors in many purported solutions to the Navier-Stokes global regularity problem that proceed through a sequence of estimates on a Navier-Stokes solution."

My proof however, does NOT proceed through a sequence of estimates of a Navier-Stokes solution. I only establish time-dependent upper bounds of the EXACT solutions u_i to the Navier-Stokes equation and there spatial derivatives to all orders. From there, I eventually show that the upper bound of the kinetic energy density cannot become infinite in finite time, thereby proving existing and smoothness.

At no point in the proof are solution estimates used in any form, but only upper bounds of EXACT solutions. Therefore, Remark (52) is not applicable to this proof.

Daniel · 7 February 2022

Dwight Walsh, you have not proved the first inequality following (6) on page 5.

Dwight Walsh · 7 February 2022

No kidding! I am only stating the main theorem on this page and not proving anything.

Daniel · 9 February 2022

I didn't mean you are meant to prove that on page 5 lol

Dwight Walsh · 11 February 2022

Then what DID you mean? I hope you didn't mean that I NEVER prove this inequality because that's what I did for most of the paper.

Daniel · 11 February 2022

Hi Dwight, yes I mean that you never proved that inequality. You use that inequality at the start of that "proof" and find it at the end of that "proof". That is not a valid proof as there is a loss of generality at the start of it.

Dwight Walsh · 11 February 2022

READ THE PAPER! IT'S IN THERE! Putting up with people who will criticize but not read gets tiring after a while.

Dwight Walsh · 7 February 2022

In view of the trouble I have had in getting my postings to render correctly, let me quickly and simply summarize the flawed logic that is plaguing most critics of my paper. It starts by ASSUMING (not proving nor even conjecturing) the existence of a finite blowup time T_b, and as a blowup time, it is possible to obtain arbitrarily large values of the fluid velocity by choosing t sufficiently close to T_b. Next, the claim is made that I failed to show that the solution is (uniformly) bounded over the semi-open interval [0,T_b). That, of course, would be quite impossible since we have already ASSUMED a blowup at time T_b. But then things get even more interesting when I show in equation (128) that no such blowup time even exists!

So, can we all say "circular logic"? You have to admit that it would be MOST CHALLENGING to prove existence and smoothness when blowup is a "given". No wonder Prof. Tao calls this an "impossible problem".

Daniel · 7 February 2022

You can't prove the form of the first inequality following (6) on page 5 with the help of assuming it.

Dwight Walsh · 7 February 2022

This mode of flawed logic is exactly how Antoine Deleforge makes his arguments in the paragraph:

“We gave you an example of a function q(x,t) that is smooth for all x\in R^3 and t in [0,T_b), that verifies eq. (109), but for which the limit when t goes to T_b of the limit when \epsilon goes to 0 of (117) is +\infty, i.e., the integral is unbounded at x=0 on the open-interval [0,T_b) (do the math, it’s pretty straightforward).”

Too bad those arguments all come tumbling down when we prove that T_b is nonexistent!

Antoine Deleforge · 8 February 2022

No, Dwight. We have pointed out to you a specific flaw in mathematical reasoning from eq. (109) to eq. (120) in your paper, where unproven statements are made, resulting in false claims and incorrect deductions. Now, either you correct those flaws, or remove that part altogether from the paper and completely restructure it accordingly. As long as you don’t do that, no one will take you seriously. Starting to point to completely different parts of your paper, mentioning “smoothness” every second sentence, or trying to dodge the problem by saying that after all that part of the paper was merely “icing on the cake” (then why is it even there?) is not helping your case. Quite the opposite.

The ability to admit that you have made a mistake is a painful but absolutely required skill for any researcher. Do you have this ability?

Dwight Walsh · 8 February 2022

WRONG!

You are the one making unproven statements. What about your presumption of the very existence of a blowup time? Doesn’t existence of a blowup time imply that there is a blowup? But you haven’t proven this. By ASSUMING there is a blowup, however, it then becomes trivial to block any proof of existence and smoothness. But it does get interesting when it is proven (without extra assumptions) that there is no blowup time.

Also, the function q(x,t) in the paper is a smooth, bounded, but otherwise unknown function, and this is the basis upon which I use this function when making conclusions. You, however, concocted your own version of this function and used it to claim that I failed to show that the solution is (uniformly) bounded over the semi-open interval [0,T_b). Now, can you prove that your version of q(x,t) is correct while mine is wrong? If not, then DON’T HASSLE ME ANY FURTHER!!

anon · 8 February 2022

Similarly, it is trivial to show that there is no blowup when you assume that there is no blowup.

Anonymous · 8 February 2022

It seems that you prefer to beleive that you can’t make mistakes.

Dwight Walsh · 8 February 2022

Where is it that I assume there is no blowup? I first PROVE that a smooth solution exists for at least a finite time in the discussion leading to equation

(79), and that this solution remains smooth for as long as it exists (ie. remains finite). This means that even if a blowup does occur, it must happen in such a manner that the solution stays smooth for t < T_b. This is often called a "smooth blowup". Finally, I use the Navier-Stokes equation itself to show that such a solution cannot evolve into infinite values, or equivalently that the conjectured blowup time T_b does not exist. So, once again, where is it that I assume there is no blowup? It seems to me that no blowup was proven, not assumed.

 **C** · 8 February 2022

First, you should have a look at the notions of « global mistakes » and « local mistake » : https://terrytao.wordpress.com/advice-on-writing-papers/on-local-and-global-errors-in-mathematical-papers-and-how-to-detect-them/ (written by a mathematician you might have some respect for).

Antoine is primarly pointing out a global mistake. In order to spot such mistakes, one isolate a segment of your argument, see which informations you take as « input » and which conclusion you get as « output ». In the segment from equation (109) to (120), you only use inequalities (109) and (111) (and smoothness until time T_b, not includes) as input, so your conclusion (120) should hold for **all** functions satisfying those assumptions (not only Navier-Stokes solutions), in particular for Antoine's proposal as q(x,t).

I hope this answer the part « why should your version of q(x,t) be correct while mine wrong? » : your conclusion should hold for all q(x,t) satisfying those two inequalities, in particular Antoine's example.

 **Dwight Walsh** · 8 February 2022

The function q(x,t) is defined in terms of the first spatial derivatives of the fluid velocity components through equations (106) and (108). It is NOT defined as the concoction of some mathematical "expert" anxious to block any proof of existence and smoothness by whatever means necessary!

 **Anonymous** · 8 February 2022

It is quite pointless to argue with Dwight. Several people on this blog have politely pointed out the error. If he continues to waste his time with this there's no one else to blame.

 **Dwight Walsh** · 8 February 2022

Well, how about that! Even you acknowledge that you can't start with one definition of q(x,t) and then change definitions midstream. Now, we are starting to make some progress!

 **Dwight Walsh** · 8 February 2022

>In the segment from equation (109) to (120), you only use inequalities (109) and (111) ...

WRONG! Equations (106) and (108) are also used to define q(x,t). As I stated above, this function is clearly defined. It is not "up for grabs" to someone who doesn't want the NS global regularity problem solved just yet.

 **Antoine Deleforge** · 9 February 2022

OK, this is my last try. If you don't make at least one step in the direction of understanding the mistake we are explaining to you after this one, I will just assume that you prefer sitting on a proof that you and you alone in the world is convinced is right rather than engaging with external reviewers to get feedback and make progress (like any true mathematician would do), and stop replying to you.

The situation is quite simple. We rigorously proved to you that the mathematical statement: "(109) and smoothness of q implies (118)" is false in general, by means of a counter example. You clearly make this statement in your paper. Hence, this part of your paper is false. This is not a matter of negotiation, debate, comparing "my q" to "your q", or any such non-sense.

It appears that you are becoming angrier and angrier, and in the process are making less and less sense in your comments. Now you are saying that (106) and (108) are also used to define q: well sure, but you never used these equations to deduce (118), so they are completely irrelevant to the discussion.

 **Dwight Walsh** · 9 February 2022

You are either mathematically illiterate or are intentionally being difficult, and in either case please keep your promise by not replying to my postings.

 **Dwight Walsh** · 9 February 2022

>Now you are saying that (106) and (108) are also used to define q: well sure, but you never used these equations to deduce (118), so they are completely irrelevant to the discussion.

WRONG! Equations (106)-(108) are used in proving inequality (109), and, although not specifically mentioned, inequality (109) is used in proving that the integrals in equations (115) and (118) are finite. I would hope, however, that this could be deduced from the context of these equations.

 **Antoine Deleforge** · 9 February 2022

Yes! It looks like we are finally agreeing on something :-) Just as you wrote: you are using (106-108) to prove (109) and then (109) to prove (118) [your words, not mine]. If we re-use Terence Tao's terminology from his post on "local vs. global errors" that was linked to you (I hope you read it?), you are using the following chain of implications:

(106-108) ⇒ (109) ⇒ (118)

Do you agree? Now, (106-108) ⇒ (109) is true, but this is irrelevant. Indeed, as was repeatedly proved to you in earlier comments, the statement "(109) ⇒ (118)" is false, which makes the entire reasoning collapse (a "local" error).

Worse than this, we proved to you that “(109) ⇒ (118)” is *globally* false, since we gave you a counter-example of it (see Prof. Tao’s post). This means that the intermediate steps “(109) ⇒ ... ⇒ (118)” that you use to prove it do not matter, we do not even need to check them [though we did, to the best of our efforts!]. Even if you provided 10 different proofs of this, they would still all be false, because of the counter example. Do you get it now?

Because this error is a global error, we believe that your proof is, unfortunately, beyond repair. Just like Gandhi Viswanathan, I feel sorry for you, because it is obvious that you are very emotionally involved and spent a significant amount of time on this. What may be frustrating to learn for non-mathematicians is that even if 99% of the statements in a mathematical proof are correct, a single error may irremediably destroy it. I wish you luck and joy in your learning path.

As a final comment, I would like to make you aware that many commenters have been kind, patient and polite with you throughout this exchange, but I don’t think the same same can be said of you, with your repeated use of insults and harsh tone. This not good form in any kind of online discussion and does not portray you in a nice light.

 **Antoine Deleforge** · 10 February 2022

I believe I can give you a simple analogy of the situation. Please take a look at the following reasoning:
– Suppose $a \geq -4$ (1)
– Therefore $a^2 \geq 16$ (2)
– In particular, $a^2 \geq 4$ (3)
– Therefore, by taking square roots, $a \geq 2$ (4)

The chain of implication of this “proof” is (1) ⇒ (2) ⇒ (3) ⇒ (4). The first two implications are true, however the last one, (3) ⇒ (4) is (obviously) false. I can prove it to you by using a counter example: $a = -3$ verifies (3) but does not verifies (4). Therefore, the entire implication is false and the proof is lost: we do not have (1) ⇒ (4).

Note that the counter example I gave in my refutation *does not verify (1)*. But it does not matter: the refutation still holds. Also, note that the fact that (1) ⇒ (2) ⇒ (3) is true is completely irrelevant to the discussion.

I hope you are able to follow the analogy with your mistake and that this clarifies things?

 **Antoine Deleforge** · 10 February 2022

I am terribly sorry, the first statement should be:
– Suppose $a \leq -4$ (1)

 **Dwight Walsh** · 10 February 2022

>Worse than this, we proved to you that “(109) \Rightarrow (118)” is *globally* false, since we gave you a counter-example of it (see Prof. Tao’s post).

Your “counter-example” is nothing more than a re-definition of the function q(x,t) such that there is now a blowup whereas before, there wasn’t. I’m sorry, but even Prof Tao’s post doesn’t license you to re-write authors’ function definitions.

 **anon** · 10 February 2022

How do you know there’s no such blowup? All you have is inequality (109), but that is not strong enough to show (118). If you want to claim that q can’t have behave like Antoine Deleforge’s explicit counterexample, you need to show it. Write (and prove…) some stronger inequality that allows one to get (118).

 **Antoine Deleforge** · 10 February 2022

@anon Yes, exactly.

Look at my analogy above Dwight. Do you agree that I am allowed to use $a = -3$ as a counter example to debunk the entire proof, even though the first hypothesis (1) says that $a \leq -4$? The situation is exactly the same with your paper: one is allowed to use any q that verifies smoothness and the inequality (109) to debunk the (109) $\Rightarrow$ (118) part of your proof, without any consideration of (106-108). If you want to make use of other properties of q to show (118), then the burden of doing this is entirely on you.

This is really basic mathematical reasoning, I am afraid I cannot explain things in a simpler way. About 6-7 people in this comment section who don’t know each other, never met before and likely work in completely different fields all agree on this. Terence Tao himself wrote an entire blogpost explaining why this is a perfectly valid way to find errors in a math paper. If you can’t get behind this, I don’t know, maybe math is simply not your thing and you should work on something else?

 **Dwight Walsh** · 10 February 2022

>Write (and prove…) some stronger inequality that allows one to get (118).

I did! I defined q(x,t) in equations (106) and (108), which I have the perfect right to do, and since the fluid velocity components are smooth, these equations imply that q(x,t) must also be smooth (or at least continuous and integrable. I realize that the absolute value signs in equation(108) precludes strict smoothness of q in the sense of equation(79)). Afterward, however, our use of q must be consistent with this definition. WE DO NOT CHANGE THE DEFINITION OF THIS FUNCTION MIDSTREAM. THAT WOULD BE MATHEMATICALLY ILLEGAL!

At this point, I make a sequence of mathematically valid steps which proves there is no blowup. Thus far, I have not encountered objections to any of these steps.

Now to me, all of this seems like basic mathematical literacy. Why is it that I have to lead “experts” by the hand through it?

Dwight Schrute · 10 February 2022

FALSE!

Brown bears are the best!

Jim Halpert · 10 February 2022

Stop telling people on the internet that you proved the Navier-Stokes problem, Dwight

Dwight Walsh · 10 February 2022

>Stop telling people on the internet that you proved the Navier-Stokes problem, Dwight

WHY SHOULD I? I started participating in this forum hoping to share my ideas concerning the Navier-Stokes Millennium Problem and maybe get some good feedback, whether it be positive or negative. My reception, however, has been generally hostile, and even the "kind" or "polite" comments have been patronizing and insulting. And at this point, the only thing that you "expert" mathematicians have been able to muster up to refute my proof is a so-called "counter-example" where the function q(x,t) [which I defined] is re-written to produce a false blowup. This, of course is not a counter-example, it's CHEATING!

Now, if you can't do any better than this to refute my proof, then you must be very desperate for a victory, and I am more convinced than ever that I have a valid proof. As such, I will continue to do my utmost to get the word out about my solution. And if you don't like it, then get out there and shoot it down, but this time don't insult my intelligence with phony counter-examples.

Dwight Walsh · 11 February 2022

>About 6-7 people in this comment section who don't know each other, never met before and likely work in completely different fields all agree on this.

SO WHAT!? Mathematics is not a democracy, and it's happened to me a bunch of times during my rather brief career when I was shouted down but later prevailed.

Dwight Walsh · 11 February 2022

>It seems that you prefer to beleive [believe] that you can't make mistakes.

No Anonymous. I acknowledge that I have my fair share of mistakes in this lifetime, but the biggest has been in trusting someone else's judgement over my own after being told that they are somehow "smarter" or "better" than me.

Dwight Walsh · 11 February 2022

>Look at my analogy above Dwight. Do you agree that I am allowed to use a=-3 as a counter example to debunk the entire proof, even though the first hypothesis (1) says that a \le -4 ?

No! Your “analogy” is total nonsense. The statement that a=-3 contradicts your original hypothesis that a<=-4, and must therefore be regarded as FALSE. Now a FALSE statement could imply a TRUE statement, but it could also imply another FALSE statement. Therefore, no conclusion can be drawn from a false statement, which is exactly what you are trying to do! This is very fundamental logic that I learned in high school.

 **Anonymous** · 13 February 2022

ehh..

 **Anonymous** · 13 February 2022

>No! Your “analogy” is total nonsense. The statement that a=-3 contradicts your original hypothesis that a<=-4, and must therefore be regarded as FALSE. Now a FALSE statement could imply a TRUE statement, but it could also imply another FALSE statement. Therefore, no conclusion can be drawn from a false statement, which is exactly what you are trying to do! This is very fundamental logic that I learned in high school.

...I'm trying to figure out if you're a troll or simply a crank. I'm also extremely uplifted by the kind replies to you on this page – there is hope in humanity after all!

 **Antoine Deleforge** · 13 February 2022

> The statement that $a = -3$ contradicts your original hypothesis that $a \leq -4$, and must therefore be regarded as FALSE.

OK great, now we know exactly what is the basic source of your misunderstanding! You seem to struggle a bit with fundamental logic and in particular the concept of implication (see e.g., this Wikipedia page if you want to learn more about it: https://en.wikiversity.org/wiki/Logical_implication). I find it surprising that you have such a gap, since on the other hand you seem to be pretty comfortable with manipulating basic integrals, derivatives, inequalities, etc., but hey, I guess different minds work differently!

Now let’s look again at the proof in my analogy and try to understand why the counterexample $a = -3$ I gave is a *perfectly valid way* to refute it. First of all, your childhood memory is correct: a false statement can imply any statement, true or false. This is known as the “principle of explosion” (https://en.wikipedia.org/wiki/Principle_of_explosion). However, this is not at all what is going on here.

Let me ask you first: how would YOU refute the false proof I gave in my analogy? Obviously, you cannot do it by refuting every single step of the proof: as we already discussed, the implications (1)⇒(2) and (2)⇒(3) are correct, so it is impossible to refute them. Hence we have to focus on the implication (3)⇒(4). Once we have chosen to focus on this implication, the rest of the proof doesn’t matter: all that matters is the “input” and the “output” statement of this implication. In that case $a^2 \geq 4$ is the input, $a \geq 2$ is the output. How would

you *prove* that this implication is false? Well, it turns out that the easiest way to disprove an implication of the form "B(a) $\Rightarrow$ C(a)" is to find a value a such that "B(a) AND not(C(a))", also known as a "counter example". In that case, the counter example $a = -3$ works. And voilà! We showed that the implication (3) $\Rightarrow$(4) is false in general. Hence, we showed that this particular proof of (1)$\Rightarrow$(4), that critically relies on the implication (3)$\Rightarrow$(4), is NOT valid.

Now the (imaginary) author of that proof could reply "But I AM THE ONE who introduced the variable a, and I DECIDED in my hypothesis (1) that $a \leq -4$, how DARE YOU changing its definition?!" But this (imaginary) author would be misguided. All we did was to rigorously prove that the particular sub-implication (3)$\Rightarrow$(4) of the proof was FALSE, purely based on its input and output, independently of the rest of the proof. Sure, we could have used another counter example such as $a = -5$, which does verify (1). But why should we bother doing this, since $a = -3$ works? Only to please the (angry) (imaginary) author? Since the proof crucially relies on the (3)$\Rightarrow$(4) step, the burden is of course on HIM to replace this false statement by a correct one!

[Note that in that particular example, we know that repairing the proof is hopeless: the (imaginary) author will never find an alternative proof of (1)$\Rightarrow$(4), because it is obvious that no one can deduce $a \geq 2$ from $a \leq -4$]

Please, take at least a little bit of time to digest this before jumping on your keyboard to write an angry reply. I am sure your are smart enough to understand this (as your paper shows, you do have some solid background in other areas of math).

Then, hopefully, you'll understand more clearly the type of refutation we gave to your proof. This is not about "your q", "our q", "changing a definition midstream" or "tampering with your proof", no more than it is about "changing the definition of a midstream" in my analogy. It is, and has only ever been, about exhibiting the falsehood of a particular logical implication.

Kind regards,

Antoine



Dwight Walsh · 14 March 2022

>The situation is quite simple. We rigorously proved to you that the mathematical statement: "(109) and smoothness of q implies (118)" is false in general, by means of a counter example. [Antoine Deleforge, 9 February 2022, 12:52 AM]

The fallacy of this argument is your "false in general" clause. As was clearly stated in both the first and last paragraphs of this section, we are working only with pre-blowup values, and your "counter-example" includes a (presumed) blowup. Whether the fluid velocity can blow up at some finite time T_b (according to the NS equation) is addressed in the next section. Meanwhile, all dependent variables are of finite value as well as smooth, including u_i and q.

Therefore, your “counter-example” is not applicable to this section, and equation (118) still stands.

Antoine Deleforge · 15 March 2022

Now you start sounding like a broken record. We explained to you that it is not a question of whether your variables “have infinite values” (I hope you realize that this is none-sense: a function does not take infinite values), but of whether q(x,t) is bounded over the semi-open interval [0,T_b). Note that this semi-open interval does *not* include T_b. We are indeed talking about the pre-blowup regime, that’s literally the *definition* of a pre-blowup regime: times t with t<T_b.

I already explained to you in full details why our counter-example invalidates your proof in my last comment (13 February, 2022 at 9:57 am), in a language that I believe should be understandable by a high-school student with a decent level of math. If you are not able to follow this, I am afraid the situation is hopeless and there is no point in continuing. Bye!

Dwight Walsh · 15 March 2022

What is it that is so difficult to understand about the fact that in this section, we are only establishing relations between certain functions for some pre-blowup value t of time. You might notice that this analysis does not consider any time value, such as a blowup time T_b, that is greater than t. Time integrals are done over the fully CLOSED interval 0 <= t' <= t t, assuming T_b exists. The q function is, of course, bounded over this CLOSED interval, which implies that equation (118) is true.

As I’ve pointed out numerous times, we are NOT considering in this section what happens as t -> T_b or if T_b even exists. Those questions are addressed in the next section. Understanding the concepts presented in the current section leading to equation (118), however, is prerequisite for understanding the arguments in the following section where we complete the proof of existence and smoothness.

Now do you understand why your “counter-example” is meaningless in regard to my proof?

Jeff Dylan · 16 May 2023

In regard to the reply from Anonymous

>…I’m trying to figure out if you’re a troll or simply a crank. I’m also extremely uplifted by the kind replies to you on this page – there is hope in humanity after all!

If you are using invalid counter-examples, circular logic, or straw-man arguments, we can be a troll and a half!

Anonymous · 10 February 2022

Rather than the article On “local” and “global” errors in mathematical papers, and how to detect them, the advise Don’t prematurely obsess on a single “big problem” or “big theory” in

the blog seems more relevant here. We have yet another example of “mathematical disease”, which is mentioned at the end of the second linked post.

Anonymous · 10 February 2022

This may be called “NS addiction”

Anonymous · 12 February 2022

In order to check your (currently disputed) claim that (118) follows from previous steps, it seems natural to formulate this claim as a separate(!) general lemma whose “input assumptions”(e.g. regularity properties of certain “input functions”) should be explicitly and precisely formulated such that: all “input assumptions” should follow from previous steps, and (118) should follow as the “output” of the lemma (based only on its “input assumptions”).

Dwight Walsh · 12 February 2022

I’m not sure I completely understand what you are saying, but I will recap the traceability of the function q(x,t) and equation (118) to see if we can find some common ground. The function q(x,t) is defined in terms of the first spatial derivatives of the fluid velocity components u_i through equations (106) and (108). Then, through the previously proven results (79) and (106), we show that q(x,t) is continuous and integrable over R^3 and therefore any sub-region. At this point, we must determine the function Lambda(x,t) by evaluating the integral on the right-hand side of equation (112). We first split this integral as shown in equation (114), and then show that both integrals on the right-hand side of this equation are finite. For the first integral, this is readily shown in inequality (115) and won’t be repeated here. The second integral reduces to an integral over a finite range of spherical coordinates used in describing a continuous function over R^3, and is therefore finite. Hence, we have proven equations (118) and (119).

Therefore, I believe this shows that all “input assumptions” of equations (115) and (118) are previously proven results.

Anonymous · 12 February 2022

In order to rigorously(!) check your claim, you need to give your (verbal) explanation a rigourous(!) mathematical form and formulate your claim on (118) as a rigorously stated lemma (as suggested above).

Dwight Walsh · 12 February 2022

COMMON NOW!!

Prof Tao’s entire article on this forum is verbal and no one has challenged him on it! If, however, a different standard somehow applies to me, then you will need to show me exactly what you mean by a “rigorous(!) mathematical form”, preferably with an example. So why don’t you take a simple theorem (eg. A = pi r^2 for a circle) and give it a “rigourous(!) mathematical form”. Meanwhile, papers consisting of equations and possibly diagrams with clear verbal explanations is the standard generally used.

Dwight Walsh · 12 February 2022

On the top line, I meant COME ON, NOW!! (— LOL)

Well, I never claimed to be infallible.

Anonymous · 12 February 2022

An extraordinary claim should have an extraordinary rigorous proof to be accepted. The basic idea is to isolate the most disputed step in your proof (i.e. your claim that (!!8) follows from the previous steps) and formulate it as a general lemma whose "input assumptions" should (rigorously!) follow from the previous steps. In this way one can check this claim independently(!) from the rest of the proof.

Anonymous2 · 12 February 2022

First: contrary to you, Terence Tao never claimed he solved a millennium problem (quite the opposite actually, if you read the blogpost!) ! Or did you seriously think he did?

Second: this is a *blogpost* for god's sake, not a mathematical research paper! What are you thinking? Rigorous math research is not published through blog posts (let alone blogpost *comment sections* (ahem)!) but through formal peer-review in mathematical journals.

Third: I don't know if you are aware but your paper looks nothing like a serious mathematical research paper. Normal math papers are structured in definition, lemmas, theorem, corollaries, proofs, precisely so that they can be *rigorously* checked easily, namely, not by having to counter-argument never-ending, rambling and hand-wavy prose (ahem!). For complex proofs, the papers generally begin with a lengthy introduction that clearly explains the structure of the proof and the different logical dependencies between lemmas and main theorems.

Your paper has none of that.

Dwight Walsh · 12 February 2022

>An extraordinary claim should have an extraordinary rigorous proof to be accepted.

All you did was rehash your posting of 12 February 2022 at 10:50 AM without answering any of my questions. Now how is it that my paper and my posting at 10:10 am falls short of the "rigorous(!) mathematical form" or "extraordinary rigorous proof"?

Dwight Walsh · 12 February 2022

>Rigorous math research is not published through blog posts.

Then how is it that numerous commenters on this forum referenced Prof. Tao's above article as justification for their tampering with the q(x,t) function (which I defined in the paper), and then claiming a counter-example to my proof of existence and smoothness based on their version of this function. It seems like this blog post research was good enough for them!

Also, you used a certain phrase in your second paragraph that I found offensive! Please don't use it anymore.



Dwight Walsh · 12 February 2022

>I don't know if you are aware but your paper looks nothing like a serious mathematical research paper.

DON'T GIVE ME THAT!! If my paper did not at least look like a serious research paper, it would have taken a typical reviewer five minutes instead of five months to reject it.



Anonymous · 12 February 2022

If you think that your proof correct and clearly written, you may try to publish it in a mathematical journal.



shantanuk · 12 February 2022

Hi everyone, have been following this interesting discussion but thought I'd intervene. I'm a recent grad from Materials science and an absolute beginner in math and very basic physics. No idea what sobolev inequalities are, or all the mathematical terms a lot of you have written. But I love to learn so all of this discussion has given me new topics to google or wiki about. I've read through and based on my very little understanding I write this. So please feel assured. And thank you all for spending the time.

First off let me start off with a few things: Hi Dwight. I really admire your passion. It's amazing to see the amount of effort you put into this. Really love that you have taken the time to challenge yourself and try to solve something and any effort on such things is commendable. You definitely have a lot more experience than me so I won't talk about math or equations here. I can only tell you what I got from my understanding of reading the comments here (in detail) and the wikis of a lot of terms I wasn't sure about. The only reason I am doing this, please be assured is not to put you down in any way. I really love the effort you're putting but just want to clarify what seems to me to be a communication issue on both sides so there's a proper direction. There's no negativity or sarcasm in any of my comments even if I use a double quote "". So please take all of this constructively, I would be the first to love to see you solve the N.S. equations (No sarcasm, only positivity). My only request here is one thing, please do read through the whole thing without trying to reply first.

So, please excuse me when I say this, I'm sure you know a lot of this but just trying to elucidate that's all not condescend since I know less than you do.
– I think smoothness and bounds are two different things. Your comment on 5th Feb at 1:57 (my time zone is PST) seems to have confused Prof. Antoine. The smoothness doesn't seem to be the focus on the problem, that's not what they seem to be arguing about. It's the boundness of equation 117. Your statement there might not hold true since smoothness doesn't always prove finiteness or boundness. To counter this, Prof. Antoine provided an example of a function q(x,t) in his comment on 4th Feb at 1:56 am where it is still smooth but makes the equation 117 unbounded which is the primary problem we're trying to solve and find a bound for.

– To this example your reply was that q doesn't obey Navier Stokes equation. But we cannot use a q that obeys Navier Stokes because that kind of q is exactly what we are looking for

which we don't have yet. We are trying to figure out what sort of q would obey it. If we already had a q which obeyed NS we wouldn't need to do this exercise. What Prof. Antoine gave an example for was a function which basically complied with your 109 inequality but still gave an unbound limit for 117. Now you can argue that you can't use any function and must only use the strongest singularity which you maintain is a Dirac delta function.. Okay let's go with that theory. If that's the case we can't just assume dirac delta is the worst case singularity we need proof.

– The other issue is that like Anonymous mentioned on 12 Dec at 11:00 am and Prof. Victor has, the limit as E->0 in 118 has been assumed to be finite by you (You've removed the e->0 term). Because while going from 117 to 118 you say we will have a smooth, finite solution over a finite range of spherical coordinates. This is where you are assuming a limit at a particular value instead of it tending to. So basically you have assumed the limit at 0 is finite implicitly. But the tends to (->) isn't exactly equal to. Your integral is actually unbound here.

– Let's also look at the problem physically, math is my weak area. If w look at the problem Prof. Tao mentioned in this article is that 1) The main thing which makes the solution difficult is the fact that there is turbulence 2) Turbulence displays highly non-linear behaviour and this is amplified in the more finer a scale you go to because it increases the higher frequency oscillations (These higher frequency oscillations are the ones we approach as our limit tends to 0 and makes things more complex to mitigate) 3) all the globally controlled quantities we are aware of are supercritical (solutions tends to Infinity as we tend to 0) but we need either critical or subcritical (equals a finite value) which means they become weaker in controlling the solution value as the scale gets finer. Prof. Tao illustrates this in the scaling paragraph where the two best quantities to evaluate this are Max Kinetic energy and Cumulative energy dissipation. Both these quantities are test q functions for example. He says they are best because the energy conservation law binds the max values they can have which may be large but finite. These two factors are also coercive i.e. they help control the solution to finiteness which even at late times still stays bound so they're the best bets for NS solutions. But even these quantities start to get affected by the scaling when we go to finer regions because of the Lambda factor. (To me this looks like the length term in the formula for a tuning fork frequency. If we make the length smaller and smaller by going to smaller scales, the frequency invariably becomes higher and higher. If length equals 0 exactly, the frequency becomes unbound. What you have done is assumed indirectly of a subcase that is satisfied until the blowup time but not at or close to it)

– Your function 117 118 have assumed a limit implicitly at a particular value and proceeded from there. So you have basically taken an asymptote and make it fixed which we cannot do basically which is why anonymous says we can't take r=0 but r^-100 or some miniscule value. Infinitesmal is the issue. You say on 12th Dec at 2:14 pm that "Therefore, the function q must still be smooth, and r^{-100} is not smooth at r=0". But the problem is we cannot set r=0, it is always Limit r->0. You are evaluating it at r=0 which means you have removed the limit and evaluated the function at a particular point in which case you are actually examining a sub-case of the real problem. The sub case being subcritical or critical while the problem that has been baffling people is one of supercriticality.

 **shantanuk** · 12 February 2022

And to add to Prof. Antoine's comment you reply at 6:28pm on 4th Feb to his assertion that you say 117 is bounded but don't give proof to which you mention equations 1) to 79) as being used as proof for smoothness and the very definitions of smooth. But they are just that – definitions for smoothness, not for finiteness. They might be smooth (continuously differentiable) but they not be finite. So like you mention there, "smoothness is proven for as long as the solution remains finite" which may be true but the inverse isn't always true. Your argument contains in itself the assertion that the time of Blowup is a fixed value and can be attained so your entire solution has turned out finite. But it's actually an asymptotic limit not an exact value. Which is what blowup means. The smooth blowup you are referring to seems to be a sub case of the more general case and a value that has been fixed.

 **Dwight Walsh** · 12 February 2022

Just to let you know I did read through your post, and while you acknowledge that you have a weak background in mathematics, I am willing to help since you are willing to learn. It might be best, however, if you first email me at mathguy@navierstokessmoothsolutions.com and we correspond by email instead of taking space on this blogpost.

I will, however, point out what I believe is your primary misunderstanding of my proof of existence and smoothness of solutions to the Navier-Stokes equation. Well, actually it is EVERYONE'S misunderstanding though few will admit it. That is, the function q(x,t) is NOT something we solve for in the NS equation. It is precisely defined in terms of the first spatial derivatives of the fluid velocity components in equations (106) and (108), and is used as a tool in the proof. From these equations and the (previously proven) inequality (79), we find that this q function is continuous and integrable over the entire R^3 space. Also, remember that there has been no blowup yet, and therefore q and the fluid velocity components are all bounded at the current time.

Now, a definition is not a statement that is proven or derived. It is a statement that unambiguously (or uniquely) describes a new entity. This new entity [the function q(x,t) in our case] can be specified any way we want, but once the definition is made, it cannot be changed.

In my proof, I DEFINED q(x,t) according to equations (106) and (108), and based on this, I proved existence and smoothness of solutions to the Navier-Stokes equation as specified in the statement (2A) of the Official Problem Description. However, Prof. Antoine (and a bunch of others) came up with there own version of q(x,t) which contains the r^{-100} factor you mentioned earlier, and with the new version of q, a finite-time blowup DOES occur. Prof. Antoine then calls this blowup a "counter-example" to my proof of existence and smoothness based somehow on the writings of Prof. Terence Tao which he never makes clear. Personally, I find nothing in the writings of Prof Tao that licenses a reader or reviewer to make unauthorized changes to a paper. Furthermore, tampering with the q function completely invalidates his proof and is GROSSLY UNPROFESSIONAL and UNETHICAL.

Anonymous · 12 February 2022

Relax Dwight and find another hobby

anon · 12 February 2022

Of course nobody can give a solution u to the Navier-Stokes equations such that blow up happens – if they did, they would have solved the problem and could collect the prize and glory!

But we can, and have, given examples of q (yes, not Q, but we can it you really need it) such that 109 is true but (119) is not bounded. This means that you need some specific feature of q coming from it being defined in terms of a solution to the Navier-Stokes that makes the (119) true. Smoothness is not enough. So what is it?

> **Dwight Walsh** · 13 February 2022
>
> >Of course nobody can give a solution u to the Navier-Stokes equations such that blow up happens.
>
> I'm not sure what you are talking about here since I didn't make such a claim.
>
> **********
>
> >But we can, and have, given examples of q (yes, not Q, but we can it you really need it) such that 109 is true but (119) is not bounded.
>
> Those examples of q are not consistent with the definition of q. As I have said a million and one times already, YOU CAN'T ALTER A DEFINITION OR YOU WILL INVALIDATE THE PROOF!! What's it going take to get that point across??
>
> > **anon** · 13 February 2022
> >
> > Once again: what property of q makes (119) bounded? It's not (109) or smoothness or being an increasing function of t, which follows from (108), because it's easy to construct counterexamples.

Dwight Walsh · 14 February 2022

>This means that you need some specific feature of q coming from it being defined in terms of a solution to the Navier-Stokes that makes the (119) true. Smoothness is not enough. So what is it?

That's a good question, and while I am not yet totally certain about it, I believe it is related to the fact that q(x,t) and Q(x,t) depend explicitly on the first spatial derivatives of the fluid velocity components u_i, but not the u_i themselves. This would tend to make these quantities much less sensitive to changes in fluid velocity over time as we see in equations (98)-(109) as well as (115) and (118).

> **Antoine Deleforge** · 14 February 2022
>
> > "and while I am not yet totally certain about it, I believe..."
>
> Yes! It looks like you are finally understanding what is missing in your proof! This makes me genuinely happy :). I was starting to lose hope.

I don't know what it is in the post of anon that made you understand, since me other posters have been trying to say exactly the same thing since the beginning, but to be honest I don't care, I am just happy this is finally resolved. Thanks anon !

I guess you will now stop claiming that you solved Navier Stokes, right?

Dwight Walsh · 14 February 2022

>I guess you will now stop claiming that you solved Navier Stokes, right?

WRONG! I still believe I have a valid proof of existence and smoothness, or at least that you have not invalidated it. It would be great, of course, if we could get some better insight as to how equations (115) and (118) come about, but that does not mean they are incorrect, and you have not refuted them.

Furthermore, you specifically designed your "counter-example" q(x,t) to blow up at the field point while still satisfying equation (109) and gave it an extremely unlikely time dependence to make it blow up at a certain finite time T. It would be an AMAZING COINCIDENT if your counter-example also satisfies my definition of q from equation (108).

Anonymous · 14 February 2022

Dear Antoine,

With all due respect, perhaps there are better things to spend your time on…

Anonymous · 14 February 2022

Dear Antoine,

With all due respect, perhaps you are not spending your time too wisely here…

Dwight Walsh · 15 February 2022

For what it is worth, I made a minor revision to the paper that I hope will clarify why the integral in equation (118) must be finite for all t>0 as well as the integral in equation (115). In both cases, it a result of inequality (109) and the fact that q is continuous and integrable over all R^3 for pre-blowup values of t. As usual, the paper is posted as the most recent version on the website http://www.navierstokessmoothsolutions.com

Dwight Walsh · 17 February 2022

I will make one more attempt to clarify what this q function is and what it isn't. I realize that probably no one else active in this forum gives a hoot, but I will make it anyway for the sake of completeness.

The function q, like the fluid velocity u, scalar pressure p, and the function Q are precisely defined in the problem statement and can only be set initially. The fluid velocity components u_i at t=0 are the initial conditions for the Navier-Stokes equation and the only actual input data for the problem. The function Q is then determined on a real-time basis from equation (13), and the scalar pressure p is obtained by solving the Poisson equation (12) for p, using the recently determined function Q as the non-homogeneous term on the right-hand side

of this equation. Finally, the function q is defined as the time integral of |Q|. To summarize, once the initial fluid velocity components u_i are selected, the values of the above mentioned quantities are determined soley by evolution of the Navier-Stokes equation for all future times t>0, and can no longer be set by the reader.

Therefore, in order to find a counter-example to my proof, you must do MUCH MORE than simply construct a function q that results in a blowup while still satisfying inequality (109). You must prove that there exists a set of fluid velocity initial values u_i(0) such that q eventually takes the form you desire while solving the Navier-Stokes equation. And if you can do that, then you will have solved the Navier-Stokes Millennium Problem by counter-example, and can collect the glory and prize money!

Anonymous · 17 February 2022

Amazing!!

Dwight Walsh · 18 February 2022

I'm glad you think so! It seems now that if you want to prove equation (118) to be wrong, you will have to do so "locally", and so far, no one has been very enthusiastic about "stepping up to the plate" for that.

Anonymous · 18 February 2022

This is truly incredible! You've overcome Navier-Stokes!

Dwight Walsh · 18 February 2022

I'm not quite sure what you mean by "overcome", but if you have finally realized that the creative re-definitions of q on the part of the reader do not constitute valid "counter-examples" to my proof, then that is GREAT news and hopefully the thumb-downers to my recent comment will come around eventually. Anyway, that is how I will view the situation unless I hear otherwise.

Anonymous · 19 February 2022

It lifts my heart to see Navier-Stokes solved! Perhaps you would consider conquering yet another Millennium Problem – the Yang-Mills mass gap? I can't wait to see what new ingenious techniques you can bring to the table!

Dwight Walsh · 20 February 2022

>It lifts my heart to see Navier-Stokes solved!

Wonderful! I take it then that you have read the paper in its entirety and agree with each of my steps. Remember that in mathematics, one small error in any one of these steps invalidates the entire proof. That's just the nature of the beast! And, of course, you realize that my struggles and sometimes fiery exchanges on this blogpost for the past few weeks have NOT been about my claim of having solved the Navier-Stokes Millenium Problem, but only that their claims of "counter-examples" that show a "global" error in my proposed proof are invalid and meaningless. Even if I "win" in these exchanges, there could still be a long ways to go in actually getting my proof accepted.

But thanks, I appreciate your confidence in me!

Anonymous · 20 February 2022

Time and again the gods bless us with oracles showing the right path toward enlightenment. Dr Walsh, I truly believe you belong to this pantheon. It is indeed a humbling experience to be interacting directly with you!

Anonymous · 18 February 2022

In order to complete your proof, you still need to show that there is no such initial data which cause a blowup. On the other hand, to show a blowup one needs to find an explicit initial data leading to a blowup. A third possibility is that the NS problem is undecidable.

Dwight Walsh · 18 February 2022

WRONG!

From equations (115) and (118), which leads to equation (119), it is shown that the time integral of |grad p| is finite over all x in R^3 and t>0. This result is then used in the following section to show that there is no finite value of t where the kinetic energy density K (at any point x) becomes infinite. (ie. where a blowup occurs). This proves existence and smoothness.

I'm afraid that if you can't find actual (or "local") error(s) in my arrival at these results, then you don't have a case against me.

Dwight Walsh · 18 February 2022

>On the other hand, to show a blowup one needs to find an explicit initial data leading to a blowup.

RIGHT!

****

>A third possibility is that the NS problem is undecidable.

This, of course, is always a possibility for any unsolved problem. For this problem, however, my "just-plain-horse-sense" tells me this is not the case, which is partly why I spent so much time and effort in pursuing my proposed proof.

Anonymous · 18 February 2022

It appears quite unlikely that the global regularity problem is undecidable. The general expectation is that it is possible in principle to construct a blow up (essentially for the same reason that proofs like dwights are likely to fail)

Dwight Walsh · 18 February 2022

>The general expectation is that it is possible in principle to construct a blow up (essentially for the same reason that proofs like dwights are likely to fail)

Where did you get this tit-bit of info? My understanding is that the general expectation is smoothness since there are (or at least seem to be) so many more attempted proofs of smoothness than attempted searches for counter-examples. Also, what is this "reason that proofs like dwights are likely to fail"?

Anonymous · 18 February 2022

At this point experts in the field (eg. Tao) expect blowup. The fact that many non-serious people have attempted proofs isn't an argument for it being true. (The existence of many religious people doesn't imply god exists either.) You'll notice that serious people are not working on it...

A reason your proof should fail is that this method of proof is ruled out by this theorem: https://arxiv.org/abs/1402.0290 Your arguments would apply to a slight modification of the averaged equation which blows up.

Dwight Walsh · 18 February 2022

>In order to complete your proof, you still need to show that there is no such initial data which cause a blowup.

I probably should have indicated this in my comment of 18 February 2022 at 3:36 PM, but inequality (6), which is part of the hypothesis of the main theorem, allows the initial values of the u_i to be any smooth function over R^3 that also satisfies the spatial boundary conditions at infinity. Therefore, the main theorem (which was eventually proven) implies that a smooth solution will result from the any smooth initial data consistent with inequality (6)

a · 18 February 2022

For the final time: your claim (118) doesn't follow from (109) and smoothness as has been demonstrated with explicit counterexamples. There must be some extra ingredient coming from the fact that your q is defined in terms of a solution of the Navier-Stokes equations. But your supposed proof of (118) never uses that fact. Do you truly not see anything wrong with your "proof"?

Dwight Walsh · 19 February 2022

I sure hope this IS the final time.

Please read my post of 17 February 2022 at 11:13 PM. That should dispel any notion that your "counter-examples" are valid. When you understand this, you will also see that inequality (109) implies that the integral in equation (118) as well as the one in equation (115) is finite for all pre-blowup values of t.

Dwight Walsh · 19 February 2022

>But your supposed proof of (118) never uses that fact.

Actually, I DO use this fact although it is not specifically stated. By allowing time evolution of the NS equation, and ONLY time evolution of the NS equation, to control the values of the u_i, we know that q and Q are determined correctly since the correct values of u_i are used. Your "counter-examples", however, disrupt this process by resetting q to who-knows-what.

Dwight Walsh · 21 February 2022

Here's another way you can view the situation of you still believe you have a counter-example to equation (118) simply by constructing your own q function such that the

solution blows up while still satisfying inequality (109). While you did recognize that (109) had to be satisfied, you seem to have forgotten the most important constraint of all, namely the NAVIER-STOKES EQUATION. And, in order to satisfy this equation, you CANNOT tamper with any of the dependent variables (including q, Q, u_i(x,t), p) since they were obtained from the time-evolution of the NS equation for all t>0. Therefore, they ARE the solution to the NS equation, and if you replace the current q with your version of it, then you will no longer have a solution to the NS equation, and your efforts will be wasted.

 **Daniel** · 18 February 2022

You are all welcome to comment on my paper "The Navier-Stokes problem" which was mentioned on this blog on 4 December 2020.

 **Anonymous** · 18 February 2022

A red flag is that you cite a book on functional analysis in order to use the triangle inequality in $\mathbb{R}^n$.

 **Daniel** · 20 March 2022

I would actually be happy if people explain errors in my NS or RH papers to me and kill my proposed proofs. No one has done that.

 **Dwight Walsh** · 23 February 2022

Recently, there has been some more controversy arising about equations (117) and (118) of my paper. I know this isn't exactly "news" anymore, but I feel it is important to get this matter cleared up ASAP, and hopefully, it won't take too long.

Examining the right-hand side of equation (117), you will see that it consists of taking the limit as \epsilon -> 0 of a certain integral (sometimes called the Poisson integral) over the "primed" coordinates covering all of R^3 except for a small sphere of radius \epsilon centered at the field point x. The value of the integral on the left-hand side of this equation is then defined as the lim \epsilon -> 0 of the integral on the right-hand side.

Now, in this exercise, the function q generally depends on the time t as well as the spatial coordinates, and we wish to construct a function q that causes a blowup at a certain finite time T_b and understand what happens as t approaches this blowup time. Therefore, we have two iterated limits to calculate, and the question is which limit should be calculated first, and which should be calculated last.

At this point, Gandhi Viswanathan, Antoine Deleforge, and Anonymous claim that the lim \epsilon -> 0 must be calculated first, and only then can we consider the "implicit" limit as t -> T_b. Now, my tendency would normally be to ask why or to even ask you to prove it. But I won't. Instead I will ask how is this possible? You need to know t in order to calculate q and the value of the function being integrated. Only then can we obtain a value for lim \epsilon -> 0. Therefore, no matter how you "cut it", the limit as \epsilon -> 0 will always be done LAST!

 **Anonymous** · 23 February 2022

Seriously you're wasting your time, Dwight. Find something constructive to do

Dwight Walsh · 24 February 2022

So, why don't you just answer the question? How do we take the limit as \epsilon -> 0 first? Then, this matter will be resolved so that no one has to waste time on it anymore.

Dwight Walsh · 26 February 2022

I have another question for all of you die-hards who insist there is an error in my obtaining equation (118) from inequality (109) and the fact that the function q is continuous and bounded over the entire sphere of integration. Why is it that you absolutely refuse to check out the steps for yourself that I made in the arguments leading to equation (118)?

The fact that q is continuous and bounded (and therefore integrable) over the sphere of integration for all pre-blowup values of t is not contested. Therefore, we would expect that the integral of the function q, transformed into spherical coordinates, over a finite range of spherical coordinates would also be finite. The only potential "catch" to this argument from the standpoint of proof is the fact that the transformation from rectangular to spherical coordinates us undefined at r = 0. But that issue is avoided by first excluding from the integration region a small sphere of radius \epsilon centered at r = 0, doing the integral, and then taking the limit as \epsilon->0. Then, since q is finite at r = 0, the contribution to the integral from this single point is zero. This is what is covered in the arguments leading up to equations (117) and (118).

So, what is so difficult about following these few steps? As expert mathematicians, I would hope that you would be able to easily pinpoint and explain any errors I might have made in these arguments.

Instead, however, you go to the trouble of resetting the q function to where you want it which then causes a blowup. You then call this blowup a "counter-example" or a "global error" to my proof of equation (118) even though it is purely an artifact of the resetting and has nothing to do with the Navier-Stokes equation. And, since it's a "global" error, it provides no specific info about the error — just that it exists. Nevertheless, everyone participating in this exchange seems willing to defend to the death their position that there is an error in my proof of equation (118), even though they do not have a clue about the nature of this "error" nor where it is.

NOW WHY IS THIS?

Anonymous · 26 February 2022

Perhaps almost all experts expect any solution to the NS problem to be much more sophisticated than your proof.

Dwight Walsh · 26 February 2022

>Perhaps almost all experts expect any solution to the NS problem to be much more sophisticated than your proof.

Unfortunately, you are probably right. Another frustration I have with them is that they refuse to recognize these same arguments, perhaps with minor modifications to the constructed q function, could conceivably block ANY proof of existence and smoothness, whether true or false.

Anyway, thanks for your reply.

Anonymous · 26 February 2022

No. Dwight Walsh. Don't take me wrong and don't waste your time. You should learn the basic PDE before you try NSE. At the minimum you should learn what the "fundamental solution" to the Poisson equation is before you fantasy anything any more here. Your draft demonstrates very limited knowledge in real analysis. It is very impolite and disgraceful posting premature comments over and over here.

Dwight Walsh · 26 February 2022

All of this, of course, is merely your opinion without an inkling of fact. Therefore, let me fill you in on a few FACTS.

(1) I DID complete the senior level PDE course an the first semester of graduate level PDEs. Also, I used them extensively in my electro-dynamics and quantum mechanics courses.

(2) I completed both semesters of senior-level real analysis with A grades.

(3) Well before there were any millennium problems, I was already somewhat familiar with the Navier-Stokes equation through my work in developing advanced coupled fluid/optics models.

(4) Your understanding of the solving the Poisson equation applied to the field of electrostatics would imply that a point charge applies a force on itself through its own self-field, which is physically impossible. Not only that but it would be an INFINITE force.

(5) You never did answer my question about how we first take the limit as \epsilon->0 of the Poisson integral.

Now, in regard to being polite or impolite, what do we call your sarcasm following my posting of February 17 2022 at 11:13 PM?

Dwight Walsh · 16 March 2022

In reviewing several different postings from the contrarians of my paper, I believe I may now have a better understanding of their primary source of contention. In their arguments, they define the pre-blowup time interval as the semi-open interval given by all values of time t such that 0 <= t < T_b. Assuming that T_b is an actual blowup time, the solution would be finite though not bounded on this interval. In examining the section leading to equation (118), however, we see that the arguments do not mention any time values greater than the time t (eg. T_b). This includes time integrals from 0 to t where we use the "dummy variable" t'. Therefore, q needs to be finite only over the CLOSED time interval defined as all times t' such that 0 <= t' <= t < T_b in order to be bounded at time t. And since it has already been proven that q must be finite for all t < T_b, it follows that q is also bounded at all times t < T_b. This implies that equation (118) is true.

Dwight Walsh · 17 March 2022

Please disregard this posting. Unfortunately, I was confused about the meaning of q at the time I wrote it. It turns out, as my paper shows, it is the time integral of |grad p| and the

fact that it remains finite for all that establishes existence and smoothness of the solution. It is NOT about q being bounded. My apologies.

 **Anonymous** · 18 March 2022

Any real number is finite. Hence being finite is not a special property of q (as long as it is well defined)

 **Dwight Walsh** · 21 March 2022

I'm not saying anything is a "special property" (whatever that is) of q. I am saying (after proving) that the time integral of |grad p| is finite for any finite value of t. This fact is then used in the next section to show that the global maximum of the kinetic energy density is also finite for all finite values of t.

 **Anonymous** · 22 March 2022

An equivalent condition for the existence of the time integral of |grad p! over an interval [0, t] is that grad p should be well-defined (almost everywhere) and (absolutely) integrable on [0, t).- so in order to show the existence of this integral for ANY positive t, you first need to show that grad p is well-defined (almost everywhere) on $[0, +\infty)$

 **J** · 17 March 2022

Are there reasons in terms of Sobolev embedding theorem(s) explaining why Strategy 2 works well for 2D NSE but not for 3D NSE? (For instance, some key global estimates work in 2D but fail in 3D due to the dimension parameter d.)

(although it may well be the case that there are interesting exact solutions from special (e.g. symmetric) data).

Do you aware of any recent results regarding this remark in the post? Are getting extra symmetries some sort of a "2D+ε" version NSE that is far from the 3D NSE without symmetries assumptions?

 **Gandhi Viswanathan** · 6 April 2022

Dear J, in 2D the total energy is critical (in the sense of Professor Tao's original post). In a 2D+ϵ version, the energy is expected to be supercritical, no matter how small $\epsilon > 0$ is made. One way to recover criticality of the energy (and hence global regularity) is to replace the Laplacian in the dissipation term with a fractional Laplacian. The resulting hyper-dissipation can be made strong enough to dominate the nonlinear term. This is the origin of the well known 5/4 exponent for the Laplacian. In d dimensions, if one uses a fractional Laplacian with exponent $(2 + d)/4$ then the energy becomes critical and one recovers global regularity. Note than when $d = 2$ the exponent is 1. In 3D, the exponent is (2+3)/4 = 5/4.

 **Dwight Walsh** · 18 March 2022

I have a question for all you folks out there who claim my proof has a "global mistake", arguing that in the segment from equation (109) to (120), I only use inequalities (109) and (111) along with smoothness in arriving at my conclusion (120), and therefore this

result should hold for ALL functions satisfying those assumptions (not only Navier-Stokes solutions). Now, in arriving at inequality (109), I used equations (13) and (108), Therefore, in using inequality (109), I also used equations (13) and (108) "automatically". So my question is this:

Can you show me how these "counter-examples", in particular Antoine's proposed q(x,t), satisfy the assumptions of equations (13) and (108)?

[Just so that you are aware, I will consider a thumb-down as an answer of "no".]

Dwight Walsh · 18 March 2022

Dear Antoine,

I am especially anxious to get your response to this question. In your post of 04 February 2022 at 1:56 AM, you stated "one can build infinitely many counter examples where q verifies (109) *while* making (117) unbounded." Now, in view of the fact that for q to verify (109), it must also verify (13) and (108), do you still support your original claim?

The ability to admit that you have made a mistake is a painful but absolutely required skill for any researcher. Do you have this ability?

Dwight Walsh · 19 March 2022

I'm afraid I have some more bad news concerning that last question I recently posted, which I see has already caused grief for at least 16 "experts". It turns out that, according to equations (13) and (108), the functions q and Q depend exclusively on the fluid velocity components u_i, and the u_i are constrained by the NAVIER-STOKES EQUATION itself! Well how about that!? — Just what I have been saying all along! Now it seems that to obtain your concocted q function, it will require nothing short of solving Navier-Stokes Millennium problem.

Now, at this point, I strongly believe that I am owed an apology from each and every "expert" who insisted my proof had a "global error" even though they could not actually identify any error, oftentimes making insulting and patronizing comments. You can either post them here or email them to mathguy@navierstokessmoothsolutions.com

Anonymous · 20 March 2022

Everyone knows the pressure depends on the velocity, Dwight. This doesn't invalidate the objection

Dwight Walsh · 20 March 2022

No kidding! I never said it didn't, but what does this have to do with validating or invalidating the objection?

anon · 20 March 2022

The point is that you don't actually use any special properties of q and Q in your supposed proof. You only appeal to properties that, as has been shown by counterexamples, are not enough to reach the desired conclusion. It could even be that the result is true, but you don't come anywhere close to showing it.

Dwight Walsh · 20 March 2022

Which proof are we talking bout here? — The Navier-Stokes Millennium Problem or the invalidity of your counter-examples?

Dwight Walsh · 20 March 2022

Nevermind — I think I follow your arguments close enough. First, I have shown that your "counter-examples" are invalid since your concocted q functions aren't constrained by the Navier-Stokes equation. Therefore, these "counter-examples" don't show anything. Second, no mention is made of any "special properties" of q or Q in your set of rules for finding proof errors, so I'm afraid your point is pointless.

Dwight Walsh · 20 March 2022

>It could even be that the result is true, but you don't come anywhere close to showing it.

Why? What's missing?

Dwight Walsh · 21 March 2022

>It could even be that the result is true, but you don't come anywhere close to showing it.

I take it you acknowledge that even if my proof is invalid, you don't have a valid rebuttal either.

Dwight Walsh · 21 March 2022

>The point is that you don't actually use any special properties of q and Q in your supposed proof.

Actually, I DO use special properties of q and Q in my "supposed" proof. Those special properties are given in equations (13) and (108) just as I have said all along. I'm afraid that if you want to find errors in my proof, you will have to do it the old-fashion way — READ THE PAPER.

Dwight Walsh · 20 March 2022

For what it is worth to you expert mathematicians, I recently posted Version 42 of my paper in which I completely removed all mention of singularities and my alternate proof of the fact that the time integral of |grad p| is finite for all values of t. Also, I do not raise the issue of a blowup time until the section where final proof of existence and smoothness is made, and then it is only to prove that it doesn't exist. So now the paper is nice and "smooth" without singularity "hiccups", and smoothly proves the main theorem.

This means, of course, that claims such as "the delta function is the wrong heuristic" are no longer applicable. Also, we have no need whatsoever for constructed q functions that are not consistent with either your rules or mine.

1011 · 19 January 2014

Reblogged this on 1011 and commented:
¿Por qué el problema de Navier-Stokes es tan difícil?
Demostración de Otelbaev (100 páginas en ruso):

Click to access Otelbaev_N-S_21_12_2013.pdf

Stephen Montgomery-Smith · 19 January 2014

Is there a blog anywhere with people trying to see if his proof is correct? So far in my reading, I am up to Lemma 6.6. I think I found a small error, but I am not sure if it is significant. The equality in the last line of the second displayed equation on page 37 seems to disagree with the definition of $\hat{C}$ on page 34 by a factor $\lambda_2^{-4\theta}$.

As yet, I don't see the main idea in his proof, and I am at the stage of checking it line by line. Also I don't know Russian, so I have to ask friends to translate for me. So right now, I don't know the statement of Lemma 6.7.

> **Terence Tao** · 19 January 2014
>
> Villatoro's blog has some detailed analysis, and has recently raised some serious issues with the paper as well, in the crucial Section 6. (It's in Spanish, but this is easy to translate online.)
>
> I can't read Russian either, so I am happy to defer the detailed checking to others, but my feeling is that this sort of abstract approach to the regularity problem, using only the energy identity and harmonic analysis estimates on the nonlinearity rather than more precise geometric information specific to the Navier-Stokes equation (e.g. the vorticity equation) is necessarily doomed to failure. I think I can formalise a specific obstruction in this regard and hope to present it here in a couple weeks.
>
> > **Stephen Montgomery-Smith** · 21 January 2014
> >
> > I think the typo is insignificant. It should read $= \lambda_2^{4\theta} \frac{\hat{C}}{250}$. And since $\lambda_2 \geq 16$, and $\theta < -\frac{3}{4}$, you can then finish off with $\leq \frac{\hat{C}}{250}$.

Andreas · 25 January 2014

Please forgive me my elementary question; I see that it is mentioned but not answered in the post-blog: What would be the impact of a negative result on the Navier-Stokes equation?

Is there any other solution than NSE does not explain physical fluids? I have searched, but could not find any texts talking about this possibility. Do you have any reference? Thank you!

Robert Coulter · 26 January 2014

There are many that do not believe that NS adequately explains the transistion from laminar to turbulent flow. This transistion is thought by some to be modeled by a mathematical bifurcation. Search the web with keywords "bifurcation", "symmetry breaking" and "fluid dynamics" to find many articles on this subject. An interesting historical note is that Hiesenberg did his dissertation on this transistion (See Orr-Sommerfeld equation). Apparently he struggled with this. Maybe it was easier to work on Quantum Mechanics!

Anonymous · 26 January 2014

The structure of NS solutions has been shown to lend itself for an satisfactory explanation of the laminar-turbulent transition. The process of the transition has nothing to do “instability” or “bifurcation”; the non-linearity in the equations of motion has an inherent capability of proliferating vorticity scales.

Robert Coulter · 27 January 2014

Are you saying that “proliferatiing vorticity scales” occur gradually as Reynolds number is increased? Or is there a point where the vortices suddenly appear — or at least rapidly increase there number? Experimental evidence does not support a positive answer to the first question. A positive answer to the second questions is admission of a transistion point.

Anonymous · 27 January 2014

The vortices emerge in large quantities in rows as the Reynolds number increases so that the flow appears to undergo a process of successive “jumps”. Depending on the geometry and test conditions of experiments, there is no particular reason that a laminar flow cannot suddenly transit into turbulence (the number of the vortices is an accumulation of those large quantities and the vorticity scales appear within a tiny margin of the Reynolds number variation); the solution of the NS in this case must be a strong non-linear function of the Reynolds number). It is extremely difficult, if ever feasible, in most high-Re experiments to detect this intricate process as the (numerous) vortices interact with each other according to the Biot-Savart relation.

Robert Coulter · 28 January 2014

The first “jump” does not appear until Re # just over 2000 (in pipe flow). Below that the flow is laminar. I have never heard of laminar flow “suddenly transit” into turbulence without increasing the Re #.

Anonymous · 29 January 2014

The critical Reynolds number (based on pipe diameter and pipe mean speed) in pipe flow experiments has been quoted from 1800 to 20000 approx., depending on test set-up and conditions. There are experimental confirmations of abrupt transition in some laminar flows within a tiny margin of the critical Reynolds number VARIATION,
for instance, in the boundary layer on a rotating circular disk.
By the way, what is a transition point?

Robert Coulter · 30 January 2014

It is generally recognized that transistion occurs above 2000. There is a wide range reported on where this transistion occurs. However, I have never seen any report on the transistioning occuring below 2000 (I will grant you the 1800 but that seems a little low.)

A transistion point –an abrupt change (1) where a mathematical model breaksdown and a new model takes over (2) where a mathematical model has an inflection point (3) where a mathematical model developes a singularity or becomes mutli-valued (any of the derivatives)

(4) a point in a mathematical model where a characteristic changes – e.g. overdamped, underdamped, subsonic flow, sonic flow

>  **Anonymous** · 5 February 2014
>
> Care to read Mullin's ann. rev. (2011)? Are we not talking about the experimental investigations any more? Hope that you find an evidence in physics for a transition point in pipe flows.

 **anonimous** · 2 February 2014

In the days following the last comment, several counter-examples have been found to some intermediate statements in the paper by Otelbaev and one conctrete gap in the proof. Details can be found on http://math.stackexchange.com/questions/634890/has-prof-otelbaev-shown-existence-of-strong-solutions-for-navier-stokes-equatio/658551#658551

Until the answers to these essential remarks are given, the result should not be considered proved.

 **Finite time blowup for an averaged three-dimensional Navier-Stokes equation | What's new** · 4 February 2014

[...] to the Navier-Stokes equations from large data and arbitrary times. Unfortunately, as discussed in this previous post, the quantities controlled by the energy identity are supercritical with respect to scaling, which [...]

 **Victor Ivrii** · 6 February 2014

I wonder what part of the proof of finite blowup for an averaged 3D Navier-Stokes equation fails in 2D case

>  **Terence Tao** · 6 February 2014
>
> The distinction shows up in the rescaling step (Section 6.4). In three and higher dimensions, the dissipation term rescales to be small (there is a factor of $(1+\epsilon_0)^{-n_0/2}$ in front of all such terms), which effectively renders it negligible in the analysis. In two dimensions, there is no such factor, and so the dissipation is of equal strength to the nonlinearity, and one cannot treat it perturbatively.
>
> The traditional viewpoint of 3D Navier-Stokes is that it has a supercritical nonlinearity – the scaling which preserves the dissipation makes the nonlinearity very large at fine scales. But one can flip this perspective around, and instead view 3D Navier-Stokes as having a subcritical dissipation – the scaling which preserves the nonlinearity (at least for the blowup ansatz in this paper – one has to take a bit of care here because there is a two-parameter family of scalings that preserve the nonlinearity) makes the dissipation very small at fine scales.

 **Rahul Suresh** · 7 February 2014

Is it right to say that, the reason the global regularity problem is hard, is because of turbulence (as mentioned early in this post)? We know that in 2D solutions always exist. But turbulence exists in 2D also. Also, have we been able to "solve" the 2D navier-stokes equation completely or do we have a good explanation (from navier-stokes) as to why turbulence happens in 2D?

Mikhail Bogovskii · 10 February 2014

To follow the "I wonder" thread, take an old-style classical form of the 3D Navier-Stokes equations

$$\partial_t u - u \times (\nabla \times u) - \nu \Delta u + \nabla \tilde{p} = f, \quad \operatorname{div} u = 0,$$

with modified pressure $\tilde{p} = p + \frac{1}{2}|u|^2$, and consider yet another "cheap-Navier-Stokes-like" version

$$\partial_t u - u \times (\nabla \times u) - \nu \Delta u = f.$$

Excluding a divergence equation $\operatorname{div} u = 0$ is enough to turn the rest into a very nice quasilinear parabolic system, while the Prodi-Serrin-Ladyzhenskaya class of solutions stays exactly the same, namely,

$$\left(\int_0^T \|u(\cdot,t)\|^r_{L^p(\mathbb{R}^3)} dt \right)^{1/r} 3, r \geqslant 2 : \frac{3}{p} + \frac{2}{r} \leqslant 1. \quad (*)$$

In this version, for the initial value problem with data $u(0,\cdot) = u^0$, the global a priori boundedness

$$\operatorname*{ess\,sup}_{0 \leqslant t \leqslant T} \|u(t,\cdot)\|_{L^p(\mathbb{R}^3)} 3, \forall\, T > 0,$$

readily follows by nearly a routine multiplying and integrating by parts, which implies the existence of a strong smooth solution of the intial value problem for the "cheap-Navier-Stokes-like" version.

And now I wonder what part of the proof of finite blowup for an averaged 3D Navier-Stokes equation fails the "cheap-Navier-Stokes-like" version that admits no blowup solutions not only in $3D$ case but even in all higher dimensions $n \geqslant 4$. For $n \geqslant 4$, the term $-u \times (\nabla \times u)$ in the "cheap-Navier-Stokes-like" version is to be substituted by the term $(u \cdot \nabla)u - \frac{1}{2}\nabla|u|^2$.

[Note: some text appears to have been lost due to being placed between < and > signs which were then interpreted as HTML by the parser. My guess though is that this cheap version of Euler/Navier-Stokes has additional cancellation conditions on the nonlinear part B (beyond that required for the energy identity) which are behind the "routine integration by parts", and which are not enjoyed by the averaged version. -T.]

Mikhail Bogovskii · 10 February 2014

To follow the "I wonder" thread, take an old-style classical form of the 3D Navier-Stokes equations $\partial_t u - u \times (\nabla \times u) - \nu \Delta u + \nabla \tilde{p} = f$, $\operatorname{div} u = 0$, with modified pressure $\tilde{p} = p + \frac{1}{2}|u|^2$,

and consider yet another “cheap-Navier-Stokes-like”
version
$\partial_t u - u \times (\nabla \times u) - \nu \Delta u = f.$
Excluding a divergence equation $\operatorname{div} u = 0$ is enough
to turn the rest into a very nice quasilinear parabolic
system, while the Prodi-Serrin-Ladyzhenskaya class
of solutions stays exactly the same, namely, the
anisotropic norm
$\left(\int_0^T \|u(\cdot,t)\|^r_{L^p(\mathbb{R}^3)} dt \right)^{1/r}$ is finite for some
$p > 3$, $r \geqslant 2 : \frac{3}{p} + \frac{2}{r} \leqslant 1.$
In this “cheap-Navier-Stokes-like “version,
for the initial value problem with data
$u(0,\cdot) = u^0$, the global a priori boundedness
$\|u(t,\cdot)\|_{L^p(\mathbb{R}^3)} \leqslant \|u^0\|_{L^p(\mathbb{R}^3)} + \int_0^T \|f(t,\cdot)\|_{L^p(\mathbb{R}^3)} dt$
for all $t \in [0,T]$
with any given $T > 0$ and $p > 3$,
readily follows by nearly a routine multiplying and
integrating by parts, which implies the existence
of a strong smooth solution of the intial value
problem for the “cheap-Navier-Stokes-like” version.

And now I wonder what part of the proof of finite blowup
for an averaged 3D Navier-Stokes equation fails in case
of the “cheap-Navier-Stokes-like” version that admits
no blowup solutions not only in 3D case but even
in all higher dimensions $n \geqslant 4$. For
$n \geqslant 4$, the term $-u \times (\nabla \times u)$
in the “cheap-Navier-Stokes-like” version is to be
substituted by the term
$(u \cdot \nabla)u - \frac{1}{2}\nabla |u|^2.$



Marcelo de Almeida · 11 February 2014

Reblogged this on Being simple.



Anonymous · 14 February 2014

Dear participants of this important discussion!
It is very useful to reconsider your proofs and conclusions about the true Navier –Stokes and also Otelbaev’s work after consideration of these brief notes:
* http://www.hrpub.org/download/20131107/UJAM1-12600416.pdf Navier –Stokes First Exact Transformation
* http://www.hrpub.org/download/20140205/UJAM3-12601996.pdf Navier –Stokes Second Exact Transformation



AeroEng · 17 February 2014

Hello,

I'm an aerospace engineering graduate starting a masters in fluid mechanics next September. Which advanced mathematics subjects should I study in order to understand this wonderful discussion?

Thanks in advance for any reply.



Anonymous · 28 February 2014

I guess asking for some basic guidance is too much to ask from the mathematical elite.



Anonymous · 15 May 2014

Dear AeroEng,

You should look for courses about functional analysis, Fourier transform, linear and nonlinear partial differential equations (PDE) first. Afterwards, ask the professors of these courses for further guidance at your university.

La mystérieuse équation de Navier-Stokes | Science étonnante · 2 March 2014
[...] Le billet de Terry Tao expliquant pourquoi le problème est difficile [...]

Shtetl-Optimized » Blog Archive » Recent papers by Susskind and Tao illustrate the long reach of computation · 2 March 2014
[...] on ideas from his earlier blog posts and papers about Navier-Stokes (see here for the gentlest of them), Tao argues that the Navier-Stokes problem is closely related to the [...]



A. A. Frempong · 28 May 2014

In 2014, the Navier-Stokes equations have been solved analytically in 3-D and 4-D by A. A. Frempong. (at viXra.org). Also solved in the same paper are the magnetohydrodynamic system of equations.



jussilindgren · 12 June 2014

Dear Prof. Tao. I think this should suffice, through an interesting lemma for perpendicular vector fields that relates elegantly to the Frobenius inner product of the Jacobians of perpendicular fields:

The solutions to the Navier-Stokes equations are known to be regular if the enstrophy of the flow remains bounded for all times.See for example

http://www.google.be/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&ved=0CCMQFjAA&url=http%3A%2F%2Fwww.researchgate.net%2Fpublication%2F228361940_The_3D_Navier-Stokes_Problem%2Ffile%2F60b7d521c85fdad922.pdf&ei=VuyZU6_zOMvLPcOIgYAK&usg=AFQjCNFflFK-4OKHoIysgpv01JjtNqztGg&sig2=UyxjjDBV1MNemaYBh-tZ5w&bvm=bv.68911936,d.bGQ

We need a proof that the enstrophy is conserved for the Euler equations with smooth and square-integrable initial data. This implies regularity for Navier-Stokes as the viscous term can only reduce the enstrophy time derivative.

The Euler equations are given by
$\frac{\partial \mathbf{u}}{\partial t} + \nabla \mathbf{u}\mathbf{u} = -\nabla p$

Taking the curl, we get the vorticity transport equation

$\frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \times (\mathbf{u} \times \boldsymbol{\omega})$

Next we take the inner product with respect to vorticity itself:
$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \boldsymbol{\omega}^T \nabla \times (\mathbf{u} \times \boldsymbol{\omega})$

Using the vector calculus identity
$\nabla \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B})$
with the identification

$\mathbf{A} = \mathbf{u} \times \boldsymbol{\omega}$

and
$\mathbf{B} = \boldsymbol{\omega}$
plugging in we have
$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \cdot ((\mathbf{u} \times \boldsymbol{\omega}) \times \boldsymbol{\omega}) + (\mathbf{u} \times \boldsymbol{\omega}) \cdot (\nabla \times \boldsymbol{\omega})$
Using the definition of vorticity rewrite the last term
$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \cdot ((\mathbf{u} \times \boldsymbol{\omega}) \times \boldsymbol{\omega}) + (\mathbf{u} \times \boldsymbol{\omega}) \cdot (\nabla \times \nabla \times \mathbf{u})$
Using the following vector calculus identity
$\nabla \times \nabla \times \mathbf{A} = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$
As the velocity field is incompressible, we have
$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \cdot ((\mathbf{u} \times \boldsymbol{\omega}) \times \boldsymbol{\omega}) + (\mathbf{u} \times \boldsymbol{\omega}) \cdot (-\nabla^2 \mathbf{u})$
taking the minus sign out
$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \cdot ((\mathbf{u} \times \boldsymbol{\omega}) \times \boldsymbol{\omega}) - (\mathbf{u} \times \boldsymbol{\omega}) \cdot (\nabla \cdot \nabla \mathbf{u})$

$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \cdot ((\mathbf{u} \times \boldsymbol{\omega}) \times \boldsymbol{\omega}) - (\nabla \cdot \nabla \mathbf{u}) \cdot (\mathbf{u} \times \boldsymbol{\omega})$

We now need the following identity for second order tensor fields $\mathbf{S}$
$\nabla \cdot (\mathbf{S}^T \mathbf{v}) = (\nabla \cdot \mathbf{S}) \cdot \mathbf{v} + \mathbf{S} : \nabla \mathbf{v}$
where we have the Frobenius inner product of second order tensor fields

$\mathbf{S} : \nabla \mathbf{v} = Tr(\mathbf{S}^T \nabla \mathbf{v})$
and making the identification
$\mathbf{v} = \mathbf{u} \times \boldsymbol{\omega}$
and
$\mathbf{S} = \nabla \mathbf{u}$

plugging in the substitution

$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \cdot ((\mathbf{u} \times \boldsymbol{\omega}) \times \boldsymbol{\omega}) - \nabla \cdot (\nabla \mathbf{u}^T (\mathbf{u} \times \boldsymbol{\omega})) + \nabla \mathbf{u} : \nabla (\mathbf{u} \times \boldsymbol{\omega})$

An important lemma for perpendicular vector fields related to the Frobenius inner product of Jacobians:

We now need the following lemma :

Given two smooth vector fields $\mathbf{f}(x, y, z)$ and $\mathbf{g}(x, y, z)$ which are always orthogonal i.e. $\mathbf{f} \cdot \mathbf{g} = 0$, then
$\nabla \mathbf{f} : \nabla \mathbf{g} = 0$

Proof. Consider the vector fields evaluated at an arbitrarily small open set around point $\mathbf{x_0}$ i.e. when $\mathbf{dx} = \mathbf{x} - \mathbf{x}_0$ with $\mathbf{x} \longrightarrow \mathbf{x}_0$

$\mathbf{f}(\mathbf{x}) \longrightarrow \mathbf{f}(\mathbf{x}_0) + d\mathbf{f}(\mathbf{x_0})$
and
$\mathbf{g}(\mathbf{x}) \longrightarrow \mathbf{g}(\mathbf{x}_0) + d\mathbf{g}(\mathbf{x_0})$

where
$d\mathbf{f}(\mathbf{x_0}) = \nabla \mathbf{f}(\mathbf{x_0}) d\mathbf{x}$
and
$d\mathbf{g}(\mathbf{x_0}) = \nabla \mathbf{g}(\mathbf{x_0}) d\mathbf{x}$
We know that the vector field are orthogonal everywhere, so that
$(\mathbf{f}(\mathbf{x}_0) + d\mathbf{f}(\mathbf{x}_0)) \cdot (\mathbf{g}(\mathbf{x}_0) + d\mathbf{g}(\mathbf{x}_0)) = 0$
So we must have
$\mathbf{f}(\mathbf{x}_0) \cdot \mathbf{g}(\mathbf{x}_0) + d\mathbf{f}(\mathbf{x}_0) \cdot \mathbf{g}(\mathbf{x}_0) + \mathbf{f}(\mathbf{x}_0) \cdot d\mathbf{g}(\mathbf{x}_0) + d\mathbf{f}(\mathbf{x}_0) \cdot d\mathbf{g}(\mathbf{x}_0) = 0$
The first term is zero by orthogonality and using the Leibnitz rule we have
$d\mathbf{f}(\mathbf{x_0}) \cdot \mathbf{g}(\mathbf{x}_0) + \mathbf{f}(\mathbf{x}_0) \cdot d\mathbf{g}(\mathbf{x_0}) + d\mathbf{f}(\mathbf{x_0}) \cdot d\mathbf{g}(\mathbf{x_0}) = 0$

$d(\mathbf{f}(\mathbf{x_0}) \cdot \mathbf{g}(\mathbf{x_0})) + d\mathbf{f}(\mathbf{x_0}) \cdot d\mathbf{g}(\mathbf{x_0}) = 0$
So that in the end we must have

$d\mathbf{f}(\mathbf{x_0}) \cdot d\mathbf{g}(\mathbf{x_0}) = 0$
which is equivalent to
$\nabla \mathbf{f}(\mathbf{x_0}) d\mathbf{x} \cdot \nabla \mathbf{g}(\mathbf{x_0}) d\mathbf{x} = 0$

which is equivalent to
$d\mathbf{x}^T \nabla \mathbf{f}(\mathbf{x_0})^T \nabla \mathbf{g}(\mathbf{x_0}) d\mathbf{x} = 0$

For every $d\mathbf{x}^T d\mathbf{x} > 0$ this will hold iff

$\nabla \mathbf{f}(\mathbf{x_0})^T \nabla \mathbf{g}(\mathbf{x_0})$
is a skew-symmetric tensor (as for such the quadratic form always vanishes). For such tensors the trace is always 0, which proves the lemma.
As the velocity field and the Lamb vector are orthogonal everwhere, that is
$\mathbf{u} \cdot (\mathbf{u} \times \boldsymbol{\omega}) = 0$
We have that
$\nabla \mathbf{u} : \nabla (\mathbf{u} \times \boldsymbol{\omega}) = 0$
So that the we have the transport equation for the enstrophy density:

$\boldsymbol{\omega}^T \frac{\partial \boldsymbol{\omega}}{\partial t} = \nabla \cdot ((\mathbf{u} \times \boldsymbol{\omega}) \times \boldsymbol{\omega}) - \nabla \cdot (\nabla \mathbf{u}^T (\mathbf{u} \times \boldsymbol{\omega}))$

Integrating over the whole space and assuming square-integrability, by using the divergence theorem we have immediately

$\frac{d}{dt} \frac{1}{2} \int_{R^3} \boldsymbol{\omega} \cdot \boldsymbol{\omega} dV = 0$

In other words the incompressible Euler equation conserves enstrophy. This automatically implies regularity for the Navier-Stokes equations as in the N-S the vorticity transport equations has only the additive viscous term which has a non-positive contribution to the time derivative of the enstrophy.

See

navierstokesregularity.wordpress.com

Best ,

Jussi Lindgren
justicus1982@gmail.com



Terence Tao · 12 June 2014

Your lemma is incorrect. For instance, if $f(x,y,z) = (x^2, yx, 0)$ and $g(x,y,z) = (-y, x, 0)$, then $f(x,y,z) \cdot g(x,y,z) = 0$ for all x, y, z, but $\nabla f : \nabla g = y$.

Note also that enstrophy production in the Euler equations is a numerically confirmed fact, see e.g. http://www.eng.warwick.ac.uk/staff/rmk/teaching/ES441/Kerr93.pdf , and in fact there is now very strong numerical evidence that the Euler equations in fact exhibit blowup, see e.g. http://arxiv.org/abs/1310.0497 (the numerics here also exhibit significant enstrophy production, which is unsurprising since bounded enstrophy implies regularity).

It's an instructive exercise for you to find out precisely where you made the error. See also: https://terrytao.wordpress.com/career-advice/be-sceptical-of-your-own-work/



Anonymous · 14 June 2014

Can one infer from the above mentioned article that for a suitable initial data NSE has (logarithmically) increasing ensthropy as well? It seems that for a positive regularity result it would be more plausible to look for an exponential or iterated exponential bound for the ensthropy or some higher order H^k-seminorm.



Anonymous · 16 June 2014

The numerics is not a simple black-and-white matter. NOT all numerical computations have been able to indicate the enstrophy production (see overhead presentation by J.D. Gibbon Levico Terme 2012?). Kerr's singularities are self-similar. Chae (http://arxiv.org/abs/math/0601661) showed a general result: there exist no self-similar singularities for the 3D incompressible Euler equations. Luo and Hou's findings (arXiv:1310.0497v2) have been disproved by Chae and Tsai (arXiv:1402.4560v1).

In fact, if the initial data has finite energy (eg smooth data with compact support in R3), Euler' solutions do not become singular in finite time. To the contrary, should the initial data contain infinite energy or a singular point itself, it is meaningless to investigate finite-time singularities.



anonymous · 2 July 2014

I have searched all over the internet looking for the proof that having enstrophy bounded implies regularity for NS and/or Euler. The closest I can find is the Beale-Kato-Majda (BKM) vorticity blowup criteria (actually mentioned in the Clay problem statement). This is a time integral of the sup of the vorticity at each time slice — not enstrophy (space integral of vorticty squared). Maybe this blog has redefined enstrophy.



Gandhi Viswanathan · 3 July 2014

I believe Leray himself proved that the L^2 enstrophy remains bounded for regular solutions of the 3-D NS system. But for the 3-D Euler system, the enstrophy can

remain bounded at the time of first breakdown. Here is an intro about the Euler eqs.: DOI: 10.1016/j.physd.2007.10.014

 **Gandhi Viswanathan** · 3 July 2014

Going back to Leray, I think he focussed on L^p norms, whereas the enstrophy is an inhomogeneous Sobolev norm. But the key point is that, unlike kinetic energy, the enstrophy is subcritical for the 3-D NS system, so as long as the enstrophy remains bounded, singularities cannot form. The Euler eqs., in contrast, have a 2-parameter scale invariance, so we cannot use concepts like subcritical and supercritical in the same sense as for the NS eqs.

 **anonymous** · 3 July 2014

I hope this clears it up.

"Similar to the situation with the energy-dissipation rate per unit mass ε(t), time averages of the enstrophy over arbitrary nonzero intervals are finite for weak solutions, but we would need to know more to confirm that solutions are regular over the interval [e.g., that the variance of the enstrophy is finite. This observation highlights the importance of intermittency in Navier-Stokes dynamics (Frisch 1995, Vassilicos 2000). Intermittency refers to the phenomena of rare but large fluctuations usually associated with turbulence. If enstrophy fluctuations are bounded at the level of the time average of the second moment, then solutions are regular. " — Doering (2009)

So, if I am reading this correctly, it is enough to have the mean enstrophy be bounded over a time interval for weak solutions. However, strong solutions required the variance of the enstrophy to be bounded.

 **Terence Tao** · 3 July 2014

Sorry, I was a bit careless in my previous comment: bounded enstrophy is known to imply regularity for Navier-Stokes (as can be seen for instance from the Prodi-Serrin regularity condition and Sobolev embedding), but this is not known for Euler.

 **victorivrii** · 20 June 2014

The following article
Taalaibek D. Omurov, "The Methods of a Problem Decision Navier-Stokes for the Incompressible Fluid with Viscosity" published at
http://article.sapub.org/10.5923.j.ajfd.20140401.03.html
claims to provide the proof of the existence and uniqueness of NS system

 **Choro Tukembaev** · 5 July 2014

For 3D Euler equations the answer see
T.D. Omurov, (2013), Navier-Stokes problem for Incompressible fluid with viscosity // Varia Informatica 2013. Ed. M. Milosz, PIPS Polish Lublin, pp.137-158.
Read pages 147-151. There assessed the proximity of solutions of the Euler and NS when the viscosity tends to zero.

Choro Tukembaev · 8 August 2014

Dear Dr. Terence Tao,

Full research of the Navier-Stokes equations is given in the following article.

Taalaibek D. Omurov, “Existence, Singleness and Smoothness in the Problem of Navier-Stokes for Incompressible Fluid with Viscosity” published at

http://literatura.kg/articles/?aid=2030

Abstract

Existence, singleness and smoothness (or conditional-smoothness) in solution of the Navier-Stokes equation is one of the most important problems in mathematics of the millennium [1], which describes the motion of viscous Newtonian fluid and which is a basic in hydrodynamics [6, 12]. Therefore in this work a nonstationary problem for Navier-Stokes of incompressible fluid with viscosity is solved [1].

Preface

The research is devoted to the development of a method for solving 3D Navier-Stokes equations that describe the flow of a viscous incompressible fluid. The study includes a requirements “Navier-Stokes Millennium Problem”, as developed method of solution contains a proof of the existence and smoothness of solutions of the Navier-Stokes equations, where laminar flow is separated from the turbulent flow when the critical Reynolds number: Re = 2300. The decision is obtained for the velocity and pressure in an analytical form, as required by the “Navier-Stokes problem Millennium”. The method of solution is supported by examples for different viscosity ranges corresponding applications.

In sections 4.3, 4.4, 7.2 and paragraphs 5, 6 new law of the pressure distribution has been found. This law is derived from the equation of Poisson type and differs from the known laws of Bernoulli, Darcy at all. Most importantly, the author has opened a special space for the study of the existence and smoothness (including conditional smoothness) equations Navier-Stokes for viscous incompressible fluid. In the case of smoothness a space with the norms of Chebyshev type has been obtained. The weighted space of Sobolev type arises in the case of conditional-smoothness. For brevity, these spaces can be called: Omurov’s spaces with different metrics.

K. Jumaliev, Academician, Director of the Institute of Physics NAS Kyrgyz Republic August 1, 2014

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The proximity of the solutions of the Euler and Navier-Stokes equations are in section 2.3. The remark at the end of paragraph 4.2 covers a limited area. The paragraph 7 is devoted to the n-dimensional case of the Cauchy problem.

Sincerely,

Choro Tukembaev

 **jussilindgren** · 11 September 2014

Dear Prof. Tao, I was wondering what would be the field-theoretical analog for the Navier-Stokes/Euler system? My understanding is that basically the Euler equation is the geodesic equation for a certain metric. What is missing is the governing equation for the metric, meaning that if one compares e.g. electromagnetism, the Lorentz force law is the geodesic equation while Maxwells equations are the Euler-Lagrange equations for the Lagrangian $F_{uv}F^{uv} + A_x J^x$ over the whole space. In similar fashion in General Relativity the Einstein Field equations are the Euler-Lagrange equations for the Lagrangian $R\sqrt{-g}$ Could there be something like this for the Navier-Stokes& Euler equations. Could the scalar curvature be substituted for a determinant of the Shape operator or some other "curvature" measure ?

If You Can Explain What Happens When Smoke Comes Off A Cigarette, You'll Get A $1 Million Prize | Business Insider · 2 November 2014

[...] mathematician and 2006 Fields Medal winner Terence Tao wrote an article on his blog about the mathematical reasons solving the Navier-Stokes equations is s.... Solving differential equations similar to Navier-Stokes frequently involves finding some [...]

随着计算流体力学的发展，如果未来计算机足够强力的话，湍流是否就不再是个问题了？ 闪新闻 · 14 December 2014

[...] Clay Institute 的某个千禧年问题（数学已经超过我的专业范围，这里只是抛砖引玉）Clay Institute 关于 NS 方程的千禧年问题并不是求解析解，而是证明给定任意光滑无散

度初值，证明或者证否三维 NS 方程始终存在光滑解。所以这是个证明问题而不是求解问题。那这个问题和 DNS 或者说数值解有没有关系呢？勉强有一点。 DNS 或者广义的 NS 方程数值模拟是求某一个弱解，NS 方程千禧年问题是证明光滑强解存在性。本质上是非常不同的问题。不过，解决 NS 问题的一种思路是先构造某种弱解，证明这个弱解的存在性，然后通过强化这个弱解，最终得到光滑的强解。法国数学家 Leray 三十年代就做了类似的工作，当然最后一步没做成，后来也没有人做成。Terry Tao 的博客里说他觉得这不是一个可行的思路。而且他今年的发表的文章似乎显示证否的可能性更大。所谓数值解的残差或者说精度，只是说给定某个初始值，如果存在光滑强解（真实值），某种数值方法得到的弱解和真实值的差在某个范围里面。所以这个概念本身并不涉及任意初值的强解存在性，也不涉及湍流。 @高阳说精度问题不是计算能力可以解决的，这个说法是错误的。凡是能用的数值方法，都必须满足收敛性。方法好坏指的是收敛快慢。换句话说，给定计算资源下谁和真实值的误差更小，如果没有计算能力限制，不存在低精度方法和高精度方法谁更准的问题。 [...]

Quora · 10 January 2015

Has the existence and uniqueness of the solutions for Navier-Stokes equations been proven for 3 dimensions by Mukhtarbay Otelbayev?

My understanding of the paper is woefully inadequate but I am sure that I came across an article where some prominent researchers pointed out conceptual mistakes in Professor Otelbayev's paper. He later acknowledged the mistakes and recanted his resul...



victorivrii · 20 March 2015

No need to refute Mr Kozachok: he is a well-known loon who made a lot of claims (mainly about Helmholz equation) which were refuted.

Yes, Otelbaev's paper contained crucial errors (and Omurov's paper too).



Choro Tukembaev · 21 March 2015

1. Шестая проблема поставлена правильно, так как четыре уравнения содержат четыре неизвестные величины, то есть три компоненты скорости и давление для трех уравнений движения плюс уравнение неразрывности. Из этих уравнений требуется найти давление и скорость, чего достаточно для решения специалисту по многомерным обратным задачам математической физики. Уравнение несжимаемости жидкости является "джокером" для выявления связи между давлением и скоростью. Тогда любое лишнее уравнение, помимо четырех уравнений шестой проблемы, нарушает постановку задачи Ч. Феффермана (Ch. Fefferman), например, как это сделал Отельбаев, придумав уравнение (1.4), а также другие.

2. "Джокером" в статье Омурова являются эквивалентные преобразования и "алгоритм пуасоннизации системы уравнений" для выявления связи между давлением и скоростью на основе уравнения несжимаемости. Это написано в параграфах 4 и 5 статьи
http://article.sapub.org/10.5923.j.ajfd.20140401.03.html



DoubtMath · 20 March 2015

Dear Prof. Tao and others who read this blog!
In the paper True Navier–Stokes Vector PDE http://www.hrpub.org/journals/article_info.

php?aid=2375 Alexandr Kozachok argues that the Navier–Stokes + continuity equations are underdetermined as vector system and require additional equations.
Whether you can refute this shocking claim?

> **Anonymous** · 7 April 2015
>
> The above mentioned article can safely be considered almost complete nonsense.
>
> > **DoubtMath** · 2 June 2015
> >
> > It seems mathematicians (except Choro Tukembaev and AnonymousMath http://en.wikipedia.org/wiki/Talk:Navier%E2%80%93Stokes_existence_and_smoothness#Yet_another_solution_proposed.3F) have not yet found the mathematical arguments to refute Kozachok's claim.

Anonymous · 19 June 2015

I don't have any real technical understanding of this topic, but here's an interesting popular article describing some apparently new explanation of the laminar-to-turbulent transition:

http://nautil.us/issue/25/water/to-predict-turbulence-just-count-the-puffs-rp

It says the critical Reynolds number is in fact around 2040, and would seem to imply that this is an actual, concretely definable mathematical constant that (like the Feigenbaum logistic constants at first) isn't known very precisely.

I'm wondering:

1) Is the approach above to explaining turbulence now an accepted thing in current NS literature? Is it likely to be of any help with the Clay problem?

2) Under the definition given, would it be feasible and interesting to compute the critical Reynolds number to high precision?

Thanks!

> **Anonymous** · 20 June 2015
>
> Interesting article! (it is still possible that the critical Reynolds number is a precisely defined mathematical constant for each application but may vary from application to application – i.e. not a "universal constant").



roland5999 · 16 October 2015

Between dealing with fully blown (compressible) Navier Stokes and other things, and not really progressing as wanted, from time to time I like to step back and re-read this blog, and always seem to learn something new.
Although I lack mathematical comprehension and aptitude, I have a physics perspective and like this blog very much because it is so very well written. Thanks again to Terry.

Taking a fresh look now, I see something that has not been seen before.
From statements like "we will not need to study the pressure here" I get the idea that indeed pressure has not yet been studied very much.

I like to come up with a suggestion on maximum kinetic energy,
combined with a very simple and obvious constraint from physics:
The pressure can not become negative!

Pressure as we understand it is a measure of inner energy of the material, so until somebody comes up with very strange new models, negative values are strictly forbidden in the physics domain.

A straightforward example is a pump which tries to suck water from a lake surface more than 10 metres below: regardless of its design, it can not work because the water column just disintegrates under its own weight – vacuum at the upper end and no flow possible.
The real pump will suck some water vapor ;-)

Basically: Yes a real fluid may be quite incompressible – that is with a positive sign (compression being a decrease of volume). But on reaching the vacuum treshold, every real fluid has no choice but to become DE-compressible i.e. react with a positive divergence, unboundedly so if necessary.

So much about the real world.

In the mean time I checked out the video of the Cafarelli lecture on the Clay Math site, which I very highly recommend for any interested subscriber of Navier Stokes:
http://claymath.msri.org/navierstokes.mov

According to him, pressure really is the culprit for creating infinite speeds which tend to blow up a solution, especially when neglecting viscosity i.e. looking at the simplified Euler equation set.

This, and the remark from above that
"Blowup can occur when the solution shifts its energy into increasingly finer and finer scales, thus evolving more and more rapidly and eventually reaching a singularity in which the scale in both space and time on which the bulk of the evolution is occuring has shrunk to zero."

convinces me that the problem is that we are indeed looking at excessively low pressures rather than high pressures, because a high pressure would direct the flow and its evolution away from the singularity and thus alleviate the situation automatically.

If you as mathematicians allow no divergence, but instead allow unbounded negative pressure, you are actually violating the conservation of energy – creating kinetic energy which the fluid just does not have, and this results in some kind of anti-black hole which cannot exist.
Thats where you have your blowup – right on the spot and quite literally so ;-)

To clarify the concept, I suggest taking a brief look at Bernoulli's principle:
https://en.wikipedia.org/wiki/Bernoulli%27s_principle

This is basically the energy conservation with respect to the Euler set of flow equations, but from the Lagrangian, "material" point of view.

I have to comment that in physics, by pressure we mean the inner energy of the volume at rest, whereas in your mathematics, the pressure appears to be some kind of "massic pressure" which is not really being used here.
The conversion ratio is material density, i.e.
"massic pressure" = actual pressure / density,
with density obviously being a constant

I dont mind going on with this “massic pressure” notion for p though.

Then, the Bernoulli principle says that
u.u / 2 + p = constant in Lagrangian coordinates, i.e. along the stream line.
In lack of a better term, let us call this the Lagrangian pressure PL.
It is then a measure of total energy of the volume (with big letter P conveniently indicating the total rather than just a fraction).

As mentioned before, p is not smaller than zero, so the upper bound for velocity is
|u max| = sqrt(2 PL)

Now, a strategy suggestion:
Assuming an initial flow field u(x,t=0), an initial pressure field p(x,t=0) has to be assumed as well. From these we simply calculate also the initial PL(x,t=0) field.

The pressure assumption can be arbitrary, as long as it is non negative and bounded. It does not need to be smooth (I think).

The only condition under which an unbounded positive pressure can be allowed is that all the volume’s mass is contracted into a single point
(and the pressure singularity occurs of course only at that point and no where else).

If there is no particular preference, I tend to suggest starting with an even pressure distribtion.

My favorite is the sea level static pressure, which is technically nicely defined as p0 = 101325 / 1.225 [Pa / (kg/m3)]

Initially, the Langrangian coordinate is the same as the Eulerian one:
xL(t = 0) = x(t = 0)

As time progresses, we need to keep track of the relevant PL distribution over space at all times, by mapping between Lagrangian and Eulerian perspectives via integration of the stream line:
xL(x, t + dt) = xL(x, t) + u dt

I admit this is not very practical.

The benefit however is that PL does not change over time, unless an external force field is present, in which case we have to add the work performed on the fluid to PL:
PL = u.u / 2 + p + f.u dt

I also have to admit that the above mentioned velocity criterion only holds roughly (on a rough scale) for Navier Stokes, but it is precisely true for the case of the Euler equations.

So I suggest starting with an attempt to prove blowup for Euler (i.e. p < 0),
and go further from there.

It is hard but will be an important outcome.

Its repercussion on physics will be that a better understanding of compressibility is needed for arriving at "THE" solution (turbulence).

Eventually, this will allow us to build better air planes so we can travel supersonically at Ryan air air fares, and that is what I am actually working on (no joke).

Ungodly Turbulence | Learning Log · 6 November 2015
[...] Fields medalist Terry Tao explains why the core problem of turbulence is so hard (from a mathematical perspective). (html) [...]

 **toumlilin** · 22 December 2015
please
wath is the notion : critical case, supercritical case in the faractional navier stokes equation

Ungodly Turbulence – Mundaneum · 3 March 2016
[...] Fields medalist Terry Tao explains why the core problem of turbulence is so hard (from a mathematical perspective). (html) [...]

 **Daniel** · 22 May 2016
Dear Prof. Tao and others,

After reading the comments on 20/7/2011 that say Burgers equation is regular I then looked at the paper mentioned in these comments for the proof. The part that I think may be wrong is that a nonlinearity is removed by considering points of peak velocity. It appears that the equation has been simplified by considering points where there is a turning point. A singularity is not necessarily found by considering points where there is a turning point. So it looks to me like what was found is based on what was assumed implying no proof. Do you think that particular proof of regularity for Burgers equation is correct? Is there a different proof?

Daniel

 **kksen** · 24 August 2016
hi,i am a newbie,can i know what level of depth of math and physics required to understand this equation and tackle it?I know i had oversimplified the situation,but at least this would give me some preparation for further self study.What things should i study specifically in order to discuss this problem?thanks in advanced.

Simulating Fire | icipriano's divulgation blog · 4 July 2016
[...] A solution of the Navier-Stokes equation is said to blow up if as i.e., at some time there is flow with arbitrarily high velocity. Whether is theoretically possible to always find global solutions that do not blow up is an open problem. You can read an explanation of why it is a complicated problem in Terry weblog. [...]

 **Choro Tukembaev** · 10 September 2016
Dear Dr. Terence Tao,
In connection with the sixth problem of the Millennium, as is known, statement of the problem was given by Charles Fefferman, please pay attention to the article
T. D. Omurov, Existence and uniqueness of a solution of the nD Navier-Stokes equation, Advances and Applications in Fluid Mechanics 19(3), 589-604 (July 2016)
A fundamental problem in mathematics is to decide whether such smooth, physically reasonable solutions exist for the Navier-Stokes equations. We restrict attention here to incompressible fluids filling all of Rn. The Cauchy problem for the nD Navier-Stokes equations is reduced to the integral equations of Volterra and Volterra-Abel, investigation of which, allows us to solve positively question on uniqueness and smoothness of the solution.

See
http://www.pphmj.com/abstract/10041.htm

Choro Tukembaev · 17 September 2016

The article

T. D. Omurov, Existence and uniqueness of a solution of the nD Navier-Stokes equation, Advances and Applications in Fluid Mechanics 19(3), 589-604 (July 2016)

is available (is free) on the website

Click to access existence_and_uniqueness_of_a_solution_of_the_nd_navier-stokes_equation.pdf

Anirudh · 25 September 2016

For a physics student,What mathematical tools he needs to learn to at least understand this problem mathematically.

Choro Tukembaev · 13 October 2016

For explanation of studies we note the following. The results are presented from simple to complex. In the second section found rigorous solution of the Navier-Stokes equations in a special class of functions with the initial condition (2.1) and with mathematical transformation (2.2). This means that in the second section investigated a minor task (1.1), (1.2), (2.1).

The original problem (1.1) – (1.3) is solved in the third section with a new transformation (3.2), which generalizes the transformation (2.2), but without taking into account (2.1) and other additional conditions. We prove the existence uniqueness of conditional-smooth solution on time on the basis of the proposed method (3.2). This transformation allows you to linearize a inertial terms of the equation (1.1) and find the connection between speed and pressure in the form (3.5).

On the basis of the method (3.2) and formulas (3.5), (3.6) we derive a system of operator equations in integral form (3.7). From the solvability of the resulting system follows the solvability of problem (1.1) – (1.3). As you know, if the studied problem is reduced to integral equations and, in addition, the parameter of viscosity is classified, we can determine the solution in different spaces.

Thus, in the third section of the article, the viscosity is small parameter, which is used for the study of turbulence. For the 3D equations this is done in [7], [8]: see list of References in the article. It should be noted, the problem is related to the regularity criterion of Beale – Kato – Majda [1], therefore the analytical solution to the 3D equations in [7], [8] satisfies this criterion. In addition, in [8] the transformation (2.2) is applied to 3D Navier-Stokes equations in a bounded region with initial and boundary conditions.

Dr. Viktor Ivanov · 23 October 2016

Dear Professor Tao,

Here are 3 counterexamples to your “supercriticality barrier”,: ‘The main purpose of’ the paper “Finite time blowup for an averaged three-dimensional Navier-Stokes equation“, submitted to J. Amer. Math. Soc.. 29 (3): 601–674.

1. Let us consider the 3D \int [(Z\cdot u)u](Z, r)A_r(Z)dZ = I, \int A_r(Z)dZ = 1, where

A_r(Z) under r \rightarrow 0 is 3D Dirac delta function, u is a square integrable solution of the 3D Navier-Stokes equations. It is clear that I = 0, but almost all averages of (Z\cdot u)u yield I \neq 0. And \int _0^t I(r)/rdr , where t C > 0} z_ju_k^2(Z, r)e^{-|z_j|^2/r}dz_j/r^3dr \leq const., where u_k \neq 0 is a component of u and t < blowup time, and hence has no blowup time, but its averages can be \infinity.

3. I(r) = \int_0^\infty \int _0^\infty \int sign (z_j)e{-|z_j|h}|u(Z, r)|^2dZ\sum _{l \neq j} s_l^2 \prod_{l \neq j}ds_l , h = (\sum _{l \neq j} s_l^2)^{1/2}, I \leq \int_0^\infty \int _0^\infty e{-Ch} \sum _{l \neq j}s_l^2\prod_{l \neq j}ds_l | |u(., r)||^2 \leq C||u(., r)||^2, where C is a positive constant. So, \int_0^t I(r)dr \leq Ctsup_r ||u(., r)||, i.e., we have no blowup time. However, for almost any averages inside of the integrals, we have I(r) = \infty.

All 3 counterexamples-types are employed in my paper 'A classical solution of the 3D Navier-Stokes and Euler problems', which I sent you and asked to endorse for ArXiv, but have no response.

Student · 29 October 2016

Professor Tao;

I know, that the strong solution of the 3D Navier-Stoke equations is unique in the class of weak solutions.

If we establish the uniqueness of strong solutions under a sufficient condition

w=u-v belongs to L^{\infty }\left ([0,T \right], H^{1}).

Is it a strong uniqueness for the strong solution of 3DNSE?

Student · 30 October 2016

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Is it a strong uniqueness for the strong solution of 3DNSE?

James Vickery, Federal Reserve Bank of New York – Princeton Bank Consortium · 12 November 2016

[...] speeds of the rotating cylinders to avoid turbulence, define the Reynolds number, monitor the Navier-Stokes simulation stability convergence, and be the primary source for mandatory quarterly historical data reporting. The interesting [...]

Anonymous · 26 February 2017

Dear Prof. Tao!

Ten years ago you announced this important topic. Therefore your conclusion is needed. Your opinion on a current updates of the Navier Stokes problem is very important. Some researchers have already published the conclusions on this problem, for example:

1).....

5) The mathematician Jean Leray in 1934 proved the existence of weak solutions to the Navier–Stokes equations, satisfying the equations in mean value, not pointwise.

6) Terence Tao in February 2014 announced a finite time blowup result for an averaged version of the 3-dimensional Navier-Stokes equation. He found the result formalizes a "supercriticality barrier" for the global regularity problem for the true Navier-Stokes

equations, and claims that the method of proof in fact hints at a possible route to establishing blowup for the true equations.

7) Terence Tao from Adelaide in 18 March, 2007 announced three possible strategies if one wants to solve the full Millennium Prize problem for the 3-dimensional Navier-Stokes equation (NSE) Why global regularity for Navier–Stokes is hard. Strategy 1 —Solve the Navier-Stokes equation exactly and explicitly (or at least transform this equation exactly and explicitly to a simpler equation) is used in works: Navier –Stokes First Exact Transformation, Navier –Stokes Second Exact Transformation.
The author of these works Alexandr Kozachok has offered (in February 2008 -Internet, in November 2013 and February 2014 – INTERNATIONAL journal) two exact NSE transformations to the simpler equations.

Conclusion: As Navier-Stokes equations are studied by European mathematician with various solutions methods like finite element method, Numerical approximations method, finite differences to get Weak solution, Exact solutions, Mild solution. But still the problem of finding smooth and regular solution to Navier- Stokes equations is open and unsolved.

Read more in " Current Updates in Fluid Dynamics with Special Reference to Europe "(by D. R. Patil ....http://sangamnercollege.edu.in/wp-content/uploads/DRP_Publications.pdf)



Alexandru Soare · 7 May 2017

Within a closed chamber of compressible fluid or gas, with fully stabilized pressure, the relation between breeze and turbulence as they affect an aircraft wing or the like is easier to obtain than if we are dealing with an external fluid or gas which is incompressible. And in fact likely the reason that such equations seem intractable when the medium is incompressible is due to some self-referential property between the forces which effect the breeze and those which make for the turbulence in question. When the breeze-turbulence system is compressed within a chamber, those elements which cause the two sets of forces to contain parts of each other cancel out relative to our analysis by way of easily-tractable equations. If we were to cause the chamber to exert less and less compressibility (not the same as compression-force) upon the fluid or gas within the chamber (that is, the sides of chamber as semi-elastic, allowing for the fluid or gas itself to partly dictate its own pressure, its own extreme motions in part effecting the changes in the positioning of the semi-elastic chamber-walls, which in turn affect the gas or fluid pressure), the forces which cancel out (and feature in both effecting the breeze and the subsequent turbulence) under compression, no longer cancel out and feature within both sets of forces as variables when written as formulas. The force which no longer cancels out within the more elastic chamber is what we will call the Velocity of the Separation of Semi-Vortexes of Turbulence (the VSVT force-variable); this is conceived of by noting that a speeding linear breeze within a fluid or gas medium will cause vortexes of the medium to either side of it (the breakdown of which is turbulence), and predicting that if the breeze is strong enough, a line of semi-vortexes will form to either side of the linear breeze and that the sides of the semi-vortexes within such a line no longer contact each other as much (vortexes spinning in the same direction whose sides contact do not cancel out as much as two cogwheels spinning in the same direction do, and these semi-vortexes we will imagine are more fully circular and cyclical than genuine vortexes happen to be, and thus will mostly cancel out their motions in

the manner of like-directional cogwheels when contacting each other upon the side.) The faster the breeze, the more quickly numbers of these semi-vortexes will separate down the line and then degenerate (almost instantly, if they exist at all outside of their potentiality) into genuine vortexes and then turbulence of the intractable sort. How quickly these semi-vortexes separate before becoming genuine vortexes is dictated by the speed of the linear breeze in question, and therefore, again, the self-referential property which causes breeze-turbulence problems to become intractable at high velocity of breeze in an incompressible fluid is this speed of the separation of semi-vortexes before they become genuine vortexes and then intractable turbulence; the breeze produces these semi-vortexes more slowly in a more compressible fluid, and they do mostly cancel out in such a situation.

If the source of breeze or propulsion could be swiveled side-to-side in such a way as to cause the line of semi-vortexes to form non-linearly and contact each other upon the sides, one will have obtained dramatic decreases in the turbulence which the system produces. The ideal swivel-rate is dependent upon the size and distance from each other of the semi-vortexes produced by a given velocity, force and volume of breeze within a certain medium.

If we were to treat the velocity of a fluid bundle as given in the Navier-Stokes equations according to the above observations and recommendations we would conclude that the ideal function-formula describing the velocity is one which alternates periodically in the following manner:

(1) increased velocity along the curvature of the upper half of a standard circle upon the x,y axis, followed by (2) 1 full diameter of the same-size circle straight-line decrease in velocity (almost instantly decreased by this much in actual physical practice), followed by (3) a deceleration along the curvature of the bottom half of a standard circle (but not necessarily below the x-axis), which is slower than the previous decrease, followed by (4) a 1 full diameter of the same-size circle as previously straight-line (that is as sudden as possible in physical practice); after which we repeat beginning at (1) again.

The size of the circle used ought not matter much in most cases, so long as the 1-diameter increases and decreases ((2) and (4)) are maintained as proportional to the circle used for measuring (1) and (3). The reason the size does not matter so much is that the semi-vortexes formed will not have a chance to break out into genuine vortexes to the sides of the waveform produced by these cyclical changes in velocity; the waveform will in fact catch and cradle any such semi-vortexes before they lose their circular curvature, and the lack of genuine vortexes and their subsequent turbulent breakdown will prevent any sudden changes in the pressure (P) variable featuring in the Navier-Stokes equations.

CK Chio · 10 July 2017

Do the solutions of the Navier-Stokes equation blow up?



Yanyou Qiao · 18 September 2017

Dear Prof. Tao:

Thanks for your detailed explanations on the N-S equation. Several months ago, I have obtained a series form solution to this equation, it's based on the Taylor expansion along time t. Although the convergence of the series still need to be proved,I think there is some new skills in my work.

My work can be found at: http://arxiv.org/abs/1706.08345, I will appreciate it very much if you can read it and make comments.

 **Turbulensi – INTEGER** · 9 October 2017

[...] https://terrytao.wordpress.com/2007/03/18/why-global-regularity-for-navier-stokes-is-hard/ [...]

Fake papers #3: Non tutto scorre liscio | Maddmaths! · 24 May 2018

[...] formare su insiemi di misura nulla! Per sapere qualcosa di più su questo problema, vi rimandiamo a questa pagina del blog di Terence [...]

Turbolenza: introduzione a questo strano e complesso fenomeno - Mathone · 13 June 2018

[...] Why global regularity for Navier-Stokes is hard [...]

 **Craig** · 3 February 2019

Is there a good heuristic probabilistic argument that the NS equations always have solutions?

 **Terence Tao** · 3 February 2019

No, especially for large data it is in fact conceivable that one could build a stable, approximately discretely self-similar solution which could withstand even a stochastic forcing term (assuming that the forcing was "less supercritical" than the nonlinearity in the Navier-Stokes equation, so that it became increasingly insignificant as the solution entered increasingly fine spatial scales). For instance I believe that the blowup solution to the averaged Navier-Stokes equation that I constructed in https://arxiv.org/abs/1402.0290 has this stability property with respect to noise, though I did not formally prove this.

 **Anonymous** · 3 February 2019

Is it possible to show that the special family of initial conditions leading to a finite time blowup, if not empty, is at least a "small" family in some quantitative sense?

 **Terence Tao** · 4 February 2019

This has not been done for three-dimensional Navier-Stokes (and, if stable blowups exist, then such statements may well be false), but for other, less supercritical, PDE there are at least two ways to obtain such a result. One is that of "almost sure global existence" in which solutions are shown to be global (and reasonably regular) for almost all data with respect to some measure (often an invariant measure for the flow, such as a Gibbs type measure, although actually constructing these measures and showing that they really are invariant can be non-trivial). In the case of 3D Navier-Stokes, this method has so far been able to produce weak solutions (with no guarantee of uniqueness) for almost all data in a very low regularity space: see this paper by Nahmod, Pavlovic, and Staffilani.

Another possibility is to show that all blowups are unstable, and from there show that the set of data that blows up is in the first category in the sense of Baire in a suitable function space. Again, this has not been done for Navier-Stokes, but in this paper of Hani it is shown that certain equations of NLS type have the property that the set of globally bounded orbits (in a Sobolev norm) is of the first category.

Craig · 4 February 2019

In your paper "Finite time blowup for an averaged three-dimensional Navier-Stokes equation", it seems to me that you have shown the new equation to be universal in the same sense that Wolfram's Rule 110 is universal, which allowed you to construct a blowup solution.

However, based on observations of liquids governed by the 3D Navier-Stokes equations, they are more similar to Wolfram's Rule 30, which appears to be not universal and is very random and chaotic. So I think you will have a much more difficult time trying to apply the same strategy to the 3D Naver-Stokes equations as you did to the averaged NS-equation, even though I agree that there is nothing stopping a blow-up of the 3D Navier-Stokes equations.

Another analogy to a finite time blowup of the NS equations is the parting of the Red Sea. I'm guessing there is no proof that it cannot happen, but it would be a miracle if it did happen.



Terence Tao · 4 February 2019

My belief is that the Navier-Stokes equations behaves turbulently for the vast majority of initial data, but there is a very small but non-empty set of initial data for which it instead behaves universally (and in particular one should be able to construct very special solutions within that set that blow up in finite time). An analogy is with the laws of physical matter: most arrangements of silicon, copper, etc. behave in a fairly chaotic fashion, but there are some very special arrangements of these physical elements that form a device capable of universal computation.

Craig · 4 February 2019

I wish you success in trying to find this initial data, because if you find it, then one could at least theoretically do all sorts of amazing things with fluids, like perhaps part the Red Sea and even this: https://en.wikipedia.org/wiki/Water_memory

But that goes against everyone's intuition that water and gas are stuff one cannot control, while only solids are controllable.



Terence Tao · 4 February 2019

If this data exists, it will likely involve regimes of velocity (or vorticity) and/or timescales that are well outside the ranges one is accustomed to experiencing with fluids (and probably well outside of the regime where the Navier-Stokes or Euler equations remain an accurate model for physical fluids such as water). I have some very vague hopes that various fluid structures such as vortex rings or vortex sheets could exhibit pseudo-solid like behaviour in certain regimes that could allow them to be controllable in a way analogous to actual solids, but I don't have a truly plausible candidate at present for such a structure.

Anonymous · 4 February 2019

In order to “design” such special fluid structures, it seems that they should have some stability, so in addition to vortex structure, they may include also soliton structures.

Craig · 6 February 2019

I was thinking about the following heuristic argument that the NS equations never blow up:

The convection term of the NS equations in Fourier space takes the form:

(I-k*k/|k|^2)* Sum_{k'+k"} [u(k')*u(k")k']

The term (I-k*k/|k|^2) has an expected value of zero, for a randomly selected k, due to cancelations. Therefore, there shouldn't be a blowup of u, as everything in the convection term is expected to almost cancel out. QED

Certainly not a proof, but this is what I think is stopping the NS equations from blowing up. I came up with this argument after reading Montgomery-Smith's paper about the Cheap Navier Stokes equations. My heuristic argument doesn't pertain to his equations, since there is no term (I-k*k/|k|^2) in his Cheap equations.

Terence Tao · 6 February 2019

Actually, if one works things out carefully, the matrix $I - \frac{kk^T}{|k|^2}$ averages out to $\frac{2}{3}I$ in three dimensions, rather than zero (note for instance that the matrix always has trace $3 - 1 = 2$, and is always positive semi-definite). In any event, the convective term is not being averaged over k (but instead over the pair k', k'', which sums to a fixed k) so the mean behaviour k does not have a heuristic relationship with the expected behaviour of the convective term.

There is certainly some cancellation present in that matrix however (which can also be viewed as the symbol of the Leray projection); it is needed for instance to ensure the energy identity (which for instance the cheap NS model does not obey).

Craig · 6 February 2019

I was actually thinking that since $I-kk^T/|k|^2$ has matrix entries of the form $k_i^2+k_j^2$ on the diagonal and $-k_i k_j$ on the off diagonal, there are 6 negative terms and 6 positive terms in the whole expression, all of basically the same form, so they should all basically cancel out. That was my heuristic reasoning.

Terence Tao · 6 February 2019

The $-k_i k_j$ terms are indefinite (they can be either positive or negative, depending on whether the k_i, k_j themselves are positive or negative), while the $k_i^2 + k_j^2$ terms are definite (always non-negative). On average, the off-diagonal

terms $\frac{-k_i k_j}{k_1^2+k_2^2+k_3^2}$ average to zero by symmetry, while the diagonal terms $\frac{k_i^2+k_j^2}{k_1^2+k_2^2+k_3^2}$ average to $2/3$, as I mentioned before.

Craig · 7 February 2019

Yes, well nothing is perfect. :-) So then I see why you think it should blowup for some input. But then the big question is why is it so hard to find a counter-example?

Is it because the problem is similar to the Collatz 5x+1 problem (the Collatz 3x+1 problem modified)? For instance, take x=9. It is odd, so (5*9+1)/2=23, (5*23+1)/2=58, 58/2=29, 73, 183, 458, ... seems to diverge to infinity (according to my spreadsheet calculations). But proving that it diverges to infinity requires computing each step, taking infinite time – there is nothing stopping the sequence from hitting say a power of 2 or some finite orbit.

Is it similar for the NS equations? Finding a counter-example isn't the hard part, but proving that the counterexample is a counter-example is the hard part?

Anonymous · 7 February 2019

It is possible that the construction of a counter-example for the NS problem is of (at least) the same difficulty as the construction a counter- example to the Collatz problem (it seems that both counter-examples should have very complicated fractal-like features.)

Craig · 7 February 2019

Anonymous, I was thinking that a possible analogy to the NS-equation is the 3x+1 conjecture, but in that conjecture there is a heuristic reason why it should converge to 1, since $\sqrt{(1/2)(3/2)} < 1$. But there doesn't seem to be a heuristic reason why the NS-equations should always have solutions, as Prof Tao said.

So that is why I suggested that a possible analogy to the NS-equations is the 5x+1 conjecture. Maybe the convection term of these equations is similar to the 5x+1 function which takes x to x/2 if x is even and x to (5x+1)/2 if x is odd? This probably leads to infinity, but one can never be sure.

Craig · 11 February 2019

I've been thinking a lot about this great Navier-Stokes puzzle more and came up with another heuristic (nonrigorous) argument that the solutions will not blow up in 3D: Put the 3D NS equations in vorticity formulation in Fourier space.

When one does this, (see https://arxiv.org/abs/math/9903042 page 12) one sees that the convective term (both the real and imaginary parts) is a sum with an equal number of positive and negative terms. If one tries to achieve a blow-up, one would need to choose the initial data so that it avoids the likely future cancellations that would result from adding these terms together. Doing this appears to be very difficult if not impossible, even though I cannot rigorously prove that it is impossible.

Notice that this argument cannot be applied to the averaged Navier-Stokes equation, since there is no vorticity formulation (as I recall was mentioned in the article. Isn't vorticity conserved in the averaged Navier-Stokes equation anyway?).

Any comments?

 **K** · 6 February 2019

I wonder: Are there generalisations of the NS Equation, which have some real parameters t. If the parameter is $t = 0$, then one has the NS Equation, and when t is some finite number, then one has an PDE that is less supercritical and for which a blowup is known or can be constructed? That sounds (for an non-expert as I am) naively like a slightly similar approach as the upper-bounding the de Bruijn-Newman constant, which would give the Riemann hypothesis for $\Lambda = 0$.

Because if that is possible, it seems that one could gradually "turn on the supercriticality" (whatever t might correspond to), and learn how to handle different amount of criticality?

 **Terence Tao** · 7 February 2019

Yes, the standard way to do this is to work with the hyperdissipative Navier-Stokes equation, see for instance https://terrytao.wordpress.com/2009/06/19/global-regularity-for-a-logarithmically-supercritical-hyperdissipative-navier-stokes-equation/ and the more recent paper https://mathscinet.ams.org/mathscinet-getitem?mr=3318746 where the exact threshold (which turns out to be supercritical, but only by a logarithmic factor) of hyperdissipation needed to ensure global regularity is established.

 **Radyadour Kh. Zeytounian** · 25 February 2019

Reply by R. Kh. Zeytounian – for instance see my 2017 new bbok <> édited by Springer International Publishing AG 2017.
The main question – for me : The <> was
resolved !?? – despite the very badli formulation of Fefferman ?
Thanks foe an answer
R. Kh. Z
25/ 02 / 2019

 **Cy** · 6 March 2012

Hi Terrence,

I don't have any education, but I like math stuff. I guess my question would be how this relates to the other Millennium Problems?

Correct me if I'm wrong, but the issue you're talking about with criticality (plus super and sub) seems like it is a 'rephrasing' or at least equivalent (for some assumptions) to renormalization.

In other words, I get the impression that chaos in calculation is at the heart of Navier-Stokes, Yang-Mills theory, and the Poincare Conjecture. At least personally, I think the key to solving the Poincare conjecture was Perelman's Entropy.

So what I'm trying to say (or ask, I guess) is how right am I in thinking that it amounts to the same problem? Renormalization in a Quantum Field Theory at least looks like turbulent vs laminar flow in that things are great within a certain bound, but then get unpredictable past a certain point. In other words, a QFT that is not asymptotically free has blow-ups similar to how Navier-Stokes has blow-ups. And a similar thing happens in Ricci flow, but Perelman got a handle on it with Perelman's Entropy. Am I way off base with these comparisons?

I know this is apples and oranges in some sense, but they're all PDE problems in 4D where we need some global invariant to wrestle with the chaos in perturbations. Right?

Anyway, I should probably go learn some more. But being a broke, nearly homeless person – maybe I have more pressing concerns, haha. Thanks for the wonderful post, it was very interesting.

-Cy

 **dcs24** · 10 April 2012

Reblogged this on Daniel C. Sutton.

 **Juha-Matti Perkkiö** · 13 April 2012

Dear Professor Tao,

I stumbled into a positive "proof" of NSE with any initial data in $\mathcal{S}(\mathbb{R}^3, \mathbb{R}^3)$ posted by A.&M.Tsionskiy into ArXiv [http://arxiv.org/abs/1201.1609] on 8 Jan 2012. I have re-checked the formal computations leading to an integral equation formulation in the space variables, but I am not quite convinced about the functional analytic argument leading to the application of the contraction principle. Has anyone else following this blog had a look on this paper?

>  **Terence Tao** · 13 April 2012
>
> The second inequality in (6.7) is incorrect; one cannot control the C^2 norm by the L^2 norm, as can be seen for instance by considering high frequency sinusoids (smoothly truncated to be of compact support).

 **Anonymous** · 7 May 2012

This article is very good quality. A knowledged person in engineering mathematics can understand better.i know integration,differention,limits. I dont know curl,legrenge transform etc. If any website first clear about smoothness.what is week solution.then what is strong solution.etc with some similar examples.then their is ability to atleast understand and admire it.

 **Betul** · 13 May 2012

I am a Physics student, my english is not very well, i am really into this topic.It is already a complicated one, besides my poor english.Can't you just write it a little simplified English so that I and students like me with poor english could understand... :(

nick · 14 May 2012

Hi Terrence,

While thinking about your article, I cam across an idea for a potential globally controlled quantity that might be coercive and subcritical.

The inspiration comes from the Heisenberg uncertainty principles. In physics these are supposed to represent the uncertainty of a measurement, but one could instead view them as curvature criteria on an imbedded space. The x,y,z momentum criteria would be on the x,y,z curvature and the E,t relationship would be analogous to the ricci flow equation, where energy is the radius of the open ball and momentum is the sectional curvature in each orthogonal direction.

One then relates the probability current to the regular current in Navier Stokes. The probability current is complex valued which is unsettling, and I dont know what to do with that. Perhaps this indicates that general solutions to these equations will take complex values, like general solutions to polynomials will take complex numbers?

In physics this relationship and the corresponding quantization of states has great coercive powers and definitely works at fine scales. Do you think such a quantity would be a possible candidate for analysis of the Navier Stokes equation?

Nick



Terence Tao · 15 May 2012

Well, a key difference here between physics and mathematics is that in the former it is often sufficient to work with reasoning which is expected to apply in almost all conceivable scenarios, but which can fail for some exceptional set of unlikely scenarios e.g. "Maxwell's demon" type scenarios which contradict the second law of thermodynamics. But for a mathematical proof one has to eliminate Maxwell demon type scenarios completely, as opposed to almost completely, which is a very different standard of proof, and which among other things tends to rule out most candidates for coercive controlled quantities in Navier-Stokes which arise from probabilistic or uncertainty-based considerations.

Anonymous · 15 May 2012

Dear Prof. Tao,

I have a problem about the global existence and uniqueness of Navier-Stokes equations. Given a good initial data $u_0 \in H^2$, consider the following mild solution form $u(t) = e^{\Delta t}u_0 + \int_0^t e^{\Delta(t-s)}u(s)\cdot\nabla u(s)ds$. By a Banach fixed point argument, we can show that there exists a local solution $u \in C([0,T], H^2)$ where T depends on $\|u_0\|_{H^2}$.

To extend the solution, we consider the energy inequality $\|u(t)\|_{L^2} \leq \|u_0\|_{L^2}$. At the time $T-\epsilon$ with $\epsilon > 0$ small, thanks to the smoothness effect of Δ, we can get an estimate of $\|u(t)\|_{H^2} \leq C$ for all $t \in (T-\epsilon, T-\epsilon+\delta]$ where C, δ only depend on $\|u(T-\epsilon)\|_{L^2}$. Since $\|u(T-\epsilon)\|_{L^2} \leq \|u_0\|_{L^2}$, we can take $\epsilon < \delta$ to get an bound for $\|u(T+\delta-\epsilon)\|_{H^2}$. From this way, it seems we can get the global existence and uniqueness.

I don't know what is the wrong in the above argument.

Thank you!



Terence Tao · 15 May 2012

The smoothing effect you refer to is only directly applicable for the linear heat equation. To obtain the analogous smoothing effect for the Navier-Stokes equation, one must also deal with the contribution of the convection term $u \cdot \nabla u$ (and also the pressure ∇p, though this can be viewed as a component of the convection term). Due to supercriticality, it is impossible to control the effect of this convection term purely in terms of the L^2 norm, and higher regularity norms such as H^1 or H^2 must be employed. This leads to a potential blowup in the H^2 norm, instead of uniform bounds and regularity.



Anonymous · 16 May 2012

Thanks! I still have some problem about this. Suppose that $x \in L^2$, writing $B(u) = \Pi u \cdot \nabla u$ and using some classical estimate of B, we have the following a'priori estimate fo the mild solution ($\latex 1/2<\alpha0$ some constants to be chosen and with the metric $d(u,v) = \sup_{0\le t\le T} t^{\alpha/2}\|u(t)-v(t)\|_{H^\alpha}$,
we can use Banach fixed point theorem to show there exists a unique local mild solution $u \in S$.

Now let $u_0 \in H^\alpha$, there exists a local unique solution $u \in C([0,T];H^\alpha)$ by Banach fixed point theorem. Thanks to the above argument and energy estimate, we can get the global solution in $C([0,\infty);H^\alpha)$ (of course $1/2 < \alpha < 1$).

I can't find any problem for the above argument.



Anonymous · 16 May 2012

The system can't show some long formula, so I write my comment again.

Thanks! I still have some problem about this. Suppose that $x \in L^2$, *writing*latex B(u)=\Pi u \cdot \nabla u$ and using some classical estimate of B, we have the following a'priori estimate fo the mild solution ($\latex 1/2<\alpha<1$): $\|u(t)\|_{H^\alpha}$ is bounded by
$Ct^{-\frac{\alpha}{2}}\|u_0\|_{L^2} + C\int_0^t (t-s)^{-1+\epsilon}\|B(u(s))\|_{H^{-2+\alpha+2\epsilon}}ds$
further by $Ct^{-\frac{\alpha}{2}}\|u_0\|_{L^2} + C\int_0^t (t-s)^{-1+\epsilon}\|u(s)\|^2_{H^\alpha}ds$.

Therefore, $t^{\frac{\alpha}{2}}\|u(t)\|_{H^\alpha} \le C\|u_0\|_{L^2} + Ct^{\frac{\alpha}{2}}\int_0^t (t-s)^{-1+\epsilon}[s^{\frac{\alpha}{2}}\|u(s)\|_{H^\alpha}]^2 ds$

Define the space $S = \{u \in C([0,T];H) : \sup_{0\le t\le T} t^{\frac{\alpha}{2}}\|u(t)\|_{H^\alpha} \le B\}$ with B,T
some constants to be chosen and with the metric $d(u,v) = \sup_{0\le t\le T} t^{\alpha/2}\|u(t)-v(t)\|_{H^\alpha}$,
we can use Banach fixed point theorem to show there exists a unique local mild solution $u \in S$.

Now let u_0 \in H^{\alpha}, there exists a local unique solution u \in C([0,T]; H^\alpha) by Banach fixed point theorem. Thanks to the above argument and energy estimate, we can get the global solution in C([0,\infty); H^\alpha) (of course 1/2<\alpha<1).

I can't find any problem for the above argument.



Terence Tao · 17 May 2012

The equation following "Therefore" is incorrect; one cannot justify the insertion of the prefactor $s^{\alpha/2}$ in the integrand because it can be much smaller than 1.

There are equations of the same strength as Navier-Stokes (in the sense that one has essentially the same energy estimates, and the same estimates for the nonlinearity and the linear evolution) but which are known to blow up in finite time (in particular, there is a dyadic model of Pavlovic and Katz which achieves this). As such, it is not possible to establish global regularity purely from energy methods and the usual linear and nonlinear estimates on the heat equation and on the nonlinearity. One can also see this from performing a scaling analysis or dimensional analysis, as indicated in the body of the main post.

Anonymous · 17 May 2012

Thank you so much for your patience and kindness, Prof. Tao! Actually I forgot the term in the integrand. I know my problem should not be a problem, but still want to find the reason.

The inequality following 'Therefore' should be:
$t^{\frac{\alpha}{2}}\|u(t)\|_{H^\alpha} \leq C\|u_0\|_{L^2} + Ct^{\frac{\alpha}{2}} \int_0^t (t-s)^{-1+\epsilon} s^{-\alpha} [s^{\frac{\alpha}{2}}\|u(s)\|_{H^\alpha}]^2 ds$. Because $\int_0^t (t-s)^{-1+\epsilon} s^{-\alpha} ds$ is integrable, we can use a Banach fixed point theorem.

Terence Tao · 17 May 2012

The expression $t^{\alpha/2} \int_0^t (t-s)^{-1+\epsilon} s^{-\alpha}\ ds$ is not uniformly bounded as $t \to 0$, and so the contraction mapping principle is inapplicable.

Due to lack of time, I will not be responding to any further arguments of this nature, and refer you to the discussion in the blog post.

Gil Kalai · 20 May 2012

Dear Terry, A few comments mentioned stochastic and quantum versions of NS and you made an appealing case that while NS might be easier for such versions and while one of such models may well be more realistic, they offer little help to the Clay Problem since proving uniform bounds for any perturbational version will translate to proving bounds to the original NS problem, while non-uniform estimates will not help.

My question is if there are variants of the NS question of stochastic (or quantum) nature which are known or expected to be more realistic, or stochastic variants where it is known or expected that the situation is easier. (Even without hope towards the Clay Problem itself.)

Terence Tao · 20 May 2012

Well, one of the obvious things to try here is to put a stochastic forcing term in the equation (say in the periodic case, to avoid some technical issues). This ought to be easier, as the stochastic noise should keep pushing one away from what we believe to be the relatively rare blowup or near-blowup scenarios, but again there could be some exceedingly perverse "Maxwell's demon" type dynamics which somehow manages to reverse all the entropy created by the noise and lead almost all of the solutions towards blowup. In the Hamiltonian case one has (in principle) the theory of invariant measures (such as Gibbs measures) to prevent this sort of entropy collapse (though this is more geared towards stochastic initial data than stochastic forcing term), but as far as I am aware there are no such invariant measures in the dissipative setting of Navier-Stokes.

Still, it is conceivable that one could do something nontrivial and rigorous with such a model, though to my knowledge the literature in this direction is quite scant.

 **Justicus** · 13 June 2012

Dear Prof. Tao, what do you make of this article? It seems that by considering the Lamb vector divergence, one can derive an analytical expression for the pressure field through a Poisson's equation which will ensure that the time derivative of the L^2 norm of vorticity is always negative and thus ensures the regularity of N-S equations with smooth, square-integrable and divergence-free initial data. This in turn implies a smooth exponential decay of the L^2 norms of vorticity and velocity.
Best regards,
Justicus

http://arxiv.org/abs/1201.1609

(please disregard the earlier versions)

 **Justicus** · 13 June 2012

Sorry, here's the correct link

http://arxiv.org/abs/1206.1281

 **Terence Tao** · 13 June 2012

The pressure in the Navier-Stokes problem is not exogeneous; it cannot be prescribed independently of the velocity and vorticity fields, but is instead coupled to those fields (as one can already see from the equation (9) in the paper, which determines the pressure as the solution to a Poisson equation involving those fields). As such, if one declares the pressure to equal something (as was done in (16), (19), (24)), there is no longer any guarantee that one can actually solve the Navier-Stokes equation with this prescribed pressure; note that all the local existence theorems for Navier-Stokes in the literature do not permit an arbitrary prescription of the pressure. (And indeed, it appears that the prescription given in the paper is inconsistent with the Navier-Stokes equations: for instance, the equations (9) and (29) will contradict each other, even at the initial time t=0, for most choices of initial velocity field u).

 **rbcoulter** · 20 June 2012

"It is a well known fact that the solutions are regular if the
enstrophy of the system stays bounded, see [3]. This means that we will
require ...."

Then it refers to the energy dissipation integral (Enstrophy). My understanding is that regular solutions meet this condition but it alone is not sufficient to ensure regularity. See Professor Tao's opening comments at the top of this blog.

 **Daniel** · 15 June 2012

Dear Terence Tao and others

This is a very interesting discussion here... Thank you..
I am not an expert in this topic...I have just one probably elementary question. Why is it the

case that we need to find an inadmissable solution of NS that has finite time blow up to prove breakdown of solutions to NS,; the actual “statements A,B,C, and D” don’t mention blow up at all ?

Daniel



Terence Tao · 15 June 2012

This is because we have local existence theorems that say, roughly speaking, that given sufficiently “nice” initial data, one can solve the Navier-Stokes equation with this initial data and with a smooth solution up to any time for which the solution has not yet blown up (say, in the sense that the velocity becomes unbounded). So if we can rule out blow up, we obtain global smooth solutions as a corollary.



Daniel · 16 June 2012

Thanks for the quick reply, it was good !



Gil Kalai · 24 June 2012

Dear Terry, I have follow up questions regarding Maxwell demons. 1) What precisely one refers to as Maxwell demons in the context of NS and other PDE. (I am sure this is clear to all experts). 2) How central is the demons issue in the global regularity question? 3) You mentioned that stochastic versions may still have some demons? Still, are there some stochastic versions which guarantee no demons, or that can be described as being conditioned on “no demons?”



Terence Tao · 24 June 2012

It’s not really a well-defined term, but basically from the local existence theory we know that if at some given time, the energy is mostly concentrated at a certain frequency scale N (or spatial scale 1/N), so that the solution tends to oscillate with frequency 1/N, then after a certain short amount of time (about time N^{-5/2}, if I recall correctly), the solution will still exist, but may now be concentrated in some combination of the frequencies N/2, N, and 2N (if we bin the frequencies in some dyadic way; this is a massive oversimplification and should not be taken too seriously). The worst case is if the bulk of the energy somehow gets transferred into the 2N frequency range, instead of the N and N/2 ranges, and if this worst case scenario happens over and over again, one could get blowup in finite time. This would be analogous to an (allgedly) pseudorandom walk favoring one direction over another so much that by time t, it has moved about O(t) away from its starting point, rather than the expected O(sqrt(t)), which is also basically the analogy behind Maxwell’s demon in thermodynamics. So any resolution of the global regularity conjecture must at some point confront and eliminate this Maxwell’s demon scenario, which is otherwise consistent with all the known conservation laws, monotonicity formulae, and local existence theory for this equation. (Actually, we can use these known facts to prevent the purely “ballistic” behaviour in which the worst case scenario happens _almost all_ of the time, analogous to a pseudorandom walk deviating about ct away from the origin in time t for some constant c > 0, but there are still intermediate scenarios, analogous to a pseudorandom walk deviating something like t/log t away from the origin, in which the worst case happens often enough to still cause blowup, which we cannot rule out from known facts.)

There are two levels of stochastic analogues of Navier-Stokes. The first is to have random initial data, but deterministic evolution; the second is to have both random initial data and stochastic evolution. (The first is like the output of a pseudorandom number generator with random seed; the second is like a pseudorandom number which also has access to additional random bits as time progresses, but those bits are incorporated into the dynamics of the (allegedly) pseudorandom number generator in a complicated and nonlinear fashion). Intuitively, this randomness (particularly in the second level) should make the demon scenario much less likely, but because everything is coupled together, and there is no obvious concentration of measure (or invariant measure) theory available, it is still theoretically possible (at our current level of understanding) that the demon scenario still occurs (much as a badly designed pseudorandom number generator may posssibly create biased output even if given perfectly unbiased random inputs).

 **Justicus** · 26 June 2012

Dear Prof. Tao,

I was wondering is the regularity issue independent of the choice of the viscosity parameter \nu ? In other words, if for some /nu the system is regular, will that imply regularity for all \nu >0 other things being equal?

Thanks for the excellent blog!

 **Terence Tao** · 26 June 2012

Yes, one can easily renormalise the viscosity to be any value one wishes (to put it another way, viscosity is not dimensionless). For instance, if $u(t,x)$ obeys the Navier-Stokes equation with viscosity $\nu > 0$ and pressure $p(t,x)$, then $\nu^{-1}u(\nu^{-1}t,x)$ obeys the Navier-Stokes equation with viscosity 1 and pressure $\nu^{-2}p(\nu^{-1}t,x)$. From these sorts of transformations it is a routine matter to deduce global regularity for arbitrary ν from global regularity for a single value of ν. (Note though that any quantitative bounds that might be associated to a global regularity result will almost certainly be affected by such renormalisations, and so one does not immediately gain much insight as to what happens in the zero viscosity limit $\nu \to 0$. In particular, there is no known formal relationship between the global regularity problem for Navier-Stokes and for incompressible Euler, though it is widely believed that the latter is less likely to be globally regular.)

 **Justicus** · 26 June 2012

Thanks!

 **Anonymous** · 16 July 2012

One of the ultimate aims to prove the global regularity of the NS Equations has something to do with turbulence. SUPPOSE that the statement (B) or (A) has been shown to be affirmative, how would the conclusion help us understand turbulence?

 **Anonymous** · 16 July 2012

A partial answer: I think it was Mandelbrot who first conjectured that the onset of turbulence happens when solutions become fractal (in the space variables). This conjecture can only be true if initially smooth solutions break down in finite time. If solutions stay smooth forever in time, then turbulence is not as Mandelbrot conjectured it to be.

 **rbcoulter** · 6 August 2012

There are some salient arguments that NS may not explain turbulence. IMO, if (B) or (A) is true, then turbulence is probably just an illusion in the sense that there would be some zoomed in micro-scale that appears “laminar”. It would be much more interesting if (B) / (A) is not true, where “turbulence” persists to any micro-scale (or singularity exists).

 **Anonymous** · 17 September 2012

Singularities may not be part of God’s mind.

How much do we know about the 2d turbulence today?
Have we learnt anything from the smooth 2d NS solution?

Clearly the regularity and the uniqueness rule out the anomaly due to bifurcations, strange attractors and scenarios of blow-up
(note that these breakdown mechanisms exist in many 1d models).
On the physics side, there are of course differences as well as similarities
between the 2d and 3d turbulence; it is hard to believe that
the 2d implication is categorically disconnected from its 3d counterpart.
One often attributes the 2d smoothness to the absence of the vorticity stretching. This argument is not entirely convincing because
the convection may induce velocity (with large gradients) which, as a result, may well tear apart vicinity flows.

 **Robert Coulter** · 16 August 2014

The 2D versus 3D situation is a good question. The mathematicians seemed to be almost completely focused on the the time evolution problem in 3D. The 2D time evolution problem does not produce singularities assuming there is not a nearby (local) zero velocity boundary condition (constraint). However, in the real world, there is almost always a nearby BC that is zero — airplane wing, pipe wall, etc. It is generally accepted by myself and others that the culprit of turbulence starts at these zero velocity boundaries creating turbulence some distance away. In short, the NS model, IMO, has some issues even in 2D if one considers real life BCs.

 **Anonymous** · 18 August 2014

The presence of a solid surface will complicate the initiation of turbulence but cannot be the core reason. (The experts must have been well aware of this fact. Hence goes the formulation of the Clay NS problem.) Consider blowing a puff of smoke into the open space. If one blows hard enough, turbulence ensues (you know it when you see it). Of course, we also observe smooth smoke rings as a result of gentle blows. With carefully controlled starting conditions in between these two extremes, a laminar puff will become turbulent.

 **Anonymous** · 20 August 2014

You might want to elaborate on what you mean by “core” reason. The point was simply that there are some intriguing issues in the 2D case in conjuction with boundary conditions. The Clay problems have been written to avoid these boundary condition issues. Why is this? Cafarelli gave some hint in his video. He was worried about infinity causing problems. Why don’t they have a case

with a BC of velocity = 0 at a finite distance? Maybe that would have been too easy — that is, to show that regularity DOES not exist in those cases. The entire universe is not an ocean nor a periodic torus ocean.

Also, the puff of smoke is air blowing past the lips of the mouth (BC velocity = 0 at the lips). Also, turbulent smoke emanating from a pipe, for instance, has air going past a bluff body, with turbulence forming downstream of the bluff body (above the pipe).

 **Anonymous** · 21 August 2014

(To Anonymous, 20 August, 2014)
The core reason for turbulence is the non-linearity term in the equations of motion (3d and 2d); the view imputing to vorticity stretching is incomplete (this is exactly why we are talking about turbulence in 2d).

The BC v->0 at infinity is implied in the Clay formulation for a class of solutions because any REGULAR solution must have bounded-energy (roughly speaking nothing happens at infinity). It is wise not to specify this decay BC simply because a solution may become singular, say in finite time as some believe (the statement C), and hence it or its influence would propagate to infinity instantaneously. (The arguments are valid for flows in R^2.)

The BC v=0 at the lips applies equally to the three cases: laminar, transitional and turbulent. Thus it cannot be the core reason for instigating turbulence.

 **G.K Chesterton** · 10 August 2012

Why is The Navier–Stokes existence and smoothness especially on turbulance still a open problem, and still incomplete.Is it true if this one is one of the hardest problmes in mathematics.regards

 **G.K Chesterton** · 10 August 2012

Dear Prof Tao : What could we do to solve Navier-Stokes Equation easily ,at last i wish to know why is the official statement of the problem was given by Prof Charles Fefferman. Because as far as i know Prof Charles Fefferman was a genius and a brilliant.He got fields medal also.By the way is there a blog(i means Mr Grigory Parelman blog

Sincerely Yours

 **Daniel** · 19 September 2012

Dear Terence Tao and others,

Following on from the comments on 15 June. I have another probably elementary question: To rule out “blow up” does one have to include the external force vector in doing so, or is it adequate to rule out “blow up” with no external force vector ?

Daniel

 **rbcoulter** · 22 September 2012

Options A and B in the Clay problem statement do not have an external force. These are the “prove regularity” options so one would not need to consider external forces for them.

The other options allow for a cleverly constrained finite force field to be applied to possibly “assist” blowup.

 **Daniel** · 22 September 2012

Thank you rbcoulter for that logical reply, I gave you a thumb up for that.

 **westy31** · 27 October 2012

I was wandering about a related problem, which I pose as a question here:
I you insert a blob of paint into a fluid,of ‘infinite Schmidt number’, does the blob ever split into separate islands of paint?
The Schmidt number is the ratio of mass diffusion and viscosity (ie momentum diffusion). If this is infinite, then you do not have diffusion. This means that the paint concentration does not smooth out: It is either zero or equal to the initial concentration.
I put in the infinite Schmidt number to make things more interesting. The blob of paint cannot smooth out its concentration, it can only deform into an ever more complex octopus-like deformed blob. So does this blob ever break up?
The second question: Suppose the blob did break up into islands, does that imply violation of regularity?

Gerard

 **timur** · 26 December 2012

As long as the velocity field stays continuous in space and time, the blob will l not break.

 **Anonymous** · 9 November 2012

Good question. I would say that Navier Stokes breaks down if the “octopus loses a leg” or develops a hole.

 **Craig** · 22 December 2012

In my humble opinion, the real reason why proving global regularity for the NS equations is hard is because it’s false. Is there a reason why finding a counter-example has been so difficult?

 **timur** · 26 December 2012

IMHO, the real reason why finding a counter-example has been so difficult is because the global regularity is true. (Sorry I could not help)

Mathematics: What progress has been made till date towards the resolution of the Navier-Stokes existence and smoothness problem? - Quora · 27 December 2012

[…] Student in MathematicsFields Medalist Terry Tao has a long, detailed writeup about the problem here:https://terrytao.wordpress.com/20…Embed QuoteComment Loading… • Just now Add […]

 **Mike** · 29 January 2013

Fairly interesting piece of work, apparently modelling of some sort of causality and the possible connection of the Navier-Stokes equations to gravitation and economic theory

Click to access fieldtheory2.pdf

 **jussilindgren** · 14 February 2013

Actually, there is a slightly improved version available, the key equation is what i call a Wick-rotated N-S-equation. It should be compatible with the relativistically invariant respective quantum equation, this is the main equation

$\frac{\partial}{\partial t}\left(\frac{\partial \vec{v}}{\partial t}+\vec{v}\cdot\nabla\vec{v}\right)=-\frac{1}{2}m_0mc^4(-\nabla p+\Delta\vec{v})$

which is a sort of nonlinear wave equation system, any ideas how to get solutions?

 **jussilindgren** · 14 February 2013

and the respective quantum equation is

$\frac{-\hbar^2}{E_0}\frac{\partial^2\psi}{\partial t^2}+i\hbar\frac{\partial\psi}{\partial t}=-\frac{\hbar^2}{2m}\Delta\psi$

 **jlgalloway** · 6 March 2013

Reblogged this on Concrete Dreams and commented:
Excellent opinion article by Terence Tao regarding why global regularity for Navier-Stokes is hard.

 Jacek Cyranka · 3 May 2013

What about computer simulations?

As I understand If a counter-example for global regularity of 3D NS exists, there should be some computer simulations which supports this. I mean initial conditions which in computer simulations seems to develop a blow up after a finite time (e.g. 1 hour of simulation) .

Of course, one can never eliminate the possibility that the blow-up was caused by a numerical method instabilities , or finite representation of numbers on a computer. But there are tools to deal with that, for instance by using interval arithmetic instead of basic fixed point arithmetic one can produce rigorous results as well.

 **David Purvance** · 3 May 2013

Numerical simulations using white noise flows in a discrete 3D periodic solution blow up. This solution and an explanation for the blow up are developed at purvanced.wordpress.com.

 **Daniel** · 6 May 2013

Hi Terrence,
Im thinking that some of the comments are against your policy and need to be deleted. I have even seen some comments that are disrespectful to you.
Daniel

 **David Purvance** · 7 October 2013

Dear Terence or other interested person,

In the development of the Navier Stokes equation in wavenumber space the projection tensor restricts flow to the PLANE perpendicular to wavenumber to maintain incompressibility. For a deterministic equation isn't this a bit arbitrary or undeterministic? There are an inifinite number of directions in that PLANE. What I have discovered is that if you require the projection tensor to be completely deterministic and restrict flow to the direction of the

initial flow, which by definition must be in the PLANE perpendicular to wavenumber, the flow remains stable and never blows up!



Juha-Matti Perkkiö · 27 November 2013

Wouldn't that imply that the standard NSE has a smooth eternal solution with that property, and by local uniqueness any smooth solution would have that property as well? That feels to funny.



David Purvance · 2 December 2013

Sorry, I don't follow your argument. What I have developed (purvanced.wordpress.com) is a regular solution for directionally stationary flows. I conjecture this solution is the general solution for flows devoid of external forces.



Robert Coulter · 14 December 2013

I would be suspicous also. Finding a way to make all initial conditions "stable and never blows up" is trivial if done with an external force acting against the flow. In other words, a force is applied to "assist" the viscosity dampening term in preventing blowups. I think the problem with have here is that many work with the pressure-free versions of NSE. The pressure must resolve to be a scalar at everypoint (i.e. not multi-valued) to ensure there is no external force. Also, the pressure (or potential field) must be constant along the boundary (even if the boundary is infinite) or one would be introducing an energy source from infinity (or boundary). If one has a proposed soluton, u = (x,t), then the pressure field p = (x,t), should then be calculated. This may not be easy since it would involve integrating the solution with various products of itself and its derivatives (the left side of NSE minus the right side viscosity term). Also, ALL paths between two points (at each point in time) must integrate to show the same delta P, and a closed path cannot have the pressure change (while integrating along it) after returning to the starting point.



Craig · 30 November 2013

Terrence, why do you say that the Navier Stokes equations exhibit pseudo randomness? You compare this problem with other famous problems, but these other problems are discrete, while the NS problem is continuous. I would really appreciate it if you could explain. Thank you.



Mickey · 6 December 2013

This isn't really a response to your post, but has anyone seen this paper by Tsionskiy-Tsionskiy on Navier-Stokes:

"Existence, uniqueness and smoothness of a solution for 3D Navier-Stokes equations with any smooth initial velocity"
Arkadiy Tsionskiy, Mikhail Tsionskiy
Electron. J. Diff. Equ., Vol. 2013 (2013), No. 83, pp. 1-17.

They say they've solved everything. I know there are many "proofs" floating around to the Clay problems, so normally I wouldn't think too much about it. But this was published in a journal with peer review, so I wonder if anybody has looked at it.

Mickey · 7 December 2013

I see now that you pointed out an error in the Tsionskiy-Tsionskiy paper in a comment on this thread in April. Please disregard my question!

susan · 11 January 2014

Eminent kazaki mathematician
Mukhtarbay Otelbaev, Prof. Dr.
has JUST now published a claimed full proof of the navier stokes clay problem.

Is it correct?

http://bnews.kz/en/news/post/180213/

He has over 200 papers, 70 phd students, is an academician of their academy of scieces and has previous papers on navier stokes and functional analysis.

SO HE IS NOT A CRACKPOT.

susan · 11 January 2014

A link to the paper (in Russian): http://www.math.kz/images/journal/2013-4/Otelbaev_N-S_21_12_2013.pdf

Sam Chio · 11 January 2014

Mukhtarbay Otelbaev published an article in which he claims to have provided a full solution to the problem with periodic boundary conditions in space variables. Is there any link to his paper in English?

Navier Stokes Solution Claimed and Twitter Algos « Pink Iguana · 12 January 2014

[...] for Wavy Vortex flow on an IBM mainframe. Can't parse the paper in Russian though. I hope Tao can read Russian, otherwise confirmation could take a [...]

Emanuela Grossi · 13 January 2014

Translation of proposed Navier-Stokes solution by Mukhtarbay Otelbaev https://github.com/myw/navier_stokes_translate

Juha-Matti Perkkiö · 13 January 2014

Dear professor Tao and all who read this blog,

assuming for a while that Otelbaev's estimate is correct, I would like to better understand his reformulation of the problem. Referring to the translation in progress from https://github.com/myw/navier_stokes_translate he wrote that

"The system of equations (1.1) and initial/boundary constraints (1.2), (1.3)
do not allow a unique solution to the pressure p(t; x). For this reason, we add the constraint

$$\int_Q p(t,x)dx = p_0.$$
"

How so? Taking the divergence of NSE we see that

$$\nabla^2 p = -\rho \sum_{j,k} \frac{\partial v_k}{\partial x_j}\frac{\partial v_j}{\partial x_k} + \nabla \cdot \boldsymbol{f}$$

and this Poisson equation most certainly has a unique solution in the torus Q. Should the integral of p over Q stay constant in the first place?

Secondly, the Millenium problem is an initial value problem. The estimate would prove it if for any smooth solenoidal velocity field v(t_0,x) one could construct a suitably smooth body force f(t,x) for 0<t<t_0 that drives v(0,x)=0 into v(t_0,x) in the whole torus Q. Is this indeed the case?

Yours,

Juha-Matti Perkkiö



uvs · 15 January 2014

As to 1: the normalization is ok. Indeed: if $p(t,x)$ is a pressure field that solves NSE, then so is $p(t,x)+g(t)$ for any function $g(t)$.

There is something else about pressure in Otelbaev's paper that bugs many people. In (1.3) periodic BCs wrt p are stated. If ∇p is periodic, then p itself must not be periodic, so potential multiple solutions are excluded, thus simplifying the problem. No BC wrt pressure are stated in the millenium prize formulation.

As to 2: Consider an IVP $v(0,x)=v_0$ and define $u(t,x)=v(t,x)-v_0$, this will satisfy a NSE with homogeneous IC and a modified force term. Solve this, so $v(t,x)=u(t,x)+v_0(x)$



uvs · 15 January 2014

Sorry, I was too fast and forgot NSE was nonlinear. The construction is indeed slightly more involved.

Start with arbitrary smooth initial data $v_0(x)$. The IVP has a unique smooth solution $v(t,x)$ locally in time, say, for $tt^*/2$. Define $u(t,x)=\phi(t)v(t,x)$. We have
$\dot{u}+u\cdot\nabla u-\triangle u=\dot{\phi}+\phi\dot{v}+\phi^2 v\cdot\nabla v-\phi\triangle v$,
which can be computed, given (v,ϕ). Thus u satisfies the original NSE for $t>t^*/2$ (as $\phi=1$) and a modified NSE with homogeneous ICs for $t0$ which concides with v for $t^*/2 <t < t^*$ then u for $0<t<t^*/2$ is that startup flow which you requested.



Anonymous · 14 January 2014

Dear All,

Here are some comments:

(1) Otelbaev's reformulation defines an over-determined pde system.
By the continuity, the NS momentum equation is a self-consistent system for the velocity field. Hence there is no need to specify any boundary conditions
on the pressure; his periodic pressure stated in (1.3) is redundant.

(2) The Clay's NS problem formulations (A to D) are mathematically consistent .

(3) The extra pressure constraint of Otelbaev (1.4) is equivalent to the pressure p in L^1(Q). But there is no justification / derivation of this constraint presented in his paper. Thus the constraint is not an apriori bound.
On the other hand, there has been no experimental evidence to indicate that the pressure

field in fluid motions is generally constrained in this manner.
(If we are free to specify extra constraints in addition to the Clay's velocity boundary conditions, the NS is trivial to solve!)

(4) Finally, his analysis does not shed any new light on turbulence.
To elucidate on the nature of turbulence in the light of NS solutions is in fact implied in the Clay formulations.

Stephen Montgomery-Smith · 19 January 2014

In the incompressible Navier-Stokes equation, the pressure is only determined up to a constant. Therefore equation (1.4) is harmless and irrelevant. Also, by the time he gets to page 12, he has rewritten the Navier-Stokes equation in a much more abstract form in equation (3.4). And the pressure term is implicitly given by equation (4.3), which is the Leray projection.

Anonymous · 20 January 2014

Equation (1.4) basically rules out the possibility of singularity (time wise). To assign the pressure in L^1(Q) means effectively that Poisson's equation for the pressure has an inverse a priori! When you say that the pressure is determined up to a constant, do you mean that the NS sees only the pressure derivative or do you mean that there will definitely be no singularity in the pressure?

Stephen Montgomery-Smith · 20 January 2014

The rest of the Navier-Stokes equations also make implicit assumptions on the solution (smooth, etc). That is why the Clay Millennium problem is stated as "existence of solutions" when in fact weak solutions are already known to exist. What the Clay problem is really asking is if there is some initial data which when the NS equations are solved, ceases to have a solution in the classical sense at some time t=a. It is well known that if one can prove that certain quantities remain bounded as t converges to a, then the classical solution extends beyond time t=a. It is in this sense that Otelbaev claims to have a solution. So implicitly stating that the pressure is in L^1 before time t=a does not effect the difficult of the problem one bit. And the main statement of Otelbaev's claimed result is that the appropriate quantities are bounded as t converges to a.

But more than that, Otelbaev never makes any use of equation (1.4) anywhere else in the paper.

Anonymous · 22 January 2014

Don't weak solutions include singularly-behaved functions?
To go beyond the time t=a, how do we know that the flow would not develop a singularity (even though the Leray type singularity has been ruled out)?
It is precisely in this sense that an analysis of the NS must make allowance for a possible singularity. Otelbaev has implicitly used BOTH (1.4) and the periodic boundary conditions (1.3); they are assumed in deriving (4.5) from (4.4).
The issue is that we are not free to apply the Helmholtz-Leray projection without constraints; any attempt to inverse the Poisson equation for the pressure must be fully justified.

Otelbaev did not derive any apriori bound on the “RHS” function, at least he did realise the constraints on the H-L projection. For a careful introduction, see Cannone’s review S1.1 in Handbook of Math Flu-Dyn (2003). An example of using the H-L projection with certain periodic BC’s is given in pp.36-39 of Navier-Stokes Equations and Turbulence by Foias etal (2000). The issue may be further clarified by considering the NS in R^3, see Lemma 1.6 of Majda and Bertozzi (2002).

 **Anonymous** · 14 January 2014

Deal all,

Another anonymous comment – not the same anonymouus as at 14-01-2014 12:25.

It seems strange to me, that Prof. Otelbaev puts $\nu = 1$ in his formulation of the problem. He writes, “we can do it without loss of generality”. I am not a specialist, but it seems very strange to me. For example, canonic Clay formulation anf formulation from Prof. Ladyzhenskaya book all contain the ν coefficients.
May be the Prof. Otelbaev is valid only for $Re=1$!? Therefore it cannot contain the turbulence considereation.

IT

 **Juha-Matti Perkkiö** · 14 January 2014

The value of the dynamic viscosity ν is totally irrelevant by rescaling. My both questions are still unanswered. The first one probably has a positive answer, provided that one can tweak the body force, but then again, prof. Otelbaev´s approach doesn’t immediately solve the problem, since one needs to play the game without arbitrary God forces.

 CK Chio · 17 January 2014

Setting $\nu = 1$ is possible because the equation can be written into dimensionless form. You can rescale the velocity and time units.

 **matthew miller** · 15 January 2014

I’m extremely excited about the prospects of a strong-solution for NS. Just wanted to convey my enthusiasm, even if it’s of no positive worth. The first time I had ever even read there were such unsolved classical problems was on this blog and I think it’d be really great if Professor Tao brought the experience full circle by giving his comments, be they speculative or definitive, with regards the news of this proposed proof. That said I’m very curious to the developments of this thread in relation to the proof. Thanks!

 **Sam Chio** · 15 January 2014

Here is link to an earlier publication M. Otelbaev et. al. 2006, in English – Existence Conditions for a Global Strong Solution to One Class of Nonlinear Evolution Equations in a Hilbert Space

Click to access Existence-Conditions-for-a-Global-Strong-Solution-to-One-Class-of-Nonlinear-Evolution-Equations-in-a-Hilbert-Space.pdf

La demostración de Otelbaev del problema del milenio de Navier-Stokes | La Ciencia de la Mula Francis · 18 January 2014
[...] exquisita el genial Terry Tao en "Why global regularity for Navier-Stokes is hard," What's New, 18 Mar 2007; también recomiendo leer a O. A. Ladyzhenskaya, "Sixth problem of the millennium: [...]

 **jj2** · 6 August 2010
No condition is violated. You may say, if you want, that the external force
is embedded in the pressure, but actually this is not the case. The external force is zet to zero and kept at that value. Pressure is eliminated by integrability conditions and the remaining equation for the velocity vector has many solutions. You cannot and should not say or think that in some
way this what appears as pressure actually is the external force. That is
what a physicist might say (as they are not too precise), mathematically
the force is set to zero, pressure is eliminated, velocoty is solved, pressure
computed from velocity and that is how it is.

 **Robert Coulter** · 29 August 2010
Redefining the pressure and external force terms on the right side of the NS equations to be pressure-only does not mean the external force is set to zero. When the global energy in the system increases, this can only come about from the application of an external force. Also, to achieve infinite global energy in finite time would require an infinite external force.

Also, the initial global kinetic energy is finite and is calculated by the global integral of the velocity squared (at t=0). Time derivatives of the velocity do not store energy. There is no potential energy term (incompressible fluid).

The apparent non-uniqueness claim also comes about because of the external force. One can arbitrarily determine multiple trajectories of the solution from a common initial condition if given latitude to vary the external force as needed.

 **Jorma Jormakka** · 29 August 2010
Professor Tao has made settings that do not allow jj2 to post any messages to this web-site. This is because I tried as jj2 to put a small announcement of a discussion in reddit.com where some students, notably one called cowgod42, tried to break my peer-reviewed Navier-Stokes paper, naturally they could not break it. First Professor Tao removed my announcement of my paper solving the Navier-Stokes problem. I managed to put it back and to have a short discussion with Robert. Now, however, the censorship on this site has become tighter and jj2 cannot post anything.

I will first see if I can post a message using another name, and them will answer to you Robert.

 **Jorma Jormakka** · 29 August 2010
Good, this new identity works, meaning that Professor Tao had put jj2 to a spam list (for simply announcing that reddit.com had not managed to break the proof and asking if the people on this site can break it, there was no spam or offence in the attempted post. I am sad that there is censorship.)

To answer the comments by Robert.

Mass in accelerating movement has kinetic energy. Fluid in accelerating movement also has kinetic energy, not potential energy. If you observe a falling mass and select the origin of the time as some point when the mass already is in accelerating movement, then surely the mass has kinetic energy stored in the time derivatives.

The external force select a trajectory of a solution requiring no external force. Physical examples you can think of is knocking an egg standing on its sharper head to one side. This knocking can be infinitesimally small and short in time. Zero energy is needed since you can always make the energy smaller by making an even smaller knock. Another example is steering a boat or a car. If you let the boat or car move where it naturally goea, you do not need to use any energy, you do not turn the wheel. Assuming that the boat can naturally move by the current either to a waterfall or to a safe harbor depending on to what direction it originally is pushed, you can steer the boat to the direction you want, stop the motor, and it will go where you wanted. The energy to select the direction where it goes can again be made infinitesimally small.

In the paper the external force is the difference of derivatives of two solutions: the selected solution having a finite-time blow up, and the currect solutiuon. By the local-in-time existence and uniqueness theorem the solution is unique if the initial conditions are fully described. In this case, the external force sets the time derivatives of velocity at t=0 and the initial conditions for velocity are fully described. Thus, the solution is unique. One solution is that the selected solution equals the currecnt solution, i.e., the force has zero value. As the solution is unique, this is the only solution. Thus, the external force has zero value from t=0 to infinity.
The energy of the external force can be obtained by integrating its value, and the energy is zero. The external force is not zero, though it has zero value. Feedback control forces still exercise control even if they are not used: if the current solution were not the selected solution, then the external force would not be zero. As it is zero, they must equal.

You see the same external control force in Internet. Even if your posts are not censured, there is control. Should you be successfully proving one of the Clay math problems, then you would see that there will be no publicity to the result even if you manage to publish it after a very long peer-review
in a respectable journal. And you may find that your identity is put to a spam list so that you could not answer to incorrect comments made against the proof. There is control even if you do not notice any force, zero value of feedback control force is not the same as zero force.

Hope these comments explain the issues you raised. The proof has been checked by over 10 mathematicians and much more have seen it. Nobody has been able to come up with a valid counterargument, so we must say that it is correct. It is so simple that any second year undergraduate can check it. I very much doubt that an undergraduate could have found it. After all, it is a nonlinear partial differential equation in three dimensions. Not at all so easy to find the solution as to check the solution.



Terence Tao · 29 August 2010

Self-promotion of one's own papers is against the stated comment policy of this blog; see

About

If you wish to discuss your own research papers, please do so using another venue, such as your own personal web pages.

That said, I will make two mathematical comments pertaining to the above discussion:

1. It is indeed true that as stated by Fefferman, the initial value problem for the periodic Navier-Stokes equation does not have unique solutions, due to the allowance of the pressure function to be unbounded at spatial infinity. It is this unbounded nature of the pressure function which also causes such unphysical effects as blowup of energy. However, if one adds the additional physical constraint that the pressure be bounded, then this lack of uniqueness disappears (it is easy to see that if the velocity field u and external force f are periodic, and the pressure p is smooth and bounded, then the pressure p is also periodic). In retrospect, it would probably have been best if the additional physical requirement that the pressure be bounded be placed in the official description of the Navier-Stokes problem, though ultimately the presence or absence of this requirement does not affect the truth of either problem (B) or (D) in that description (because the unphysical solutions $(\tilde{p}, \tilde{u})$ with unbounded pressure are just (variable velocity) Galilean transformations of a physical solution (p,u), at least when f=0; indeed it is not difficult to show that one must have $\tilde{u}(t,x) = u(t, x - x_0(t)) + x_0'(t)$ and $\tilde{p}(t,x) = p(t, x - x_0(t)) - x_0''(t) \cdot x + p_0(t)$ for some smooth functions $x_0 : \mathbf{R} \to \mathbf{R}^3$ and $p_0 : \mathbf{R} \to \mathbf{R}$. In particular, one has global regularity for at least one of these solutions if and only if one also has global regularity for the physical solution.).

Note also that one does not need any particularly fancy solutions to demonstrate non-uniqueness in the case of unbounded pressure. For instance, one can simply take $u_1(t,x) = a(t)$, $u_2(t,x) = u_3(t,x) = 0$, $p(t,x) := -a'(t)x_1$ for any smooth function a to show non-uniqueness in this setting. (Among other things, this makes Theorems 2.2 and 2.3 of the EJDE paper trivial to prove.)

2. Even without the imposition of the physical condition that the pressure be bounded, the proof of Theorem 2.4 in the EJDE paper, which would answer part (D) of the official description of the Navier-Stokes problem, is incorrect, due to a circular definition of f. To prove Theorem 2.4, one must provide data u^0, f with the property that there are no global solutions (p,u) to the Navier-Stokes problem with this given data; or equivalently, every local solution (p,u) to the Navier-Stokes problem with this given data develops a singularity in finite time. Note in particular that f has to be specified prior to u. However, in the claimed proof of Theorem 2.4, f is not defined independently of u, but is instead defined in terms of u. This is not a valid definition of f for the purposes of demonstrating Theorem 2.4, since u only appears in the statement of that theorem *after* f has been selected; hence any definition of f that involves u would be circular. Using a control equation does not eliminate this circularity, because the solutions u appearing in the statement of Theorem 2.4 are not forced to obey any control equation, but are instead arbitrary solutions to the (uncontrolled) Navier-Stokes equation with the specified data.

In more detail: the purported proof of Theorem 2.4 defines f in terms of a putative solution (p,u) to a controlled Navier-Stokes equation by adding the control equation

$$f = \partial_t u - \partial_t U.$$

However, once f is chosen in this fashion, there is nothing preventing the existence of a second solution $(\tilde{p}, \tilde{u})$ to the original (uncontrolled) Navier-Stokes equation with this forcing term f, due to the lack of uniqueness. (Indeed, if one sets f=0 and u=U as indicated in the purported proof of Theorem 2.4, then there are plenty of such second solutions $(\tilde{p}, \tilde{u})$, thanks to Lemma 2.1.) Such solutions would contradict the conclusion of Theorem 2.4, but are not ruled out by the arguments in the purported proof of that theorem because f is controlled by u, not by $\tilde{u}$:

$$f \neq \partial_t \tilde{u} - \partial_t U.$$

I have no interest in discussing this paper further. I will permit exactly one final response on this topic; however, all subsequent posts that are in violation of the comment policy of this blog will be deleted.



Jorma Jormakka · 29 August 2010

Dear Professor Tao,

I was not doing any self-promotion by posting to your site a message: In case you are interested you find the solution to the problem you are discussing in this site in a published paper. Instead, as you were discussing the problem, you should have been expected to be interested in published results concerning the problem. You can see that I have made no home page and do not run a blog. People who do self-promotion have always hopepages and blogs. Those who do not have, like me, kind of dislike being in publicity. I thought you would be delighted that there is a new result on your field.

As there were posts from me and Robert Coulter responded to them, I think you should not have put my identity to a spam list. That is certainly a questionable practise. If you wanted me to stop, why did you not simply write a post explaining your policy rules.

As for your comments, they are wrong.

Feffermain in his problem statement says clearly that the solutions are unique up to some finite T. They are not, so Fefferman does not state correctly but definitely incorrectly.

There could indeed have been a condition that the pressure be bounded. Instead, for the nonperiodic case, Fefferman explicitely writes that no conditions are set to pressure. Thus, you agree that conditions must be set to pressure.

You state that the external force must be set first as a point function, not as a feedback control function. This is not stated in the problem statement. Thus, you agree with me that this a necessary condition to be demanded. You must alos agree that it is nowere given in the official problem statement. This is discussed in my paper in section 4.

Your last comment is clearly wrong. When the force is defined in the way the solutions become unique because of the time derivatives for the velocity vector are defined.

> If you do not allow me to make posts, do also not allow people like Robert Coulter to comment to my existing posts. Otherwise I cannot respond to incorrect claims against the paper.
>
> If you think my paper is incorrect, then submit a paper to EJDE explaining your arguments. They are incorrect and posting clearly incorrect arguments with the authority of a leading figure is highly questionable.



Jorma Jormakka · 29 August 2010

Robert,

Mathematically there is no such thing as apparent non-uniqueness that is not real non-uniqueness. Nor are there such things as pressure term which is actually a force, and force that is actually pressure. Mathematically, pressure is pressure, and force is force.

The solutions are non-unique because there is a family of solutions allowing a free function g(t), such that g(0)=g'(0)=0. Thus, the solutions are mathematically non-unique and not in some way apparently non-unique.

Because the space-periodic problem is posed on R^3 x [0,infty) there are no conditions that deny these solutions. If the space-periodic problem were posed on R^3/Z^3 x [0,infty), we could argue that the presented solution is not acceptable since the external force does not exist if it is defined using the velocity vector that does not exist in torus times time.

However, this would not help. Statement B is proven in exactly the same way as Statement D. Notice that no similar definition of a nicer space can be done in the non-periodic case. The same construction (the transform in the beginning of Section 3) can be made to any solution for zero force. Let us assume we have one such solution and it is smooth in the whole space-time. We do the transform to this solution and get a new solution with finite-time blow-up. Even if we try to demand that in acceptable solutions the pressure must have finite energy, the new solution that has been obtained by the transform is in this case a perfectly valid solution in R^3 x [0,infty), i.e., it exists and thus the external force also exists. The new solution does not satisfy the conditions that we imposed, so it is a counterexample: for those initial values and external force there does not exist a smooth solution satisfying what we demanded since the unique solution that in this case exists, does not fill the demands we posed.

Posing more conditions is not a way to exclude counterexamples. In some way we must exclude the existence of the feedback force. In the space-periodic case this was possible by posing the problem in a torus, in the non-periodic case it is not possible.

The only way to correct the Clay problem setting is to impose periodicity of pressure in the space-periodic case, growth conditions to pressure in the nonperidic case (finite energy etc.), and in both cases to exclude feedback forces as external forces.

The motivation why should anybody want to exclude feedback forces and the mechanism how to do it so that feedback forces cannot in some way be smuggled in, requires careful thinking. Basically, we would like to allow any external forces in NSE, also feedback forces. They appear in many technical applications, like a propulsion system for a ship is feedback control by the captain. There is no good reason to exclude these forces. If these forces are

excluded, it is difficult to think we are talking any more of the original problem where the external force was general. The restricted problem without feedback forces would be more limited in applications.

Notice also that the uniqueness of solutions in this case was strongly believed in the field of PDEs, and apparently proved by a theorem e.g. in Temam. It is wrong in this case. Remember also that people in PDEs like to call people from other fields as cranks and crack-pots, as is shown by the discussion in reddit.com. These people certainly need a small lesson.

 **Robert Coulter** · 30 August 2010

Per Professor Tao's instruction, this is my last post concerning this paper:

The right side of the NS equations are the "sum of the forces". The pressure gradient term is one of these forces along with the viscous drag term and external force (if present). If the total kinetic energy of the system increases over any delta t then the external force is present. I cannot see anyway around this. Per my interpretation of the problem, If the external force is present then it must comply with the Clay problem's bounds on the force (it must be finite).

It seems like the main thrust of this strategy is to somehow get this external force under the umbrella of the pressure term, maybe because Fefferman simply forgot to forbid it. Maybe he also forgot to forbid converting the fluid mass to energy per E=mc2.

It seems like the strategy here is to construct a trivial blowup solution and then look for weasel words in the Clay NS problem statement to allow it. This certainly doesn't advance or enlighten anyone on the mysteries of the NS problem.

Unified Discussion There is no god · 6 September 2010

[…] 2) The insolvable paradoxes contained in the Navier Stokes equations: https://terrytao.wordpress.com/2007/0…#comment-35308 (This arises from an assumption of non-discreteness. Noones ever simulated or worked with a truely […]

 **Thierry** · 27 September 2010

Professor Tao

As you mentionned the molecular limit for the physical NS problem, you might be interested by the fact that there are papers published on molecular simulations of the Rayleigh Taylor instability at scales < μm.
The results show that the molecular simulation explains the phenomenon better than NS which begins to break down at these scales.
It is not surprising to find such a result given that the NSE were established for a continuum.
When this fundamental hypothesis breaks down, e.g when the spatial scales are such that it appears that the fluid is discrete rathen than continuous, the NSE must be replaced by another theory which deals with discrete systems.

This suggests by analogy with the infinite potential of pointlike particles that when a mathematical description of a physical theory (here NSE) is extrapolated to a domain where it is physically invalid, the mathematical results are pathological too. This pathology manifests itself generally by an appearance of singularities.
I am well aware that this is no mathematical demonstration but I am convinced that the NSE

always blow up at small scales because that is the clearest signal the Nature can send to tell us that something is wrong with the theory at these scales.

I have also a question. Do you believe that a stochastic interpretation of NSE could be possible by proving some kind of generalised ergodic theorem?



Terence Tao · 27 September 2010

It is true that physics no longer follows the Navier-Stokes equations at microscopic scales, but this does not seem to shed much light either way on the regularity of the Navier-Stokes equations, other than to say that physical considerations cannot be used to justify regularity. For instance, in a two-dimensional universe one would encounter the same physical breakdown of Navier-Stokes at microscopic scales, yet the Navier-Stokes equations are known to enjoy global regularity in two dimensions.

The general question of rigorously deriving macroscopic equations of motion for continuous media from the microscopic equations of motion of N-body systems is an excellent one in general, and one for which we still have very few actually rigorous results (as you suggest, one often has to make a lot of unjustified assumptions about "mixing" before one can derive a macroscopic model, lest one run into Maxwell's demon or some other similar obstruction). Typically, though, one has had to first understand the regularity theory of the macroscopic equations quite well before one could ensure convergence of the microscopic equations to the macroscopic model, so this task may be even harder than the global regularity problem (unless one worked locally in time, of course). But it is perhaps conceivable that Navier-Stokes with a stochastic noise component could be viewed, conjecturally at least, as the limit of some stochastic particle model.



Robert Coulter · 28 September 2010

Based on Professor Tao's suggestion at the opening of this blog, the following is my current opinion about the implications of a Navier Stokes resolution on conduit flow.

Note: In this post "perfect fluid" is a fluid that has no molecular bumpiness (i.e. a perfect continuum), but may have any viscosity.

Dimensionless quantities in fluid flow (I will speak only in terms of flow in conduits since this is what I know, not airplane wings, e.g.) are the Reynolds number, the friction factor (Darcy or Fanning), and relative roughness. The relationship of these three quantities is depicted very well in the Moody diagram – see wiki link: http://www.mathworks.com/matlabcentral/fx_files/7747/1/moody.png (For those that prefer equations, see Colebrook - White, link: http://en.wikipedia.org/wiki/Darcy_friction_factor_formulae
The relative roughness (on the right side of the diagram) is the degree of roughness on the inside wall of the conduit relative to the conduit diameter. The Reynolds number (at the bottom) is a measure of the degree of mixing/turbulence occurring inside the conduit. The friction factor (on the left) indicates the conduit's resistance to flow and is usually the key parameter engineer's desire in using the diagram. Upon examining this diagram, one minor peculiarity is that the velocity is embedded twice — in the
Reynold's number and the friction factor. This makes for some minor hassles in that the velocity cannot be solved for explicitly from knowing the other quantities. An iteration

process is needed which is can be started by guessing
the friction factor, solving for the Reynold's number, and then using the diagram to fetch a new friction factor and repeating the process until there is convergence at the correct point on the diagram.
The most striking oddity in the diagram, and one that has fascinated me for many years, is that even though it is a log/log plot of two carefully chosen dimensionless quantities, the curves are still highly unlinear. Also, at Reynolds number, 2000 – 10000, there is a huge undefined gap between where laminar flow breakdown and transition flow begins. This is called the "critical region". Another important characteristic is that the friction factors in the laminar region only depend on the Reynold's number. The friction factors in the fully turbulent region only depend on the relative roughness.

Note that the Moody diagram / Colebrook equation are based on our experience with REAL fluids and REAL conduits. Real fluids have "bumpiness" at the molecular level, and there are no perfectly smooth conduits (so the bottom curve may not represent a perfectly smooth conduit).
What would the Moody diagram of a perfectly smooth fluid look like? Would the perfect conduit line move? The resolution of the NS problem may help in this regards.

In summary the main oddities, IMO, in Moody's list are:

1. Turbulent curves are highly unlinear for decreasing Re number.
2. Discontinuity (of friction factor) between laminar and turbulent regions. (As far as I know there is no known mathematical model for this region.)
3. Position and shape of the smooth conduit line (for perfect pipe/ perfect smooth fluid).

In regards to #1, the Colebrook equation illustrates the nonlinearity of the friction factor in the turbulent region. The friction factor is implicit in the Colebrook equation and must be solved by iterative methods.
.
The major oddity is #2 (above). This is referred to as the "critical" region where laminar flow breaks down. At this point, there seems a complete absence of models/understanding. Engineers are instructed that this flow may be laminar or turbulent, but there is no guide for any given situation. Some versions of Moody's diagram show the curves in this region. I think this is done only to provide a "number" as most engineers need one. The Colebrook equation appears to breakdown here (does not converge). Of course, it was not meant to represent this region. Does the breakdown of the Colebrook equation offer any insight here? Will resolving NS in either direction help here? I believe so.

The start of this "critical" region is generally understood to be where laminar flow is no longer "guaranteed". It is generally accepted as a Reynolds number of 2000. Below this value, laminar flow will be maintained regardless of any protrusions or interruptions to the laminar flow regime. The flow "dampens out" the interruptions. This is generally attributed to the viscosity of the fluid. Beyond Re 2000, there is the first onset of instability in the flow. An interruption may send at least some portion of the fluid into an "oscillatory" or "sinusoidal" state.

At this point it may be useful to define laminar and turbulent flow. To engineers, the definitions are based on how the friction factors are calculated. Laminar flow IS when

the laminar flow equation for the friction factor is applicable. Turbulent flow IS when the Colebrook equation is applicable. These definitions; however, are not very descriptive of the fluid flow behavior. I will try to make some attempt at a descriptive definition as follows:

For purposes of these descriptive definitions imagine fluid flowing in a circular conduit and an arbitrary radial line from the center of the fluid flow to the inside wall of the conduit. We assume that the flow is at steady-state and sufficiently downstream of any "macro" disturbances (Fluid is given time to recover.):

1. Laminar flow: There are no flow reversals along the radial line.

2. Turbulent flow: There are flow reversals along the radial line. By extension, one can suggest that more turbulence corresponds to a greater degree/measure/magnitude of flow reversals along the radial line.

3. Critical Region: We cannot predict with sufficiently accuracy the degree (if any) of flow reversals that occur along the radial line for a given situation.

Another key point about the Moody diagram needs to be made. The curve at the very bottom is for an ideal perfectly smooth conduit. This is a conduit with, at least in theory, no protrusions (disturbances) on the inside wall. Of course, perfectly smooth conduits do not exist. Also, there are no fluids that
are perfectly "smooth" in the sense that they do not depict a perfect continuum because of molecular interaction at the very fine scales (I will call these perfectly smooth fluids "perfect' from here out). If a perfectly smooth conduit and a smooth fluid existed, where would this curve be?
Would it be near this one or further down? Would it be an extension of the laminar line with no discontinuity? This will discuss further below. I believe that a resolution to the NS problem with have the greatest impact on the position of this curve.

Warning: From here on out, there will be my opinion and conjecture. Do not accept as fact!

Upon learning about the NS Clay prize problem, I was very skeptical that any resolution to any supposed mathematical "difficulties" with the NS equations would have any impact on the practical matters of fluid flow analysis and engineering. Part of this belief, or maybe bias at the time, is that theoretical
models have little applicability to fluid flow with the exception of the laminar regime. The more I looked into it , the more I came to believe that the mathematicians may have something to contribute here. Part of my conversion
is based on:

1. NS appears to be a very good model within the context of its assumptions. More to the point, even the engineering models (Colebrook, Darcy, etc.) above make assumptions. For example,a Newtonian fluid is assumed in both cases.

2. I now believe that the NS model is sufficient that regardless of the eventual outcome of the resolution of the NS singularities (Singularities exist – positive or Singularities – negative), I believe that something will be learned in either case that will be applicable to our understanding of fluid behavior.

I will venture to how a resolution of the NS equations may affect our understanding of real fluid flow. It is likely that a resolution in either direction will significantly improve the ability to model perfect and real fluids. The bottom line questions is what a perfect fluid (no molecular bumpiness) would do in perfect conduit (inside wall perfectly smooth) after moving past a disturbance for a sufficient length. The following, I suggest, is that in NS positive singularity case is turbulence occurs. In the NS singularity negative, case it does not.

Case #1: NS develop singularities. That is, the flow (velocity) or its derivative are not defined at some point. IMO, this would be the most exciting development. At this point there could be defined a new regime of flow. Maybe it could be called "hard" turbulence. Hard turbulence, I conjecture, would correspond to the velocity achieving infinity along the radial line or experiencing a discontinuity or step. At this point, a number of questions will come up:

1. Where along the Moody diagram will hard turbulence appear? in the critical region? At the beginning of turbulence? Or much further to the right in what is currently called full turbulence.

2. Depending on the type of singularity, will Lagrangian particle trajectories be conserved?

3. If Lagrangian particle trajectories are not preserved, what happens?. Does the particle split? Or maybe does it become like quantum mechanics where it will not be possible to define a particular particle trajectory with all certainty — just the probability of a specific path.

4. Will there be a new quantity, maybe call it gamma, that indicates the rate of singularity formation per unit volume of flow. Will this in time replace the Reynolds number as the de facto measure of turbulence. Will the gamma value depend on Reynold's number, relative roughness or both?

5. Once disturbed, always disturbed (critical velocity, shear rate, etc.) if flow precedes into a perfectly smooth conduit (the bottom curve)?
No extension of the laminar to high Re# for perfectly smooth pipes. The discontinuity stays in place for the perfectly smooth pipe / perfect fluild. Once turbulence starts in can be maintained by holding the velocity constant even in perfectly smooth pipes.

6. Turbulence can be caused by mathematical fact based on Re # components (visc,vel,dens,char. leng) in addition to obtrusions/disturbance (Pipe roughness, molecular "bumpiness"). Perfect fluids can develop turbulence.

7. Critical region exist for perfect conduits, perfect fluids. It can be defined for certain Re# and roughness per NS.

8. Whole nature of fluids changes. Speculation about behavior of perfect fluids in stars?

9. Since perfect fluids can develop hard turbulence, how close are they to real fluids in behavior. Will the perfectly smooth line move any, or is it already fairly close to the real world "smooth" conduit and real world fluids?

Case #2: NS does not develop singularities.

From my readings, this seems to be what most mathematicians want. Unfortunately, IMO, it would have little impact outside the mathematical community. Better algorithms for CFD

may come available, but that
is about it.

Concerning the Moody diagram above:

1. Turbulent flow will be interpreted as simply highly involved, complex, or disturbed laminar flow. With “soft” turbulence only maintained by roughness in the conduit (for a perfect fluid).

2. If disturbed to any degree, the turbulent flow will recover if all disturbance removed in a perfectly smooth pipe with perfect fluid (velocity held constant). Laminar line extends to high Re#
for perfectly smooth pipes and perfect fluids. Note: May take a very long length of perfect pipe to dampen a a perfect fluid if velocity if Re# is very high. Nether the less, this will occur in a finite length for any nonzero viscosity. Perfect fluid in perfect pipe for inviscid may oscillate indefinitely (after a disturbance). We assume no slip at wall in all cases.

3. Turbulence will only be a macro phenomenon. Basic nature of fluids will not change much.

4. The Moody diagram for a perfect fluid would show that the roughness curves spread out more with the curves for smooth pipes approaching the laminar line at the bottom.

5. Critical region does not exist for perfect conduits, perfect fluids. Critical region still holds for certain ratio of Re# to pipe roughness. Point where obstructions are likely to create oscillatory flow.



Lizhi Du · 25 October 2010

Tao is the only smart guy in this world! Only you can solve Hard Problems, other people never can! So, reject him even without any reason!



Robert Coulter · 31 December 2010

Fefferman statement A, when applied to Euler equations, appears true if one is allowed to arbitrarily specified the initial time derivative of the velocity field. The simple case is to set all the initial velocity time derivatives equal to zero. This creates the steady-state case for all time. The pressure field is solved from the remaining convective terms. This would appear to be true for any velocity field meeting condition (4).

I assume, though, that the time derivatives cannot be arbitrarily specified. One must show that statement A holds irregardless of the value of the initial velocity time derivatives.

In the NS case, the steady state condition cannot be maintained without an external force to exactly counteract the drag term on the right.



rbcoulter · 24 May 2012

I now have reluctance on the above post. The pressure field can be recovered from any arbitrarily chosen velocity field through the pressure Poisson equation. However, the pressure cannot be multivalued at a point which, I believe, could happen from the integration along different spatial directions. The only way to “force” the pressure to be single valued at every point is to allow the time derivatives to be nonzero. Because of this, I suspect must Euler flow fields are unstable. If any one else knows otherwise, please let me know.

 **Hafid** · 29 January 2011

Dear Terence,
If we have
1). An equation (E) wich the existence of solution is satisfied but we can't prove the uniqueness.
2). A perturbed equation (Ep) of (E) wich the existence and the uniqueness of solution are satisfied.
3).The solutions (Sp) of (Ep) converge strongly to the solutions (S) of (E).
Question:
Are that (2) and (3) implies the uniqueness of solution in (E)?

 **Terence Tao** · 29 January 2011

Unfortunately, no; there can exist "ghost solutions" that solve the original problem, but are not limits of the perturbed problem.

For instance, consider the ODE $\dot{u} = 2|u|^{1/2}$ with initial data $u(0) = 0$. It has at least two solutions, $u(t) = 0$ and $u(t) = t^2$. But if one considers the smoothed ODE $\dot{u} = 2(\varepsilon + |u|)^{1/2}$ for any $\varepsilon > 0$ with the same initial condition $u(0) = 0$, there is only one solution (the trivial zero solution), by Gronwall's inequality. Thus we see that uniqueness is lost in the limit due to the presence of the "ghost solution" $u(t) = t^2$.

 **Hafid** · 2 February 2011

Under what conditions do I get the uniqueness of solution in (E)?

 **Santiago** · 6 February 2011

Prof. Tao,

What is your opinion on Dr. Amador Muriel's "An Exact Solution of the 3D Navier-Stokes Equation"

 **Robert Coulter** · 7 February 2011

It appears that the main argument of the Muriel paper is that some
other fluid flow model (referred to as "TE" in the paper), which has its roots
from various non-continuum theories (kinetic theory, Liouvilles's equation, etc.), exactly solves the NSEs. The only supposed "proof" of this is that
this "TE" model's pressure calculation, from a certain flow condition,
is "the same" as the pressure that would be calculated from the NSEs.

It appears that the author does not understand the Clay prize problem.
Clay is not looking for models comparable or superior to NSs. They are looking for inconsistencies within NS itself.

IMO, the author's only intent here is to some how validate a
model by comparing it to NS and then argue that NS is solved, or that
NS is lacking..
The logic seems to go around in a circle.

 **Gandhi Viswanathan** · 9 February 2011

Dear Professor Tao,

I understand the need for a (conserved or monotone)
quantity which is subcritical or critical (under scaling)
to show global-in-time regularity. However, what
about for proving blow-up? Do you really need a
subcritical or critical quantity ?

I would guess that the answer is yes, but I am unable
to see this clearly. For instance: imagine we find some
time dependent functional of the velocity W(t,u)
(e.g., some Sobolev norm) which blows up at some
finite time T* (where of course the finite time singularit
obeys the BKM criteria for blow ups and so forth).
Then, even if this quantity W is supercritical, are we
still not able to show blow up?

SO, it seems to me that to show blow up, we do not
require subcritical or critical quantities. Is there something wrong with this reasoning?

If you could give a (even if brief y/n) reply, I would
very much appreciate it.

Thank you.

 **Terence Tao** · 9 February 2011

Interesting question! It is in principle conceivable that one could establish blowup via a monotonicity formula that gives lower bounds for the growth of a supercritical quantity. However, in practice, in order to establish such a monotonicity formula, one often has to obtain upper bounds on the growth of error terms that are just as supercritical as the main quantity in the formula, and this is unlikely to be possible for the reasons discussed in the main post. (This is particularly the case given that we know that there are many solutions to Navier-Stokes that do _not_ blow up, and so any such monotonicity formula must somehow use special properties of the initial data, and then must work hard to show that these properties are somehow preserved by the evolution, and then we are back to the problem in the main post.)

From a more heuristic perspective, the blowup mechanism (if such exists) is likely to come from the delicate nonlinear behaviour of the high frequency modes, and a supercritical quantity (which is dominated by low frequency behaviour) is unlikely to be capable of capturing this behaviour properly. While one cannot rule out the possibility of an algebraic miracle that somehow makes the supercritical quantity monotone and forcing blowup, it is still very strange that a property of the low modes is responsible for making the high modes blow up, and so I would doubt that this possibility actually occurs.

 **Gandhi Viswanathan** · 9 February 2011

Dear Professor Tao,

Thank you for the prompt reply, which goes straight to the heart of the matter.

David Purvance · 23 February 2011

Correct me if I am wrong, but it appears that a Fourier-Laplace analysis shows that 3D solutions only have to be integrable, not necessarily smooth: http://purvanced.wordpress.com

Gandhi Viswanathan · 23 February 2011

Dear David,

Your quantity J has a u inside it. This masks a nonlinearity in u.

The crucial point is this: the nonlinearity can excite higher frequency modes (because the quadratic nonlinearity becomes a convolution in the Fourier domain). This exciting of higher frequency modes may go on uncontrolled (the so-called direct cascade). Since the time scales become smaller for the higher frequency modes, what we cannot rule out is the Fourier transform of the velocity field acquiring a power law tail in finite time. Such a power law tail automatically guarantees the breakdown of smoothness, because all smooth functions have rapidly decaying Fourier transforms. If you go back to your own results, you will see that you cannot rule out the possibility of this type of finite time singularity.

Moreover, the addition of dissipation (via Laplacians) is unable to control the cascade unless a power 5/4 of a Laplacian is used. Professor Tao has improved this 5/4 result in the margins, via an additional logarithmic factor. Excepting such marginal improvments, the 5/4 result, originally due to Lions, is still the best result to date. (I have reviewed in RSTA recently this 5/4 result of Lions: doi: 10.1098/rsta.2010.0257.)

It is perhaps due to the nature of these obstructions that Professor Fefferman has suggested the investigation of weak solutions. Another intriguing problem is that there appears to exist a time T* such that no matter how small the (positive) viscosity is made, yet half the energy is dissipated away by this time T*. This result is really amazing and suggests new questions: How can half the energy be dissipated away independently of the size of the viscosity? We need to understand how so much energy can be dissipated independently of the viscosity. There is something interesting going on…

> **David Purvance** · 23 February 2011
>
> Dear Gandhi,
>
> Rather than here, perhaps Prof. Tao would prefer that we carry on this conversation at my blog, but…
>
> My solution fully recognizes that the Navier-Stokes equation creates a nonlinear eigenproblem, precisely for the reason you cite. Many nonlinear eigenproblems are documented in the literature, and, in the case at hand it only means that Navier-Stokes' eigenvectors have a relationship more complicated than traditional linear eigenproblems.
>
> **Anonymous** · 16 September 2013
>
> Dear Gandhi and Terry,
> In this paper mentioned above
> http://rsta.royalsocietypublishing.org/content/369/1935/359.abstract

The L^1 estimate on frequency side is interesting to me. But I don't see how the statement right after (4.10) works. Can any of you give a bit more clarification?

Thanks a lot!

bk



Gandhi · 16 September 2013

Dear Anonymous, thank you for pointing this out. You are right. TMV and I must see how much is salvageable. There is a gap and we will see if it can be fixed. As soon as we figure it out, we will make public any correction and take any other necessary steps.



Robert Coulter · 23 February 2011

"How can half the energy be dissipated away independently of the size of the viscosity?"
Gandhi Viswanathan

Isn't this really more about why the "mean" Laplacian of the velocity increases as the viscosity decreases. Isn't this intuitive from what we know of turbulence?
The system adjusts to lower viscosity by decreasing the length of the mixing scale.

If you meant something else, please let me know.



Robert Coulter · 24 February 2011

It has occurred to me that it may be possible to construct a trivial blowup solution that stays within the constraints of Fefferman conditions (4),(5) and (7). Although the force appears to be sufficiently constrained the "power draw" from the surroundings appears not to be. The local power draw density can be expressed as:

PDD = Fe * v

where Fe is the applied force and v is the local velocity. As long as the external force is present. the power draw can be made to be large as needed
by increasing the local velocity. One may imagine that a singularity could possibly be created this way — maybe some type of energy focusing scenario. The singularity would have a local infinite power draw from the surroundings (not the flow field). The total bounded energy, condition (7), could be maintained, if need be, by applying a modest negative force to the rest of the flow field.

If I am missing something here, please let me know.

Existence And Smoothness of The Navier-Stokes Equation « Hezhigang's Blog · 11 March 2011

[...] Why global regularity for Navier-Stokes is hard [...]



JUSTICUS · 31 March 2011

Can anyone comment on the total kinetic energy, what is its relation with regularity?



Robert Coulter · 31 March 2011

Apparently, the finiteness of the initial kinetic energy is enough to show that smooth initial conditions stay smooth for all time in 2D. In 3D, this is not sufficient (See Professor Tao's discussion above) because of the energy scaling being different.

 **JUSTICUS** · 5 April 2011

But basically if the solution blows up, the total kinetic energy must increase as function of time (normally one has that dE/dt=-vZ, where Z is the enstrophy and v is viscosity?

 **Robert Coulter** · 5 April 2011

No. In fact the total KE must decrease monotonically (assuming no external force and a nonzero initial velocity field) for NS.. This is because the viscosity term is always removing energy from the system, and no new energy is being added.

This does not exclude the energy from concentrating in small regions of the flow field (at the expense of reducing it in other regions). Note that the total energy expression is an integral over the entire field and therefore cannot in itself exclude a point (velocity) having an infinite value.

 **Hafid** · 23 April 2011

1) . When u(0,x)=0 , is the solution u=0 unique ?
2) . Has u(t ,x)=0 the conditions which are claimed by the Clay-Foundation?
Thank you .

 **Robert Coulter** · 23 April 2011

No. There are many exactly solvable solutions to the Navier Stokes Equations.
See “Vorticity and Incompressible Flow” by Majda , Bertozzi for many examples.
Also see the Taylor-Green Vortex, http://en.wikipedia.org/wiki/Taylor–Green_vortex. The zero solution is the most trivial of them all.

The Clay Foundation is looking for either a blow up solution that starts from a non-blowup state, OR they are looking for a proof that the Navier Stokes equations do not “blow up” for reasonable starting conditions. (Note that the Clay problem defines what they (Fefferman) consider to be reasonable starting conditions..)

 **Hafid** · 23 April 2011

Thank you for your reply.
What is your opinion on this paper?
http://arxiv.org/abs/1104.3255

 **Terence Tao** · 23 April 2011

The deduction after (3.48) is incorrect. The supercritical expression $\int_0^t \|A_l u_\varepsilon\|^2$ appearing on the right-hand side in (3.48) has not been demonstrated to be bounded uniformly in ε, so one has not demonstrated that $\|\xi\|$ converges to zero as $\varepsilon \to 0$. As discussed in the main blog post, control of a supercritical expression is in fact the principal difficulty in using standard techniques (such as energy estimates) to control Navier-Stokes solutions in three and higher dimensions.

Also: it is against the policy of this blog to use the comment space to promote one’s own research: see https://terrytao.wordpress.com/about/
Such comments will be subject to deletion.

Hafid · 23 April 2011

Thank you for your reply.
Theorem 3.1. means that u saisfies this condition.
All my respect Professor Tao.
I thought the site's goal is to enrich the ideas.

> **Terence Tao** · 23 April 2011
>
> Theorem 3.1 shows that each u_ε lies in $L^2 V_l$ for each l and ε, but does not bound the norm of u_ε uniformly in ε, which is what is needed for the argument after (3.48). Instead, it provides a bound which depends on ε in an unspecified manner (and, for supercritical l, the bound will blow up as ε goes to zero).
>
> Your approach is an instance of Attempt 1 of Strategy 3 as listed in the blog post (Using weaker or approximate notions of solution), and as mentioned in the discussion, the error commonly made in this approach is to mistake qualitative control (in this case, the assertion that u_ε lies in $L_t^2 V_l$ for each ε) for quantitative control (in this case, the assertion that the $L_t^2 V_l$ norms of u_ε are bounded uniformly in ε).
>
> As stated in the blog policy at https://terrytao.wordpress.com/about/ , the comments of this blog are not to be used for promotion of one's own research.

Hafid · 23 April 2011

Thank you for your useful comments, I will send you the second version.

> **Anonymous** · 26 April 2011
>
> There are at least 2 typos in the abstract of your paper. How seriously do you expect to be taken?
>
> Life is difficult for everybody, I sympathize if you really suffer from the situation, but try to reconsider your goals in life, reality is the best you will have. You will (most probably) gain nothing from taking time from Terence without putting serious work to make such interactions efficient.
>
> Good luck.

Hafid · 28 April 2011

Dear Professor Tao;
As I promised here is the second version v2:
http://hal.archives-ouvertes.fr/hal-00586485

> **Gandhi Viswanathan** · 28 April 2011
>
> Dear Hafid,
>
> If you add hyperviscosity, it becomes a completely different problem. One can then show that finite time singularities are prohibited.
>
> Lions showed more than 40 years ago that hyperviscosity leads to global regularity.
>
> This happens precisely due to issue of supercriticality and subcriticality discussed by Professor Tao. With the 5/4 Laplacian of Lion, the energy is critical.

> The challenge is to study the standard parabolic Navier-Stokes system, for which the energy is supercritical.
>
> Lion's 5/4 result is the best to date (excepting marginal logarithmic improvements to the 5/4 result, due to Professor Tao).
>
> Find us a critical or subcritical conserved or monotone globally controlled quantity for the standard parabolic N-S system and you will catch people's attention!

 **Hafid** · 28 April 2011

Dear Gandhi Viswanathan,
Professor J. L. Lions did not prove the strong convergence.
The solution is in Roma.

>  **timur** · 28 April 2011
>
> In your cited work [9] it appears that you mistook subsequential convergence as sequential. When you have a compact sequence you can only deduce that a subsequence converges, so in Theorem 3.3 of the current paper, which is recalled from [9], you cannot say the whole sequence u_epsilon converges to a weak solution of the NSE.

 **Hafid** · 28 April 2011

Thank you for your interest in my paper. But it is also true for a sequence!

 **Hafid** · 29 April 2011

Standard methods from PDE appear inadequate to settle the problem. Instead, we probably need some deep, new ideas. Charles L. Fefferman.
http://arxiv.org/abs/1104.3255

>  **Terence Tao** · 29 April 2011
>
> The inequality (3.17) is incorrect. (3.11) bounds the (supercritical) L^2 norm of u_ε, but in (3.16) it is the (subcritical) L^∞ norm of u_ε that needs to be bounded instead.
>
> I strongly recommend that you learn the distinction between subcritical, critical, and supercritical quantities, as discussed in my blog post above. Without this understanding, any attempt to resolve the Navier-Stokes global regularity problem is certain to fail (and the error can usually be located by finding the first step in the argument in which a uniform bound on a critical or subcritical quantity is claimed).
>
> To be blunt, your paper does not contain the deep, new ideas that Fefferman was alluding to. Please do not post here again.

 **Robert Coulter** · 29 April 2011

This may be a trivial question for the mathematicians participating on this blog:

Much reference is made to scale invariance and its role in possible blowup of NS. When I look at Professor Tao's explanation of this property, is the possible blowup occurring on constant re-scalings of the original solution moving backward in time? In other words, if one has a solution to time, T, and notices something odd at T and then rescales, is the oddity magnified but at a previous time because the rescaled solution (say *) has the oddity

occurring at T* = T/lambda^2 which for lambda > 1 seems to be previous to the original solution?



Terence Tao · 29 April 2011

It depends to some extent on how one normalises things, but if a solution $u(t,x)$ does something bad (e.g. blows up) at time $t = T$, then the rescaled solution $u^{(\lambda)}(t^*, x^*) := \frac{1}{\lambda} u(\frac{t^*}{\lambda^2}, \frac{x^*}{\lambda})$ blows up at time $t^* = T\lambda^2$, not $t^* = T/\lambda^2$. (As mentioned in the blog post, for $\lambda > 1$, one should view $u^{(\lambda)}$ as a "magnification" of u.)



Umesh · 5 May 2011

"Thus, for instance, if a unit-scale solution does something funny at time 1, then the rescaled fine-scale solution will exhibit something similarly funny at spatial scales 1/ \lambda and at time 1/\lambda^2.."

Excuse my stupidity Prof. Tao, but you just corrected this statement in your previous post, and now my initial confusion has cleared, but can you please also clarify, why do you say in the above statement "...rescaled fine-scale solution will exhibit something similarly funny..." does it actually mean "..rescaled coarse-scale solution..". I know I am muddling up something very basic here, but a little time from you to clarify this will help me greatly. Thank you for your time Sir.



Terence Tao · 5 May 2011

In that particular context, the unit scale solution is $u^{(\lambda)}$ and the fine scale solution is u (or, if one wishes, one could call the unit scale solution u and the fine scale solution $u^{(1/\lambda)}$; they are both rescalings of each other, one by λ and one by $1/\lambda$.)



Umesh · 6 May 2011

Thank you Prof. Tao. I have another question regarding the Navier-Stokes equation problem. What exactly is meant by 'arbitrarily large initial data u_0 ?



Terence Tao · 13 May 2011

This is not a precise technical term, but in the current discussion, this basically refers to data u_0 for which is not know to be bounded in some function space norm by a fixed quantity.



Hafid · 29 April 2011

I took my students that you are a good example but you lack the education.
Tchao



Hafid · 29 April 2011

You are the best.



Robert Coulter · 30 April 2011

But if u is finite, say u(T,x) =a, then $u^{(\lambda)} = a/\lambda$ at $T^* = T\lambda^2$?



Robert Coulter · 30 April 2011

It seems to me all the scale invariance implies is that later solutions can be "mapped" from earlier solutions. However, the later solutions have magnitudes reduce by the factor lamba and are further away from the origin. Earlier mappings have magnitudes increased by the

factor lambda and are closer to the origin. This seems to make sense when imagining a decaying vortex (e.g.). Anyway, just having trouble visualizing how scale invariance may depict blowup later in time.



Gandhi Viswanathan · 30 April 2011

Dear Robert,

When you refer above to 'mapped' solutions to earlier times, etc. we must take care about whether we are talking about mapping to the SAME solution or a NEW solution.

Scale invariance can refer to a single (same) solution, in which case the solution is said to be self-affine. Initially smooth self-affine blow-up solutions have been ruled out for the 3-D Navier-Stokes and Euler equations (I forgot the refs.). On the other hand, scale invariance can refer to the PDEs themselves. In other words, given a solution, we can create a whole family of (distinct) rescaled solutions. The latter is the type of scale invariance which is relevant to the Navier-Stokes system. (Notice the superscript in $u^{(\lambda)}$.)

If there is supercritical scale invariance AND nonlinearity, very interesting things can happen in principle, for the following reason. In some short period of time T, the quadratic nonlinearity can excite higher frequency Fourier spatial modes. So the Fourier transform can become 'wider' or 'fatter' (due to convolution). If this were the only effect, it would not be a big problem. But the supercritical scale invariance allows the same type of process to occur again, but now at a much faster NEW time scale. This can be repeated over and over, faster and faster, such that in finite time the Fourier transform may acquire a power law tail, i.e. it may become genuinely 'fat tailed' and no longer decay rapidly. If this happens, then a singularity has developed, because the Fourier transform of a smooth solution must decay more rapidly than any power law.

This is more or less my understanding of why Professor Tao states in his original post that the obstruction is supercriticality.

Indeed, as expected, when the Navier-Stokes equations are perturbed to make them critical (or subcritical), the 'cascade' process is disrupted because as the energy cascades to the higher spatial frequency modes, the time scales cannot not get faster and faster. So it takes an infinitely long time for the Fourier transform to sprout a power law tail. In my understanding, this fact alone allows one to prove global-in-time regularity for these perturbed problems (such as hyperviscosity models).

About 'visualizing' scale invariance in blow-ups, etc.: one suggestion is to understand fractals, because they are very visual and and fluid turbulence has much in common with fractals and power laws, e.g, Kolmogorov's 5/3 power law spectrum. (There is actually a somewhat close relationship between scale invariance, fractals and power laws.)



Robert Coulter · 1 May 2011

Gandhi –

Thanks for your detailed and descriptive explanation. I understand the point now.

Leray had conjectured a (backward) self-similar singular solution. This was disproved by Necas-Ruzicka-Sverak (1996).

 **Robert Coulter** · 3 May 2011

The meanings of supercritical, critical and sub-critical appear to have different meanings to mathematicians. Am I interpreting this correctly in saying that the scaling of the velocity (as it relates to NS) is sub-critical (1/$latex\lambda$)? However, the energy scaling is supercritical (($latex\lambda$). Is this because the spatial realm (after rescaling) increases by the cube of x (length) (and this effects the calculation of the total energy?)? If this is not correct, can someone provide the explicit expression showing why the energy scaling is supercritical?

 **gandhiviswanathan** · 5 May 2011

Dear Robert,

The terminology is explained in PDE books etc., for instance see this online wiki:

http://wiki.math.toronto.edu/DispersiveWiki/index.php/Critical

In the context of the Navier-Stokes system, we say that energy is supercritical because it is not suitable for bounding or controlling the fine scale behavior. In other words, given an initial condition with finite fixed energy E, the solution in time can evolve ever finer-scale structures, possibly even leading to a finite time singularity. Although energy is a conserved quantity, it becomes useless at infinitely small scales.

 **Robert Coulter** · 5 May 2011

Ghandi:

Thanks. That looks like an excellent link explaining these concepts.

Navier-Stokes 方程简介 « Fight with Infinity · 6 June 2011
[...] Tao Why global regularity for Navier-Stokes is hard [...]

 W Ethan Eagle · 5 July 2011

Hello,

This may appear to be stream of consciousness, but since this is a blog and not a referred journal perhaps I'm ok with that. I am a PhD student at Michigan in the area of experimental fluid mechanics. I stumbled across this blog, and wanted to add some thoughts focused on (my) questions about strategy 2, given that my thoughts are not yet fully developed. I study the effect of vorticity in high speed flows with the application of flow breakdown paramount to questions about pathology in the NS which may be avoided through engineering control. Therefore the idea that topological considerations a-la vorticity have some part yet to play in the unfolding drama of the NSE intrigues me.

Background
In particular I am interested in how development and competition between vortex stretching terms which exist as strain terms compete with viscous terms. Using a simple 1-D argument, a length scale is defined via this competing effect for a given initial condition (e.g. Burger's Equation)

In 2-D NS, I am aware of an oddity commonly referred to as the 'inverse energy cascade' which allows for energy containing structures to grow unboundedly in the fluid domain,

until they reach the bounds(physical or computational)... and yet formalized 'regularity' of these solutions apparently have been given. (I will admit that this was a TIL moment)

This is curious to me in a number of respects, the first of which is that vorticity production in a 2-D flow seems inherently bound to point sources and is irrational for any closed path that does not contain such point sources.

The 'topology' of the 3-D analog is much more exotic, but can be simply thought of as a closed path (bounded survace) contains a magnitude of 'circulation.' (and that a 2-D 'line' vortex which extends to infinity is unphysical) – I frequently like to make the connection to Physics via EandM that there are no magnetic monopoles.) I will forgo looking up the formal definition and procede colloquially (while crossing my fingers this doesn't ruin my argument).

Method(?)
Vorticity 'closure' would appear to me to be a strategy of type 2.

The curiosity of the second problem is the energy transfer mechanism which occurs in the NS equations is dependent on the gradient, and is (might be?) dominated by high gradient sources(Rapid strain part). Thus the vorticity is most likely to be contained in small (self-organizing?) regions in the flow. As the flows organize, then other mechanisms compete to dissipate and diffuse vorticity.

Results(?)
My essential conjecture is a production term which provides for an inverse energy cascade in some (small) spatial/temporal range near high gradients. (but perhaps such a conjecture is like Maxwell's demon and is entropy destroying?!?) In any case, away from the boundary (or away from production regions of high gradient) this threshold value reverses and dissipation and diffusion eventually matches(and excede) production.

My physical intuition of the problem suggests that at a boundary the flow behaves like its 2-D, and only after reaching some threshold does the flow recover its 3D entropy-increasing behavior.
(this seems proof positive via an entropy argument over similar large scales)

Conclusion(?)
Given an energy source (high gradient) can entropy generation be negative(integrally) in a small region so long as it is positive(integrally) in some larger region? – how would this manifest physically?

If so, would this explain the vulgarity of the NS by the attempts to find smaller and smaller regions over which entropy is still found to be positive (because of scale similarity?) when this is impossible due to the 3D region locally appearing 2D at the smaller scale – think 'earth is flat' – and the relatively non-isotropic distribution of vorticity? The only(?!) way to avoid such a 2D vs 3D effect would be to go to infinite reynolds number, but this would seem to violate the premise of starting with finite bounds.

Discussion(?)
Sorry for the confusion apparent in these thoughts. Any discussion which attempts to clarify them (if that is even possible) would be appreciated.

P.S. Do the problems with NSE formalism also occur for decaying flow in the absence of boundaries?

 **Robert Coulter** · 5 July 2011

"Given an energy source (high gradient) can entropy generation be negative(integrally) in a small region so long as it is positive(integrally) in some larger region? – how would this manifest physically?"

Yes. There is often a good bit of misunderstanding about entropy. Thermodynamic theory only states that system+surroundings must increase in entropy. The "system" may "choose" one or the other — which ever one causes the greatest increase in TOTAL entropy (system+surroundings). This is at the heart of why some chemical reactions are exothermic and others are endothermic.

 **Gandhi Viswanathan** · 20 July 2011

Dear Professor Tao,

I have been trying for a long time to get a deeper understanding of the importance of scale invariance and have a question about supercriticality in the simpler viscous 3-D Burger's equation:

$\frac{\partial u}{\partial t} + u \cdot \nabla u = \Delta u$

Here the viscosity parameter is set to 1 without loss of generality. The only difference compared to the 3-D Navier-Stokes equations is the lack of pressure, so there are no inverse Laplacians and no singular integrals, etc.

As far as I can tell, the scale-invariance argument you use for Navier-Stokes also applies to the above viscous 3-D Burger's equation. Yet, we know that the viscous 3-D Burger's equation has global-in-time regularity. This has been shown using a maximum principle for the kinetic energy density (an explanation is given in doi:10.1088/0951-7715/21/12/T02). Specifically, the 3D viscous Burger's equation has the same supercriticality issue, yet it is regular.

The only way that I see how to reconcile the two facts is to assume that there must be a critical or subcritical monotone or conserved quantity for the Burger's case which is absent for the NS case. Is this correct? And if so, is this monotone quantity the kinetic energy density u^2 ? And if so, for the NS case, is the nonlocal dependence of pressure on velocity the reason why the energy density is no longer the required monotone quantity? Finally, if all the above are true, then does this not mean that in addition to supercriticality, an equally important obstruction for the NS case is the nonlocal property of the kernel of the inverse Laplacian (in the pressure).

If you have time, I would greatly appreciate quick answers or hints.

Your blog is terrific.

-Gandhi

 **Terence Tao** · 20 July 2011

Yes, in the case of Burgers there is another monotone quantity, namely the supremum of the kinetic energy density (or equivalently, the L^∞ norm of u), which makes the problem subcritical in this case.

 **Gandhi Viswanathan** · 20 July 2011

Thanks!

 **Localisation and compactness properties of the Navier-Stokes global regularity problem « What's new** · 4 August 2011

[...] the lack of a controlled coercive quantity that is either subcritical or critical, as discussed in this post – because this norm is supercritical. Actually, one can guess at the existence of such a [...]

Why is Navier–Stokes existence and smoothness difficult to prove? - Quora · 22 August 2011

[...] prove? Ethan Eagle A much longer and in-depth look at this topic can be found here:https://terrytao.wordpress.com/20...This answer .Please specify the necessary improvements. Edit Link Text Show answer summary [...]

 **The problem with open science | The Great Antidote** · 12 September 2011

[...] is) as the crux of his 'open science is better science' conjecture. He also mentioned Terence Tao's blog, and his own personal use of sites like delicious and friendfeed. Then at some point, I realized [...]

 **David Brown** · 21 January 2012

The Navier-Stokes equations are presumably a maximal likelihood estimator for some set of equations that are fully compatible with the Standard Model of particle physics and general relativity theory. By looking at this vastly more complicated Markov branching process, mathematicians and theoretical physicists might stumble upon new and promising approaches to the Navier-Stokes equations.

 **tmr** · 21 March 2009

Probably that is a Koch-Tataru paper.

 **Student** · 22 March 2009

Dear tmr:

Thanks. That's kind of what I thought, but I was wondering if there are any others that can be recommended.

 **Anonymous** · 26 March 2009

Dear Terry,

Did you used to have notes on the Nash-Moser iteration scheme somewhere? I looked on your website and could not find it. Thanks!

 **Terence Tao** · 29 March 2009

Dear anonymous,

The Nash-Moser notes are at http://www.math.ucla.edu/%7Etao/preprints/Expository/nashmoser.dvi

Anonymous · 29 March 2009

Dear Prof. Tao,

Excuse me my question is not related to the NS equations but since I do not know under which post we should write our general questions, I am writing it here.

My question is that is it possible to have two dependent random variables such that one has binomial distribution and the other has normal distribution?

thanks

Student · 17 April 2009

I would really appreciate if you could explain why

"Riesz transforms are bounded in BMO, but not in L^oo"

a claim made by "Limiting Case of the Sobolev Inequality in BMO, with Application to the Euler Equations" by Kozono and Taniuchi.

Best regards,



Terence Tao · 17 April 2009

Dear Student,

If you test the Riesz transform R_j against the indicator function $1_{x_j>0}$, you will obtain a logarithmic divergence. (This is easiest to see in one dimension, i.e. applying the Hilbert transform to the Heaviside function.) So R_j is not bounded on L^∞. The bound on BMO is more complicated to show, and basically requires the use of Calderon-Zygmund theory; I would suggest Stein's "Harmonic analysis" as a reference.

anonymous · 18 April 2009

I have a question about strategy 2, discovering globally controlled quantities. Controlled quantities seem to be connected to invariance of the equations. For example, with the (linear) Schrodinger equation a phase rotation corresponds to the conserved quantity of mass, a space translation corresponds to momentum, etc. However, these invariances are precisely the defects to compactness of the linear Schrodinger operator, in the sense of the profile decomposition. In fact, one can think of the profile decomposition as the statement that the known invariance of the Schrodinger operator are the only (at least, noncompact) ones. It seems reasonable that we might be able to reverse this logic, and conclude that the known conserved quantities are the only ones. I don't know very much about Navier-Stokes, but would it be plausible to attempt to classify the conserved quantities in a similar manner?



Terence Tao · 18 April 2009

Dear anonymous,

For Hamiltonian equations (such as the Schrodinger equation), Noether's theorem provides a very close connection between conserved quantities and invariances of the

equation. But conservation laws aren't necessarily the only ways to get control, and in any event Navier-Stokes is not a Hamiltonian system (it is not time-reversible).

Another obvious source of controlled quantities are monotonicity formulas – a formula that demonstrates that some coercive quantity is non-increasing in time. For instance, energy is a monotone quantity in Navier-Stokes. But unlike the situation with conservation laws of Hamiltonian systems, where we have a nice characterisation, there is no known general way to systematically classify all monotone quantities associated with an equation such as Navier-Stokes. But if one could find more such monotone quantities (and if they had a favourable scaling), then this would be major progress on the Navier-Stokes problem. (The solution to the Poincare conjecture, incidentally, was only made possible by the discovery by Perelman of certain scale-invariant monotone quantities for Ricci flow; one could conceivably imagine a similar miracle leading to a breakthrough for Navier-Stokes, though given that this equation is much less geometric in nature than Ricci flow, such miracles are perhaps less likely.)



Student · 24 April 2009

Dear Professor Tao:

Thank you very much for your reply above. I have the follow-up questions.

I understand that this improvement below was made (Sorry for not using LaTeX; I'll retry if it's not acceptable):

From Beale-Kato-Majda (1984):

||grad u||_{L inf}
<=
C(1 + ||vort u||_{L inf}(1 + log^{+}||u||_{W^{s+1,p}) + ||vort u||_{L^2})

To Kozono, Taniuchi (2000):

For 1 < p n/p, with C(n, s, p)

||f||_{L inf}
<=
C(1 + ||f||_{BMO}(1 + log^{+}||f||_{W^{s,p}}))

In words of Ohkitani Koji (2008), that is,

"At the level of the BKM theorem which deals wit ha sup norm of the vorticity, it was not possible to rule out a possibility that vorticity blows up mildly as a logarithmic function of space variables. According to the recent updates using a BMO norm, we can safely exclude such possibilities"

So what is the next step? Unfortunately, Kozono and Taniuchi did not really leave any thought on the subsequent possible advancement. Looking back, it seems, at least to my Professor, BMO was the obvious choice. But he thinks it's hard to see any more improvement possible beyond.

Elaboration would be very much appreciated. I will reread Tataru's paper you mentioned; it seems to be connected somewhat.

Finally, please share the source of the paper you are talking about, in reply to a comment by Katz below:

“Thanks to the recent work of Escauriaza-Seregin-Sverak and others, we know that blowup solutions to Navier-Stokes must in fact blow up in the critical norm L^3({\Bbb R}^3). “



Student · 6 May 2009

“Try a topological method. This is a special case of (1). It may well be that a primarily topological argument may be used either to construct solutions, or to establish blowup; there are some precedents for this type of construction in elliptic theory.”

Please give us your specific recommended references (books, articles) to examples of these “precedents.”

Best regards,



Terence Tao · 7 May 2009

Dear Student,

One example of such methods that comes to mind is Gromov’s topological method to construct J-holomorphic curves in his celebrated paper on non-squeezing. There are also a number of methods for elliptic variational problems with a somewhat topological flavour (e.g. the mountain pass lemma).



NSE Beginner · 18 May 2009

Dear Professor Tao:

1. Aside from using Leray-Projection operator, one way to get rid of the pressure is to work on the vorticity equation. Solving this vorticity equation should be equivalent to solving the Navier-Stokes Equation. Is that correct? And if so, any pros and cons for this idea?

2. Beale-Kato-Majda showed in 1984 that if the vorticity does not blow up, then the solution can be extended beyond time T. It seems obvious to me that if the gradient, first derivative, of the vorticity does not blow up, then the same result applies (I am asking this question because showing that the gradient of the vorticity does not blow up seems easier from looking at the vorticity equation). Is that correct?

Thank you.



Terence Tao · 19 May 2009

Dear NSE beginner,

Well, the vorticity equation still involves the velocity u, which requires the Biot-Savart law to recover from the vorticity, so one is pretty much back to where one started. But it does seem clear that the dynamics of the vorticity (and in particular, the behaviour of vortex stream lines, vorticity shearing, etc.) is going to be an important aspect of the problem.

At fine scales (which is where all the difficulties to the problem lie), the derivative of vorticity is going to be much larger than the vorticity itself, so it is extremely likely that the BKM result extends to the derivative. By the same token, though, this derivative is going to be harder to control than the raw vorticity; in particular it will be sensitive to

shearing effects caused by irregularity in the velocity field u, which as mentioned earlier cannot be so easily eliminated from the vorticity equation.

 **Student** · 23 May 2009

Dear Professor Tao:

There was nowhere else for me to write this; at least there was somebody else who asked you about the Nash-Moser Notes on this blog.

Below I am writing about your Nash-Moser Notes

http://www.math.ucla.edu/%7Etao/preprints/Expository/nashmoser.dvi

On page 3, in the middle you state

"we conclude from (5) (with x = u_n ..."

but I don't see any x in (5). You mean u = u_n?

On page 4 bottom, (11), I think the power of N should be 40, not 20.

In the middle page 6, you define the Littlewood-Paley operators, the case for N. I think there is one extra f^ that you do not need, for the case of > N that is.

Finally, my question: on the bottom of page 6 you claim a certain identity "Suppose for now that u_1, . . . , u_{n+1} \in 5B. Then from the identity..."

I do not know how you obtain the third equality; i.e. going from second line to third. Please explain.

 **Terence Tao** · 23 May 2009

Thanks for the corrections! In the application of (5), x and h should be u and v (similarly the h in (5) should also be a v).

On page 6, on the third line of the equality, $P_{N_{n+1}}$ (you can see this also by looking at the next displayed inequality). The point is that $\Phi(u_n)$ is equal to $-\Phi'(u_n)v_n$.

 **PDEbeginner** · 26 May 2009

Dear Prof. Tao,

I read your nice note on the Nash-Moser iteration scheme http://www.math.ucla.edu/%7Etao/preprints/Expository/nashmoser.dvi as being shown above. But it is a little abstract, could you give some nice literatures having some examples? (I tried to find some examples in your lecture notes on Nonlinear Dispersive and Wave Equations at New Mexico State University, but failed).

Thanks in advance!

 **timur** · 26 May 2009

Dear PDEbeginner,

To me, the notes you mentioned was the most readable account on Nash-Moser, the others being very complicated, abstract, long, or the main idea was buried in details. If you think

you lack intuition on what is “frequency” of a function you might want to look at first couple of the following notes (which are again by prof. Tao):

http://www.math.ucla.edu/~tao/254a.1.01w/

 **PDEbeginner** · 26 May 2009

Dear Timur,

Thanks a lot for your kindness and your reference! I have downloaded part of the lecture notes provided by you and plan to read them (especially on Littlewood-Paley projection part).

 **Anonymous** · 27 May 2009

Hamilton’s paper, referenced in Tao’s notes, has a number of examples of how the Nash-Moser inverse function theorem can be used.

 **hegel triad** · 3 June 2009

Hi Prof. Tao,
I would like your advice on the following. I am at a conference in France, and the famous Prof. Temam has just shared with his audience a model used in oceanography – the Lagrangian-averaged Navier-Stokes-alpha. He said that it regularizes the NS by modifying the nonlinearity so that scales smaller than alpha are swept by larger scales, while preserving their fundamental inviscid transport properties such as convection of vorticity. I have misgivings as the audience was shell-shocked. Please express your opinion on the model.

The Vlasov-Poisson system « Hydrobates · 4 June 2009

[…] of the equations. For a discussion of the significance of scaling properties in general see Terry Tao’s post on the Navier-Stokes equations. In the case the potential and kinetic energies satisfy an inequality of the form and this plays […]

 **Student** · 13 June 2009

Dear Professor Tao:

You stated:

“There are a couple of loopholes that one might try to exploit: one can instead try to refine the control on the “waiting time” or “amount of mixing” between each shift to the next finer scale, or try to exploit the fact that each such shift requires a certain amount of energy dissipation.”

I’m sorry but I do not have a good grasp of what you mean here. It seems as if there is some periodic finer rescaling from time to time and shift between each. I can see why if there is such thing, it would be a problem but how do you know that there IS such periodic rescaling and how should it be described mathematically? Or are you suggesting to suppose that there is such periodic rescaling and try to overcome it?

And why does this shift require any energy dissipation? I think of the Laplacian in the NSE when you say dissipation, but I’m afraid that’s not what you mean.

Any elaboration would be appreciated.

 **Terence Tao** · 13 June 2009

Well, if I if I could prove that a solution could periodically shift its scales to higher scales indefinitely, then I would have disproven the regularity conjecture; my point is that this is an "enemy" that one will have to address at some point if one is to solve the conjecture. I also don't know whether the dissipation effect of the Laplacian will be strong enough to prevent an infinite number of such shifts, but this is one potential way to stop such a cascade from occurring. I used some ideas distantly related to this when establishing global regularity for the logarithmically supercritical wave equation, but this equation was only barely supercritical, and is unlikely to be adapted as is for the Navier-Stokes equation (though perhaps some sort of "logarithmically supercritical Navier-Stokes equation" is within reach by these sorts of methods).

 **Global regularity for a logarithmically supercritical hyperdissipative Navier-Stokes equation « What's new** · 19 June 2009

[...] supercritical and thus establishing global regularity beyond the reach of most known methods (see my earlier blog post for more [...]

Liveblogging Science 2.0 | Serendipity · 29 July 2009

[...] to support community approaches to mathematics. In particular, he deconstructs one particular post: Why global regularity for Navier-Stokes is hard. (https://terrytao.wordpress.com/2007/03/18/why-global-regularity-for-navier-stokes-is-hard/), which [...]

hyfen.net » Doing it in the open: Michael Nielsen at Science 2.0 · 30 July 2009

[...] a lot of time thinking about open science and mass online collaboration. He began by highlighting a post on mathematician Terence Tao's blog that outlined the Navier-Stokes problem and sketched a [...]

 **Sudent** · 9 September 2009

Dear Professor Tao:

As you mentioned in the discussion above, taking \alpha as the power of Laplacian,

\alpha = 1/2 + d/4

implies global existence of strong solution (cf. Jiahong Wu, "Generalized Incompressible NSE in Besov Sp.").

Hence, with \alpha = 5/4 in the case of 3-d, the case is closed. Naively it seems in 2-D, the best we can hope for is 2-D NSE. But I heard a rumor that 2-D Euler is also done (or Nets Katz stating "2-D problem for Euler which is critical"). Yet, 2-D Euler is basicaly 2-D NSE wth \alpha = 0. So I guess it required a somewhat different proof or some unique advantage of low dimension or something.

Does global strong solution to 2-D Euler Eq. exist? And if so, how do we prove it? Could you give me some reference to a textbook or paper?

Thank you.

 **timur** · 10 September 2009

Local well posedness of the Euler equations for initial velocity in H^s with s>n/2+1 has been proven by Lichtenstein, Guynter, and Ebin-Marsden. Correct me if I am wrong, but it seems there is no local existence result for initial velocity in H^s with s<=n/2+1.

The global well posedness theory in 2D is as follows:

* Global well posedness for initial vorticity in L^∞ [Wolibner '33], [Yudovich '63], [Kato '67].
* Global existence of weak solutions for initial vorticity in L^p, $1 < p \leq \infty$.
* The above is extended to p=1 and measure-valued vorticities [Delort].

Global well posedness for Euler in 2D is possible because in vorticity formulation it becomes a transport equation.

 **Student** · 12 September 2009

Dear timur:

thank you very much. I found your information very helpful.

 **Student** · 14 November 2009

Dear Professor Tao:

Above you described the rescaling of NSE and stated that

“Strictly speaking, this scaling invariance is only ... with the non-periodic domain {\Bbb R}^3 rather than the periodic domain {\Bbb T}^3”

I’m sorry if it’s obvious but it is not clear to me why the rescaling must change in T^{3}. If you could explain, I would appreciate it.

 **Terence Tao** · 14 November 2009

When one rescales a function which is periodic with period 1, one gets a function with a different period. Since functions on the torus $\mathbb{T}^3$ only correspond to functions which are periodic with period 1 (in all three cardinal directions), this shows that scaling by an arbitrary real parameter cannot be done while still staying in the standard torus. (For scaling by an integer parameter, one can still rescale, but one should caution that integrated quantities such as energy rescale with a different factor than their non-periodic counterparts. Alternatively, one can scale the length of the torus along with the solution, thus working with tori such as $\mathbb{R}^3/(\lambda\mathbb{Z})^3$ for various $\lambda > 0$.

 **Student** · 14 November 2009

Dear Professor Tao:

I understood. Thank you very much.

 **Anonymous** · 23 November 2009

dear PRof Tao can I get the formula for the viscosity term in the NSE in Lagrangian co-ordinates?because I think attacking NSE in Lagrangian co-ordintes would be a step towards arriving at the solution but the viscosity term is what i cant figure out

 **Student** · 24 December 2009

Dear Professor Tao:

You stated

"It has long been known that by weakening the nonlinear portion of Navier-Stokes (e.g. taming the nonlinearity), or strengthening the linear portion (e.g. introducing hyperdissipation), ..."

I am aware that by raising the power of Laplacian, e.g. from 1 to any real number equal to or bigger than 5/4 in the 3-D case, one can obtain the global regularity of the Navier-Stokes Equation (NSE). But what is the explicit modification on the Nonlinear portion,

u \cdot \nabla u,

that will achieve the global regularity, even with the power of Laplacian is 1, in the case d = 3?

We certainly cannot make the \nabla part of Nonlinear portion hyperdissipative because it will screw up the divergence-free advantage in many ways. E.g. dotting NSE by u and integrating by parts no longer allows us to get rid of the nonlinear portion if the \nabla is made hyperdissipative.

Has such modification been done?



timur · 25 December 2009

For example, you can use $(Du)\nabla u$ where D is some smoothing operator like $D = (I - \Delta)^{-1}$. $u\nabla(Du)$ is also possible. The key words that might be useful are Leray regularization, Leray-alpha, modified Leray-alpha, Navier-Stokes-alpha, Simplified Bardina model, etc.



Student · 24 December 2009

Dear Professor Tao:

You stated

"most nonlinear PDE, even those arising from physics, do not enjoy explicit formulae for solutions from arbitrary data (although it may well be the case that there are interesting exact solutions from special (e.g. symmetric) data)."

Also,

"It may well be that there is some exotic symmetry reduction which gives (1), but no-one has located any good exactly solvable special case of Navier-Stokes (in fact, those which have been found, are known to have global smooth solutions)."

Could you give some reference to any work done on these ideas, on NSE or other PDEs. Any elaboration will be appreciated too.

Thank you very much.

Critical wave maps « Hydrobates · 20 January 2010

[...] of activity at the moment. For more information about this classification and its significance see this post of Terence Tao. Actually there are a couple of results around on problems which could be called marginally [...]



Amplitude-frequency dynamics for semilinear dispersive equations « What's new · 2 April 2010

[...] analogous in many ways to the even more famous global regularity problem for Navier-Stokes (see my previous blog post on this [...]

 **Frank Trunk** · 22 April 2010

The problem with the Navier – Stokes equations is with the physics. They don't define the physics of 3D fluid flow completely. Part of that problem is with Newton's second law. I have done some work on the problem, but unforntunately is has been classified by the Dept. of the Navy. Define the proper physics and you will have the solution to your problem.

 **Robert Coulter** · 23 April 2010

I find this post by by Professor Tao to be very enlightening. I have been fascinated with this issue since about 2002 when I learned of the Clay Millenium prize.

Concerning the previous comment:

I believe from the mathematician's viewpoint that this is mainly an issue about this class of equations. The fact that they have something to do with fluid flow is secondary.
That being said, I believe that having a great insight into fluid behavior may offer some aid in solving the problem.

 **Robert Coulter** · 23 April 2010

Hello Professor Tao:

You stated earlier:

"(Note also that it is known that singularity can only occur if the velocity goes to infinity; in practice, of course, relativistic effects (among other things) would kick in long before then.)"

Is the above correct? Does the vorticity blowing-up necessarily imply the velocity blowing up also? If infinite velocity is required for the singularity, then should one rule out searches for other types of singularities (e.g. infinite gradients (velocity step changes))

 **Terence Tao** · 25 April 2010

Oops, you're right, the Beale-Kato-Majda criterion is with respect to the vorticity, not the velocity.

Nevertheless, an L^infty bound on the velocity does imply regularity for Navier-Stokes (though not for Euler, as far as I know) through energy arguments (at least in the case when the energy is finite, which is the usual hypothesis for the global regularity problem). Indeed, one can show inductively (using energy estimates and Gagliardo-Nirenberg, and the bounded velocity hypothesis) that the H^k norm of the velocity does not blow up in finite time as long as the solution stays smooth. In fact one can even relax L^infty to L^3 (thanks to the Escuriaza-Seregin-Sverak result).

 **Robert Coulter** · 25 April 2010

Professor Tao:

Thanks for your response. I will try to look up some of those sources.

 **Robert Coulter** · 26 April 2010

Concerning NS and the initial conditions of the Clay problem:

It appears that it has been proven that if the velocity becomes infinite it is limited to a Hausdorff dimension zero, which I believe that this means that it has no metric scale — no length, no area, no volume, etc — Just a set of points, no more — Is this correct??

It seems like an earlier post said that a group of singularities might be possible along a "segment". Did I misread this? Would this be Hausdorff dimension one?

Also, has it been proven that if the singularity exists that it has no time dimension measure?



Terence Tao · 26 April 2010

I believe that the best partial regularity result known for 3D Navier-Stokes is that the singularity set has Hausdorff dimension at most 1 in space, a result of Caffarelli-Kohn-Nirenberg, building on earlier work by Scheffer. There are some variants of this result by Fang-Hua Lin and by Katz-Pavlovic, and there may also be some subsequent literature in this direction (although the dimension 1 is very natural for scaling reasons and it would be a major breakthrough to lower this number). I believe there are also spacetime versions of this (using the usual heuristic that one time dimension has the same scaling as two spatial dimensions for parabolic equations, I would expect the best upper bound on the dimensions of times where the solution is singular to be 1/2).



Robert Coulter · 4 May 2010

Concerning the Clay Problem Fefferman Statement:

At (5) on the first page there is a specification for the forces allowed. Can someone explain the mathematical notation here? It says for any K. Is it implied that K must be positive? Doesn't a negative K imply force can grow large with increasing x and t?



Terence Tao · 4 May 2010

The bound (5) is required to hold for *all* K, both positive and negative. But in this case, the inequality for positive K is stronger than that for negative K, which thus becomes redundant.

This condition is basically asserting that the force lies in the Schwartz space,

http://en.wikipedia.org/wiki/Schwartz_space



Robert Coulter · 5 May 2010

Professor Tao:

Thanks for your response.

Also, again, thanks for having this blog.
I look forward to any other information you may want to share on this subject.



Bob · 13 June 2010

An insightful web-paper, thank you very much. I have two general questions in this regard:

You said that: "There are countless other global regularity results of this type for many (but certainly not all) other nonlinear PDE"

It somewho implies to me that: there are some similar problems with such a desired theory but NS leaves as unresolved (though, prior to read your post i thought that NS is one of the

simplest possible prototype in the field of nonlinear PDE for which we do not know desired theory).

1. You just count 2d NS as a sample from mentioned "countless" examples. But It seems that (i'm not expert) 2D sitution is very different from 3D for nonlinear vector PDEs, my question is that are there some similar nonlinear hydrodynamics 3D vector PDEs for which we have uniquness and global in time regularity results (e.g. they meay be compressible, and/or non-isothermal or coupled with some scaler transport, and/or have some additional terms, etc.)?

It is very helpful if you give us some reference in this regard, if above issue has positive answer.

2. Also, are there some modifications for 3D NS which follows the desired regularity (if positive, could you please state theoreticaly minimal ones)?



Robert Coulter · 13 June 2010

I believe the ones that provide regularity strengthen the viscous drag term – e.g. Maddingly-Sinai (2003).



Robert Coulter · 13 June 2010

Also, there were discussions along this line in earlier posts.



Bob · 2 July 2010

Robert,

thanks for hint. I implied something more than what already stated above, in particular more realistic case (in fact goal is to know that these set of eq. are hard or other alternatives are also hard problems).

e.g. is there requlrity results if we ignore incompressibility and consider 3d compressible viscous flow?

or if we add limited copressibility; commonly by a term porportianal to time derivative of pressure field to continuty eq.?



PDE Beginner · 2 July 2010

Some comments and a question to Bob's comments:

"It somewho implies to me that: there are some similar problems with such a desired theory but NS leaves as unresolved"

I think similarly difficult problems that have caught much attention recently and remains basically unsolved in the supercritical regime include Quasi-geostrophic Equation, Porous Media Equation, Cordoba-Cordoba-Fontelos (CCF) Model. The relation among these equations can be simplified as follows: Take the 2-D NSE, and its curl to get a vorticity equation. The nonlinear term looks like

u\cdot\nabla v

where v is the vorticity of u. u here is now \grad^{\perp}(-\triangle)^{-1}v.

The QG is the case where u is \grad^{\perp}(-\triangle)^{-1/2}, and hence the Riesz transform of v. The Porous Media is slightly complicated but can be simplified to a very similar form to

the QG (identity plus a singular integral operator. cf. the work by Castro, Cordoba, Gancedo, Orive on this). The CCF model is the case where Riesz transform of QG is changed to Hilbert transform. Finally, there is a modified QG which interpolates between the vorticity Euler Equation and the QG (cf. work by Constantin, Iyer and Wu on this).

For all these equations above, the nonlinear term is more complicated than that of Burger's; while the best known conserved quantity for these is the sup norm making the case of the power of fractional Laplacian on the dissipation term 1/2 the critical case, almost (cf. work by Luis Silvestre on slightly supercritical Quasi-geostrophic if you wish) nothing is known in the supercritical regime. I find the case of the NSE similar. The best conserved quantity there is different making the power of the fractional Laplacian that makes it critical at 1/2 + d/4 (d is dimension) (and similarly for the MHD Equations cf. work by Wu on this) but essentially we have no result in the supercritical regime (cf. work by Terence Tao on the logarithmically supercritical).

I now have a question for your question: You stated

"if we add limited copressibility; commonly by a term porportianal to time derivative of pressure field to continuty eq."

Can you give some reference where such method of reducing compressibility by a term proportional to time derivative of pressure field? You said "common," so I would imagine there are examples. I am new to such idea.



Bob · 2 July 2010

PDE Beginner: first thanks for comments, though i did not still get an effective response; ignoring adding higher order smoothing term (is lot); i look for some known theory for 3d related problems.

your answer: in practice sound speed in any media is finite, so in principles incompresibility constraint is unphysical; of course plays rule in small scales where the time scale of vortices are comparable to that of sound speed.

anyway (i do not know theoric work, it was my question here) there are some computational worksd on this method; e.g.:

Adding limited compressibility to incompressible hydrocodes, Journal of Computational Physics, Volume 34 (1980) 390-400,



Robert Coulter · 2 July 2010

Has anyone done any work on the action of entropy in relation to NS and/or Euler?
My hunch is that in the NS stokes case that the "process" would tend to evolve in the direction of a lower free energy state. Maybe this would be the Helmholtz Free Energy = A = U − TS. In Euler there is no energy dissipation, so it appears only the TS term is active (process seeks higher S always). In NS the process "arrow" is decided by competition of seeking a lower energy state (lower U via viscosity removing energy) versus seeking higher entropy (higher S − nonlinearity term randomizing particle trajectories?). Both terms seek to diminish the macro world's ability to recover energy/work from the system (left unattended).

 **Robert Coulter** · 2 July 2010

In addition to my previous post:

In the case of NS there appears to be a pathway to T (temperature) in that there is an implicit assumption the the viscous term removes the kinetic energy and "magically" removes it from the system. Practically this is not hard to visualize if the system isothermally maintained with the Q (heat) passing into the external heat reservoir.

In the Euler case there is (from what I can tell) no easy visualization. There is no obvious pathway to T (T interpreted to be random kinetic energy) . The energy stays in finite discrete "velocity bundles" (for lack of a better term). Do the velocity bundles progress to so much smaller scales that the Lagrangian particle trajectories collide providing the only mechanism in the Euler case of dispersing the kinetic energy to T?

In the NS low viscosity case the action is probably similar. The "entropy" increasing effect is much stronger than the viscous (U decreasing action).

 **jj2** · 4 August 2010

In case you are interested, you find the proof of Statement D of the Clay Math. Institute Navier-Stokes problem in Electon. J. Diff. Equ. Vol 2010(2010), No. 93, pp. 1-14 at http://ejde.math.txstate.edu. The official problem statement did not require the pressure to be space-periodic and did not exclude a feedback force as the external force. Thus, a simple blow-up solution could be explicitly constructed. Changes to the problem that are needed so that the presented example is not any more valid are not small modifications.

 **Robert Coulter** · 5 August 2010

Blow up term g is only a function of the time variable. Looks like all velocity vectors go to infinity across the entire space at t = a. Doesn't this imply infinite energy at blowup? If so, how did the energy enter the system? My hunch is that an infinite external force is in the math somewhere; otherwise, one would have to explain how the energy appears.

 **jj2** · 5 August 2010

The energy comes from the time derivatives of the velocity vector at t=0.
The external force does not have infinite energy for t=0 to t=infinity.
The official problem setting does not require solutions to be physically meaningsful as blow-up solutions in general are not physical. The physical
problem of the pressure growing to infinity when space dimensions grow to infinity is only in finding an infinite physical container. Gime me such and I give you the solution. Physically you can make a finite volume fluid behave in this way if you first speed the higher order time derivatives of velocity to suitable values. Then apply zero force, the fluid will follow the solution in Lemma 1.

 **jj2** · 5 August 2010

Is is possible to recover the "original" problem by a small modification? Several modifications are needed, they are described at the end of the article, they are not small. What is the "original" problem? Fefferman explicitely says that he does not give the whole NSE problem but four very small subproblems as the millennium prize problem. Only Fefferman has ever

defined these problems, so they must be the original problems. The
whole issue of modifications was if the stated problem is poorly defined and needs to be modified, but it is well defined and solved in the paper.
If you pose a million dollar question, try to check before posing it that it
is the question you want solved.

 **jj2** · 5 August 2010

See also arxiv:0809.4935. This paper is 2.5 years old and was in journal review for 15.5 months, presently submitted in a journal. So far no serious errors found in 2.5 years, only small issues like a few typos.

 **jj2** · 5 August 2010

Why the initial values of the time derivatives of the velocity vector were not defined at t=0? There was an old theorem stateing that the solutions were unique. As the solutions were unique, feedback forces did not need to be considered (there is nothing to select if there is only one solution).
Unfortunately, the proof of this old theorem implicitly assumed the pressure to be periodic. Proofs which implicitly assume something are wrong. This therem seemed to be known and believed by many people.

 **Robert Coulter** · 6 August 2010

I suspect that condition (9) of the Fefferman paper is violated. It appears that the external force is embedded in the pressure equation.

 **Terence Tao** · 22 March 2008

Dear Dave,

Unfortunately there is an error in deriving (26) in your paper. You seem to be treating various monomials involving the U_n as if they were non-negative numbers, when in fact they are matrices of indefinite signature. As such it is not necessary the case that a difference is bounded in magnitude by a sum; for instance, $\|U_1 - U_2\|$ might be larger than $\|U_1 + U_2\|$ (e.g. consider the case when $U_2 = -U_1$), and similarly $\|S_n - T_n\|$ can be larger than $2\|S_n\|$. Also, it is not true that matrix product order has no influence on the "bound" of the product: for instance, it is entirely possible for $\|U_1 U_2\|$ to be different from $\|U_2 U_1\|$. (Both are bounded above by $\|U_1\|\|U_2\|$, but this is not the type of expression you have on the right-hand side of (26). Similarly, $\|U_1 - U_2\|$ is bounded above by $\|U_1\| + \|U_2\|$, but this is again not the type of expression you have on the right-hand side of (26), in which the norm is outside the sum and product, rather than inside.)

More generally, one cannot control a non-commutative flow by its commutative counterpart, because cancellations in the latter do not necessarily force cancellations in the former. It's easiest to illustrate this phenomenon with a discrete equation $u(t+1) = \exp(U(t))u(t)$ rather than a continuous equation $\frac{d}{dt}u(t) = U(t)u(t)$. It is an easy matter to find non-commuting matrices A, B whose commutator $\exp(-B)\exp(-A)\exp(B)\exp(A)$ has an eigenvalue larger than 1. If we then let U(t) vary periodically between A, B, -A, and -B in turn and let u(0) be an eigenvector of $\exp(-B)\exp(-A)\exp(B)\exp(A)$ corresponding to that eigenvalue, then $u(t)$ grows exponentially in time, even though the abelianisation $\sum_{t=0}^{T} U(t)$

stays bounded. It is easy to convert this discrete-time example to a continuous-time one but I will leave this as an exercise.

[The one time in which one can safely control a non-commutative flow by a commutative one is if the latter has no cancellation in it, in which case Gronwall's inequality applies.]

 **DavePurvance** · 18 April 2008

Dear Terence,

While it may turn out that "commutative flows", as you term them, may be the only flows physically possible, I did not mean my mathematical bounds to be derived from a commutative flow. My bounds, which I now denote as $\mathbf{b_n}$ with corresponding matrices B_n, use the commutative solution's matrix exponential function, but this function is evaluated with a noncommutative solution's U_n. $\mathbf{b_n}$ are therefore not solution coefficients to the Navier-Stokes matrix differential equation or any other differential equation that I know of, unless, of course, U_n become commutative.

Relating to your criticism that the inequality
$\|S_n - B_n\| \leqslant 2\|B_n\|$ (1)
is not established, isn't its opposite
$2\|B_n\| \leqslant \|S_n - B_n\|$ (2)
contradictory? For instance, in the commutative case when $S_n = B_n$ with $\|B_n\| > 0$ this latter inequality asserts
$2\|B_n\| \leqslant 0$. (3)
S_n and B_n and the contested inequality (1) must accommodate both commutative and noncommutative cases.
Finally, the numerical tests. The 10th order B_{10} is comprised of 512 matrix product terms and each matrix is 1029×1029. Using numerous "white-noise" initial flows, the sum of these matrix product terms always agrees with the bounds derived from (1). If (1) is not true, then how does one explain these (near miraculous) numerical results?

 **Jonas** · 18 April 2008

Dave,
As far as I can tell there is no miracle going on here. That the inequality holds for white noise initial data says very little. There are numerous examples of inequalities in mathematics which hold for a typical random sample of the objects in question, however it is also often the case that the examples where the inequality is false are highly structured objects which a random sample have essentially no chance of ever turning up. A proper test of the bound would e.g. start out wit ha single initial data and then try to modify it locally to make the inequality come closer to equality, then modify this data and repeat this until, possibly, data is found which violates the inequality.

In you discussion above about inequalities (1)+(2) you miss the alternative that in general none of the inequalities need to be true. There could very well be matrices for which (1) holds and others for which (2) holds.
For matrices in general let A be the 3X3-matrix which has a zero in the upper left corner and the rest is 0, and let B be the matrix which has -1 at the last two position in the top row the

the rest is 0. If we let A be B_n and B be S_n then inequality (1) fails, since the norm of A-B is 3^0.5 > 2, and the norm of A is 1.

 **DavePurvance** · 19 April 2008

Dear Jonas,

Your comments on my numerical tests may be true, and I would love to have the computing power to try what you describe as a valid numerical test. In fact, I hope someone is interested enough to try, even though if nothing negative is found, someone else might criticize that you just didn't probe hard enough.

One point that I believe adds credence to my numerical tests is that both solution matrices S_n and bound matrices B_n are made from the same system matrices U_n, and, U_n are made from the same flow coefficients $\mathbf{u_n}$. So , if S_n is random, so is the bound B_n, and if S_n is structured, so is B_n.

Also, I'm sure one can always produce simple 3×3 matrices that support any argument. The challenge is to make your argument with matrices that represent the periodic 3D incompressible Navier-Stokes equation.

 **285G, Lecture 7: Rescaling of Ricci flows and kappa-noncollapsing « What's new** · 21 April 2008

[...] to begin the surgery program discussed above. (In contrast, the main reason why questions such as Navier-Stokes global regularity are so difficult is that no controlled quantity which is both coercive and critical or subcritical is known.) The [...]

Lower and Upper Bounds for Periodic Navier-Stokes Flows « Smooth Chaotic Fluids · 24 April 2008

[...] differential equation also has a general noncommutative time series solution. These all have been recognized by the world-renown expert Terry Tao (read in his blog's comments). Terry also happens to be a very decent and fair person in my [...]

 **DavePurvance** · 9 May 2008

Terence et al,

I seemed to have messed up the links.

My proposed solution to the periodic 3D Navier-Stokes problem will always be arXived here. My WordPress blog discussing this solution will always be here.

Your criticisms and comments are always welcome.

 **DavePurvance** · 30 May 2008

Dear Terence,

Thanks for acknowledging in your 22 March 2008 comment above my commutative $\mathbf{v}(t) = \sum_{n=0}^{\infty} \mathbf{v}_n t^n$ and noncommutative $\mathbf{u}(t) = \sum_{n=0}^{\infty} \mathbf{u}_n t^n$ solutions to my periodic Navier-Stokes matrix differential equation

$d\mathbf{u}/dt = \left(\sum_{n=0}^{\infty} U_n(\mathbf{u}_n)\right)\mathbf{u}$, (1)

having initial condition $\mathbf{v}(0) = \mathbf{u}(0) = \mathbf{u}_0$.

In the same comment, though, you misunderstood the pair of matrix polynomial inequalities

$\|S_n - B_n\| \leq 2\|S_n\|$

$\|S_n - B_n\| \leq 2\|B_n\|$ (2)

I used to put bounds

$|\mathbf{b}_n| \leq |\mathbf{u}_n| \leq 3|\mathbf{b}_n|$ (3)

on the general noncommutative solution coefficients $\mathbf{u}_n$. I take full responsibility for this misunderstanding because I didn't really prove (2). My arXiv paper now does.

At the time of your comment I denoted in (2) B_n as T_n and in (3) $\mathbf{b}_n$ as $\mathbf{v}_n$. Perhaps this notation was the source of the misunderstanding. Whatever the case, I am sure you understand that this is an important issue and I ask you for the opportunity to counter your criticism of (2).

The inequalities in (2) do not mean that a commutative flow $\mathbf{v}$ of one initial condition $\mathbf{u}'_0$ somehow can control another noncommutative flow $\mathbf{u}$ of another initial condition $\mathbf{u}_0$.

Matrix polynomials $S_n(U_0, .., U_{n-1})$ are defined by

$\mathbf{u}_n = S_n(U_0, .., U_{n-1})\mathbf{u}_0$. (4)

Bound coefficients $\mathbf{b}_n$ are the Taylor time expansion coefficients of the matrix exponential

$\mathbf{b} = \exp\left(\sum_{n=0}^{\infty} 1/(n+1)U_n t^{n+1}\right)\mathbf{u}_0$ (5)

assuming the same noncommutative U_n as in (4) and resulting in matrix polynomials $B_n(U_0, ..., U_{n-1})$ with

$\mathbf{b}_n = B_n(U_0, ..., U_{n-1})\mathbf{u}_0$. (6)

Again, as noted in (4) and (6), S_n and B_n are functions of the same $U_0, ..., U_{n-1}$ and therefore both are a function of the same initial condition $\mathbf{u}_0$. The difference between $\mathbf{v}$ and $\mathbf{b}$ is that $\mathbf{v}$ is a solution to (1) because $\mathbf{v}$ assumes commutative U_n. $\mathbf{b}$ is not as solution to (1) because $\mathbf{b}$ assumes noncommutative U_n. They both, however, are stable functions of U_n independent of their commutative properties and S_n and B_n simplify to the same polynomial when U_n commute.

Inequalities (2) are generalizations of a pair of equivalent commutator inequalities, which for illustration, I will derive here for the two simple matrix monomials making up the matrix commutator

$[X, Y] = XY - YX$. (7)

XY is the first monomial and YX is the second. The reasoning for putting bounds on (7) is the same as the reasoning used to derive (2). The Navier-Stokes matrix polynomials S_n and B_n, however, are much more complicated than the monomials XY and YX used here. So to get the full derivation of (2), you will have to read my paper.

By the the definition of any matrix norm

$\|XY\| \leq \|X\|\|Y\|$

$\|YX\| \leq \|X\|\|Y\|$. (8)

Adding the two halves of (8) gives

$\|XY\| + \|YX\| \leq 2\|X\|\|Y\|$. (9)

Again by the definition of any matrix norm

$\|XY - YX\| \leq \|XY\| + \|YX\|$. (10)

Adding (9) and (10) and simplifying gives

$\|XY - YX\| \leq 2\|X\|\|Y\|$. (11)

Doubling both halves of (8) and subtracting the result of each from (11) gives

$\|XY - YX\| - 2\|XY\| \leq 0$

$\|XY - YX\| - 2\|YX\| \leq 0$. (12)

which upon rearrangement yield the desired inequalities

$\|XY - YX\| \leq 2\|XY\|$

$\|XY - YX\| \leq 2\|YX\|$. (13)

So actually, inequalities (13) are just two operations away from the well known matrix commutator inequalities (11). And again, the reasoning for deriving the Navier-Stokes (2) parallels the reasoning provided in (7)-(13).

Inequalities (13) can be easily checked numerically for arbitrary square matrices X, Y. I used random complex matrices and MATLAB's "gallery" function to mix and match different types of X, Y. The experiments ranged in dimension from 2 to 2000 and the inequalities (13) never failed. And, of course, as detailed in my paper, I have also numerically checked the Navier-Stokes (2) through time order 10 using both random and structured initial flows. These checks have never failed either.

Sorry for the confusion and thanks for letting me clear it up.

lib.mexmat.ru/forum · 5 June 2008

Tao https://terrytao.wordpress.com/2007/03/18/why-globa.....ment-26446 писал(а):
At present, all known methods for obtaining global smooth solutions to a (deterministic) nonlinear PDE Cauchy problem require either {В настоящее время, все известные методы получения глобальных гладких решений (детерминированных) нелинейных PDE (Cauchy проблема) требуют то или иное (также?)}
1. Exact and explicit solutions (or at least an exact, explicit transformation to a significantly simpler PDE or ODE); -{Точные и явные решения (или по крайней мере точное, явное преобразование к значительно более простому PDE или ODE)}
2. ;Perturbative hypotheses (e.g. small data, data close to a special solution, or more generally a hypothesis which involves an somewhere); or
3. One or more globally controlled quantities (such as the total energy) which are both coercive and either critical or subcritica.- {Одно или более глобально управляемые количества, (типа полной энергии), которые являются принудительными и также критическими или подкритическими.}

Или, может быть, самого автора стоит об этом спросить, воспользовавшись окном для комментариев его блога по указанному в цитате адресу. Возможно, он, действительно, как и Вы, почему-то молчаливо подразумевает п. 1 в обязательном сочетании с п. 3, хотя из приведенной цитаты это, кажется, никак не вытекает. При этом заодно можно предложить Tao и его коллегам по обсуждению сформулировать свое отношение к высказываниям Л.Д. Ландау и М. Клайна, о чем мы здесь столько спорим.

Shock waves « Hydrobates · 12 June 2008
[...] If viscosity is included in the description of fluids then the Euler equations are replaced by the Navier-Stokes equations. There is reason to suspect that in this case shock waves are smoothed out and a smooth initial configuration remains smooth in the course of the evolution, for all time. There are simple examples where this can be seen but there is still no global regularity result for Navier-Stokes (and no counterexample). The Clay Foundation has offered a prize of one million dollars for the solution of this problem in either direction. The fact that the prize has not yet been collected is a sign of the difficulty of the problem. For a discussion of this question and its broader mathematical significance I recommend the excellent account of Tao. [...]


Anonymous · 16 July 2008
I am looking for more information about the use of Strichartz estimates in solving wave equations. Can anyone provide me some reference? I am not looking for research papers yet. I want to grasp the main ideas. The info on DispersiveWiki is not particularly useful. And I don't seem anything revelant except a lot of research papers which I want to avoid at the first stage. Thanks.


Terence Tao · 17 July 2008
Dear Anonymous,

You might try Chris Sogge's book on nonlinear wave equations, or my own on nonlinear dispersive equations.


Anonymous · 17 July 2008
Thanks. I'll have a look at both.


Global existence and uniqueness results for weak solutions of the focusing mass-critical non-linear Schrödinger equation « What's new · 19 July 2008
[...] smooth solutions to the Navier-Stokes equation is one of the Clay Millennium problems that I have blogged about before, but global existence of weak solutions is quite easy with today's technology and was first [...]


Claes Johnson · 22 July 2008
A resolution of the Clay Navier-Stokes problem is proposed in the article
Blowup of Incompressible Euler Solutions, published online July 19 2008
in BIT Numerical Mathematics.


Anonymous · 22 July 2008
To Claes Johnson

I don't think this paper is called "resolution"... it is just some numerical evidence. For incompressible Euler, finite time blow-up or not, has been a long time controversial question. Both sides have numerical "evidence"...

For incompressible Navier-Stokes, physicists tend to believe no finite time blow-up.

Claes Johnson · 23 July 2008

To Anonymous:

This discussion is important. I hope e.g. Terence will take part.
You express a common misconception, addressed in the article, that computational evidence is not mathematical evidence.
The central concepts are wellposedness and turbulence, and in this context a computational solution is as much a solution as anything.
Please read the article with an open mind. Looking forward to your comments.

Claes

Anonymous · 29 July 2008

Claes,

The issue is that the NS equation is highly numerically unstable, so simulations showing blow up tell us very little; the blow up may be an artifact of the simulation rather than an indication that NS actually blows up.

[different anonymous from 7:20pm above]

Claes Johnson · 29 July 2008

To Anonymous 2:

You are right that an unstable numerical scheme for a stable problem can give artifacts. But our computational Euler solution is validated by a posteriori output error estimation by Euler residuals multiplied by stability factors obtained by solving dual linearized problems.
A computed Euler solution is thus a representative solution and not an artifact. Since the computed solution shows blowup and is representative, there is blowup.

Correspondence (with e.g. Terence) on the blowup problem is published on http://www.nada.kth.se/~jhoffman/pmwiki/pmwiki.php?n=Forum.Clay

Claes

Anonymous · 30 July 2008

Claes, take a look at [[Goodstein's theorem]] in Wikipedia for how fast a function can grow and still be considered bounded in a mathematical sense. Double exponential is not bad at all. Triple or quadruple or 57-times-iterated exponential is only a little bit worse. Then you get functions like Ackermann's, which are worse than n-times-iterated exponential for arbitrary large n, but for which you can still write down an formula in terms of induction on more than one variable. And then there are functions that grow so fast that no expression can even

be written down–one can only prove indirectly that they are still finite. Goodstein's theorem says that starting with any n, if you iterate a certain calculation for enough steps you'll eventually reach zero. But it turns out (proved by Kirby and Paris) that as a function of n, the number of steps before you hit zero is so large, that it's impossible to write down a formula for it. Yet that it's always finite (the process never goes on without bound) is a remarkable fact of pure mathematics, even though even for values as small as n=4, f(n) could not be written down as a decimal number if you could engrave a billion digits on each electron in the physical universe.

A more practical example: suppose you have a computer program with array references like a[x*y + z]. It is provably impossible in general for a compiler to tell without running the program whether there will ever be a subscript-out-of-range error. Suppose instead the references are like a[37*y++19*z+3*w]–that is, you never multiply two variables together in a subscript, you can only add them or multiply by constants. Then compile time checking is possible (this is called Presberger arithmetic). The running time formula is a tower of iterated exponentials in the size of the expressions, which is absolutely intractable in the worst case, but it's a remarkable theoretical result that it's decidable at all. And it turns out that for most real practical cases, the worst case exponential tower is avoided. And even without that, the decidability is of interest to pure mathematicians. Maybe they're not of interest to anyone else, but pure mathematicians went into that field precisely because they're into that sort of thing. It's not up to anyone else to talk them out of it.

Who knows, maybe global stability will involve growth functions like that. If so, the Clay math people want to know about it. In a physical sense it's as bad as a singularity but from a pure math point of view it's not the same at all. And as Terence and others have said, reformulations of the problem may be very interesting from a physics perspective, but the pure math question posed as a Clay problem is very specific.

 **Anonymous** · 30 July 2008

Re the above: an amusing question occurs to me for the math folks here (I'm just a computer guy). What happens if someone turns in a proof of NS global stability, that is rigorous but depends on something like a large cardinal axiom? Is that even imaginable, mathematically speaking?

 **Claes Johnson** · 30 July 2008

To Anonymous:

You touch an essential point: There is a big difference between
100 and 10^100 = googol. If you do not make this distinction
in a quantitative analysis, it is not an analysis. I am surprised
to see that some (pure) mathematicans do not make this distinction,
and I wonder what mathematical tradition it can reflect.

Claes

 **westy31** · 8 August 2008

Hi turbulence friends,

I happened to be thinking about the Navier Stokes equation and turbulence, when I found this discussion. I have some questions and comments.
I recently put a free n-dimensional Navier Stokes simulator on the web at:
http://www.xs4all.nl/~westy31/CellFlow/CellFlow.html
One question I am working on is how turbulence scales in n dimensions. It was remarked by Terrence in this discussion that 3D turbulence is supercritical, while 2D turbulence is critical. This I would like to understand.

First, I will comment on scaling. I think the scaling
$u \to \frac{u}{\lambda}, x \to \frac{x}{\lambda}, t \to \frac{t}{\lambda^2}$
can be considered a special case of scaling behaviour that is most easily understood in terms of dimensional analyses. If you are given data about a fluid flow, without mention of the units used, you would not be able to deduce the correct units. This is because classical physics has a symmetry that corresponds to the fact that there are no natural length, time and mass scales. The scaling law for a quantity is easily written down by just writing out the dimension of the quantity:
$x \to x\ m$
$t \to t\ s$
$u \to u\ \frac{m}{s}$
etc,

The λ scaling is a special case: $m = \frac{1}{\lambda}, s = \frac{1}{\lambda^2}$.

Deriving an energy spectrum from dimensional analyses only, was just what Kolmogorov did in his 1941 paper. The outcome, which is easy to check yourself, is
$E(k) = C\epsilon^{\frac{2}{3}}k^{\frac{-5}{3}}$
See also the Wikipedia page:
http://en.wikipedia.org/wiki/Turbulence

This outcome is independent of the dimension of the flow, (Note that k stands for the norm |k|, not the vector) which seems to contradict the claim that 3D and 2D have a different criticality. Personally, I have a hard time believing that the energy spectrum does not depend on space dimension, so I would welcome any arguments to that end. I heard (see Stephen's comment) that in 2D, you have 'Kraichnan power of [minus] 3 law', but it is unclear to me how this relates to Kolmogorov's theory. Clarification would be welcome here.

While thinking about the -5/3 law, I decided they need to be modified anyhow, to meet some consistency requirements. If you try to integrate the dissipation ($\epsilon(k) = E(k)k^2\nu$) over k, to check that you retrieve ϵ, you run into a diverging integral. To solve this, I propose:
$E(k) = C\epsilon^{\frac{2}{3}}k^{\frac{-5}{3}}exp(-(\frac{\nu^{\frac{3}{4}}}{\epsilon^{\frac{1}{4}}})^2)$
Note that this formula is scale invariant.
The double exponential cut-off arises because for wave number higher than the Kolmogorov scale $\frac{\epsilon^{\frac{1}{4}}}{\nu^{\frac{3}{4}}}$, dissipation takes over. With the ultraviolet exponential cut-off, we can integrate the dissipation and get a consistent answer.

Next, try to integrate E(k) to get the total turbulent kinetic energy. Again, we get infinity, but now because of the infrared limit. This is solved by an abrupt low wave number cut-off k_0,

which is justified because we have a fluid in a finite container.
Interestingly, we can now for a dimensionless number out off the ratio between the high and low wave number cut-offs: Re= $\frac{k_0 \epsilon^{\frac{1}{4}}}{\nu^{\frac{3}{4}}}$.

This dimensionless number can be interpreted as the Reynolds number, if we insert the Kolmogorov velocity for velocity, and 1/k_0 as the scale in the regular expression for the Reynolds number.
The idea of this Reynolds number makes sense. After all, a laminar flow is quite smooth, and this should somehow be reflected in the energy spectrum formula. The formula I propose automatically makes a smooth (=steeply declining in wave number) energy spectrum if the Reynolds number is low.

Because of the viscous cut-off, I personally believe that the Navier Stokes equation does not blow-up, ie has zero energy in infinite wave numbers.

Gerard

PDEbeginner · 23 August 2008
Dear Prof. Tao,

This article is very nice and very helpful for me to understand the last term's lecture on NSE I attended. As for the supercritical and critical things (I didn't understand it in the lecture), I was wondering if my present understanding is right:

When the energy transfers to the finer and finer scales, the velocity will concentrate on smaller and smaller space. This will possibly make the critical and subcritical norm of velocity blow up.

On the other hand, some people works on partial regularity of the solutions, suitable solutions, and considers Hausdorff dimension of singular points. I was wondering if this is helpful for solving the problem.

Thanks a lot!

All the best,

PDEbeginner

Alberto G. P. · 24 August 2008
Hi,

The String 2008 conferences were held at CERN this week. The talk by Minwalla was about 'Nonlinear Fluid Dynamics from Gravity' and seem me very interesting:

(pdf here: http://indico.cern.ch/getFile.py/access?contribId=24&resId=0&materialId=slides&confId=21917)

I have read the Terence's post and I think that he didn't include this technique in the Strategy 3. Unfortunately I am not an expert on this subject, more exactly, I know very little about physics and mathematics. Nevertheless I am very interested in the following questions:

1) Could we know something about Navier Stokes equations (NSE) by means of the proposed duality between NSE and EFE (Einstein Field Equations)?.

2) We could discovery analytic EFE solutions in AdS space and then obtain his dual NSE solutions (?).
3) Is the global regularity problem for EFE resolved? If so it can be use the proposed duality to resolve the global regularity problem for NSE.

Can somebody help me? I asked about Minwalla's talk in some physics's blog but nobody reply me. Perhaps I am doing senseless questions (:-)sorry.

 **Pedro Lauridsen Ribeiro** · 25 August 2008

Hi Alberto,

The aforementioned transparencies are a bit hard to read,
as some of the math symbols there are truncated. However,
I can make some comments on your questions proper.

First, let's set the stage with a bit more of detail. Minwalla's
proposed duality uses a particular instance of the
Maldacena-Witten AdS/CFT correspondence which relates a
particular class of solutions of Einstein's equations with negative cosmological constant in d+1 dimensions which possess "locally" a conformal infinity of anti-de Sitter (AdS) type with solutions of
"generalized" Navier-Stokes equations in d dimensions. For that,
he uses the Fefferman-Graham asymptotic expansion of the
conformally rescaled bulk metric in a collar neighbourhood of
conformal infinity, whose coefficients satisfy, due to the Einstein
equations, a family of recursion relations which constitute a non-linear
Fuchsian system (i.e., a system of differential equations whose
coefficients present isolated algebraic singularities for one or
more coordinates, here the distance to the boundary in geodesic
normal coordinates for the conformally rescaled metric). Such
system determines uniquely the asymptotic expansion of the
bulk metric near infinity (hence, its _long-distance_ or
long-wavelength behaviour) from the boundary metric and
a symmetric rank-2 tensor on the boundary called the
boundary stress-energy tensor. When conserved and traceless,
this tensor reduces to the rescaled electric part of the bulk
Weyl tensor, but this is not always the case – in general, the
divergence and the trace of the boundary stress-energy tensor
give rise to conformal invariants at the boundary which can often
be computed exactly, due to the Fefferman-Graham equations.
It's the inhomogeneous conservation law of the boundary
stress-energy tensor that gives rise to the "generalized"
Navier-Stokes equation of Minwalla, for a particular class of
bulk metrics.

Now, about your questions (in some places without pretenses to rigour):

1.) The AdS-CFT correspondence exchanges large scales in the bulk with small scales in the boundary, but knowing the behaviour of the boundary conservation law in the large involves geometric information deep in the bulk. The class of solutions of the bulk Einstein's equations which give rise to the "generalized" Navier-Stokes equations has a small-codimension singular locus (a "black brane") deep inside. In this case, one may expect to obtain blowup for these "generalized" Navier-Stokes equations, but Minwalla presents only circunstantial evidence to the conjecture that the "generalized" Navier-Stokes equation always arises from the procedure above – this involves proving global existence and uniqueness for the Fefferman-Graham system for the bulk metric.

2.) Using the Fefferman-Graham formulae for the boundary stress-energy tensor in terms of the bulk metric, one can obtain the actual form of the coefficients of the "generalized" Navier-Stokes equations, but not its actual solution, unless the form of the resulting equations is, of course, sufficiently simplified. Otherwise, the equation is as hard to solve as ever.

3.) When the boundary metric and the boundary stress-energy tensor are real-analytic, the Fefferman-Graham system admits a unique solution and, moreover, the asymptotic expansion actually converges in a sufficiently small collar neighbourhood of the conformal infinity (this expansion is a power series in the geodesic distance to the boundary if and only if the boundary stress-energy tensor is conserved and traceless, otherwise the expansion involves logarithmic terms as well). This was proved by Kichenassamy. As far as I know, almost nothing is known in general about the maximum domain of convergence. For non-analytic boundary data, there seems to be so far no sufficiently strong estimate of the solution in terms of the boundary data. This question was studied by Michael T. Anderson in several papers, almost all of them in Euclidean signature, because then one can use techniques from nonlinear boundary value problems (Leray-Schauder degree, etc.), but the problem is quite different from a PDE perspective, and Wick rotation (to move back and forth from Lorentzian to Euclidean signature) is a very tricky procedure in curved spacetimes, which globally in general cannot be done, specially if the metric is not stationary. Only very few of Anderson's papers deal with Lorentzian signature. In one of these, he shows that a sequence of Cauchy data for geodesically complete, asymptotically AdS spacetimes (i.e., subject to the boundary condition that the conformal infinity should be that of AdS, or, more precisely, of its universal cover) which become asymptotically stationary at infinite times actually give rise to a _globally_ stationary spacetime. This is rather different from the asymptotically flat case, since nonstationary (albeit sufficiently small) perturbations of the

Cauchy data for Minkowski spacetime give rise to nontrivial, smooth
and geodesically complete spacetimes, as proven by Christodoulou
and Klainerman. The difference is that the boundary conditions needed
for the Einstein equations in the case of a negative cosmological constant
(as here the "unperturbed" solution – i.e. AdS – is not globally hyperbolic)
do not let the nonlinear effects disperse at large times – they are focused
back in an "almost-perodic" manner. The result of Anderson gives evidence
to the possibility that AdS spacetime is globally nonlinearly unstable – according to his result, nonstationary perturbations will eventually lead
to geodesic incompleteness. But here there's a caveat: to prove his result,
Anderson needs a "unique continuation" property for the rescaled linearized Einstein equations across the boundary. Here he borrows his intuition from the Euclidean case, where such property holds even in the nonanalytic case. However, the linearized Einstein equations are of wave type (modulo
some gauge fixing), and in the nonanalytic case (which is even beyond the
reach of Kichanassamy's approach), there are counterexamples to
unique continuation by employing nonanalytic perturbations of
the coefficients of lower order (Cohen, Hörmander, Alinhac-Baouendi)
if, in particular, the boundary is totally geodesic (i.e., gliding – null geodesics tangent to the boundary always have infinite order contact), which is
always the case for conformal infinities. A proof of global nonlinear
instability of AdS would probably involve proving that such perturbations always lead to geodesically incomplete bulk metrics, which might be the
case for all we know, but it remains an open (albeit perhaps solvable in
this form) problem. Now, to move from this back to Minwalla's "generalized"
Navier-Stokes equations, one needs estimates relating boundary quantities to bulk quantities (which is also needed if one wants to extend
Kichenassamy's existence result to the nonanalytic case and prove
on the other hand the _stablility_ of AdS, which does hold locally in time in
4 dimensions, as proven by Friedrich). Such estimates exist in the
case of Euclidean signature, but it's hardly likely they carry through
to Lorentzian signature, specially in nonanalytic cases. Since the boundary stress-energy tensor involves the Weyl tensor, and so do the estimates obtained by Christodoulou and Klainerman in their proof of the global nonlinear stability of Minkowski spacetime, there may be such estimates
in spite of that.

You see, it's not that your questions are senseless, it's just that they are
too hard, but nevertheless always worth asking...;-)

Cheers,
Pedro



Pedro Lauridsen Ribeiro · 25 August 2008

Ah, yet on Alberto's third question: for large asymptotically
flat data, Christodoulou recently proved that, under some
reasonable technical assumptions, solutions of Einstein's

equations develop trapped surfaces in finite time, which implies gravitational collapse by Penrose's singularity theorem. For those brave souls who wish to endeavor going through the proof (594 pages!!!), the preprint can be found at http://arxiv.org/abs/0805.3880.



Anonymous · 11 August 2011

Klainerman and Rodnianski found a simplification of Christodoulou's proof. This paper is 117 pages long.



Alberto G. P. · 25 August 2008

Let me make some comments about I have understood

We can start with a bulk metric that define an AdS space. The bulk metric is an exact solution of EFE with negative constant. There are an unique asintotic expansion of the bulk metric valid only in a certain region belongs to AdS (near infinity). The coefficients of this expansion are used to buid a new tensor defined on the boundary of AdS (stress-energy tensor). If it is imposed some conservation law to this tensor then the generalized NSE arise. But although we know the stress-energy tensor we don´t know dual generalized NSE solutions (i.e.it hasn´t got the solutions yet).
The global regularity for EFE is an open problem. Moreover It can not be used the AdS/CFT correspondence to solve global regularity for generalized NSE because nobody know how to transform the bulk quantities to boundary quantities.

As thanks let me conclude with a quotation from Don Quixote:

"nunca fuera caballero
de damas tan bien servido
como fuera Don Quijote
cuando de su aldea vino
doncellas curaban dél;
princesas de su rocino,"

;-).



Terence Tao · 25 August 2008

Dear PDEBeginner,

You are correct that the difficulty in the Navier-Stokes regularity problem is in preventing passage of energy to higher scales, which will cause subcritical norms to blow up (and critical norms to either blow up, concentrate, or otherwise develop a singularity). Thus far, the partial regularity results that are known do not prevent this blowup from occurring.
To oversimplify, the above blowup scenario is consistent with concentration on any set of (parabolic) dimension 1 or less, but not on dimension higher than 1. And indeed, the known partial regularity theory, in particular the result of Kohn, Caffarelli, and Nirenberg excludes singularities with dimension higher than 1. (I am not sure if singularities of dimension exactly 1 are known to be excluded. Given that in two dimensions no singularities occur, this suggests that 1-dimensional singularities do not occur in three dimensions, though this is nowhere near a rigorous argument.)

 **westy31** · 26 August 2008

Hi again,

Question:
Would the Energy spectrum contain enough information to decide if the equation blows up?
For example, say the energy spectrum is k^-5/3. This would mean, I presume, that the velocity does not fluctuate infinitely fast in space, because high frequancy components of the field get progressively smaller. Not so for higher derivatives of the field though, since these get multiplied by k each time you differentiate.
As a second example, take k^-5/3*exp(-k/k_c).
This one stays small for large k, for all derivatives. It would seem to me that this guarantees smoothness.

Gerard

 **Terence Tao** · 26 August 2008

Dear Gerard,

Yes, smoothness of the velocity field is equivalent to the energy spectrum decaying faster than any power of k as $k \to \infty$. So if the Kolmogorov $k^{-5/3}$ power law managed to extend itself to arbitrarily high frequencies then regularity of Navier-Stokes would break down; but I believe the expectation is that the viscosity term in Navier-Stokes should cause one to leave the Kolmogorov regime at sufficiently fine scales. (The derivation of the Kolmogorov law implicitly assumes that energy is spread evenly throughout space at each frequency scale, and that the rate of energy flows remain constant, whereas it is known that blowup only occurs if the energy concentrates on a low dimensional set, and if the rate of energy flow becomes infinite. So blowup is not expected to be caused by the same mechanism that gives the Kolmogorov law, or at least the two phenomena should occur at different scales.)

On the Convergence of Periodic Navier-Stokes Flows « Smooth Chaotic Navier-Stokes Flows · 28 August 2008

[...] On the Convergence of Periodic Navier-Stokes Flows Filed under: Mathematics — DavePurvance @ 2:26 pm It's not that the 3D spatially periodic, incompressible Navier-Stokes equation can be posed as a nonlinear matrix differential equation. It's not that this matrix differential equation has a smooth and bounded matrix exponential solution, found when the flow in this differential equation is Taylor expanded in time and the resulting system matrices are all commutative. And, it's not that this time expanded matrix differential equation also has a more general time series solution. It is called the noncommutative solution because its coefficients involve noncommutative matrix polynomials. This matrix differential equation and its commutative and noncommutative solutions have been recognized by the world-renown expert Terence Tao. Read Terence's 22 March 2008 comments under his blog Why global regularity for the Navier-Stokes is so hard. [...]

 **Claes Johnson** · 28 August 2008

Dear Gerard:

It is necessary to give a quantitative meaning "smooth", which means that the size of Sobolev norms must be taken into account. If the C^1 norm of a function is a number of moderate

size, say 10^2, you can say that the function is C^1-smooth, but if the C^1-norm is of size 10^100 = googol, then it is not C^1-smooth. The reason to introduce Sobolev spaces with norms is to measure norms, and to measure norms is to make a distinction between 10^2 and 10^100. It is very strange that this aspect seems to be completely lost in the discussion of the Clay problem. In order to answer if a smooth solution exists, you first have to define what "smooth" is. This is not done now and no progress is being made. So Terence: what is your definition o "smooth"? Is the size of Sobolev norms not relevant? If not, how do you make a distinction between a smooth and a non-smooth function? What is then the meaning of Sobolev norms?

 **Bill Layton** · 15 September 2008

Hi All,

This is a very interesting discussion that I've enjoyed reading!

Might helicity play a key role in NSE uniqueness / non-uniqueness and deserves much more study?

It is the other key invariant of the 3d Euler equations and a control quantity for the 3d NSE. Of course it is much harder than enstrophy in 2d because it has 2 signs and dissipation reduces helicity mode by mode rather than by global magnitude.

The plan is one that surely many have thought about. If the NSE nonlinearity is zero (or sufficiently small) uniqueness is trivial. If the helicity is zero or sufficiently small, then the flow is also unique and smooth (a recent result of Berselli). The "only" problem is then to interpolate between these two cases locally.

This is surely a technically intricate problem. But, to me it seems like one with more hope of success for the first few steps than squeezing a "new" a priori bound on ||grad u|| from what are in many ways essentially the same estimates that Leray derived in his seminar papers.

From the computational point of view (and computational insight is very useful since it's generally easier to prove something that is true and understood at some level), helicity is a scalar so much easier to visualize.

So, ...
Any ideas on this path among the many possibilities??

Bill

 **David Collins** · 26 September 2008

Dear Terence,

I am a physics layman and math undergraduate from germany on a ODE level. I have a bulk of wild ideas, some probably persued before and many false. Some at least seem to fit to some problems outlined above.

Smooth ... ? ... Discrete

a) Instead of only using discrete models in order to find smooth limit tactics, maybe Navier Stokes equations lead in the other direction to a 'self-quantization'. A superposition of data ensembles might 'wash out' the many probable topological knots of vortex filaments

('Shaking Strings in a box leads to Knot formation' http://www.pnas.org/cgi/content/abstract/104/42/16432) (long shots 1 & 5)

b) Velocity field modeled as a rapid growth phenomena.
A toy model which would be less rigid and would allow a bit more complexity than triangular or cubic lattices, and that is appears frequently in nature could be a recently further generalized model of primordia growth, the Snow map http://maven.smith.edu/~phyllo/Assets/pdf/snow.pdf which allows continuous diffeomorphism via irregular rhombic tilings (which are periodic weak attractors) between disjunct discrete lattices (strong asymptotic attractors). In it's current form it is of course only an analog to laminar flow of molecule ensembles. But there is a striking interplay between random divergence angles in local topology & pseudorandomness in global topology. What happens when taking the limit in this model? (long shots 2, 5, 6)

c) With the unproved assumption that nature follows a 'least crypticity princible' and thus the simplest mathematics possible one could heuristicly ask 'If a smooth continuous solution to fluid dynamics does exist, why does nature bother with quanta (x > 0)?'

d) Maybe the mathematical superposition of all possible solutions lead to temporal and spacial aliasing effects, thus leading to quasilattice geometry. Thus blowups or quasi-blowups in Ackermann/Goodstein scenarios. (Very wild guess)

Dimensional analysis shows that fluid entities have analogs in Plancks Laws:

Dynamic Viscosity n (eta)[kg/ms] ~ Spectral Energy Density u
Surface Tension s (sigma) [kg/s²] ~ Spectral Radiance per Frequency I(f)
Pressure Change dP/dt [kg/ms²] ~ Spectral Radiance per Wavelength I(r)

Dimensional analysis also shows possible uncertainties or complements:

Dyn. Viscosity & Volume: n * V > h
Kinematic Visc. & Mass v * m > h

(measuring Volume of superfluid H3 with exact viscosity n = 0 ?)

Again a very wild guess:
A 'bubble' or more precisely a 'Spherical Hill Vortex' might be the missing link between 'Hollow black body' and 'Fluid', having n, s, dP/dt aka u, I(f), I(r).

h) A vector that keeps appearing in this dimensional analysis and in Navier Stokes is w := 1/rt or [1/ms]. For instance h=mw and Du = w in the viscose term.

i) First attempt to interpretate w: This could be interpreted either as rate of curvature change (dK/dt with K=1/r) or as change of spacial frequency. A 3-d Snow model might show which and when is the appropriate interpretation via the parameters D (spacial frequency, periodic attractors), the 'crookedness' of the primordial front (local discrete curvature) i.e. the divergence angles and/or the mean parastechy angle (global curvature). Actually a common formulation of NS has the ('fluid kinematic energy') dimensions

F/V = mw² (Navier Stokes)
n = mw (Dyn. Viscosity as 'fluid impuls')

U ~ mw (Spect. Radiance per frequency)
P= mw/dt (Pressure as ‘fluid force’) etc.

j) Second attempt: After asking my calculus professor about the nature of this ‘dual’ vector, she gave me the hint to look up the Poisson resummation formula which relates a summation over the numbers x_n = n, to a summation over the numbers y_n = 1/n .
This in turn can be generalized via Pontryagin duality to dual lattices, and more general via Selberg trace leads directly into the heart of pseudo/randomness, Riemanns Zeta hypothesis.

Thanks for your time and patience

Dave

Turbulence « Unruled Notebook · 28 October 2008
[...] If you can handle comfortably some mathematical language, read the excellent post titled “Why Global Regularity for Navier Stokes is Hard” by Terrence [...]

 **isaac** · 15 November 2008
Dear Terence,
What do you do during your free time? Any hobbies?

 **Liangyu** · 2 January 2009
Hi, terry, if the solution of the Navier Stokes equation can develop sigularity in finite time then the Navier Stokes equation surely cannot be an accurate model of 3D fluid dynamics. Which would implie that some physical assumption in the formulation of the Navier-Stokes equation is wrong?

 **Terence Tao** · 3 January 2009
Dear Liangyu,

The derivation of the Navier-Stokes equations from more fundamental laws of physics (e.g. Newton’s laws of motion) involves a number of simplifying assumptions (most notably, treating a fluid as a continuum rather than as consisting of a large number of atoms). Presumably, if these equations lead to singularity, then these assumptions would break down before the actual singularity is reached. (Note also that it is known that singularity can only occur if the velocity goes to infinity; in practice, of course, relativistic effects (among other things) would kick in long before then.)

 **Jonas** · 29 July 2009
Dear Terence,
How does one show that a singularity can only occur if the velocity is unbounded? Who proved this first?

 **Terence Tao** · 8 August 2009
This was first shown in a paper of Beale-Kato-Majda (the result was initially for the Euler equations, but the same arguments apply to the Navier-Stokes equations). I believe the method is basically an energy method, controlling the growth of Sobolev norms via integration by parts and standard inequalities (Holder, Sobolev, Gagliardo-Nirenberg, etc.).

Michael Nielsen » Doing science online · 26 January 2009
[...] information and insight. To understand how valuable Tao's blog is, let's look at a example post, about the Navier-Stokes equations. As many of you know, these are the standard equations used by [...]

 **km** · 27 January 2009
Hi,

In the following link:

http://knol.google.com/k/claes-johnson/the-clay-navier-stokes-millennium/yvfu3xg7d7wt/14#

a comment says that a blowup solution in Navier-Stokes means turbulence and "A turbulent solution is a non-smooth solution, and blowup from smooth data initial data is the same as transition from laminar to turbulent flow".

So, if the above is true, then Navier-Stokes can be used to describe turbulence only if it does give non-smooth solution. Is that right?

 **tmr** · 27 January 2009
Is it known for blow up to occur in 4 dimensional Navier-Stokes?

 **Terence Tao** · 28 January 2009
Dear km: Unlike blowup (which means that the solution ceases to be smooth after a finite amount of time), which is a precise mathematical concept, turbulence (which means that energy shifts to higher and higher frequencies, or equivalently to finer and finer scales) does not have a canonical mathematical definition. But, at an intuitive level at least (and oversimplifying somewhat), one can think of blowup as an infinite amount of turbulence: a significant portion of the energy moves into infinitely fine scales within a finite amount of time, causing singularity. So if there is no blowup for the Navier-Stokes equation, it means that the solution can become somewhat turbulent for a finite amount of time (as is of course seen in real-life fluids), but eventually the viscosity effects dominate and damp out the turbulence. If there was blowup, then the mathematical Navier-Stokes equation would become singular after some time, though in the physical setting what would happen instead is that the solution becomes so turbulent at so fine scales that the simplifying assumptions used to derive the Navier-Stokes equation (e.g. assuming that a fluid is a continuum, rather than being made of molecues) break down, and some other dynamics take over. This does not seem to occur in practice (fluids do become turbulent, but not all the way to the molecular scale; at some point, viscosity effects assert themselves to remove the turbulent energy from the system). Whether this is always the case is, literally, a million-dollar question.

Dear tmr: No blowup result is known for any higher-dimensional Navier-Stokes equation as far as I am aware. Heuristically, the higher-dimensional equations should be more unstable and so would have a better chance of exhibiting blowup, but by the same token the instability may make it even harder to *rigorously* demonstrate blowup in higher dimensions than in lower ones. But perhaps some low-dimensional reduction of a high-dimensional Navier-Stokes equation might be amenable to study. (In the periodic

setting, one can view a low-dimensional Navier-Stokes solution as a solution of a high-dimensional Navier-Stokes equation also, simply by extending the solution trivially in some additional spatial dimensions, so in that case, at least, global regularity of high-dimensional NS implies the same for low-dmiensional NS.)

Mathematics, Science, and Blogs « Combinatorics and more · 28 January 2009
[...] and blogs. Michael's primary example is a post over Terry Tao's blog about the Navier-Stokes equation and he suggests blogs as a way of scaling up scientific conversation. Michael is writing a book [...]

 **km** · 28 January 2009
Hi Terence,

Thanks for the explanation. So, reality suggests that Navier-Stokes equation should not give blow-up solutions if we believe Navier-Stokes equation describes it accurately.

 **big bangs spectator** · 29 January 2009
The Laplacian is physically problematic, however, that we haven't observed a blowup doesn't mean it can't occur. E.g., there have been reports of strange new experiments, e.g., a hexagonal pattern in a rotating bucket of fluid. Ps a quick off-topic perspective statistic: the Physical Review Series database gives 4844, 3271, 1997, 299, 804, 564, 44 citations with the words Schrodinger, Josephson, Ginzburg-Landau, Landau-Ginzburg, sine-Gordon, Navier-Stokes, KdV in the Title/Abstract.

 **km** · 29 January 2009
First, I have to say that I am an novice in turbulence or Navier-Stokes equation (NS equation).

Second, I think it is impossible to observe blowup or infinite amount of turbulence in real life. An object in real life with singular properties is probably the black-hole. I can think of no other physical phenomena in reality that possess singularities like infinite velocity or energy.

Third, your comment suggests a possibility that not observing a blowup or infinite amount of turbulence in real life doesn't mean NS equation can't give blowup solution.

 **tmr** · 29 January 2009
Thanks a lot prof Tao for your reply!

 **naresh** · 11 February 2009
heloo frnds

i think everyone over here are sticking themselves to naviers stoke equation , now lets think different, is navier stokes equation true????????????? i dont think so specially for a 3d case, reason is
as far as my knowledge, in an flow over an aircraft wing huge turbulance exist, first of all we need to understand how viscosity varies in 3 dimension with speed, below mach 1 its a very simple case we can say viscosity is proportional to speed but after mach 1 shock waves begin to form so here comes the case , hence if we understand the variation of viscosity with speed keeping the surface in mind then turbulance can be solved,

naviers stokes equation is wrong because!!!!!!!!!!!!!!!!!!!

it relates velocity with pressure only what about temperature?????
for god sake viscosity of air changes with temperature hence we need to consider it/// please think about this hence we shldnt find the grad of kinematic viscosity instead we should use the variation of temperature with space coordinates in terms of viscosity, next thing is co effcient of viscosity must be considered , but it is nt done in navier stoke equation
so my opinion is navier stokes equation is absolutely wrong and hence it is unprooved

frnds and professors ive got the solution for this problem but i dnt know how to move about pls help me

thanks a lot for sparing your precious time sin reading my opinion

thankyou and please reply me ur questions and comments

im eager to hear from you
thanks again



Shree Nidhi · 27 April 2014

NS equation is wrong! Great discovery!



Tarun · 7 March 2009

Thanks for the great run down, Professor Tao.

I was reading about the function space BMO (Bounded Mean Oscillation) when I came across a paper which said that the regularity of BMO is ideal for Navier-Stokes solutions. I was wondering why this was and what role does (or maybe, should) Fefferman duality ($(VMO)^* = (BMO)$ and $(BMO)^* = H^1$) play in the regularity of Navier-Stokes solutions?

Thanks a lot,

Tarun



Terence Tao · 7 March 2009

Dear Tarun,

The space BMO (or more precisely the space $\nabla BMO := \{f : \nabla^{-1} f \in BMO\}$) is generally believed to be the weakest function space in which one can plausibly hope to (locally) solve the Navier-Stokes equation by perturbation theory methods, at least without first applying some further renormalisation of the equation. Basically, in order for the equation to make sense even as a distribution, the velocity field u should be locally square-integrable in spacetime; and in particular the effect of the nonlinearity $(u \cdot \nabla)u$ (viewed as a forcing term) should have this square-integrability. If one follows the perturbation theory philosophy of assuming that the nonlinear solution u behaves like its linear counterpart $e^{t\Delta}u_0$, and then one inspects the characterisation of BMO functions in terms of Carleson measure properties of their heat extensions, we see that what one is asking for is essentially that $\nabla^{-1}u_0$ lies in BMO.

This seems to not be too directly related to the Fefferman-Stein duality between BMO and H^1, although of course the Carleson measure characterisation of BMO dualises to give a version of the atomic decomposition for H^1.

Student · 20 March 2009

Dear Professor Tao:

I am a student studying stochastic differential equation and am a novice with physics or applied math. Some elaboration on your sentence below would be very much appreciated:

"Things look better on the stochastic level, in which the laws of thermodynamics might play a role, but the Navier-Stokes problem, as defined by the Clay institute, is deterministic, and so we have Maxwell's demon to contend with.)"

Dear Tarun:
I would be interested to know the paper which argued that the space of BMO is ideal for NSE solution.

Thank you.



Greg Kuperberg · 18 March 2007

This is an interesting-looking review of a topic that I don't know beans about. But I notice that you do not mention models of fluid flow with quantized vortices. These models approximate continuous 3+1-dimensional fluid equations by non-local 1+1-dimensional equations. It seems conceivable, although I know nothing about the analytic issues, that you could hope for some kind of convergence as the quantization parameter goes to zero.

My wife, Rena Zieve, studies fluids with quantized vortices. In her case, the motivation is not that the model approximates classical fluids, but that it is a true description of superfluid helium.



Terence Tao · 18 March 2007

Dear Greg,

It is indeed the analytic issues (and specifically, the establishing of estimates which control the solution, or convergence of approximate solutions to the true solution) which are the heart of the matter. Suppose for instance that we do manage to construct a global solution $u_\hbar$ to a quantised Navier-Stokes for every $\hbar > 0$ (this is certainly conceivable, as one can do similar things for other relaxations, regularisations, or discretisations of Navier-Stokes). Now, as you say, one now needs to establish convergence in some topology of these quantised solutions in the limit $\hbar \to 0$. In order to get any useful sort of limit, as a bare minimum one needs the sequence $u_\hbar$ to enjoy some sort of bound in some function space norm, uniformly in $\hbar$, otherwise all sorts of bad things could happen (e.g. breakdown of the energy conservation identity). The key phrase here is uniformity in $\hbar$. If one has some argument that bounds solutions to the quantised Navier-Stokes uniformly in $\hbar > 0$, then by taking limits one expects the same argument to work directly in the limit $\hbar = 0$. In other words, one can dispense with the quantised Navier-Stokes and work with the original Navier-Stokes equation directly.

Thus, for the point of view of establishing estimates (which is the key problem), perturbing the equation by varying a small parameter does not really help. Such tricks are however useful for more *qualitative* or *formal* aspects of the theory. For instance, to justify things like the energy conservation identity for very rough solutions (for which one cannot directly

justify things like differentiation under the integral sign), a typical trick is to first relax the solution to a smoother solution (and perhaps a smoother equation), establish the relevant identity in that smooth setting, and then take limits to recover the identity in the original solution. These sorts of perturbations are also useful for a number of simple topological arguments, such as an application of the continuity method (showing that all solutions obey some property P by showing first that the set of solutions obeying P is open, closed, and non-empty, and then using the connectedness of the solution space).

From a physical viewpoint, it may well be that one of these modified equations is in fact a more realistic model for fluids than Navier-Stokes. But for the narrow purposes of solving the Clay Prize Problem, we're stuck with the original Navier-Stokes equation :-) .

 **Greg Kuperberg** · 18 March 2007

You make it sound like it could be worthwhile to try to disprove the conjecture.

 **Nets Katz** · 19 March 2007

Terry,

I think you are a little pessimistic about strategy 1 for proving blow-up – working with ensembles. For that strategy to work it is not absolutely essential that the typical solution blow up. In order for Navier Stokes to blow up, you just to have each scale "activated", that is have energy flow sufficiently into each scale.
Suppose you could prove with positive probabiity that energy cascades into a generic situation in which it is one scale higher. Generic has to mean that we get positive probability of flowing to the next scale. You might get a measure zero set of solutions with blow-up but nevertheless use a probabilistic argument
to prove that it occurs.

 **Terence Tao** · 19 March 2007

Greg, I certainly think one should pursue the blowup direction as well as the regularity problem. As noted in my main post, though, our technologies for establishing blowup are rather limited at present. Which brings me to Nets' interesting idea. This idea may at least be able to establish a "norm inflation" scenario, in which some high regularity Sobolev norm is shown to increase quite rapidly in a bounded amount of time. That would be enough to disprove a fairly strong version of global regularity, namely that one can bound some critical global spacetime norm uniformly by, say, something depending only on the L^3 norm of the inital data. (This would imply as a corollary by standard persistence-of-regularity arguments that any Sobolev norm of the solution at a late time is controlled by the Sobolev norm at time zero, and the L^3 norm of the initial data.) I suspect that this type of global regularity result, while very common for critical problems, is probably false for supercritical problems, and might be disprovable by some sort of contradiction argument (e.g. by Bourgain's induction on energy method). If one was extremely optimistic one might then hope to run a Baire category type argument to create an actual blowup, but this looks somewhat unlikely to me.

 **Terence Tao** · 19 March 2007

Dear Nets,

I thought about it a bit more, and this strategy to establish blowup may run up against the same supercriticality issues which plague the global regularity problem.

Thanks to the recent work of Escauriaza-Seregin-Sverak and others, we know that blowup solutions to Navier-Stokes must in fact blow up in the critical norm $L^3(\mathbb{R}^3)$. This rules out self-similar type solutions in which the solution shifts all of its energy from one scale to the next while keeping critical norms under control. It is plausible that these results can be localised, leading one to also rule out "self-similar + radiation" solutions (such as those appearing in the recent works of Merle and Raphael for NLS, as well as even more recent work on wave maps by Krieger-Schlag-Tataru and Rodnianski-Sterbenz) in which a lot of L^3 mass is radiated away to coarse scales but the concentrating portion of the solution stays bounded in L^3. If these scenarios are ruled out, then the blowup solution must become increasingly large (and hence increasingly nonlinear) in critical norms at fine scales as one approaches the blowup time. This creates a lot of scope for an anti-Maxwell's demon (Maxwell's angel?) to cause trouble – one may start with a promisingly randomly distributed ensemble at coarse scales, but as one pushes the ensemble into fine scales with large critical norm, and the evolution becomes very nonlinear and unpredictable, Maxwell's angel could conceivably sneak in and drain a lot of entropy out of the ensemble, and eventually nudge the entire ensemble into states in which the energy all dissipates harmlessly and one has global regularity. (Now, if we could obtain some sort of rigorous version of the second law of thermodynamics here, one might be able to prevent this from happening, but this looks very remote at present.)

 **Nets Katz** · 19 March 2007

Terry,

That's really unfortunate. For this to work, nonlinearity needs to be a friend and not an enemy. In fact the test problem should first to prove blow up for 3D Euler with finite energy.

Some advantages there: Energy never dissipates away, cascading always occurs exactly by vortex stretching so that some structure seems to be preserved (for instance the locations of zeroes just convect)

Of course the passage to high scales should happen at time scales so startlingly fast that the linear term is having very little effect at all so that the blow up for Euler should indicate the blow up for Navier Stokes.

 **Nets Katz** · 20 March 2007

In fact, so that I can understand your objection better, let's restrict our attention to 3D Euler. The critical norm which you say we should be desperately clinging to is L^{infty} of vorticity. We don't need a recent result here. Beale-Kato-Majda already prove that if we have blow up of Euler then the L^{infty} of vorticity must blow up. Does that mean the equation becomes "increasingly unstable"?

By contrast let me mention a 2D problem for Euler which is critical. There global solvability is guaranteed since vorticity is convected so that L^{\infty} of vorticity is conserved. Beale-Kato-Majda show that the growth of

Sobolev norms in time is doubly exponential. The question, perhaps somewhat less glamorous than global solvability, is whether this estimate is sharp and double exponential growth actually occurs. As we've discussed in the past, if one looks at the highest order part of the growth exclusively, in a certain position, one is forced to consider a system of ODE for SL(2) valued variables, one variable for each active scale. The criticality enforces that different scales could be of equal strength and that many different scales have to work together to achieve double exponential growth. More over if we look at this system of N equations which governs the growth of the frequencies that are like 2^{-N}, this system is highly nonlinear, in that, in the relevant time scale which is 1/N, you could have all of the N/2 highest scales change completely. You can make very precise that this critical setting becomes increasingly unstable as the N gets larger. In particular the system has increasing numbers of variables more remote from each other in scale, all of which interact. That is tough going.

On the other hand, in the supercritical case, if you posit any reasonable worst cascading of energy, what happens is that the highest order scale dominates exponentially the lower order scales. The cascading process should indeed be nonlinear, but it seems self-similar as frequencies increase. In the time we can activate the N+1st scale, the Nth scale might get all messed up – that's the nonlinearity – but we don't care so much what happens at slightly lower scales. The problem is essentially local in scale.

Now you might argue that this local process could have a derandomizing effect on the ensemble. (And the more times you do it, the more derandomizing it gets.) But if you could show that were true, it would probably be the most amazing theorem in fluid mechanics ever. In any case, in this setting, it seems to me the supercritical problem might be easier to understand than the critical one. Am I wrong?



Terence Tao · 20 March 2007

Dear Nets,

It's a bit trickier to use scaling analysis for Euler, since it has a two-parameter scaling symmetry rather than a one-parameter one; the spatial scale, temporal scale, and velocity magnitude are related only by a single constraint rather than two. Nevertheless I agree that the problem of improving the double-exponential Beale-Kato-Majda bound for 2D Euler is a great problem – it is perhaps the one place where the wall separating the impossibly supercritical problems from the feasible critical problems is thinnest. It is clear that some improvement should be possible, and whatever technique is used to accomplish this will undoubtedly be interesting.

I also agree that it is against all physical intuition that fluid equations could derandomise ensembles of solutions – but we don't seem to have any way at present to make this intuition rigorous. If we have to admit the existence of derandomisers to move the solution to either the best-case scenario or worst-case scenario at whim, then any supercritical equation could

conceivably do just about anything from generic blowup to global regularity, depending on the mood of the derandomiser. This is why I feel that "understanding pseudorandomness" needs to be part of the solution (unless we find a new monotonicity formula or something, which lets us gain more control on either the best case or worst case and allows us to establish results even if the equation tries its best to derandomise the flow).

 **sdenisov** · 4 February 2012

Hi Terry :). There is a simple perturbative argument which shows that double exponential growth for, say, higher Sobolev norms can not be significantly improved (although I do not have the sharp estimate yet). This low bound is true for arbitrarily long but finite time and assumes that the Sobolev norm is already large for t=0. More interesting question is the lower bound which is valid for all time. In case of 2d Euler on the torus, the best estimate is superlinear growth of the vorticity gradient. However, if one considers the dynamics of patches (a very different problem) then there is a mechanism for INFINITE in time double exponential rate of merging and singularity formation (with a regular strain present). Anyhow, the double exponential bound for Euler might be improved by only very little.

Ars Mathematica » Blog Archive » Why Navier-Stokes is Hard · 20 March 2007
[...] Tao has a thoughtful post that explains why proving existence results for Navier-Stokes equations is so [...]

Not Even Wrong » Blog Archive » All Sorts of Links · 23 March 2007
[...] each have fascinating blog entries on Millenium problems. Terry Tao writes a long explanation of Why Global Regularity for Navier-Stokes is Hard. He also comments about the recent New York Times piece about him and about math education issues. [...]

 **Gil Kalai** · 29 March 2007

Dear Terry

This is a wonderful post. I have a (rather generic) question: is looking at the problem at high dimensions (even asymptotically somehow) can make things easier regarding the negative direction or even regarding some aspects of the positive direction?

 **Terence Tao** · 29 March 2007

Dear Gil,

Intuitively, the equation should behave worse in higher dimensions, because the relationship between the energy and the scaling becomes increasingly unfavourable. Certainly the equation is more unstable; but it is hard to convert instability to actual blowup (it's the "Maxwell's angel" problem discussed earlier – the instability might have the improbable effect of always rescuing the solution just before it blows up). Conversely, in two dimensions, where the energy becomes critical, the result of Beale-Kato-Majda shows that one has global smooth solutions for Navier Stokes.

Another popular model to study is the *hyperdissipation* model, in which the dispersive factor $-\Delta$ in the Navier-Stokes equation is replaced by a power $(-\Delta)^\alpha$, where $\alpha > 0$ is a parameter which serves a similar purpose to dimension, in that it determines the relationship between energy and scaling. The level $\alpha = 5/4$ is the threshold beyond which the energy becomes subcritical and one has global regularity. (I'm not sure what happens exactly at 5/4,

but perhaps Nets does.) In this paper it is shown that as α moves from 5/4 continuously down to 1, the (upper bound on the) dimension of the singular set at blowup time moves continuously from 0 to 1.



Nathan Reynolds · 30 May 2015

Hi, I am a student that is studying the Navier-Stokes equations applied to ocean forces and motions. What are the Criticisms and Assumptions of the Navier-Stokes equations?



Bruce Coulter · 31 May 2015

You probably got the thumb downs because it sounds like a homework question. See http://en.wikipedia.org/wiki/Navier%E2%80%93Stokes_equations



Nathan Reynolds · 3 June 2015

Hi, Wikipedia is not the answer in the topic of the Navier-Stokes Equation. It is not yet mentioned in Tao's blog the Assumptions and Criticisms with relation to smoothness and existance



Hongjie · 31 March 2007

Dear Terry,

3D Navier-Stokes equation is globally well-posed when \alpha=5/4. More generally, n D NSE is globally well-posed when \alpha=(n+2)/4. One proof of this can be found, for example, in the paper http://arxiv.org/abs/math/0104199 you mentioned. But it is not explicitly stated there...



Y. Charles Li · 4 April 2007

Hi, Terry

This is a very nice article, will be helpful especially to fresh Ph.D.'s.

About the scaling argument, simple as it is, it does dictate important estimates, e.g. for 3D NS, Leray obtained: For any initial condition u(0), when

t > T = C \nu^-5 || u(0) ||^4_L^2

no more singularity. Using the scaling,

|| u(0) ||^4_L^2 —> \lambda^2 || u(0) ||^4_L^2 , T —> \lambda^2 T

as dictated by the scaling t —> \lambda^2 t. Pretend this is 2D (of course, 2D global regularity has easier argument),

|| u(0) ||^4_L^2 —> || u(0) ||^4_L^2 , T —> T

while t —> \lambda^2 t by scaling. So any local solution can be rescaled to global.

I'm also glad you start to appreciate "turbulence". I wrote a thing on Mathematical Intelligencer:

Click to access Nature-T.pdf

It should have appeared in hard copy. It is a follow-up of the famous article by Ruelle and Takens:

http://www.springerlink.com/content/h1760361517x10h2/

I hope it brings chaos "closer" to turbulence. How close? it is hard to judge.
From a different perspective, the case is like the "global warming" problem,
the debate of whether turbulence is chaos is more or less over, the question is:

What can one do about chaos (turbulence) in terms of physical "reachable" description?

I wrote something along this line which is too immature to make public. I asked
Joel Lebowitz to read it for me and he is reading it.

Best Regards

Charles

 **Y. Charles Li** · 4 April 2007

Ruelle and Takens paper link is not working properly:

D. Ruelle, F. Takens, On the nature of turbulence, Comm. Math. Phys., 20 (167-192),
23 (243-244), 1971.

Y. Charles Li

 **Dave Purvance** · 7 April 2007

In the time evolution of a viscous, incompressible flow's Fourier modes convection's quadratic nonlinearity is expressed as a convolution of an unknown flow unshifted and shifted in wavenumber. When the unshifted portion is left to be some unknown function and its shifted counterpart is assumed to be a time series of finite order, and, when a discrete span of wavenumbers are considered simultaneously, then the three-space Navier-Stokes evolution equations become a large matrix differential equation. As the order of the time series and the number of wavenumbers considered get large, this Navier-Stokes matrix differential equation becomes ever closer to the continuous three-space Navier-Stokes evolution equations. Using conjugate symmetry and similarity transforms, it is argued at "http://arxiv.org/ftp/math/papers/0610/0610086.pdf" that the exact solution to the Navier-Stokes matrix differential equation is a stable matrix exponential and that the assumed time series is its truncated Taylor expansion in time.

Care to comment?

 **Terence Tao** · 7 April 2007

Dear Dave,

This is an instance of the type 1 of Strategy 3 listed in my post above: "Using weaker or approximate notions of solution". A localisation of the frequency space is essentially equivalent via the uncertainty principle to a discretisation of physical space. As discussed above, there is indeed no difficulty creating a global solution for the approximated equation. When however one tries to take a limit to create a global strong solution for the original equation, it is necessary to obtain a uniform subcritical or critical bound on all approximate solutions (i.e. to achieve Strategy 2) in order to extract a smooth limit. Otherwise, the best one can achieve are the global weak solutions of Leray.

 **Dave Purvance** · 8 April 2007

Dear Terence,

Thanks for your reply.

Unless something strange happens to a matrix eigendecomposition as the matrix size becomes infinitely large, convection remains oscillatory (purely imaginary eigenvalues) for all time. Convection is therefore absolutely bounded by the magnitude of the initial condition. Viscous shear (negative real eigenvalues) diffuses these convective oscillations to zero as time goes to infinity. The only problem I see in taking a limit here is that convective oscillations may become extremely (infinitely?) fast.

Is this the limit problem you are talking about ?

 **Terence Tao** · 8 April 2007

Dear Dave, this is part of the problem, but more importantly is the fact that as one transitions from finite-dimensional state spaces to infinite-dimensional ones, being bounded in one norm no longer implies being bounded in another. Thus for instance it is conceivable that limit remains bounded in energy, but is no longer smooth, which is the whole point of the Navier-Stokes global regularity property. (Bounded-energy global weak solutions were constructed all the way back in 1933 by Leray, essentially by a variant of the method you describe, but it is known that weak solutions need not be smooth.) In particular, the energy may concentrate in finer and finer length scales until a singularity occurs in finite time. This singularity is not entirely visible from the finite dimensional approximations, as they only have finitely many scales in the first place; it is a phenomenon which emerges in the limit.

In order to obtain regularity control on the limit, as opposed to merely a weak solution, it is not enough to bound the energy (which is a supercritical quantity); one also needs to bound a critical or subcritical quantity, and no such global bounds for arbitrarily large data are currently known at present (this is the "Strategy 2" I discuss above).

 **Simons Lecture III: Structure and randomness in PDE « What's new** · 8 April 2007

[...] For , one has global smooth solutions for small data with either sign. For large data in the focusing case, finite time blowup is possible. For large data in the defocusing case, the existence of global smooth solutions are unknown even for spherically symmetric data, indeed this problem, being supercritical, is of comparable difficulty to the Navier-Stokes global regularity problem. [...]

 **One side of "our" difficult problem (i.e. turbulence, did you think about something else?) « Alex's Blog** · 9 April 2007

[...] Why global regularity for Navier-Stokes is hard « What's new [...]

 **Dave Purvance** · 10 April 2007

With the best of respect, I really believe the new bound provided by the Navier-Stokes "matrix" differential equation is the [-1,1] bound on convection (purely imaginary eigenvalues). Convection "chirps up" in frequency in all powers of increasing time, but stays bounded in magnitude by [-1,1]. Viscous shear (real negative eigenvalues) smoothly diffuses these highly nonlinear oscillatory chirps to zero as time goes to infinity. There needs to be no mention of energy.

And with this I'll quit babbling...

sgrajeev · 15 April 2007

What a thoughtful and inspiring discussion!
Some questions

Are there other examples of supercritical PDEs that are better understood, or are the difficulties you mention generic to all such cases? e.g. \waveoperator\phi+\lambda\phi^p=0 for p big enough.

The scaling arguments you start with also appear in
quantum field theory where supercritical means something like `non-renormalizable'. Is there more to this analogy? If there is, it is bad news for proving regularity.

There is some geometry in the Euler equations: the fluid follows a geodesic w.r.t. the L^2 metric on the group of volume preserving Diffeomoprhisms (Arnold). Does this suggest anything for Navier-Stokes?



Terence Tao · 15 April 2007

Dear sgrajeev,

Most supercritical evolution equations are just as poorly understood as the Navier-Stokes equations, unfortunately, with the notable exception of the supercritical elliptic and parabolic equations for which there is a favourable sign which allows for some sort of maximum principle or comparison principle to take hold; basically, in such cases, the nonlinearity is extremely strong, but it always acting in a favourable direction, using its strength to reduce the size of the solution rather than increase it. However, in oscillatory evolution equations such as wave or Schrodinger equations it appears that there is no similar phenomenon taking place; the nonlinearity can "try" to reduce the size of its solution, but it is so strong that it can "overshoot" and end up changing the sign or direction of the solution and making it much larger at the same time.

There do seem to be analogies between classical PDE and quantum field theory (which can be viewed as a kind of quantum PDE) but this is definitely an underexplored area of study. For instance, the Cauchy problem for quantum field theory has not been studied much, even in linear models (perhaps it is a bad question to ask).

It is true that the Euler equation has some nice geometrical structure, and I would indeed think this structure will be key in the future understanding of this equation, and thus indirectly to Navier-Stokes. There is however a major obstruction with using the diffeomorphism group structure to understand Navier-Stokes, namely that this equation contains the Laplacian (via the viscosity term), which relies on the Euclidean (or Riemannian) geometry of space. This type of geometry is not preserved at all by diffeomorphisms. Because of the presence of two very different types of geometry in Navier-Stokes it seems significantly less likely that we get the same type of "geometric miracles" (e.g. unexpected monotonicity formulae) that we do in, say, Ricci flow, which only involves one type of geometry. But there could still be many surprises in this equation.

Rajeev · 15 April 2007

Thanks.

The Euclidean metric on the domain does go into the Euler equations as well; e.g., in determining the metric on the group of volume preserving diffeomorphisms. The length^2 of a tangent vector to the group is just the length^2 of the velocity field w.r.t. the Euclidean metric integrated on the domain of the fluid (which is also the kinetic energy $H_0 = \int |v|^2 dx$). Euler equations are the geodesic equations for this metric, according to Arnold.

Viscosity introduces the Laplacian in a more direct way. The viscous force is the gradient of the function

$H_1 = \int |\nabla v|^2 dx$. Often dissipative equations can be thought of as a Hamiltonian system with a complex valued hamilton $H = H_0 + iH_1$. (Sorry to plug my work here, but it is relatively recent, so may not be known: arXiv:quant-ph/0701141).

Not claiming that this gives a better control on the scaling properties of Navier-Stokes. But, it looks like Navier-Stokes has a geometrical meaning too, using complex geometry. Might be of independent interest.

Rajeev's Journal » Blog Archive » Fuzzy Fluids · 13 May 2007

[...] Terrence Tao has made some deep observations on why the regularity of three dimensional Navier-Stokes is such a hard problem. He has gone on to many other equally fascinating topics, I remain fixated on his main point there: that Navier-Stokes is `supercritical'. The nonlinearities become stronger at small distance scales, making it impossible to know (using present techniques) whether solutions remain smooth for all time. Thus it is crucial to understand the scale dependence of non-linearities in fluid mechanics. [...]

Smooth Solution to the 3-Space Navier-Stokes Equations by David Purvance « Smooth Chaos in Fluid Dynamics · 22 May 2007

[...] post argues that viscous shear is both the coercive and critical quantity that assures smooth solutions to the 3-space-periodic Navier-Stokes [...]

DavePurvance · 22 May 2007

This post argues that viscous shear is both the "coercive" and "critical" quantity that assures smooth solutions to the 3-space Navier-Stokes equations. Isn't this what physicists have long been expected?

math student · 23 May 2007

Hi Prof. Tao,

I am curious that if we assume the solution is periodic , any partial result ?

Terence Tao · 23 May 2007

Dear math student,

The difficulty in global regularity for Navier-Stokes lies at the fine spatial scales – in particular, scales much smaller than the period L. Intuitively, we expect the behaviour at such scales to not "see" the periodicity, and so whatever difficulties which are present in the non-periodic case will also be expected to be present in the periodic case.

Dear Dave,

There is an error in your post when you attempt to derive (42) from (40), trying to diagonalise different matrices A_n simultaneously. The rotation matrices T_n used to diagonalise A_n depend on n, so you cannot deduce (42) from (7) by conjugating by a single matrix.

More generally, there is no quantitative advantage gained in discretising the problem by introducing some truncation parameters (such as the parameters N and M in your post). The only bounds in the discretised model which would be of use to the original continuous model would be those bounds which are uniform in the truncation parameters, since these are the only bounds which will survive the passage back to the limit. But if there was a bound in the discretised model that was independent of those parameters, one could also have proven it directly in the continuous model simply by taking the limit of the _proof_ rather than of the result. Truncated models are useful for justifying some qualitative statements (e.g. ensuring that all sums and integrals converge, etc.) but do not progress towards the heart of the matter, which is to establish global bounds on solutions to the continuous equation.

 **DavePurvance** · 25 May 2007

Let's try this again...

Dear Terence,

Even though it only becomes clear in the paragraphs below (47) of my post, convective matrix A_{n+1} becomes known only after $A_n, A_{n-1}, ..., A_1, A_0$ are known for $n = 0, ..., N$, so a simultaneous diagonalization of all convective matrices is not possible. I don't see how diagonalizing convective matrices one at a time in increasing time-order n invalidates the Navier-Stokes matrix differential equation (7) or its matrix exponential solution (45). Could you explain? Also, the coercive, critical (Gaussian low-pass) nature of viscous shear holds independent of M and N and therefore should remain in the limit as $M, N \to \infty$.

I hope I am understanding your welcomed criticisms.

 **Terence Tao** · 25 May 2007

Dear Dave,

Each time you apply one of the transformations T_n to diagonalise one of the matrices A_n in (7), you undo the diagonalisation of the preceding matrices. Thus, as I said above, the formulation (42) of your truncated Navier-Stokes equation is not valid (indeed, as you have just commented, the matrices A_n are not simultaneously diagonalisable in general, so there is no way that (42) can be deduced from (7)).

 **Frederick** · 28 May 2007

Dear Terence,

Even one puts N-S equation in a cubic box, the essential difficulty remains, because the high frequency (ultra-violet) part remains. Can we truncate the ultra-violet part? Numerical (lattice) simulation tries to smear it, so lattice simulation is still an ostrich policy.

I guess it is the meaning of some of your meaning. I think this is right and the main difficulty of nonlinear analysis. Actually I think the whole subject of nonlinear equations is try to find some ways to deal with the ultra-violet part. Because in a nonlinear equation, one could not omit even the 1 billionth Fourier mode. Such as x^2, even x is a periodic function, whose property is known, nearly all the information about x^2 is lost. This is

mystic to me, why even one nonlinear term could destroy all the information and brings lots of randomness? There seems no efficient method to calculate Fourier coefficients of x^2 and to understand property of x^2.

The above is my understanding of nonlinear equations from the point of Fourier analysis.

My question are:

1. Is there some other Fourier series other than sine, cosine that is friendly to nonlinear terms? I mean, when one expands x in some new Fourier series, then x^2 could be easily calculated. If one expands x in terms of exp(i*n*omega*t), then one has to take into consideration of all the terms such as exp(i*(m-n)*omega*t), even m=1 billion+1 and n=1 billion. Could we expand x in terms of NewFourier(n, omega, t), when we calculate x^2, maybe only several m and n that are needed to calculate accurately x^2? Have we find such new Fourier series? Or there is just no such series at all? Just as quintic polynomial equations have no general solution in radicals according to Abel–Ruffini theorem.

2. Someone saw a analogy between ultra-violet part of N-S equation and ultra-violet divergences of quantum field theory. Is the analogy proper and accurate? Surely if it is accurate, renormalization group is quite a great tool to extract information.

Frederick

DavePurvance · 29 May 2007

Dear Terence,

I have added two extra steps to equation (42) to make clear that T_n is used as a similarity transform on A_n to change, if necessary, its wavenumber flow basis from V_n to V_0. T_n does not effect any lower order matrix $A_{n'}$ for $n' < n$. For one to argue that there is something wrong with (42), I would think one would have to argue that the 3-space Naiver-Stokes equations and its solution cannot be described in terms of a single wavenumber flow basis. I haven't seen anyone argue this yet. Have you?

You are right, though. These similarity transforms cannot be "deduced" from (7) . They don't invalidate (42) either.

DavePurvance · 30 May 2007

Dear Terence,

I might also add that the similarity transform $V_0 D V_0^H$ is necessary in my post's (42) to keep the flow from "diffusing" into the one forbidden flow direction $\mathbf{v}_\perp$ as a passive scalar can do in the advection-diffusion equation.

(Excuse my less-than-stellar grammer. I believe in my last comment "effect" should read "affect". Also, WordPress promises me that sooner or later they will get all of the bugs out of rendering latex in blog comments.)

2006 ICM: Étienne Ghys, "Knots and dynamics" « What's new · 3 August 2007

[...] have no discernible structure other than that of a general diffeomorphism. As I discussed in my own post on Navier-Stokes, the discovery of a new conserved quantity for fluid equations could potentially be extremely [...]

 **Stephen Montgomery-Smith** · 16 August 2007

My feeling is that there aren't any undiscovered globally controlled quantities of the type described in strategy (2). I admit that my reasoning is full of holes. But the energy estimate for the Navier-Stokes equation gives rise (via a heuristic argument) to the Kolmogorov 5/3 power law for the spectrum. However in 2D we have enstrophy estimates giving rise to the Kraichnan power of 3 law. My understanding is that both experiment and numerics confirm this, at least to some extent.

Similarly, if there were some strange, new, undiscovered globally controlled quantity, perhaps one might expect this to also give rise to a law different to the 5/3 law. But then experiment would have picked this up.

I do admit that one of the huge holes in my argument is that energy and enstrophy in 3D and 2D come equipped with well defined dissipation rates. So in a way, it is just a thought. But, for example, if one could make helicity somehow "monotone", then this would definitely qualify.

 **Terence Tao** · 17 August 2007

Dear Stephen,

That's a fairly plausible argument. (One could perhaps hypothesise that there are quantities which are globally controlled, but only start becoming strong enough to affect ensemble distributions such as the power law at very small scales, beyond what experiment and numerics can detect. Of course, if that was the case, then it would be a mathematical artefact rather than a physically or computationally useful quantity, but I presume it would still be technically legal to use it in order to claim the Millennium prize. :-))

There is also the possibility that there is some sort of "adaptive" controlled quantity, which relies on the prior history of the solution as well as on the current state, and so is not cutting down the energy surface in a way which would distort the power law. This seems to go against thermodynamical intuition though; fluids don't seem to have much "memory". On balance I would agree with you that Strategy (2) is unlikely to work, though it can't be ruled out completely, and it's not as if the other strategies have a significantly higher probability of success right now anyway. :-)

 **Stephen Montgomery-Smith** · 17 August 2007

I know that the Navier-Stokes equation is "memoryless" but whenever one actually looks at numerics, the solution really does look like it has a lot of memory. I have observed this quite dramatically with the 2D Euler equation. The vorticity scalar is pushed around by the flow, and after a certain amount of time seems to arrange itself in "puff pastry" like thin layers of positive and negative vorticity, to the point that one could look at an instance of a flow, and make a rather good guess as to how long the flow has evolved.

From watching the Science Channel, I get the impression that a similar thing happens with the solar magnetic field. Presumably this satisfies some kind of memoryless hydromagnetic equation. But according to the T.V. shows I watch, the magnetic fields get all twisted up around each other. At some point they "snap" (presumably get stretched so much that some

kind of dissipation kicks in), and this causes release of huge amounts of energy, giving rise to solar flares that cause inconvenience to man made communications satellites.

And I know that you have spent some time with the "magnetization variable" formulation of the Navier-Stokes equation. While everyone who seems to have worked on it seems to have got no-where, nevertheless the magnetization variable differs from the velocity field by the gradient of a scalar field, and perhaps that scalar field codes in a kind of abstract memory.

(For those of you unfamiliar with the magnetization variable formulation, I did a short write up at http://www.qeden.com/wiki/Navier-Stokes_Existence_and_Smoothness).

And I guess that this paper:
http://www.iumj.indiana.edu/IUMJ/fulltext.php?artid=42034&year=1993&volume=42
is an example of what an adaptive controlled quantity would be.



Stephen Montgomery-Smith · 18 August 2007

I was looking back at the old comments (I only found this page a few days ago). I saw Hongjie's comment from March 31st that the hyperviscous NS is globally well posed for alpha=5/4. Actually this has a rather short proof, because in this situation we have the boundedness of
int_0^infty || (-Lap)^{5/8} u ||_2^2 dt.
Then the proof proceeds following the same lines as, say, the proof that the Prodi-Serrin conditions are sufficient for global well-posedness for regular NS.



"Math Doesn't Suck", and the Chayes-McKellar-Winn theorem « What's new · 20 August 2007

[...] In that case, Markov chain theory lets one conclude that if the solution started out at a fixed total energy E, and the system S was isolated, then the limiting distribution of microstates would just be the uniform distribution on the energy surface ; every state on this surface is equally likely to occur at any given instant of time (this is known as the fundamental postulate of statistical mechanics, though in this simple Markov chain model we can actually derive this postulate rigorously). This distribution is known as the microcanonical ensemble of S at energy E. It is remarkable that this ensemble is largely independent of the actual values of the transition probabilities; it is only the energy E and the function H which are relevant. (This analysis is perfectly rigorous in the Markov chain model, but in more realistic models such as Hamiltonian mechanics or quantum mechanics, it is much more difficult to rigorously justify convergence to the microcanonical ensemble. The trouble is that while these models appear to have a chaotic dynamics, which should thus exhibit very pseudorandom behaviour (similar to the genuinely random behaviour of a Markov chain model), it is very difficult to demonstrate this pseudorandomness rigorously; the same difficulty, incidentally, is present in the Navier-Stokes regularity problem.) [...]



Stephen Montgomery-Smith · 7 September 2007

I am having doubts about my earlier remark that the Kolmogorov 5/3 law suggests the non-existence of undiscovered global controlling quantities. The reason I say this is because in the 2D case we have the Kraichnan power of 3 law that comes from the enstrophy. But in the 2D case, there are also better controlling quantities than the enstrophy, namely the L_p norms of the vorticity for all p greater than 2.

Notes : Wiki page on NS equations « jtstnsp · 2 October 2007
[...] Now ,the next job is to take a look at Terry Tao's article on NS equations. [...]



A quantitative formulation of the global regularity problem for the periodic Navier-Stokes equation « What's new · 9 October 2007
[...] Tuesday, October 9th, 2007 in math.AP, paper Tags: compactness, Navier-Stokes equations I have just uploaded to the arXiv my paper "A quantitative formulation of the global regularity problem for the periodic Navier-Stokes equation", submitted to Dynamics of PDE. This is a short note on one formulation of the Clay Millennium prize problem, namely that there exists a global smooth solution to the Navier-Stokes equation on the torus given any smooth divergence-free data. (I should emphasise right off the bat that I am not claiming any major breakthrough on this problem, which remains extremely challenging in my opinion.) [...]

nOnoscience » Blog Archive » Turbulence · 29 December 2007
[...] If you can handle comfortably some mathematical language, read the excellent post titled "Why Global Regularity for Navier Stokes is Hard" by Terrence [...]



cgjoh@csc.kth.se · 16 March 2008
For computational evidence of blowup of incompressible Euler solutions, see the recent article (to appear in BIT Numerical Mathematics)

Blowup of incompressible Euler solutions

available on http://www.csc.kth.se/~cgjoh



DavePurvance · 20 March 2008
Dear Terence,

In your comments above I believe you have confirmed my argument that for any spatially periodic flow $\mathbf{u}$ of initial finite value $\mathbf{u_0}$ the incompressible Navier-Stokes equations can be posed as the nonlinear matrix differential equation
$d\mathbf{u}/dt = U(\mathbf{u})\mathbf{u}$. (1)
This equation is nonlinear because matrix U is a function of the unknown flow $\mathbf{u}$. When $\mathbf{u}$ in $U(\mathbf{u})$ is expanded as a time series
$\mathbf{u} = \sum_{n=0}^{\infty} \mathbf{u}_n t^n$ (2),
then (1) becomes
$$d\mathbf{u}/dt = \left(\sum_{n=0}^{\infty} U_n(\mathbf{u}_n) t^n \right) \mathbf{u}. \quad (3)$$
When U_n have a common flow basis, i.e., when U_n commute, then the solution to (1) is the stable matrix exponential
$\mathbf{u} = \exp\left(\sum_{n=0}^{\infty} 1/(n+1) U_n t^{n+1}\right)\mathbf{u}_0$. (4)
However, as you correctly pointed out in your comments above, commutativity of U_n must be assumed a priori and, therefore, it is a mistake to assert (4) is a general solution to the Navier-Stokes (1).
To remedy this commutativity problem, I have recently argued in this arXiv paper that using the time expanded flow (2) everywhere in (3) gives the equation
$\sum_{n=0}^{\infty} n\mathbf{u}_n t^{n-1} = \sum_{p=0}^{\infty} \sum_{q=0}^{\infty} U_p \mathbf{u}_q t^{p+q}$. (5)
Matching coefficients in (5) when $t^{n-1} = t^{p+q}$ solves for the unknown flow coefficients $\mathbf{u}_n$. For $n > 0$ they are

$\mathbf{u}_n = 1/n \sum_{p=0}^{n-1} U_p \mathbf{u}_{n-1-p}$. (6)

Without assuming commutativity, let the Taylor expansion of the matrix exponential function (4) be

$\mathbf{v} = \sum_{n=0}^{\infty} 1/n! \left(\sum_{p=0}^{\infty} 1/(p+1) U_p t^{p+1} \right)^n \mathbf{u}_0 = \sum_{n=0}^{\infty} \mathbf{v}_n t^n \mathbf{u}_0$. (7)

This is a stable function of U_p and $\mathbf{v_n}$ converge even when U_p do not commute. Obviously, when U_p commute,

$\mathbf{u_n} = \mathbf{v_n}$. (8)

What I have also argued in this same paper is that when U_p do not commute, the illusive bound to the periodic Navier-Stokes flow is

$|\mathbf{u}_n| \leqslant 3|\mathbf{v}_n|$ (9)

The paper numerically confirms this bound (9) for $n \leqslant 10$ using a white-noise initial flow. These numerical results have given me the courage to ask you again for your opinion of my recent findings (even though it was a humbling experience last time). Thank you.

Dave Purvance

A bound for periodic Navier-Stokes flows « Smooth Chaotic Fluids · 20 March 2008

[...] as Terry Tao correctly pointed out, commutativity of must be assumed a priori and, therefore, it is a mistake [...]



DavePurvance · 20 March 2008

Dear Anonymous,

It's only a mistake to hold that my matrix exponential function is a general solution to the periodic Navier-Stokes equation. The bound that I have identified is this same matrix exponential function, but expanded without the commutativity assumption. It is NOT claimed to be a solution, just a bound.

Dave Purvance